
**ARMY TECHNIQUES PUBLICATION FOR CASUALTY
RESPONSE, TACTICAL COMBAT CASUALTY CARE,
AND FIRST AID**

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Preface

Army Techniques Publication (ATP) 4-02.11 serves as a primary doctrine reference for holistic casualty response. Implementation of the techniques presented in this publication enable Service members to execute casualty response, render first aid, and prevent greater harm to injured Service members.

The principal audience for ATP 4-02.11 is Army personnel as well as employees of civilian agencies executing and supporting operations. Commanders, staffs, and subordinates ensure their decisions and actions comply with applicable United States, international, and in some cases host-nation laws and regulations. Commanders at all levels ensure that their Service members operate in accordance with the law of armed conflict and the rules of engagement. Refer to Field Manual 6-27/Marine Corps Tactical Publication 11-10C for more information on *The Commander's Handbook on the Law of Land Warfare*.

This publication implements or is in consonance with the following North Atlantic Treaty Organization (NATO) Standardization Agreement (STANAG) and the standards of the American, British, Canadian, Australian, and New Zealand (ABCANZ) standards:

TITLE	NATO STANAG	ABCANZ
Aeromedical Evacuation	3204	
CBRN First Aid Handbook	2358	
Coalition Health Interoperability Handbook		256
First Aid Dressings, First Aid Kits, and Emergency Medical Care Kits	2126	
Land Operational Reports	2020	
NATO Land Urgent Voice Messages (LUVN) Pocket Book	2627	
Pre-Hospital Combat Casualty Care Standards for ABCANZ Soldiers		2121
Requirement for Training in Casualty Care for All Military Personnel	2122	
Tactical Combat Casualty Care Card		2136

ATP 4-02.11 uses joint terms where applicable. Selected joint and Army terms and definitions appear in both the glossary and the text. Terms for which ATP 4-02.11 is the proponent publication (the authority) are bold and italicized in the text and are marked with an asterisk (*) in the glossary. For other term definitions shown in the text, the term is italicized, and the number of the proponent publication follows the definition.

This publication applies to the Active Army, Army National Guard/Army National Guard of the United States, and United States Army Reserve unless otherwise stated.

Throughout this publication, as appropriate, reference is made to the major policy guidance affecting each specific topic. These references should not be considered as the only policy guidance available. When issues arise that require consideration of policy guidance, the issue should be thoroughly researched and, as appropriate, coordinated with the supporting staff judge advocate, governmental or nongovernmental agency involved.

The Army proponent and preparing agency of ATP 4-02.11 is the United States Army Medical Center of Excellence. Send comments and recommendations on Department of Army Form 2028 (*Recommended Changes to Publications and Blank Forms*) to Commander, United States Army Medical Center of Excellence, ATTN: ATMC-FD (ATP 4-02.11), 2377 Greeley Road, Suite B, JBSA Fort Sam Houston, TX 78234-7731; by e-mail to usarmy.jbsa.medical-coe.mbx.ameddcs-medical-doctrine@army.mil; or submit an electronic Department of the Army Form 2028.

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Introduction

ATP 4-02.11 provides discussion on casualty response and first aid procedures for nonmedical personnel in environments from home station to combat situations. This publication is meant to be used by trainers and individuals being trained based on common first aid tasks. These tasks are meant to reinforce and maintain proficiency in the correct procedures for providing first aid throughout a Service member's time in Service.

ATP 4-02.11 is designed to facilitate training and first aid competencies by bridging first aid training across the spectrum of assignments from initial military training to permanent duty station and deployment. ATP 4-02.11 introduces Tactical Combat Casualty Care (TCCC) with first aid tasks and procedures associated with combat situations. Training scenarios can be completed by combining individual and multiple first aid tasks with collective tasks.

ATP 4-02.11 consists of three parts, 20 chapters, and two appendixes.

- Part One, Casualty Response
 - Chapter 1, Casualty Response
 - Chapter 2, Casualty Extraction and Movement
- Part Two, Fundamentals of Tactical Combat Casualty Care
 - Chapter 3, Tactical Combat Casualty Care
 - Chapter 4, Massive Bleeding Control
 - Chapter 5, Airway Control
 - Chapter 6, Respiration and Breathing Control
 - Chapter 7, Circulation Control
 - Chapter 8, Hypothermia Control
 - Chapter 9, Secondary Injury Assessment Using Pain, Antibiotics, Wounds, Splinting and Mechanism of Injury, Injury Type, Signs and Symptoms and Treatments
 - Chapter 10, Eye Trauma
 - Chapter 11, Head Injury
 - Chapter 12, Burns
 - Chapter 13, Fractures and Splinting
 - Chapter 14, Expedient Techniques for Other Secondary Injuries
 - Chapter 15, Casualty Monitoring and Evacuation Preparation
- Part Three, First Aid Procedures for Non-Tactical Combat Casualty Care Injuries
 - Chapter 16, Bites and Stings
 - Chapter 17, Climatic and Environmental Injuries
 - Chapter 18, Sickle Cell Trait
 - Chapter 19, First Aid in a Chemical, Biological, Radiological, and Nuclear Environment
 - Chapter 20, Combat and Operational Stress Control
- Appendix A, First Aid Case and Kits, Authorized Medical Allowance List
- Appendix B, Rescue Equipment

The United States Army Medical Center of Excellence (MEDCoE) is reorganizing the placement of terms and definitions found in proponent publications within Army 4-02 series publications. Based on current doctrinal changes, certain terms for which ATP 4-02.11 is the proponent that has been added, rescinded, or modified for purposes of this publication. The Glossary has acronyms and defined terms. See introductory table-1 on and introductory table-2 on for specific term changes.

Introductory Table-1. New Army terms

<i>Term</i>	<i>Remarks</i>
care under fire	establishes term definition and proponent publication
casualty response	establishes term definition and proponent publication
tactical evacuation care	establishes term definition and proponent publication
tactical field care	establishes term definition and proponent publication

Introductory Table-2. Modified or Rescinded Army terms

<i>Term</i>	<i>Remarks</i>
combat lifesaver	modifies term and changes proponent publication to ATP 4-02.11
first aid	modifies term and changes proponent publication to ATP 4-02.11

PART ONE

CASUALTY RESPONSE

Casualty response is of paramount importance to military units and leaders in winning our Nations's wars and providing a critical role in ensuring the welfare and effectiveness of the armed forces. In the unpredictable and high-stakes environment of warfare, casualties are an unfortunate reality. Whether it is injuries sustained on the battlefield or the loss of fellow Service members, casualty response encompasses the immediate and long-term actions taken to mitigate the impact of these incidents. Effective casualty response saves lives but also boosts the morale of troops, enhances unit cohesion, allows leaders to maintain operational readiness, and win our Nations's wars.

Chapter 1

Casualty Response

CASUALTY RESPONSE OVERVIEW

1-1. **Casualty response is the immediate and organized care, evacuation and reporting of those injured or killed in action.** Casualty response is a leader driven system designed to synchronize the planning, rehearsal, and execution of individual and collective tasks required to complete the mission while improving casualty outcomes. Due to the nature of military operations both at home and in combat, nonmedical Service members must sometimes rely on themselves and other Service members to render first aid until medical assistance arrives. Self-aid, buddy aid, and combat lifesaver (CLS) support is normally the first care a casualty receives in an operational setting and is provided until medically trained personnel (combat medic, corpsman, or medical technician) arrive to begin treatment.

1-2. First responder care includes self-aid, buddy aid, CLS tasks, and Tactical Combat Casualty Care (TCCC). Care provided by first responders is not limited to Service members but is provided to all those requiring initial treatment, stabilization, and when necessary, movement to higher roles of care for further treatment.

1-3. First responder care varies by Service, with the core training focused on TCCC. The application of TCCC occurs anywhere in the pre-hospital setting and can be tailored by the Joint Trauma System (JTS) to support Service-specific mission requirements. The TCCC training has four individual tiers or sets of responsibilities (see table 1-1 on page 2):

- Self-aid or buddy aid.
- Combat lifesaver.
- Combat medic or corpsman.
- Combat paramedic and provider.

1-4. Department of Defense Instruction (DODI) 1322.24, directs standardized combat casualty care training for all Service members and Department of Defense (DOD)-expeditionary civilians. According to DODI 1322.24, all Service members will be trained and proficient in basic lifesaving TCCC skills. The TCCC skills replace legacy military first aid skills. This evidence-based approach to combat casualty care reflects the lessons of nearly two decades of war and has the potential to reduce preventable prehospital trauma-related deaths.

Table 1-1. Medical Tier Responsibilities

<i>All Service Members (Nonmedical Personnel)</i>	
Care Under Fire	Tactical Field Care
• Ensure scene safety. • Move the casualty to safety. • Identify and control life-threatening bleeding.	• Rapid casualty assessment. • Follow casualty treatment procedures to render any treatments indicated. • Seek help as directed by unit standing operating procedure.
<i>Combat Lifesaver (Nonmedical Personnel)</i>	
Care Under Fire	Tactical Field Care
• Suppress hostile fire to minimize the risk of injury to personnel. • Minimize additional injury to previously injured Service members. • Assist in providing self-aid and moving casualties, if feasible.	• Maintain security and situation awareness. • Perform casualty assessments. • Follow casualty treatment procedures to render any treatments indicated. • Support the combat medic or corpsman, as directed.
<i>Combat Medic or Corpsman (Medical Personnel)</i>	
Care Under Fire	Tactical Field Care
• Suppress hostile fire to minimize the risk of injury to personnel. • Minimize additional injury to previously injured Service members. • Assist in providing self-aid and moving casualties, if feasible.	• Assume the primary role for casualty assessment and treatment. • Manage and direct the casualty response. • System using all available responders.
<i>Tactical Evacuation Care (TACEVAC)</i>	
In TACEVAC the Combat Medic or Corpsman:	
• Reassess casualties for the arrival of evacuation assets. • Communicate findings to the TACEVAC medical personnel. • Ensure Proper staging and support loading. • Secure casualties on the evacuation platform.	
<i>Combat Paramedic or Provider (Medical Personnel)</i>	
Care Under Fire	Tactical Field Care
• Suppress hostile fire to minimize new or additional injury to casualties. • Assist in providing self-aid and moving casualties, if feasible.	• Provide casualty assessment and treatment when functioning as the unit medic. • Manage and direct casualty response efforts utilizing all available responders.
<i>Tactical Evacuation Care</i>	
In TACEVAC the Combat Paramedic or Provider	
• Reassess casualties for the arrival of evacuation assets. • Communicate findings to the TACEVAC medical personnel. • Ensure proper staging and support loading. • Secure casualties on the evacuation platform.	

1-5. The JTS established the standard of care for TCCC or prehospital medicine in the operational environment for the DOD. The JTS is part of the Defense Health Agency (also known as DHA) providing clinical practice guidelines (CPG)s and performance improvement for all levels of military trauma care. The Committee on Tactical Combat Casualty Care (CoTCCC) also supports the JTS. The committee is composed of representatives from the Army and Air Force and are responsible for developing CPGs for the delivery of TCCC in the operational environment.

TERMINOLOGY

1-6. A thorough understanding of the fundamentals of casualty response can save a life, prevent permanent disability, or reduce long periods of hospitalization. Knowledge of key terms used in Army and joint doctrine helps Service members better understand and appreciate the role they play in providing first aid in tactical and nontactical environments. Knowledge of the fundamentals of casualty response and key terminology also helps Service members understand when and how to provide first aid in all settings.

1-7. A *casualty* is any person who is lost to the organization by having been declared dead, duty status – whereabouts unknown, excused absence-whereabouts unknown, missing, ill, or injured. (Joint Publication [JP] 4-02). The casualty classification is given to a Service member prior to treatment by medical personnel. A Service member who is wounded and conducts self-aid or is given treatment by a CLS is still classified as a casualty. That Service member is then moved to a casualty collection point (CCP) where medical personnel conduct triage and treatment of that casualty. Once treated by medical personnel, the Service member is now classified as a patient and enters the roles of medical care. Once a casualty is deceased, they become human remains and should be separated from the patient flow. The requirements for handling human remains are different and specifically explained in ATP 4-46. Service specific procedures should be followed when providing aid, evacuation, and reporting of combat casualties.

1-8. A *casualty collection point* is a location that may or may not be staffed, where casualties are assembled for evacuation to a medical treatment facility with medical personnel (ATP 4-02.2). The CCPs are normally predesignated along the axis of advance or evacuation routes. Forward of the Role 1 battalion aid station (BAS), the combat medic or corpsman, CLS, and Service members take casualties to the CCPs. A *battalion aid station* is the forward-most medically staffed treatment location organic to the maneuver battalion (ATP 4-02.4). These points facilitate acquisition of casualties by supporting ambulance teams and reducing evacuation time. When used by the BAS, CCPs help preserve BAS mobility, preclude carrying casualties forward, and reduce evacuation time to the sustainment area.

1-9. *Casualty evacuation* is the unregulated movement of casualties aboard any vehicle. Also called CASEVAC (JP 4-02). The Army defines *casualty evacuation* as the movement of casualties aboard nonmedical vehicles or aircraft without en route medical care (ATP 4-02.13). If a casualty is moved to a medical treatment facility (MTF) in a nonmedical vehicle without en route care provided by a medically trained Service member (such as a combat medic or corpsman) then the TCCC should be continually performed while the casualty is being transported. Casualties transported in this manner do not normally receive en route medical care. If the casualty is acquired by a dedicated medical vehicle with a medically trained crew, the role of first aid ceases, and the casualty becomes the responsibility of the Military Health System chain and is then referred to as a patient.

1-10. The core principles of casualty response include self-aid or buddy aid (first responder care), medical treatment (provided by combat medics, corpsmen, or medical technicians), and leader responsibilities (such as security, scene control, establishing a CCP, monitoring personnel, managing casualty movement, fulfilling reporting requirements, and coordinating evacuation). These principles must be thoroughly rehearsed and executed to maximize survivability in any operational environment. While these principles remain constant, the specific conditions in which they are applied can vary widely and are relevant to both military operations and civilian incident responses. This adaptability ensures that the response strategies are effective in different settings and enhances the overall efficacy of emergency management.

1-11. Joint Publication 4-02 defines *combat lifesaver* as a Nonmedical Department of Defense person who has received additional trauma training and equipment, providing enhanced medical treatment beyond self-aid or buddy aid. (JP 4-02). The United States (U.S.) Army defines a ***combat lifesaver as a Soldier trained and equipped to provide enhanced first aid as a secondary mission, known as CLS***. Enhanced first aid is the measures which require an additional level of training above self-aid and buddy aid. The CLS provides

this level of enhanced first aid and treats the most common causes of death on the battlefield, which are massive hemorrhage, airway and/or respiratory problems. The CLS also taught the skills to prevent complications and treat others associated but not life-threatening injuries. A minimum of one individual per squad, crew, team, or equivalent-sized unit should be trained. The primary duty of this individual does not change. The additional duty of a CLS is to provide enhanced first aid for injuries based on their training before the combat medic or corpsman arrives.

1-12. First aid is the immediate lifesaving measures performed on casualties or military working animals. Every Service member is trained to be proficient in a variety of specific first aid procedures on injured personnel. This training enables the Service member or a buddy to apply immediate first aid measures to alleviate a life-threatening situation on an injured person. Select Soldiers are trained to perform first aid on military working animals. For additional information on first aid for military working animals refer to the deployed medicine website.

1-13. A *first responder* is anyone who provides initial and immediate treatment to self or others. (JP 4-02) First responders play a crucial role in casualty response by providing immediate aid, including hemorrhage control and airway management, and improving the chances of survival for wounded personnel in the operational environment.

1-14. *Medical evacuation* is the timely and effective movement of the wounded, injured, or ill to and between medical treatment facilities on dedicated and properly marked medical platforms with en route care provided by medical personnel. Also called MEDEVAC (ATP 4-02.2).

1-15. A *medical treatment facility* is a facility established for the purpose of furnishing medical and/or dental care to eligible individuals. Also called MTF (JP 4-02). The Army defines a *medical treatment facility* as any facility established for the purpose of providing medical treatment (FM 4-02). This includes battalion aid stations, Role 2 facilities, dispensaries, clinics, and hospitals.

1-16. Medical treatment is the care and management of wounded, injured, or ill Service members by medically trained, certified or credentialed personnel. It may include emergency medical treatment (EMT), advanced trauma management, and resuscitative and surgical intervention.

1-17. Joint services define a *patient* as a sick, injured, wounded, or other person receiving medical care, dental care, or other medical treatment (JP 4-02). The U.S. Army defines a patient as a wounded, ill and injured Soldier who receives medical care or treatment from medically trained personnel (ATP 4-02.10). Once the casualty has entered the medical continuum of care (combat medic, corpsman, evacuation crew, or MTF), the role of first aid in the care of the casualty ceases and the casualty becomes the responsibility of the Military Health System. Once the casualty has entered the Military Health System they are referred to as a patient.

1-18. *Tactical Combat Casualty Care* is the deliberate integration of tactics and medicine to treat casualties during combat. Also called TCCC (JP 4-02). The unit standard operating procedure (SOP) and operation order should address TCCC procedures and casualty evacuation (CASEVAC) in detail to include aid for chemical casualties with particular emphasis on lifesaving tasks. The SOP should cover the duties and responsibilities of essential personnel, the evacuation of chemical, biological, radiological, and nuclear (CBRN) contaminated casualties (on routes separate from non-contaminated casualties), and priority for operating essential weapons and positions. The SOP should specify preferred and alternate methods of evacuation, and the SOP should make provisions for retrieving and safeguarding the weapons, ammunition, and equipment of casualties. Slightly wounded personnel are treated at the appropriate role of care and returned to duty as soon as possible. CASEVAC and medical evacuation (MEDEVAC) should be rehearsed like other critical parts of an operation.

1-19. *Triage* is the system of sorting and prioritizing casualties based on the tactical situation, mission, and available resources (FM 4-02). Casualties are sorted and categorized in one of the treatment categories (immediate [I], delayed [D], minimal [M], and expectant [E]) prior to being taken to the treatment areas. This sorting occurs at the triage point and is called the "IDME" or the "DIME." See paragraph 3-87 on page 38 for additional information on the triaging of casualties.

CASUALTY COLLECTION POINT OPERATIONS

1-20. When designating a CCP, the unit commander determines if medical personnel will be at the CCP. This decision is based upon the assessment of risk versus the availability of personnel. Medical personnel may not be available to staff these points, and CLS and ambulatory patients may be required to perform self-aid, buddy aid, or CLS support. The CCPs should be identified on watercraft, operational overlays, and planned by phase for operations. The CCP planning considerations include site security, proximity to the landing zone (LZ), cover and concealment, and access to evacuation routes. Leaders should provide guidance on aid and litter teams, distribution of casualty equipment, and proper avoidance of choke points. Medical personnel are responsible for activity inside the CCP, and the supported unit leaders are responsible for activity outside the CCP.

1-21. It is essential for units to meticulously plan the movement of casualties from the point of injury (POI) to supporting Role 3 MTF. Medical staff and unit leaders must be well-versed in the flow of casualties, extending their understanding at least two levels above their own. This encompasses knowledge of medical regulating, casualty tracking, and hospitalization protocols. For instance, platoon leaders and their medics should clearly know the path a casualty takes once they depart from the company CCP and BAS. Similarly, company leaders and senior medics should be familiar with the trajectory of a casualty beyond the BAS and through the Role 2 MTF.

1-22. Moreover, it is vital to consider potential changes in the designated Role 3 MTF, which may occur when crossing a phase line or when the capacity of a facility is reached. Medical personnel must stay informed about these variables and regularly update themselves on their current status. This will ensure that the casualty care and evacuation chain remain efficient and responsive to dynamic operational conditions.

1-23. Leaders must be able to answer the following questions to establish the tactical casualty flow:

- Where are the casualties being evacuated to?
- Will ground, air, or water assets conduct evacuation to the CCP?
- How will evacuation be conducted to the CCP?
- What are the distances and times of travel?
- Will expected casualties be able to make it that far? If not, what parts of the plan need to be corrected?
- Who will evacuate the casualties (CASEVAC or MEDEVAC)?
- Are medical assets positioned to ensure continuity of care?

SITE SELECTION

1-24. The CCP layout (see figure 1-1 on page 6) site is addressed during the planning phase of the military decision-making process, or troop leading procedures (commonly known as TLPs). Tactical battle plans will vary depending on the type and number of units incorporated in the plan. Mission variables are always a factor for CCP site selection during operations.

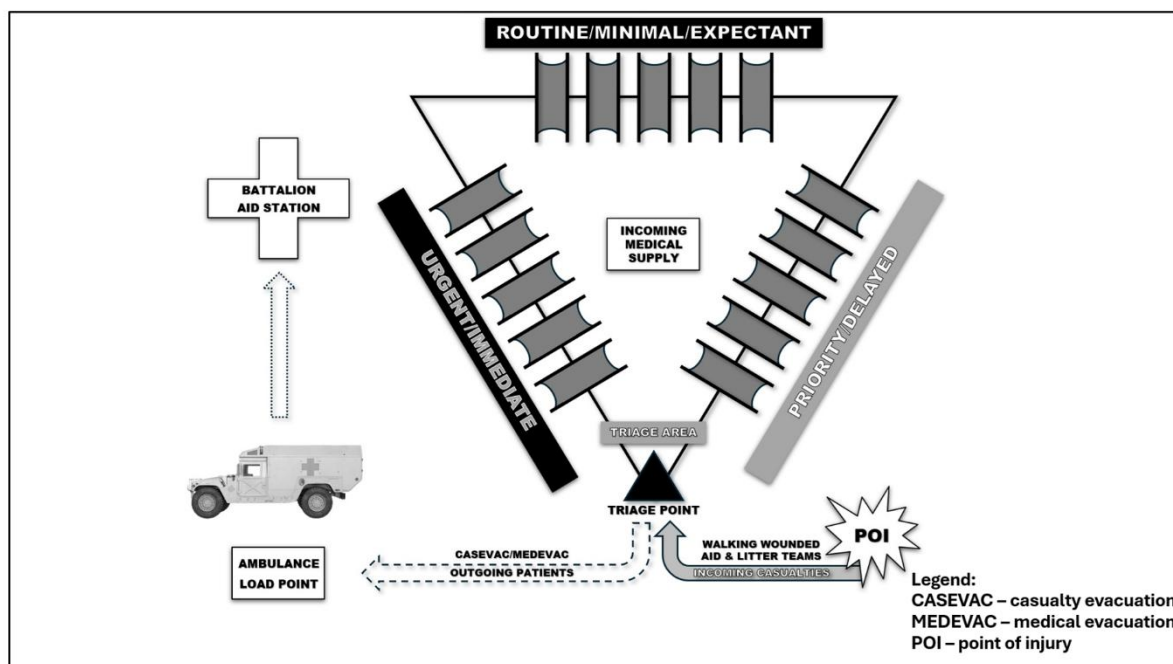


Figure 1-1. Casualty collection point layout in an open area or field

1-25. The location of the CCP will depend on the unit location, the tactical situation, and the number of casualties to be evacuated. The unit first sergeant determines the location of the company CCP, or the platoon sergeant for the platoon CCP with guidance from the respective combat medics. Battle drills and the tactical SOP should be established on how casualties will be moved from the POI to the CCP.

1-26. The company CCP should be in a covered and concealed position in the company trains area while the platoon CCP should be located at the platoon's rear. The CCP should be:

- Away from natural or enemy choke points.
- Covered and concealed.
- Expandable if casualty load increases.
- In a building or on a hardstand (exclusive CCP building limits confusion).
- In an area with good drainage.
- Inside the perimeter and allow for passive security.
- Near lines of patient drift, or natural routes along which wounded Soldiers may be expected to go back for medical care from a combat position.
- Near and have access to evacuation routes (foot, vehicle, and aircraft).
- Near templated areas of expected high casualties.
- Next to objective choke points.
- Reasonably close to the fight.
- Trafficable to evacuation assets.

Note. "Lines of Drift" are paths of least resistance that offer the greatest ease while considering obstacles and modes of transit to the objective.

1-27. Establishing a CCP in a building (see figure 1-2 on page 7) may be the only choice when conducting operations in urban areas. After the unit ensures that the building is cleared and secured, leaders—

- Determine the structural, environmental, medical, and CBRN hazards prior to receiving casualties.
- Determine location of control point and triage.
- Minimize congestion.
- Remove or relocate furniture or obstructions.

- Consider rigid litter or non-rigid pole-less litter for movement.
- Use largest rooms possible.
- Separate treatment categories by room. If triage categories are separated by walls, it is more difficult to monitor all the casualties.
- Color-code and mark rooms by treatment categories.

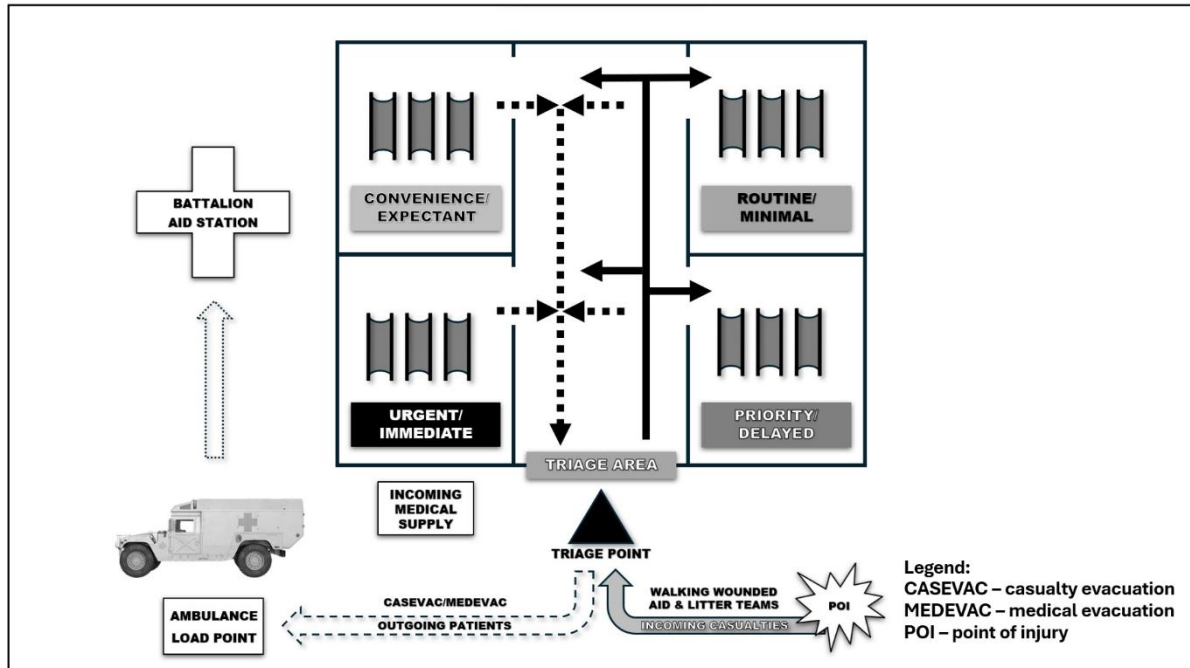


Figure 1-2. Casualty collection point in a building

CASUALTY COLLECTION POINT SITE SELECTION

1-28. The CCP site selection is addressed during troop leading procedures and the CCP SOP should be rehearsed. The CCP should be established with:

- A day and night marking system.
- A triage point for entry into and exit from the CCP to track casualties and control access.
- Properly marked evacuation categories for day and night (non-tactical or tactical) operations.
- A killed in action (KIA) area away from the other casualties.
- A medical resupply area.
- A helicopter landing zone (commonly known as HLZ).

1-29. The first sergeant or platoon sergeant is responsible for casualty flow and everything outside the CCP. They—

- Provide for CCP structure and organization (color coded per tactical SOP for both day and night).
- Maintain command and control and situational awareness.
- Control aid and litter teams.
- Ensure security is established and maintained.
- Remove, collect, identify, organize, and maintain casualty equipment outside of the treatment area.
- Track casualties and equipment into and out of CCP.
- Establish accountability and send reports to higher.
- Move casualties through CCP entrance and exit control point.

1-30. Medical personnel are responsible for operations inside the CCP. Medical personnel assigned to the CCP include the:

- Triage officer or noncommissioned officer who sorts and organizes the casualties into appropriate treatment categories.

- Medics, corpsman, or medical technicians who organize medical equipment and supplies.
- Medical personnel, CLS, and aid and litter teams who assist with treatment and packaging of casualties.

1-31. Once triaged, casualties categorized as "Minimal" should ideally stay with their assigned unit or assist with security at the CCP, if possible. Deceased personnel are designated as KIA. Their remains should stay with the assigned unit until mortuary affairs personnel arrive, or until the remains are transferred to a morgue. The mortuary affairs SOP should clearly outline the handling of the remains, considering theater-specific requirements, environmental factors, and the religious beliefs of the Service member.

1-32. It is also important to note that the remains of deceased personnel should not be moved with patients on MEDEVAC ambulances. This practice helps maintain the dignity of the deceased and ensures the focus remains on providing urgent care to the living casualties during MEDEVAC.

CASUALTY RESPONSE FOR UNITED STATES ARMY UNITS

1-33. The treatment and evacuation of casualties are difficult tasks for an organization to execute due to the operating distance away from its higher headquarters. Units should allocate adequate assets to the platoon to assist in casualty treatment and evacuation. Platoon medics evaluate and treat sick and injured Soldiers and assist the unit in the evacuation of casualties.

CASUALTY CARE

1-34. When casualties occur, the platoon must provide aid to those wounded in action. Combat medics, who are assisted by nonmedical personnel, provide EMT to alleviate potentially life-threatening injuries and to ensure maximum survivability. This support is commonly provided by enlisted personnel and includes first aid (self-aid or buddy aid), enhanced first aid by the CLS, and EMT by the platoon medic.

Wounded in action

1-35. The ability of fellow Soldiers to provide aid to wounded comrades is heavily influenced by the tactical situation. Minimizing casualties is more achievable when Soldiers concentrate on defeating or neutralizing the enemy responsible for causing the injuries.

1-36. During combat, it is crucial to move casualties to a covered and concealed position where first aid can be administered safely. Once the immediate threat subsides, squad leaders should coordinate the evacuation of the wounded to the platoon CCP, typically located to the rear of the platoon's position. At the CCP, the platoon medic will perform triage, stabilize the injured, and initiate their transfer to more comprehensive medical facilities. For detailed operations at the CCP, refer to paragraph 1-20 on page 4.

1-37. Before evacuating casualties to the CCP or further, leaders must ensure that:

- The casualty's personal protective equipment, such as protective masks, accompanies them.
- All vital operational, sensitive items, and equipment are removed from the casualty.
- Medical documentation and identification tags are securely attached to each casualty to streamline medical treatment and tracking processes.

1-38. Additionally, the removal of equipment should include items like automated network control devices, global positioning systems, maps, position-locating devices, and laser pointers. Each unit should have an SOP for managing the weapons and ammunition of its wounded personnel, ensuring these measures are systematically applied, and adhere to safety and operational security standards.

Killed in Action

1-39. The platoon leader designates a location for the collection of KIAs. Personal effects remain with the human remains. While treating the human remains with dignity and respect, squad leaders remove, and safeguard all equipment, and government issued items until they can turn the equipment over to the platoon sergeant. The platoon sergeant turns over the human remains to the first sergeant. The platoon should not transport human remains on the same vehicle as wounded Soldiers. The human remains are normally transported to the rear on empty resupply trucks but are dependent on unit SOP.

1-40. Commanders and first sergeants establish command procedures to ensure the Soldier's next of kin are properly notified in accordance with (IAW) ATP 4-46/MCRP 3-40G.3/NTTP 4-06/AFTTP 3-2.51. The potential for unofficial communications exists with the KIA and casualty operations. The use of cell phones and computers in proximity to the area of operation enables many Soldiers to contact their home station regarding the casualty. This kind of communication is unofficial and unacceptable. The next of kin for Soldiers wounded in action or KIA should not receive notification through unofficial means. Therefore, the use of cell phones and computers in proximity to the area of operation should be subject to a communication blackout until the next of kin is notified by a casualty notification officer. No internet or phone calls home should be permitted during the communication blackout.

EVACUATION CONSIDERATIONS

1-41. Conducting CASEVAC is a fundamental unit responsibility and a key aspect of sustainment operations. To effectively execute CASEVAC, unit leaders should consider the following points to optimize their planning and response:

- Know the Evacuation Platform:
 - Determine if a medical provider is onboard.
 - Check for the presence of necessary monitoring equipment.
 - Ascertain the capacity of the platform regarding how many ambulatory and litter casualties it can evacuate.
- Understand Platform Specifications:
 - Large Platforms: Evacuate multiple casualties, with routine cases loaded first, followed by priority, and critical (urgent) cases last, ensuring they are first off at the destination.
 - Small Platforms: Prioritize evacuation of critical (urgent) and priority cases first, with routine cases evacuated as space allows.
 - Verify that all securement devices are functional and appropriate for the type of litter and evacuation platform being used.
- Packaging Requirements for the Evacuation Platform:
 - Identify the types of litters available.
 - Determine if the litter is stirrup-based versus non-stirrup-based pole-less and is compatible with the evacuation platform.
 - Ensure there is enough equipment to prevent hypothermia and keep the casualty warm and dry.
- Secure the Casualties and Determine the Loading Sequence on the Evacuation Platform:
 - Implement procedures for tying down or securing litters to the vehicle, ensuring safety during transportation.
 - Ensure a clear and prioritized loading sequence to streamline the evacuation process.
 - Plan and coordinate the movement of casualties to the evacuation platform efficiently.

1-42. By addressing these areas and integrating enhancements such as leading in conducting CASEVAC and ensuring all litters are securely tied down, leaders can improve the efficiency and safety of evacuation operations, ultimately enhancing casualty care during critical situations.

Medical Evacuation

1-43. A MEDEVAC may be provided by dedicated air or ground ambulances. The MEDEVAC includes the provision of en route medical care, whereas CASEVAC may not provide medical care during movement. For more information on MEDEVAC refer to ATP 4-02.2. If dedicated MEDEVAC platforms (ground and air) are available, casualties should be evacuated on these conveyances to ensure they receive proper en route medical care. If both MEDEVAC and CASEVAC are used, the most seriously injured should be evacuated on MEDEVAC platforms while the less seriously injured are moved by CASEVAC.

1-44. When possible, medical platoon ambulances provide evacuation and en route care from POI or the platoon or company's CCP to the BAS. The ambulance team supporting the company works in coordination with the senior combat medic supporting the platoons. In mass casualty situations, nonmedical vehicles may be used to assist in CASEVAC as directed by the company commander or leader. However, plans for use of nonmedical vehicles to perform CASEVAC should be included in the unit SOP and briefed at the sustainment and combines arms rehearsals.

1-45. Ground MEDEVAC is the primary means of evacuation during large-scale combat operations due to the operational environment and the greater availability of ground ambulances. Aeromedical evacuation is the preferred method of evacuation due to the greater speed and range of air ambulances. Mission variables should be considered before executing aeromedical evacuation. When selecting a site for a LZ, the LZ should be relatively flat and free of hazards such as power lines. Loose debris should be removed to avoid objects being sucked into engines or causing injury due to the rotor wash. The platoon provides local security of the LZ until the evacuation is complete.

1-46. The senior military person present determines whether to request evacuation. The decision to request MEDEVAC is based on the advice of the senior medical person at the scene, the patient's condition, the tactical situation, and availability of a dedicated medical platform. Casualties will be picked up as soon as possible, consistent with available resources and pending missions. The following are evacuation priority categories of precedence and criteria used in their assignment:

- Priority I – Urgent is assigned to emergency cases being evacuated as soon as possible and within a maximum of one hour to save life, limb, or eyesight; to prevent complications of serious illness; or to avoid permanent disability.
- Priority IA – Urgent Surgical is assigned to patients that should be evacuated as soon as possible and within a maximum of one hour who must receive far forward surgical intervention to save life, limb, or eyesight and stabilize for further evacuation.
- Priority II – Priority is assigned to sick and wounded personnel requiring prompt medical care. The precedence is used when the individual should be evacuated within four hours or if the medical condition could deteriorate to such a degree that the casualty becomes urgent, or whose requirements for special treatment is not available locally; the individual will suffer unnecessary pain or disability.
- Priority III – Routine is assigned to sick and wounded personnel requiring evacuation, but condition is not expected to deteriorate significantly. The sick and wounded in this category should be evacuated within 24 hours.
- Priority IV – Convenience is assigned to patients for whom evacuation by medical vehicle is a matter of medical convenience rather than necessity.

Note. The North Atlantic Treaty Organization (NATO) Standardization Agreement (STANAG) 3204 has deleted the category of Priority IV - CONVENIENCE. There is a requirement for it in an operational environment and is still included in the United States Army evacuation priorities.

Medical Evacuation Request

1-47. Immediately following an attack and upon learning of a casualty within the platoon, the vehicle commander, squad or section leader, and platoon sergeant will start to prepare an evacuation request. There is no difference between a ground or air evacuation request. Units transmit a 9-line MEDEVAC request, report number M030 IAW FM 6-99 to the command post. Units use Graphic Training Aid 08-01-004, *MEDEVAC Request Card*, for the correct reporting lines and brevity codes for United States Forces. Units use this report to request evacuation of sick and wounded personnel. This report is like the NATO MEDEVAC request (9-liner MEDEVAC) IAW STANAG 2020 and STANAG 2627. The NATO 9-liner MEDEVAC request is used only when specifically directed in the orders process.

1-48. Along with the standard 9-line MEDEVAC request format, the higher role of care may require a mechanism of injury, injury type, signs and symptoms, and treatments (MIST) report, which should be provided by the platoon medic or CLS on-site. The Department of Defense (DD) Form 1380 (*Tactical Combat Casualty Care (TCCC) Card*), when complete, is the MIST report. For more information on MEDEVAC request process see paragraph 3-62 on page 34, or refer to ATP 4-02.2.

1-49. For casualties in areas not supported by MEDEVAC or when MEDEVAC assets are unavailable, various multimodal patient movement (MMPM) or CASEVAC assets are employed to conduct the evacuation mission, with MMPM being the preferred method of evacuation.

Casualty Evacuation

1-50. Service members may have to be available to perform or assist in an extraction, rescue, and evacuation of casualties. There are multiple extrication devices, tactical situations, environmental situations, and maneuvers to accomplish appropriate movement for a casualty. For detailed information on casualty movement, refer to ATP 4-02.13.

WARNING

Casualties transported in this manner may not receive proper en route medical care or be transported to the appropriate MTF to address the patient's medical condition.

If the casualty's medical condition deteriorates during transport, or the casualty is not transported to the appropriate MTF, an adverse impact on the casualty's prognosis and long-term disability or death may result.

1-51. CASEVAC is the responsibility of the unit to transport less severely injured casualties when MEDEVAC resources are limited. The planning of CASEVAC is synchronized with existing medical support plans to ensure a streamlined process. This involves organizing, preparing, coordinating, and practicing the evacuation procedures in advance to guarantee efficiency. The plan ensures that teams trained in first aid are adept at handling stretchers, working around medical posts, and correctly assessing triage categories. Whenever possible, non-medical vehicles, aircraft, and watercraft used for transporting casualties should be supported by CLS or medical staff to provide necessary care during transport.

1-52. Sufficient space may not be available to permit a caregiver to accompany or assess the casualties. Refer to ATP 4-02.13 for more information. The type of en route monitoring and medical care or first aid provided is limited by the following factors:

- Skill level of the individual providing care. The combat medic is qualified to provide EMT, while the CLS is trained to provide enhanced first aid. The combat medic can provide emergency medical interventions, whereas the CLS can only monitor the casualty and ensure the basic lifesaving first aid tasks are accomplished.
- Medical equipment available.
- Number of casualties being transported.
- Accessibility of casualties. If a nonstandard evacuation vehicle is loaded with the maximum number of casualties, the combat medic or CLS may not be able to attend to the casualties while the vehicle is moving. If the condition of a casualty deteriorates and emergency measures are required, the vehicle may need to be stopped for the combat medic or CLS to provide additional care.

MULTIMODAL PATIENT MOVEMENT

1-53. When medical personnel provide care during the transport of casualties on non-medical platforms, it's called MMPM. MMPM involves moving casualties efficiently and providing care en route using available transportation methods, as outlined in JP 4-02. MMPM platforms can include maritime, land, or air vehicles, and they may serve as the primary or secondary mode of transportation between care facilities. However, utilizing these platforms means accepting increased risks for both patients and transport teams due to their combatant status. To ensure safe patient movement, MMPM platforms may carry standardized medical kits, emphasizing the importance of thorough planning and preparation.

UNIT REPORTING

1-54. A casualty report is filled out when a casualty occurs, or as soon as the tactical situation permits. The casualty report is usually completed by the Soldier's squad leader and turned into the platoon sergeant, who forwards it to the first sergeant. The report should include a brief description of how the casualty occurred (including the place, time, and activity being performed) and who or what inflicted the wound is included. If

the squad leader does not have personal knowledge of how the casualty occurred, they get this information from Soldiers who have the knowledge. For further discussion on human resources support to casualty operations, refer to ATP 1-0.1.

1-55. As casualties occur, the nearest observer informs the platoon sergeant who then informs the first sergeant via the most expedient method available (such as a free text within Mission Command Systems or a radio voice). The first sergeant submits a personnel status report to the Battalion S-1 section. This report documents duty status changes on all casualties. Casualties are taken to the CCP for classification of injury type (urgent, priority, routine, or return to duty), evacuation, and integration into the medical treatment system.

Medical and Personnel Accounting

1-56. When a Soldier becomes a casualty, the platoon combat medic or senior combat medic records the medical treatment the Soldier receives on the DD Form 1380. The BAS and the Role 2 read the Soldier's DD Form 1380, when they treat the Soldier. The Battalion S-1 should electronically receive a notification message to update the Soldier's patient tracking status. In turn, this message should be forwarded to the Soldier's company. In this manner, the casualty's location can be determined, and the Soldier can be properly accounted for by their unit.

Personnel Reports

1-57. Units employ personnel reports to inform the command group about casualties and to initiate requests for replacement personnel. Typically, a request for replacements is made immediately following a casualty report. This systematic approach ensures that the command group remains fully informed about personnel status and can quickly respond to the need for additional forces to maintain operational effectiveness. The prompt communication of these details is crucial for strategic planning and the swift adaptation to evolving battlefield conditions.

Personnel Battle Loss Report

1-58. Units transmit a Casualty Report (known as CASREP), report number C001, also known as the personnel battle loss report, IAW FM 6-99 to the command post as casualties occur. This report is like the NATO notification of casualty (stage 1) (NOTICAS1) IAW STANAG 2020. The unit completes a DA Form 1156 (*Casualty Feeder Card*) with witness statements and submits the entire packet to the first sergeant. The personnel battle loss report is an interim report to update information sent in the last personnel daily summary report.

1-59. The DA Form 1156 is used to report those Soldiers who have been killed and recovered or those who have been wounded. This form is also used to report Soldiers who are KIA, missing, captured, or not recovered. The Soldier with the most knowledge of the incident should complete the witness statement. This information is used to inform the Soldier's next of kin, and to provide a statistical base for analysis of friendly or enemy tactics. For more information on the Army casualty program refer to Army Regulation (AR) 638-8.

1-60. Once the casualty's medical condition has stabilized, the company commander may write a letter to the Soldier's next of kin. During lulls in the battle, the platoon forwards casualty information to the company headquarters. The first sergeant ensures a completed DA Form 1156 is forwarded to the Battalion S-1, who then enters the data into the defense casualty information processing system.

CASUALTY RESPONSE REHEARSALS

1-61. During the troop leading procedures, the organization rehearses the CCP plan per unit SOP. Unit leaders and planners include the leaders from the medical platoon in the military decision-making process, orders production and in CCP rehearsals. This ensures the evacuation plans mutually support the unit's mission and the medical tactics, techniques, and procedures (TTPs) are understood. Some areas to concentrate on when conducting a casualty response rehearsal are:

- Alerting aid and litter teams, and conduct:
 - Alert and movement.
 - Evacuation equipment preparation.

- Clearing and securing weapons.
- CCP members conduct:
 - Clear and secure CCP location.
 - Triage Point.
 - Marking and tagging.
 - Accountability and reporting.
 - Equipment removal, tagging, and consolidation.

1-62. Casualty response is a portion of the company's sustainment plan that the platoon and squad must plan, prepare, and execute. Concurrently with other operational planning, the platoon develops its sustainment plan during mission analysis and refines it in the war gaming portion of the troop leading procedures; and should describe the duties and responsibilities of personnel in relation to casualty response. Rehearsals are normally conducted at each echelon of command to ensure the smooth, continuous flow of materiel and services.

DUTIES AND RESPONSIBILITIES

1-63. Sustainment responsibilities for the maneuver platoon and squad include requesting support requirements through the company and ensuring sustainment operations are properly executed when support elements arrive in the platoon area. The platoon sergeant is normally in charge of these functions, with guidance and oversight provided by the platoon leader. The platoon sergeant must submit accurate personnel and logistical reports, along with any other necessary information and requests.

Note. The duties and responsibilities of leaders and platoon members must be executed even in the absence of a particular leader to ensure mission accomplishment is IAW the commander's intent.

Medical Planners

- 1-64. During the planning phase, battalion medical personnel and planners play a critical role by:
- Providing recommendations and guidance to leaders on the Army Health System support plan.
 - Collaborating with unit leaders to coordinate various aspects, including:
 - Identifying CCP locations for subordinate units throughout different phases.
 - Organizing medical tasks and their distribution.
 - Developing plans for on-objective ground evacuation and off-objective air and ground evacuation.
 - Ensuring security measures for HLZ and evacuation platforms.
 - Assessing the need for additional support from subordinate units.
 - Establishing primary, alternate, and contaminated ground evacuation routes.
 - Determining patient decontamination requirements.
 - Coordinating external medical support provided to the battalion.
 - Identifying instances during the operation when CASEVAC may supplement MEDEVAC ambulances.
 - Establishing connections with tactical operations to ensure seamless coordination.

1-65. During the execution phase, battalion medical personnel and planners carry out essential tasks with efficiency and care. They prioritize patient care by triaging, treating, monitoring, and preparing individuals for evacuation as needed. Delegation of treatment is managed effectively to ensure resources are used optimally. In situations requiring additional support, they can request assistance from other medical or platoon assets. Furthermore, they provide valuable guidance and recommendations to leaders on casualty management, aiding in informed decision-making during challenging circumstances.

Platoon Leader

1-66. The platoon leader is responsible for their platoon's execution of the sustainment plan at platoon level. They plan and relay support requirements for mission accomplishment to the company headquarters where it is consolidated and passed on.

1-67. The platoon leader, assisted by the platoon sergeant, is responsible for developing their sustainment plan. This starts with checking their current Class VIII supplies in order to find out exactly what is available. Knowing what they have on hand helps them accurately estimate their support needs. It's crucial for the commander to be aware of these supplies so they can prepare for treating expected casualties and identify critical supplies. This verification is key not only to confirm the sustainment plan's accuracy but also to ensure that the platoon's support requests are made as early as possible. Based on this inventory and their maneuver plans, the platoon leader can then create and submit their sustainment execution plan and support requests.

1-68. The sustainment plan should address key operational considerations, including:

- Types of support: Determine the specific medical support needed based on operational requirements and tactical factors.
- Quantities: Assess the quantities of support required, including potential needs for Class VIII resupply or adjustments to preconfigured Class VIII packages.
- Threat assessment: Analyze the composition, disposition, and capabilities of the expected enemy threat, considering factors such as anticipated contact locations, expected casualties, and the impact of enemy special weapons capabilities.
- Terrain and weather: Evaluate how terrain and weather conditions will impact the sustainment plan, including considerations for securing maintenance areas and CCPs, determining CASEVAC routes, and planning for contaminated personnel and equipment evacuation.
- Time and location: Identify when and where sustainment support will be needed, including—selecting optimal locations for CCPs, enemy prisoner of war collection points, and determining travel times to medical treatment facilities.
- Requirements: Define support requirements by element and type, including prioritization of medical support and emergency resupply.
- Risk assessment: Assess whether lulls in the battle will provide opportunities for support elements to conduct resupply operations safely.
- Resupply techniques: Select the most suitable resupply technique based on information gathered during the sustainment planning process.

1-69. During the planning phase, unit leaders undertake various crucial tasks to ensure effective medical support:

- Allocate aid and litter teams and CASEVAC equipment to facilitate timely response to casualties.
- Conduct rehearsals for casualty responses to ensure readiness and effectiveness in handling medical emergencies.
- Perform precombat inspections of medics, platoon casualty response kits, and CLS bags to verify equipment and readiness. Casualty response kits are unit dependent tailored to their specific mission that would include appropriate litters, litter straps, panel markers, hypothermia prevention kits or extra medical supplies.
- Establish an accountability and reporting plan to track medical resources and casualties accurately.
- Determine the CASEVAC plan for each phase of the operation, including CCP, helicopter landing zone, and ambulance exchange point locations.
- Organize the gathering and distribution of casualty equipment and sensitive items as needed throughout the operation.
- Establish accountability procedures and submit reports to higher commands as required.
- Rehearse the proper management of KIA remains to ensure respect and dignity in handling.
- Identify and ensure the security of CCPs, HLZs, and ambulance exchange point locations to prevent disruptions to medical operations.
- Determine the task organization and distribution of medical personnel to optimize their effectiveness and coverage.
- Prepare evacuation requests and establish CASEVAC link-up points to facilitate smooth patient transfers.
- Assist medics with EMT augmentation and provide guidance in directing aid and litter teams to maximize medical support capabilities

Platoon Sergeant

1-70. The platoon sergeant holds a pivotal role as the primary sustainment planner, coordinator, and operator within the platoon, reporting directly to the platoon leader. They execute logistical plans, relying on established SOPs from both the platoon and company. Working closely with the platoon leader and squad leaders during mission preparations, the platoon sergeant determines specific support requirements aligned with the tactical plan and ensures their proper arrangement. Additionally, the platoon sergeant:

- Coordinates human resources support with the company first sergeant, covering personnel accountability reports, casualty reports, replacement operations, and other administrative tasks.
- Supervises the evacuation of casualties, unprivileged enemy belligerents, and damaged equipment.
- Oversee the collection, initial identification, and evacuation of human remains to the mortuary affairs collection point.

1-71. As the second-in-charge, the platoon sergeant primarily follows instructions from the platoon leader. However, they typically establish and operate the unit's CCP, which involves directing aid teams and managing the CASEVAC process during operations. They also maintain the platoon's strength level information by consolidating casualty reports and orienting replacements.

1-72. In the event of wounded soldiers requiring evacuation, the platoon leader or platoon sergeant can take various actions:

- Coordinate ground evacuation with the nearest company.
- Request attachment of a dedicated ambulance from the company or battalion for operations forward of the larger element.
- Use organic platoon assets for evacuating wounded soldiers.
- Coordinate for Army aeromedical evacuation and aerial CASEVAC through the company or battalion channels.

Squad Leader

1-73. It is the squad leaders' responsibility to ensure casualties receive immediate TCCC and then immediately notify the platoon leader or platoon sergeant of all casualties. The use of CLS is critical. As a minimum, one member of each squad should be trained and equipped as a CLS. The squad leader is responsible for ensuring the casualty feeder card is filled out properly and provided to the platoon sergeant, as well as requesting evacuation for their personnel.

Combat Medics

1-74. EMT procedures performed by combat medics encompass critical interventions such as airway management, establishing intravenous or intraosseous access, hemorrhage control, shock prevention or treatment, splinting fractures, hypothermia management, and medication administration for pain or infection.

1-75. Combat medics undergo training under the supervision of the battalion surgeon or physician assistant, focusing on medical tasks, while the medical platoon leader and platoon sergeant concentrate on tactical training. The platoon combat medic assumes responsibilities such as triaging injured personnel, conducting sick call screening, assisting in evacuations, training CLS, requisitioning medical supplies, recommending CCP locations, providing guidance on operational public health, and supporting CLS as needed.

1-76. Combat medics are assigned to the battalion medical platoon and attached to the maneuver companies, typically with one combat medic per platoon and one senior combat medic per company, strategically located for rapid medical treatment of casualties.

1-77. The company senior combat medic is positioned with the company trains, located 500 to 1000 meters. or one terrain feature to the rear of the company, ensuring immediate recovery and medical support, while remaining concealed from direct enemy fire. The company's first sergeant or executive officer supervises evacuation activities and sustains operations within the platoon.

1-78. During the planning phase, the senior combat medic or platoon combat medic provides recommendations and plans for medical support, Army Health System support, casualty response, and

evacuation. They also coordinate CCP locations, medical task organization, evacuation plans, and conduct precombat inspections.

1-79. During the execution phase, the senior combat medic or platoon combat medic implements triage, treatment, and monitoring, delegates treatment, requests assistance from other medical or unit assets, and offers guidance on casualty management and evacuation to support effective medical care in combat situations.

Combat Lifesavers

1-80. Immediate, far-forward medical care is critical on a widely dispersed and dynamic battlefield to prevent casualties from succumbing to wounds. Due to the unpredictable nature of combat, medical personnel may not always reach and treat wounded soldiers promptly. CLS play a crucial role in bridging this gap, as they can stabilize various types of casualties and mitigate the deterioration of a wounded soldier's condition until medical personnel arrive. Acting as CLS is a secondary mission conducted when the tactical situation allows, and they render enhanced first aid when a casualty occurs, either through self-aid, buddy aid, or CLS intervention.

1-81. The CLS performs tier 2 TCCC skills, serving as a link between the tier 1 skills taught to all Soldiers and the advanced medical training, provided by combat medics or higher-level medical providers. This capability is organic and aims to fill the gap between basic life-saving skills and advanced medical care. Training CLS should be a priority for platoons to ensure that they can replicate essential medical techniques in the absence of a combat medic. Each squad or equivalent-sized unit should have at least one certified CLS, who must recertify annually according to AR 350-1.

1-82. The primary role of a CLS is to address the most common causes of death on the battlefield, such as a massive hemorrhage, airway, and/or respiratory issues. They are trained to provide immediate care to save lives, including controlling bleeding and performing needle chest decompression for tension pneumothorax. The CLS may assist combat medics in patient care, evacuation preparation, or support medical personnel during prolonged care situations when not engaged in combat duties.

1-83. A significant advantage of the CLS is their proximity to injured squad or crew members. During large scale combat operations, the battlefield is often spread out and fast-moving, making rapid casualty treatment challenging. CLS ensure that squad CLS bags and litters are properly maintained, identify medical supply shortages, participate in casualty treatment drills, use enhanced first aid skills in the field, and are familiar with CCP locations and tactical SOP for establishing them.

CONTAMINATED CASUALTY CARE

1-84. In a CBRN environment, Soldiers may need to employ self-aid, immediate lifesaving interventions, and decontamination measures. This includes administering correct antidotes for nerve agent exposure and assuming the appropriate Mission-Oriented Protective Posture level. Further guidance on contaminated casualty care can be found in Chapter 20 of this publication and refer to ATP 4-02.7/AFTTP 3-42.3.

1-85. Buddy aid is crucial when individuals are unable to provide self-aid. In a CBRN environment, buddy aid actions involve:

- Treating immediate life-threatening injuries.
- Identifying the type of contamination (chemical, biological, or radiological).
- Administering appropriate medical CBRN defense material (prioritizing the casualty's own supplies).
- Decontaminating exposed skin and contaminated gear using various methods such as Reactive Skin Decontamination Lotion (RSDL) or the M291 Skin Decontamination Kit, which includes dry earth, bleach and water for chemical contamination, soap and water for chemical, biological, and radiological contamination, and brushing or vacuuming for radiological contamination.
- Assisting the casualty to do the remaining protective clothing, if required.
- Promptly moving the casualty to a decontamination point for further care.

1-86. For comprehensive guidance on patient decontamination procedures, refer to ATP 4-02.7/AFTTP 3-42.3.

CASUALTY RESPONSE FOR UNITED STATES MARINE CORPS UNITS

1-87. United States Marine Corps (USMC), Marine air-ground task forces require unit SOPs and operation orders address TCCC procedures, CASEVAC, and in-detail aid for chemical casualties with particular emphasis on lifesaving tasks. Wounded personnel should be treated at the appropriate role of care and returned to duty when possible. Hospital corpsmen evaluate sick and injured Marines, provide treatment, and arrange evacuation if necessary. CASEVAC and MEDEVAC are rehearsed like other critical parts of any operation. Casualty management begins at the POI or onset of illness throughout triage, treatment, and transport to the next role of medical care to include outside Service capabilities as necessary.

CASUALTY CARE

1-88. Health Service Support (HSS) is the employment of medical forces in support of the warfighter to deliver on-demand healthcare capabilities and ensure a healthy, fit, and medically ready force; counters the health threat to the deployed force; provide critical care for and management of combat casualties.

1-89. The HSS directly supports the National Military Strategy of forward presence and power projection. It also strengthens the warfighting commander by providing essential care in theater and MEDEVAC of casualties across the continuum of care to MTFs in the continental United States for definitive care without sacrificing the quality of care.

Mission

1-90. The HSS mission is to minimize the effects that wounds, injuries, and diseases have on unit effectiveness, readiness, and morale. The mission is accomplished by an aggressive and proactive preventive medicine program that safeguards personnel against potential health risks and by establishing an HSS system that provides appropriate care from the POI or illness to the appropriate taxonomy of care.

Principles

1-91. The HSS principles are guides for planning, organizing, managing, and executing HSS. Seldom will all principles exert equal influence in each situation and the HSS is most effective when it can identify the priority principles and bring them to bear in support of the commander's mission. Refer to JP 4-02 for more information. The Marine Corps HSS system encompasses:

- Conformity. The medical plan must integrate and comply with the commander's plan.
- Proximity. The medical plan must provide HSS as close to combat operations as the tactical situation permits.
- Flexibility. The medical plan must shift HSS resources to meet changing requirements.
- Mobility. The medical plan must anticipate requirements for rapid movements of HSS units to support combat forces during operations.
- Continuity. The medical plan must provide optimum, uninterrupted care and treatment to the wounded, injured, and sick.
- Coordination. The medical plan must ensure that HSS resources in short supply are efficiently employed and used effectively to support the planned operations.

Casualty Care and Management

1-92. Marine Corps HSS deploys through task-organized units of employment to provide care throughout the continuum of healthcare. The taxonomy continuum of healthcare includes the following capabilities:

- Policy and resource acquisition.
- Prevention and protection.
- First responder.
- Damage Control Resuscitation.
- Damage Control Surgery.
- Theater hospitalization.
- Definitive care.

- En-route care.

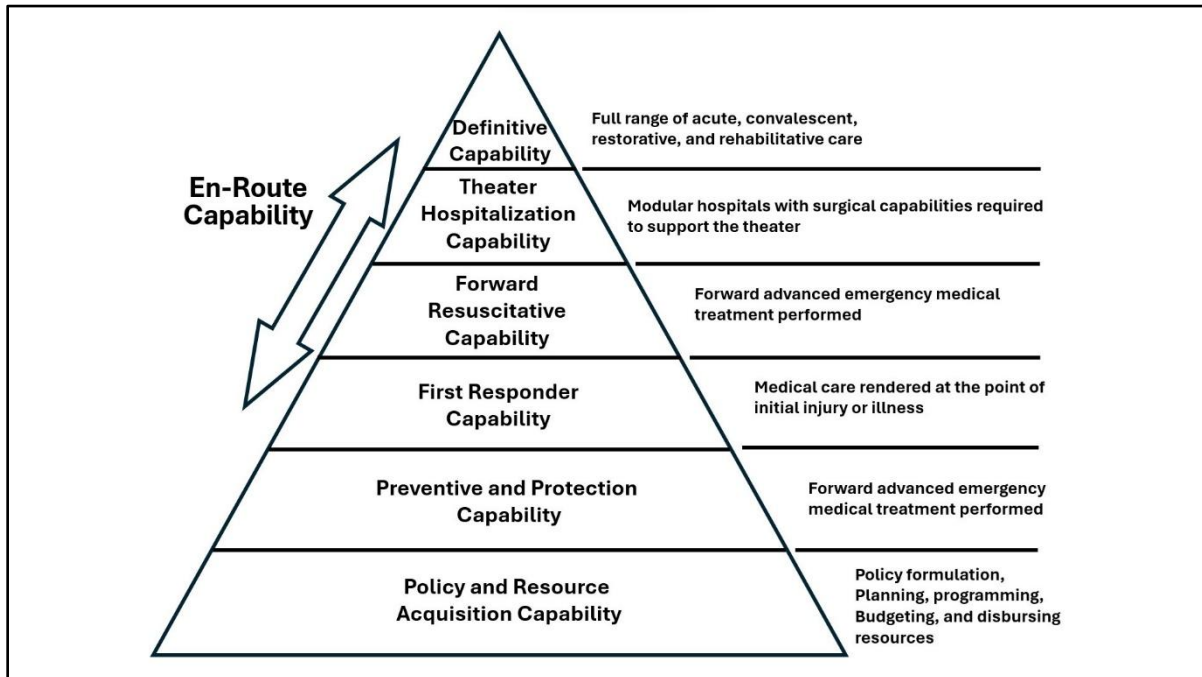


Figure 1-3. United States Marine Corps Taxonomy Continuum of Health Care Capabilities

Roles of Care for Marine Corps

1-93. Health Services Support within an area of operations is organized into roles of care that extend from forward to the rear of the area of responsibility. Each role of medical capability is designed to provide the mobility and capability that is required to meet basic health care needs of the supported operational units, phased treatment, casualty holding, and preparation for the evacuation of the sick, wounded, or injured.

1-94. Marine Corps operating forces' HSS is flexible to meet the demands of the mission, enemy forces, terrain, and other tactical situations. Furthermore, each role of care provides treatment capabilities in support of the role of care below it. Each capability of the HSS system is limited by four interacting factors:

- Urgency of the patients' needs.
- Unit mobility.
- Capabilities of HSS personnel, equipment, and supplies.
- Workload of each HSS unit is related to its treatment capacity.

Point of Principal Treatment

1-95. The HSS system provides a continuum of care that begins at the POI and ends at the appropriate treatment facility. Casualties are evacuated through the HSS system until they arrive at a facility with the capability, time, and bed capacity to begin definitive intervention to return them to duty or prepare them for further evacuation. This is known as the point of principal treatment. The site of the principal treatment is determined simultaneously by the patient's diagnosis, HSS unit's capabilities, and workload.

Role 1 Care

1-96. Role 1 care is the first medical attention that military personnel receive and is also referred to as unit level medical care. This role of care includes immediate lifesaving measures, disease and nonbattle injury prevention and care, combat and operational stress preventive measures, and patient location and acquisition (collection).

First Responder Self/Buddy Aid

1-97. The first responder provides unit level medical care. This is comprised of personnel from a unit and its organic BAS or squadron aid stations. In the case of organizations without organic medical elements, unit level care is provided by medical elements at the regimental, group, or support squadron level or other designated medical elements. TCCC has three phases and occurs during a combat mission and is the military counterpart to prehospital EMT.

1-98. TCCC in the military is predominantly administered by enlisted personnel, encompassing a range of roles such as first responders, self-aid, buddy aid (first aid), CLS, unit corpsmen, independent duty corpsmen, and other medical personnel. TCCC emphasizes addressing the most probable threats, injuries, and conditions encountered in combat, focusing on a concise selection of interventions aimed at mitigating the most critical of these threats and conditions.

Combat Lifesaver for Marine Corps

1-99. The CLSs for the Marine Corps are nonmedical military personnel selected by their unit commander for additional training beyond basic first aid procedures. A minimum of two Marines should be trained per table of organization corpsmen. The primary duty of a CLS does not change. The additional duty as a CLS is to provide enhanced first aid for injuries, based on his training, before the corpsman arrives or to assist the corpsman. CLS training is normally provided by medical personnel organic to sustainment units. The senior medical person designated by the commander manages the CLS training program.

Unit Hospital Corpsman

1-100. The company or squadron corpsman represents the first point where a sick, injured, or wounded Marine receives care from trained medical personnel. If emergency or lifesaving measures are required before a Hospital Corpsman (HM) arrives, they must be performed by fellow Marines trained as a CLS, first responder self/buddy aid. Care from unit HM includes primary and secondary assessments, followed by emergency or lifesaving measures (such as establishing and maintaining an airway, control of bleeding, cardiopulmonary resuscitation (CPR), treatment for shock, and fracture stabilization). The HMs duties include basic medical skills, use of medical equipment and supplies, and initiation of requests for assistance and evacuation.

The Aid Station

1-101. The aid station is the Role 1 for the Marine Corps. Treatment at the aid station is distinguished by the skills of a physician, physician assistant, and an independent duty corpsman. Treatment at an aid station is provided based on a more comprehensive evaluation and treatment plan, which may include restoration of airway, use of intravenous fluids, antibiotics, and application of splints and bandages. These elements of medical management prepare patients for return to duty or evacuation to the appropriate role of care.

AMPHIBIOUS OPERATIONS

1-102. The stages described in the following paragraphs represent only notional phasing. Other variations and combinations resulting from such factors as threat level, mission, terrain, geography, weather, force at risk, and opposing forces are possible.

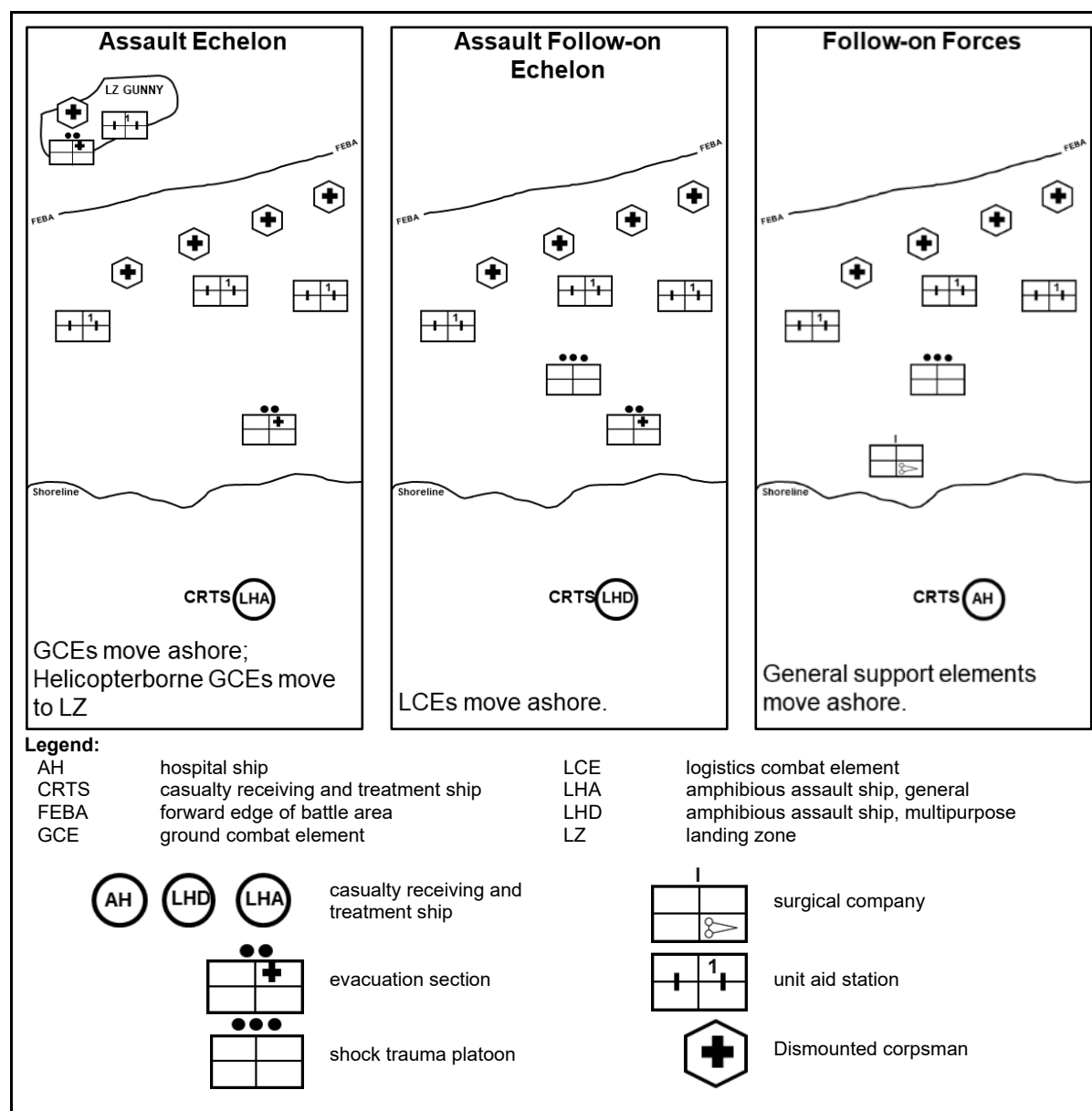


Figure 1-4. United States Marine Corps notional phasing of health care support

Assault Echelon

1-103. During the assault phase, HSS capabilities ashore are limited to organic elements within the Marine Air-Ground Task Force. First response medical care for assault forces is provided by self-aid, buddy aid, and hospital corpsmen of the landing force rifle platoons. When the tactical situation permits, BASs are established, and additional medical professionals provide care. The BAS is normally divided into two sections, with assigned battalion nonmedical litter bearers divided between them. One section lands with the battalion combat train and provides in-close support to the assault force. The second section lands with the field train and establishes interim evacuation stations until relieved by follow-on HSS elements.

1-104. Evacuation stations are typically expanded and staffed by support and medical battalion personnel, drawing assets from the shock trauma platoon or triage/evacuation platoons of the surgical companies. When established with the landing force support party (also known as LFSP), the supporting medical battalion

constitutes the beach evacuation section(s) of the landing force support party. The primary role of a BAS is to treat or evacuate assault force casualties to a designated casualty receiving and treatment ship. If the supporting medical battalion personnel come ashore with the landing force support party, they may supplement the beach evacuation section(s).

1-105. When evacuation stations attached to the landing force support party become operational ashore, established BASs are relieved to conduct their primary missions in supporting their parent battalions. The expansion of the HSS facilities follows the landing of supporting evacuation stations. The HSS assets are typically established at the beach service support area and follow-on combat service support areas, logistic support areas or forward arming and refueling points.

Assault Follow-on Echelon

1-106. While most logistic support capabilities during the assault follow-on echelon continue to be sea-based, projected HSS capabilities ashore expand along with the logistics combat element. Capabilities could be additional forward resuscitative surgery systems or shock trauma platoon with mobile combat logistics companies. When progress of assault units is such that the beachhead is relatively secure, HSS is enhanced from follow-on forces.

Follow-on Forces

1-107. The HSS shifts its posture to achieve shore-based health care consistent with the expected combat intensity and duration of sustained operations ashore, independent of sea-based facilities. This phasing is achieved by upgrading capabilities ashore by consolidating HSS capabilities once they come ashore. If a sustained land campaign is envisioned, additional HSS will normally be provided by theater hospitalization (expeditionary medical facilities, hospital ships, or other Service-equivalent facilities such as combat support hospitals and expeditionary medical support).

Casualty Evacuation

1-108. Timely movement of casualties from the operational environment is important for safety, care for the wounded, and troop morale. Unit leaders are responsible for coordination of CASEVAC from the POI to the appropriate role of care. The owning commands are responsible for CASEVAC from POI or illness to casualty treatment stations, and for casualty movement from the first responder capability to forward resuscitative care (also known as FRC) capabilities.

1-109. CASEVAC can be accomplished through dedicated, designated, or opportune ground or air transportation provided by a combination of litter carries, manual carries, ground transportation, and limited air transport (fixed- or rotary-wing).

1-110. When conducting operations ashore, the patient evacuation team (PET) coordinates patient movement of the patient with requesting unit by ground or air transport or a combination of the two, from POI to the appropriate forward resuscitative care and between various roles of care. The direct air support center and PET receive patient evacuation requests via the 9-line request (see paragraph 3-62 on page 34) over the tactical air request, helicopter request net, or alternate communication means. The PET then determines the appropriate means of patient movement and designated HSS facility.

1-111. If air is the appropriate evacuation means, the direct air support center coordinates air support. If ground evacuation is required or more appropriate, the PET informs the requesting unit to coordinate patient movement with the logistics staff section or logistics combat element. The PET will continue to assist as necessary and track all patient movement through completion of the mission. Depending on the size of the operation, the PET might only assume responsibility for patient movement from POI to forward resuscitative care, and the Marine logistics group HSS medical support operations center or higher regulating authority will assume movement above forward resuscitative care. This should be decided prior to deployment and written into the "Annex Q" (Medical Services) of the operational plan or operation order.

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Chapter 2

Casualty Extraction and Movement

CASUALTY EXTRACTION

2-1. Transportation of casualties away from danger or to evacuation vehicles is a critical aspect of first aid, requiring careful attention to avoid exacerbating injuries. For comprehensive guidance on casualty transport and evacuation procedures, refer to Soldier Training Publication (STP) 8-68W13-SM-TG, ATP 4-02.13, and ATP 4-02.2. Risk assessment is essential during casualty movement, and for detailed information on this process, consult ATP 5-19.

WARNING

In cases where a casualty is involved in a vehicle crash, there is a potential for spinal injury. Unless facing an immediate life-threatening scenario (fire or explosion), they refrain from moving a casualty suspected of a back or neck injury. Seek guidance from medical personnel on appropriate transportation methods.

2-2. Service members and CLS should adhere to the unit SOPs when extracting casualties from vehicles. Extraction methods will vary based on mission requirements and vehicle configurations in the operational area.

2-3. Service members should familiarize themselves with safety principles, casualty assessment, and proper training procedures. For instance, if a casualty is on fire, extinguish the flames, address life-threatening bleeding with a tourniquet (TQ) if necessary, and promptly seek cover.

2-4. To remove a casualty laterally from a vehicle—

- With the assistance of another Service member, grip the casualty's arms and legs.
- Stabilize the casualty's head and neck as much as possible.
- Lift them clear of the vehicle and transfer to a safe area on the ground.

Note. If available, medical personnel may use a specialized board or splint to stabilize the casualty's head, neck, and upper body.

2-5. To remove a casualty upward from a vehicle—

- Use a pistol belt or similar material to encircle the casualty's chest, aiding in their upward movement from the vehicle.
- With assistance from another Service member inside the vehicle, elevate the casualty using the pistol belt or by grasping their arms.
- While stabilizing the casualty's head and neck as much as possible, lift them free of the vehicle and position them on the topmost side.

Note. If medical personnel are present, they may employ a specialized board or splint to stabilize the casualty's head, neck, and upper body.

- Depending on the circumstances, relocate the casualty from the topmost side of the vehicle to a secure area on the ground.

WARNING

Avoid manual carries for casualties with suspected neck or spine injuries, unless immediate life-threatening hazards are present. Seek medical advice on appropriate movement and transportation protocols.

CASUALTY MOVEMENT

2-6. Timely evacuation of casualties from the operational environment is crucial not only for their safety and care but also for maintaining troop morale. Squad leaders bear the responsibility for CASEVAC from the POI to the platoon CCP. At the CCP, the senior medic assists the platoon sergeant or first sergeant in coordinating MEDEVAC by ground or air ambulance, or by CASEVAC. It's imperative for leaders to minimize the number of Soldiers needed for CASEVAC.

CASUALTY MOVEMENT ROUTE

2-7. Casualties with minor wounds might still be able to walk or help carry more seriously injured comrades. In the field, Soldiers can create improvised litters by threading poles—such as tent poles or antenna poles—through the sleeves of zippered combat uniform blouses or ponchos. Another option is to use a travois or skid, which allows a wounded individual to be strapped on and pulled by one person. Casualties can be evacuated from the POI to the CCP or directly to the BAS by litter teams. Depending on the situation, casualties may stay with the unit until transportation is available, or they might be left at a designated location for later retrieval. For more information on making and using improvised litter, refer to ATP 4-02.13.

2-8. From the platoon area, casualties are typically evacuated to the company CCP and then to the BAS. The company first sergeant, with support from the platoon sergeant, is usually tasked with moving casualties from the platoon CCP to the company CCP. Unit SOPs should cover this process, including casualty marking during limited visibility conditions. Small chemical lights can be effective for this purpose. Once casualties are gathered, assessed, and treated, they are transferred to the company CCP. Upon arrival, this process repeats until they are evacuated back to the BAS.

2-9. In cases of wide dispersion within the company, casualties may be evacuated directly from the platoon CCP using ground or air platforms. Aeromedical evacuation could be limited due to weather, terrain, or enemy presence. Sometimes casualties must be moved to the company CCP or battalion combat trains before rotary-wing aircraft evacuation becomes feasible. If there aren't enough battalion ambulances, supply vehicles may transport casualties to the BAS after delivering supplies. Urgent casualties are prioritized for MEDEVAC, while less critical cases may be moved by other means. Occasionally, the platoon sergeant may direct aid and litter teams to carry casualties to the rear.

CASUALTY HANDLING

2-10. Casualties evacuated manually must be handled with care to avoid exacerbating their injuries. Rough handling could cause further harm, so movement efforts should be organized and methodical. Each movement should be deliberate and gentle, considering the casualties' injuries and providing necessary first aid or TCCC as appropriate. Immediate movement may be necessary in certain safety-critical situations, such as a burning vehicle or enemy fire. However, manual carries should not be used for casualties with suspected neck or spine injuries unless there is an immediate life-threatening hazard, and medical advice should be sought.

2-11. Various methods for carrying and dragging litters can be used, depending on the casualty's condition, the terrain, and the level of enemy threat. When planning how to move a casualty, factors to consider include cover and concealment, the weight of the casualty versus the size and strength of the rescuer, nature of the casualty's wounds, the distance to cover, the carrying technique, environmental hazards, and enemy activity. Rescuers must be prepared to adapt their plans quickly based on the changing dynamics of the situation to ensure the casualty is moved swiftly and safely. Special care is also necessary when moving casualties near

different aircraft, vehicles, and ships, and responders should be knowledgeable about the specific hazards each platform presents.

2-12. Once a casualty has been treated with a hasty TQ, if the tactical situation allows, priority shifts to getting them to cover and out of enemy fire. Carries and drags enable responders or the CLS to accomplish this swiftly without worsening the casualty's condition. When providing first aid and considering casualty transport, responders must carefully assess the situation, particularly in scenarios like motor vehicle accidents where spinal injuries are a concern. In such cases, moving the casualty may only be advisable if there's an immediate threat to their life, with trained medical personnel ultimately directing transportation decisions.

Manual Carries

2-13. Manual carries pose physical strain on the bearers and carries the risk of exacerbating the casualty's injuries. However, they can be crucial in saving a casualty's life when other means of transport are unavailable or impractical due to terrain or tactical considerations. Depending on factors such as the bearers' strength and endurance, the casualty's weight, nature of injuries, and obstacles encountered, the distance a casualty can be manually carried varies. Carries can be employed for both conscious and unconscious casualties by one or two bearers. When executed correctly, carries offer the casualty greater protection from additional harm compared to drags and can facilitate movement over longer distances, typically ranging from 50 to 300 meters. For more information on manual carries refer to ATP 4-02.13.

Drags

2-14. Drags, on the other hand, are used when swift removal from dangerous or hostile environments is imperative. They allow the bearer and casualty to maintain a low profile, utilizing cover and concealment to evade hazards. Drags are generally employed for shorter distances, typically up to 50 meters. It's important to exercise caution during casualty handling to avoid causing further harm. For more information on drags refer to ATP 4-02.13.

LITTER EVACUATION

2-15. Whenever possible, transporting a casualty on a litter is preferred over manual carries. Litter transport offers greater comfort to the casualty and reduces the risk of worsening their injuries. It facilitates easier and faster evacuation and enables the casualty to be transported over longer distances compared to manual carries. Ideally, a standard litter should be used whenever available. In cases where standard litter is not accessible, an improvised litter can serve as a suitable alternative until the casualty can be transferred to a standard one. It's crucial to ensure that standard litter is always maintained in a serviceable condition. For further guidance on litter evacuation, refer to ATP 4-02.13.

EVACUATION PLATFORMS

2-16. In operational scenarios, ambulances may be insufficient in number to meet evacuation demands or may not be suitable for transporting patients over certain types of terrain. In such situations, many vehicles commonly used by units can be repurposed to transport casualties without significant modifications. Units should prioritize planning and training on how their existing vehicles can be used to transport casualties, both with and without litter. Leaders must assess not only their organic platforms but also explore the potential use of civilian, other Services, and coalition platforms. For detailed guidance on evacuation platforms and CASEVAC, refer to ATP 4-02.2, ATP 4-02.13, or STP 8-68W13-SM-TG.

2-17. It's important to note that the current Standard Training Publication contains tasks and training guidance that are not aligned with the current standards for continuity training across the operational force. Once the revised STP 8-68W13-SG-TG is approved, it will be available for download on the Army Publishing Directorate and Central Army Register websites. In the meantime, for the most up-to-date Combat Medic Specialist tasks, refer to the Individual Critical Task Lists posted on the Central Army Register website. You can find these by searching for "68W ICTLs" in the search window. This ensures that you are using the most current and applicable training materials available.

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PART TWO

TACTICAL COMBAT CASUALTY CARE

TCCC is crucial for ensuring the safety and health of Service members in combat-related situations. Specifically designed for battlefield conditions, TCCC provides a comprehensive approach to casualty care, focusing on immediate treatment and management. In the fast-paced environment of warfare, adherence to TCCC principles can significantly improve survival outcomes. It equips military units and leaders with the necessary knowledge and skills to provide timely and effective care to injured personnel, thereby reducing fatalities and enhancing survival rates. This publication highlights the vital importance of TCCC in military operations, emphasizing its essential role in saving the lives of Service members and enhancing the overall effectiveness of units.

Chapter 3

Fundamentals of Tactical Combat Casualty Care

TACTICAL COMBAT CASUALTY CARE OVERVIEW

3-1. In accordance with DODI 1322.24, medical readiness training (also known as MRT) encompasses all facets of medical support across the competition continuum and forms the cornerstone of effective force health protection. TCCC serves as the DOD standard of care for both nonmedical and medical first responders. The TCCC-All Service member course supersedes previous trauma skills taught in self-aid and buddy aid courses, mandating that all Service members receive role-based TCCC training and certification aligned with their assigned skill level as outlined by the JTS. DODI 6040.47 designates the JTS as the DOD's Center of Excellence for trauma management.

3-2. Service members performing self-aid, buddy aid, and CLS support prioritize their primary combat mission, rendering aid as a secondary mission when feasible. While these roles do not replace the specialized skills of combat medics, corpsmen, and healthcare technicians, they may assist in providing far-forward care and preparing casualties for evacuation if mission and time permit.

3-3. Self-aid, buddy aid, and CLS support are integral components of a comprehensive plan aimed at caring for wounded Service members. A significant advantage of individual Service members providing such support is their proximity to injured squad or crew members. This proximity allows for more immediate care compared to waiting for a combat medic, especially in situations where the medic may be occupied with treating other seriously wounded personnel. First responders and CLS personnel are trained to administer critical immediate care, including addressing severe bleeding and performing needle chest decompression for tension pneumothorax, potentially saving lives in the crucial moments following injury.

SERVICE SPECIFIC SKILLS FOR TACTICAL COMBAT CASUALTY CARE

3-4. Individual medical readiness training is built upon a foundation of knowledge, skills, and abilities tailored to operational medicine. Operational medical readiness training focuses on developing a core set of validated, operationally unique skills and abilities necessary to accomplish mission objectives. These core competencies are expanded as needed to address specific Service missions, informing unit readiness reporting as part of medical specialty team readiness. DODI 1322.24, published by the JTS, outlines basic TCCC skills

for all Service members, allowing individual Services to augment these requirements based on their specific needs.

3-5. TCCC skills are categorized into four tiers, spanning from all Service members to credentialed providers, with each tier building upon the knowledge and skills of the preceding tier. Each tier represents a progression in the complexity and scope of medical interventions, enabling Service members to provide increasingly advanced care to injured personnel. The tiered approach ensures that personnel are trained to the appropriate level based on their roles and responsibilities within their respective units or specialties.

3-6. Tier 1, or All Service member skills, encompasses fundamental skills taught to all Service members, focusing on basic hemorrhage control and recognition of serious injuries. This tier aims to equip every Service member with essential lifesaving capabilities, enabling them to provide immediate care to injured comrades in the battlefield environment. Training in Tier 1 skills is critical for ensuring that all Service members are prepared to respond effectively to medical emergencies, regardless of their primary occupational specialties.

3-7. Tier 2, or CLS skills, involves more advanced care aimed at addressing the most common causes of battlefield fatalities. This tier emphasizes recognizing, preventing, and communicating life-threatening complications to medical personnel. CLS training provides select Service members with enhanced capabilities to stabilize casualties and facilitate their evacuation to higher levels of care, thereby improving overall survival rates in combat environment.

3-8. Tier 3 or Combat Medic and Corpsman skills contains the skills that are much more advanced and focus on invasive care. This tier of skills requires significantly more medical knowledge and skills than self-aid, buddy aid, or CLS.

3-9. Tier 4 or Combat Paramedic and Provider skills are for combat paramedics and advanced providers. This tier provides the most sophisticated care to keep wounded Service members alive and get them to definitive care.

ADDITIONAL REQUIRED SKILLS FOR ARMY PERSONNEL

3-10. The U.S. Army has proactively expanded upon the foundational skills outlined in DODI 1322.24 by identifying additional Tier 1 and Tier 2 competencies critical for its personnel. This initiative highlights the Army's commitment to enhancing the skill sets of its members, ensuring they are well-prepared for the challenges they may face. The tracking and documentation of these competencies are meticulously managed through the Digital Training Management System. This system serves as the official record for all training activities within the Army, making it imperative that every course and task completion is recorded accurately in the Digital Training Management System. This process will continue until the Digital Training Management System is replaced by the forthcoming Army Training Information System, which promises to further streamline and optimize training management and record-keeping within the Army.

Tier 1 Skills for Army Personnel

3-11. The MEDCoE consistently evaluates and updates all relevant initial entry training and initial military training medical lesson plans to ensure compliance with TCCC-All Service member requirements. Soldiers who are not certified during initial training will receive certification upon arrival at their first assigned unit. This ensures that all personnel are adequately trained in TCCC before deployment.

3-12. According to DODI 1322.31, the frequency of TCCC-All Service member training is determined by the Military Service, with the condition that all personnel receive training prior to deployment. Additionally, recertification and refresher training for Component 1 (Active Army) are required annually, while Component 2 (Reserve) and Component 3 (National Guard) require biannual recertification. Detailed task requirements and standards for the Army can be accessed through Central Army Registry website.

3-13. In alignment with DODI 1322.24, the U.S. Army has integrated Warrior Tasks and Battle Drills for *Perform Tactical Combat Casualty Care*, into its training protocols. More information on Warrior Tasks and Battle Drills can be found in STP 21-1-SMCT and on the Central Army Registry website. This initiative ensures that Soldiers are proficient in providing critical medical care during combat situations, enhancing overall readiness and effectiveness in the field.

Tier 2 Skills for Army Personnel

3-14. Immediate and forward medical care is crucial in dynamic battlefield environments to prevent fatalities among wounded Soldiers. Given the challenges of reaching all casualties promptly, CLS plays a vital role in providing advanced lifesaving interventions. CLS training equips Soldiers with necessary skills and knowledge to stabilize various types of casualties and to mitigate the worsening of injuries until medical assistance arrives.

3-15. It is essential to have at least one CLS per squad, crew, team, or similar-sized unit, ensuring readiness for medical emergencies. Recertification of CLS personnel every 12 months at the unit level is mandated by AR 350-1, ensuring proficiency and readiness. Additionally, comprehensive CLS and CASEVAC training for all Army personnel is imperative for small-unit operations with widely dispersed forces.

Combat Lifesaver Training

3-16. The U.S. Army MEDCoE developed the CLS course to ensure comprehensive medical support for Army personnel. This course is used by both active duty and reserve component Soldiers across all units throughout the force. Implementation of CLS training within corps, divisions, brigades, and battalions is overseen by designated credentialed providers, who supervise CLS programs and ensure adherence to established standards.

3-17. Training support packages, containing student and instructor materials, facilitate CLS training at the unit and organizational levels. Training and testing are conducted based on the tasks, conditions, and standards set by the United States Army Training and Doctrine Command (also known as TRADOC). Proof of CLS course completion is documented IAW AR 600-8-104, with certificates issued to Soldiers who successfully complete the training, and records maintained in the Army Training Requirements and Resources System.

3-18. Locations conducting CLS training are responsible for documenting completion through local systems or, if necessary, submitting records to the U.S. Army MEDCoE Instructional Technology Support Services Help Desk. The CLS course allows for retests following failed examinations, with a minimum of one retest opportunity provided. However, course managers may establish guidelines for retesting based on individual circumstances.

3-19. Instructors responsible for CLS training must meet specific qualifications outlined in the individual critical task list Task: 081-68W-2003. Leaders are tasked with ensuring adequate training opportunities for nonmedical personnel within their units, coordinating with medical personnel for remedial training as needed. Enrollment procedures, facility requirements, course materials, and equipment procurement are essential considerations for successful CLS training implementation, ensuring Soldiers receive effective medical training and support.

Combat Lifesaver Recertification

3-20. CLS skills are perishable and must be recertified every 12 months to ensure proficiency in the performance-tested tasks. Unit instructors utilize the standards provided in the training support package for recertification purposes. Personnel staff, rather than medical platoon leaders, are responsible for updating personnel records after CLS training completion.

3-21. Leaders should maintain records of CLS certification for all units under their supervision and notify unit commanders about upcoming certification expiration dates. Proper planning for recertification testing involves ensuring instructors have appropriate guides and tests, access to necessary facilities, personnel, equipment, and supplies. Personnel records must be updated following successful CLS training, a task entrusted to the personnel staff.

3-22. To sustain skill proficiency, CLS training should be integrated into home station training activities, including field-training exercises, and deployment preparations. Additionally, CLS skills should be exercised during deployment for training, such as rotations through combat training centers. This ongoing training ensures that CLS personnel remain competent, and ready to provide lifesaving care in diverse operational environments.

Combat Lifesaver Bag

3-23. Each certified CLS should ideally be equipped with a CLS aid bag, with its contents carefully regulated to prevent unauthorized alterations. While exceptions exist, for The United States Army Special Operations Command surgeons, modifications to aid bags must adhere to validated mission requirements and receive approval from The United States Army Special Operations Command Defense Collaboration Services, Surgeon. All necessary Class VIII supplies and materials for CLS operations are procured through standard supply channels, with detailed information available in Appendix A regarding CLS bag supplies.

3-24. CLS aid bags are allocated to CLS personnel solely upon deployment, both for training exercises and actual missions. It remains the responsibility of each CLS to maintain their aid bag according to the prescribed packing list, ensuring all items are serviceable, and within their expiration dates. In combat scenarios, where medical supplies may be rapidly depleted, emergency resupply options include combat medics, ambulances, BAS, or nearby MTFs. However, it's important to exhaust supplies from the casualty's first aid kit before tapping into resources from the CLS bag.

ADDITIONAL REQUIRED SKILLS FOR MARINE CORPS PERSONNEL

3-25. The Marine Corps has expanded its training curriculum to include additional Tier 1 and Tier 2 skills, building on the foundation set by DODI 1322.31. This strategic enhancement ensures that Marines are equipped with a broader set of capabilities, tailored to meet the evolving demands of modern military engagements and operational responsibilities. The identification of these extra skills underlines the commitment of the Marine Corps to maintain superior readiness and effectiveness in all aspects of its mission.

Tier 1 Skills for Marine Corps Personnel

3-26. In accordance with TCCC-All Service Member Course curriculum developed by the JTS, the USMC will focus on the skills prescribed for All Service Members and the following additional skills described in subsequent chapters:

- Apply Rigid Eye Shield (paragraph 10-1).
- Perform Casualty Burn Care (paragraph 13-9).
- Record Treatment on a DD Form 1380 (paragraph 16-13).

3-27. See Appendix A for details on the USMC first aid kit.

Tier 2 Skills for Marine Corps Personnel

3-28. In accordance with TCCC CLS course curriculum developed by the JTS, the USMC will focus only on the skills prescribed for CLS already described in this publication.

3-29. A CLS kit is issued to CLS-trained Marines to assist the Navy Corpsman in multiple casualty situations or act as the primary first responder when HMs are not available, as in small unit operations.

3-30. The CLS kit is a portable lightweight bag containing medical supplies that enables CLS-trained Marines to treat extremity hemorrhaging (treat one double limb and two single limb casualties), sucking chest wounds, and clear a wounded Marine's airway.

3-31. Using units are required to replenish the CLS Kit through standard supply (military standard requisitioning and issue procedures) requisitions; however, during combat, medical supplies can be quickly depleted. Additional supplies can be obtained from BASs or other nearby MTFs or in Line 4 of the 9-line MEDEVAC request.

Note. No Service manuals are required for the CLS Kit.

ADDITIONAL REQUIRED SKILLS FOR AIR FORCE PERSONNEL

3-32. The USAF assigns the same level of importance to TCCC training as its fellow Services, ensuring a standardized approach across the military. The USAF has not distinguished additional skills within the tier 1 and tier 2 categories of TCCC competencies. For comprehensive TCCC information and specific training requirements, USAF personnel are advised to refer to DODI 1322.24, DODI 1322.31 and the Deployed

Medicine website. It is also recommended that members familiarize themselves with any relevant Service-specific TCCC guidelines. The information provided in this publication is invaluable for general understanding and can be applied to the TCCC procedures, use of the expedient techniques detailed in each section, and for the administration of first aid for injuries not specifically covered under the TCCC protocols.

TACTICAL COMBAT CASUALTY CARE TRAINING

3-33. Continuous evolution characterizes TCCC, CLS, medical treatment, and interventions, aligning with shifts in medicine and operational contexts. To stay informed of the latest changes and updates, individuals should regularly consult the deployed medicine website and the United States Army Medical Material Agency website. Service members and Department of Defense-expeditionary civilian personnel, whether augmentees to operational units or individual augmentees, undergo comprehensive medical readiness training tailored to their deployment circumstances. This is done by encompassing triage, treatment, and management of various health hazards, including CBRN threats.

3-34. A profound grasp of TCCC fundamentals can significantly impact outcomes by conserving fighting power, mass, and momentum to support the current fight and potentially saving lives, averting permanent disability, and minimizing hospitalization durations. Emphasizing what actions to take, what to avoid, and when to act. TCCC aims to treat casualties effectively, prevent further harm, and uphold mission completion. Central to TCCC's objectives is addressing and mitigating the causes of preventable death on the battlefield, with training targeting specific areas such as bleeding control, including junctional and noncompressible bleeding, as well as addressing tension pneumothorax and airway complications.

3-35. Leveraging technology for enhanced accessibility and continuous learning, the deployed medicine website and mobile application by the Defense Health Agency offer invaluable resources. These platforms furnish on-demand multimedia content, facilitating lifelong learning and real-time updates in TCCC training and education. Among the resources provided are training and education materials, the TCCC-All Service member curriculum, updated videos, podcasts, and downloadable CPGs, empowering personnel with comprehensive support for optimal preparedness and skill development.

TACTICAL COMBAT CASUALTY CARE PHASES

3-36. The TCCC framework encompasses three critical phases of care tailored to the dynamic challenges of battlefield medicine. These phases, structured to address varying levels of urgency and complexity, are Care Under Fire (CUF), Tactical Field Care (TFC), and Tactical Evacuation Care (also known as TACEVAC). Mastery of these phases equips military units and leaders with the proficiency needed to deliver prompt and efficient medical assistance to injured Service members in combat scenarios. Such skills play a pivotal role in preserving life, mitigating the potential for long-term disability, and optimizing the prospects of full recovery for wounded personnel.

CARE UNDER FIRE

3-37. **Care under fire is the first phase of care in tactical combat casualty care where very limited care is provided for the casualty while the enemy threat is still active. Also known as CUF or care under threat.** During CUF, the primary focus is on providing immediate life-saving care while still under enemy fire. This phase involves skills such as returning fire, moving the casualty to a safer location, and controlling severe bleeding using techniques like TQs. The goal is to stabilize the casualty quickly to prevent further injury or death. In CUF, available medical equipment is limited to what is carried by individual responders or casualties, and the only medical priority is addressing massive extremity bleeding due to the ongoing threat in a potentially chaotic and dangerous situation.

3-38. The order of actions during CUF is mission-variable dependent, with scene safety being paramount. For instance, if the threat is suppressed and the tactical situation allows massive life-threatening bleeding and extremity bleeding should be addressed prior to patient movement from the POI. However, each scenario is different, and the ordering of actions could vary accordingly. It's crucial to remember that in the combat environment, there is no "time-out" when casualties occur. Rapid medical and tactical decisions must be made simultaneously, as good medicine can sometimes be bad tactics. Potentially, endangering the lives of responders and jeopardizing the mission's success.

3-39. The tactical situation dictates the initial actions during CUF, with little time available for casualty care while under effective enemy fire. Prioritizing suppressing hostile fire and gaining fire superiority is crucial to minimizing the risk of injury to personnel and preventing further harm to the casualty. Wounded Service members should be directed to continue returning fire, moving quickly to nearby cover, and performing self-aid if able. It's essential not to become a casualty oneself, assessing the situation, suppressing enemy fire, and developing a plan before moving to care for a casualty under fire.

3-40. In CUF, early control of severe bleeding is the number one medical priority, as extremity hemorrhage is the most frequent cause of preventable battlefield deaths. Major vessel injuries can quickly lead to shock and death, making it crucial to identify and address life-threatening bleeding promptly. Only massive bleeding warrants intervention during CUF, with treatment of non-life-threatening bleeding deferred until the TFC phase. Despite the grave danger posed by enemy fire, applying a hasty TQ quickly and effectively remains essential in this phase (if this tactical situation allows) to save lives and prevent further casualties.

Scene Safety

3-41. CUF is defined as the aid rendered at the trauma scene while there is still an active threat. The scene in CUF is never safe, however, the active threat can be reduced. The responder can help others to reduce the threat (return gunfire, or firefighting) and to enable rescuers to reach the casualty and move the casualty to cover. The responder should not risk their personal safety or the safety of the casualty if the threat has not been reduced.

Control of Massive Bleeding

3-42. In the CUF phase, the primary focus is on controlling massive bleeding to prevent hypovolemic shock, a critical and potentially fatal condition. Effective management of severe bleeding is essential to stabilize the casualty and ensure survival until further medical treatment can be administered. For detailed strategies and techniques on controlling massive bleeding, see Chapter 4, which provides comprehensive guidance on the procedures and equipment necessary for this vital emergency intervention.

Casualty Movement

3-43. If a casualty is unresponsive or unable to move, a rescue plan should be devised and executed if tactically feasible. Do not put two people at risk if it can be avoided. If cover is not available or the wounded Service member cannot move to cover, they should lie flat and motionless. For more information on casualty movement techniques see chapter 2, paragraph 2-6 of this publication.

Proper Communication and Documentation

3-44. Communication is vital in providing effective care for casualties, involving interaction with the casualty, unit leaders, and on-scene medical personnel. Rescuers should communicate with the casualty as soon as they can safely reach them, by assessing their level of consciousness and orientation. It's crucial to reassure conscious casualties and request assistance from other responders or medics when available, as rescuers are rarely alone in such situations.

3-45. Documentation of all care provided to casualties is essential, primarily utilizing DD Form 1380. This form should include details such as the casualty's name, sex, time of injury, the nature of the injury, along with findings from rapid casualty assessments and the medical aid provided. This information is critical for the continuity of care and should be securely attached to the casualty or recorded directly on them if the form is unavailable.

3-46. Securing DD Form 1380 on the casualty, by attaching it to their belt loop or placing it in their pocket ensures it accompanies them to the next level of care. If the form is unavailable, pertinent information can be directly written on the casualty's body or on a strip of medical tape. This documentation is vital for ensuring comprehensive and effective care throughout the casualty's treatment process. (See figure 15-2 on page 154 for an example of a completed DD Form 1380).

Combat Lifesaver Role in Care Under Fire

3-47. In the context of CUF, the CLS assumes a role similar to individuals providing self-aid. Their primary objective is to suppress hostile fire to minimize the risk of injury to personnel and further harm to wounded Service members. The CLS utilizes available medical equipment, which is typically limited to what the individual responder, casualty's first aid kit, or CLS bag carries. It's emphasized that the first responder and CLS should prioritize the use of the casualty's first aid kit before resorting to other resources. See Appendix A, paragraph A-7 for further discussion on the CLS bag.

3-48. Given the urgency of the situation, the sequence of actions in CUF is dictated by the tactical environment. With limited time available for casualty care amidst enemy fire, suppressing hostile fire and gaining fire superiority are prioritized to protect both personnel and casualties. This may involve personnel assisting in returning fire instead of immediately attending to casualties, especially if the casualty can still engage in combat. Wounded Service members exposed to enemy fire are directed to seek cover, perform self-aid if possible, and continue returning fire while moving to safety.

3-49. In CUF, the primary focus is on recognizing and halting massive bleeding while seeking cover to shield oneself and the casualty from hostile fire. Signs of massive bleeding, such as pulsing or steady bleeding from wounds or traumatic amputations, must be promptly addressed. The immediate application of a hasty TQ is recommended if indicated, as injuries to major veins or arteries can lead to shock or death within minutes. Extremity hemorrhage, a common cause of preventable combat death, underscores the critical role of TQs in saving lives during CUF scenarios.

3-50. Prompt application of TQs to halt life-threatening bleeding not only saves lives but also enables injured Service members to continue fighting while awaiting further medical care and evacuation. It's emphasized that all Service members with access to a first aid kit should be trained in TQ use, highlighting the importance of preparedness and training in combat lifesaving techniques.

TACTICAL FIELD CARE

3-51. ***Tactical field care is the second phase of care in tactical combat casualty care. Care provided once the threat has been neutralized, the scene is safe, or the casualty has moved or been moved out of the immediate threat situation. Also known as TFC.*** This phase permits more extensive medical interventions, including the assessment and management of airway, breathing, circulation, and other injuries. Essential skills such as wound dressing, pain management, blood or blood product resuscitation, and fluid resuscitation for burns are pivotal during this phase. TFC also encompasses scenarios in a combat or tactical environment not involving direct enemy fire, such as parachute injuries in a combat zone.

3-52. Transitioning from CUF, TFC provides the opportunity for a more deliberate approach to casualty assessment and treatment, given the time and relative safety available. However, the tactical situation remains dynamic, and the potential for reverting to a CUF scenario always exists. In TFC, both the casualty and the first responder or CLS should maintain cover and concealment. The primary management plan involves maintaining tactical situational awareness, triaging casualties as needed, and conducting a thorough casualty assessment.

3-53. Mission personnel are tasked with establishing a security perimeter according to unit tactical SOP or battle drills, maintaining security, and remaining vigilant for potential threats from hostile forces. Casualties in shock or under the influence of controlled substances pose risks to themselves and others. Hence, those with altered mental status should have their weapons cleared and secured, communications secured, and sensitive items redistributed.

3-54. During the TFC phase, responders have more time and relatively greater safety to administer further medical care. They may need to continue treatment until medical personnel arrive, assisting them with continued treatment and preparation for evacuation. However, the anticipated time for evacuation to the next level of care may vary based on the operational situation, requiring CLS to remain prepared to reengage the enemy and resume the mission.

3-55. While medical personnel prioritize casualty treatment, coordination between responders and unit leaders is crucial for requesting evacuation assets based on the operational situation. Despite the expanded scope of care in TFC, available medical equipment remains limited to what is carried into the field by the

casualty's first aid kit, the CLS, and other responders' first aid kits or supplies in a medic's aid bag. It's emphasized that supplies within the casualty's first aid kit should be used first before resorting to other available supplies.

TACTICAL EVACUATION CARE

3-56. ***Tactical evacuation care is the third phase of care in tactical combat casualty care. Care provided during transportation from the battlefield to a medical treatment facility.*** Tactical evacuation care is the phase of TCCC focused on preparing the casualty for evacuation to a higher role of care, typically an MTF. This phase encompasses both preparing the casualty for evacuation and executing the evacuation process while ensuring continued monitoring and treatment of the casualty's condition. Skills required during tactical evacuation care include managing the casualty's airway and breathing, maintaining circulation, and providing ongoing medical interventions as necessary.

3-57. The NATO term of tactical MEDEVAC refers to the intra-theater movement of patients between MTFs within the joint operational area. This process involves transferring patients from one MTF to another using ground, rotary, or fixed-wing assets, along the lines of communication within the joint operational area. In the U.S. Army, this includes MEDEVAC operations between different roles of care, from Role 1 through Role 3, supported by both ground and air ambulances. However, the term "TACEVAC" should not be used to describe the tactical evacuation care phase of TCCC and to prevent confusion between the evacuation process itself and a specific phase of TCCC.

3-58. The primary principle in tactical evacuation care is to continue monitoring and providing appropriate assessment and treatment until the casualty is transferred to medical personnel or reaches a higher level of care. During this phase, first responders or CLSs conduct a deliberate assessment to identify and treat previously unrecognized life-threatening injuries. While similar to TFC in many aspects, the presence of additional medical personnel and equipment on the evacuation asset may enable the provision of more extensive care and casualty management.

3-59. Non-medical first responders or CLSs are not expected to provide care during evacuation. However, if they do, the skills used in CUF and TFC still apply. Pre-evacuation procedures involve documenting all assessments and care provided during CUF and TFC on DD Form 1380. Additionally, an evacuation request using the 9-line format is initiated according to unit SOP to initiate MEDEVAC.

3-60. During staging, each casualty is reassessed, and their medical condition is monitored to ensure they receive necessary care while waiting for evacuation. This step is vital for maintaining the stability of casualties and adjusting medical interventions as needed before they are transported to a higher level of medical care. Communication plays a key role during this phase, as it helps reduce the casualties' anxiety and provides them reassurance about the care, they are receiving. While, keeping the casualty informed and as calm as possible under the circumstances.

3-61. Overall, tactical evacuation care ensures that casualties receive continuous monitoring, assessment, and treatment during the evacuation process until they reach a higher level of care. It involves coordination between medical personnel, tactical leaders, and other responders to ensure a smooth and efficient evacuation process in challenging tactical environments.

3-62. The proper procedure for performing a 9-line MEDEVAC request is critical in tactical environments to ensure timely evacuation support for casualties. Here's a detailed outline of the steps involved:

- **Collect Information:** Gather all necessary information required for the MEDEVAC request, including the casualty's condition, location, and other pertinent details, using authorized brevity codes.
- **Initiate Contact:** Clearly announce "I HAVE A MEDEVAC REQUEST" over the radio frequency designated for communication with the unit controlling evacuation assets. Wait for one to three seconds for a response. If there is no response, repeat the statement.
- **Transmit MEDEVAC Request:** Contact the unit controlling evacuation assets using the proper call sign and frequency assignment as specified in the signal operating instructions from an automated net control device.
- **Sequence and Format:** Transmit the MEDEVAC information in the proper sequence, stating all line numbers clearly. Line numbers 1 through 5 must be transmitted within 25 seconds during the initial

contact with the evacuation unit. Lines 6 through 9 can be transmitted while the aircraft or vehicle is en route.

- Follow Procedure: Adhere to the procedure provided in the explanation column of the MEDEVAC request format, ensuring all required information is transmitted accurately.
- Radio Procedures: Pronounce letters and numbers according to appropriate radio procedures to ensure clarity and understanding.
- Use "BREAK": Use "BREAK" on long lines of information to break up the transmission and reduce its length, facilitating better reception.
- End Transmission: Conclude the transmission by stating "OVER" to indicate the completion of your message and readiness to receive further instructions or acknowledgment from the evacuation unit.
- Maintain Radio Communication: Keep the radio on and listen for additional instructions or contact from the evacuation unit, remaining available for any further communication regarding the MEDEVAC request.

3-63. By following these steps and adhering to proper radio protocols, Service members can effectively transmit a 9-line MEDEVAC request, facilitating prompt and efficient evacuation support for casualties in tactical environments. For additional information on requesting MEDEVAC and the MEDEVAC standard format refer to ATP 4-02.2.

TACTICAL COMBAT CASUALTY CARE DURING OPERATIONS

3-64. During operations, the battlefield is often widely dispersed and rapidly moving, posing challenges for the rapid acquisition of casualties and medical treatment. As a result, medical platoon personnel may face limitations in providing immediate, far-forward medical care. To address this, the plan is to integrate CLS into the system of providing care for wounded Service members. The three primary goals of TCCC are to treat casualties, prevent additional casualties, and complete the mission.

3-65. When a nonmedical Service member encounters an unconscious or injured casualty, they must quickly assess the situation and casualty's condition to determine if it's safe and if immediate first aid measures are necessary to prevent further injury or death. They should ascertain if trained medical personnel are available or direct others at the scene to call for or locate trained medical personnel. Once first aid is initiated, efforts should continue without interruption until medical personnel take over. Service members may also need to rely on their own first aid skills for self-aid. The CLS serves as a link between the self-aid or buddy aid skills taught to all Service members and the medical training provided to medical personnel.

3-66. Immediate medical care in far-forward positions is crucial in a dispersed and dynamic operational environment to prevent casualties from succumbing to wounds. In situations where medical personnel may not reach all wounded Service members promptly, the CLS plays a vital role in providing advanced lifesaving measures beyond basic self-aid or buddy aid. A well-trained CLS can stabilize various types of casualties and mitigate the deterioration of a wounded Service member's condition until medical personnel arrive. However, functioning as a CLS is a secondary mission undertaken when the tactical situation allows.

3-67. One significant advantage of the CLS is their proximity to injured unit members. While medical personnel may take several minutes or longer to reach casualties, especially when attending to severely wounded individuals, the CLS can provide immediate lifesaving care, such as controlling severe bleeding and performing needle chest decompressions for tension pneumothorax.

3-68. When not engaged in combat duties, CLS personnel may also assist medical personnel in providing care and preparing casualties for evacuation. The role and responsibilities of the CLS are delineated in different phases of care:

- In CUF, the CLS responds by suppressing hostile fire to minimize the risk of injury to personnel and prevent additional harm to previously injured Service members.
- In TFC, the CLS maintains security and situational awareness while attending casualties and preparing them for evacuation.
- In Tactical Evacuation, support the Combat Medic or Corpsman, as directed.

3-69. CLS personnel are specifically trained to tackle the primary causes of battlefield fatalities, which include massive hemorrhage and airway or respiratory issues. They possess the skills necessary to prevent complications and manage other injuries that, while not immediately life-threatening, could become serious

if left untreated. This training ensures that CLS personnel are well-prepared to provide critical first aid in combat situations, enhancing the survival chances of those injured in the field.

KEY FACTORS INFLUENCING TACTICAL COMBAT CASUALTY CARE

3-70. Prehospital care in the combat setting differs significantly from the civilian setting due to various factors that influence TCCC. These factors include:

- **Hostile Fire:** Combat environments are characterized by the presence of hostile fire, posing an immediate threat to both casualties and responders. This necessitates the integration of tactical considerations into casualty care protocols.
- **Tactical Considerations:** Tactical priorities and objectives often dictate the approach to casualty care. The safety of personnel and the completion of the mission take precedence, influencing how and when medical interventions are performed.
- **Environmental Considerations:** Combat operations can occur in diverse environments, ranging from urban areas to remote and hostile terrain. Responders must adapt their care techniques to address environmental challenges such as extreme weather conditions, limited visibility, and difficult terrain.
- **Wounding Patterns:** In combat, injuries are often caused by explosive devices, firearms, and other weapons unique to warfare. Understanding these specific wounding patterns is crucial for effective casualty management and prioritization of care.
- **Equipment Constraints:** Medical resources and equipment may be limited or unavailable in the combat setting, requiring improvisation and prioritization of essential interventions. Responders must work within the constraints of available resources to provide optimal care.
- **Delays Reaching Higher Roles of Care:** Evacuation to higher levels of care may be delayed in combat situations due to operational factors, such as ongoing enemy engagement or logistical challenges. This necessitates prolonged on-site care and management of casualties by first responders.
- **Level of First Responder Training and Experience:** The training and experience level of first responders, including CLS and medical personnel, significantly impacts the quality and effectiveness of prehospital care in combat. Adequate training ensures responders are proficient in TCCC principles and can adapt to dynamic battlefield conditions.

3-71. Overall, these factors shape the unique challenges and considerations involved in providing prehospital care in the combat setting, highlighting the importance of specialized training and protocols like TCCC for managing battlefield casualties.

FIRE SUPERIORITY IN TACTICAL COMBAT CASUALTY CARE

3-72. Fire superiority is paramount on the battlefield, serving as the best medicine to mitigate threats and protect personnel. In the midst of combat, all individuals should prioritize returning fire and seeking cover. The sequence of initial actions must align with the tactical circumstances at hand. Given the limited time available amid effective enemy fire, providing casualty care becomes a secondary consideration.

3-73. Suppression of hostile fire and establishment of fire superiority are critical objectives aimed at reducing the risk of harm to both personnel and casualties while accomplishing the mission. In certain situations, individuals may be required to engage in returning fire rather than immediately attending to casualties, including wounded personnel who retain combat capability. Those injured and exposed to enemy fire should be instructed to continue engaging the enemy, seek cover promptly, and administer self-aid measures if feasible. These actions are essential for maintaining operational effectiveness and safeguarding lives amidst combat conditions.

CASUALTY SELF-AID

3-74. The skills outlined in this publication equip Service members to administer initial medical care at the POI, sustaining casualties until further assistance arrives. Responsive casualties should be directed by the first responder or CLS to return fire, apply self-aid, re-engage, and seek cover if feasible.

3-75. If cover is inaccessible or the wounded Service member cannot move to safety, they should lie flat and remain still. If self-aid is beyond the casualty's capability or cover is unattainable, implementing a rescue

plan, if feasible, becomes imperative. A hasty TQ should be applied promptly and tightly to stop the bleeding (ideally within one minute), as the tactical situation dictates. The urgency of a TQ application is underscored by the potential for a casualty to succumb to exsanguination in as little as three minutes.

3-76. Initial medical care within the military context is administered by Service members leveraging their trained basic skills, encompassing self-aid or buddy aid. These skills encompass immediate life-saving interventions, as well as treatment for non-combat related injuries or conditions affecting functional capacity.

3-77. The CoTCCC has identified five critical life-saving skills, namely rapid casualty assessment, TQ application, wound packing with hemostatic dressing, application of pressure bandages, and basic airway maneuvers for airway management.

3-78. Vital to sustaining life is maintaining an adequate blood volume to facilitate oxygen transport to tissues. Halting bleeding and ensuring open airways are pivotal first aid measures, as oxygen deprivation swiftly precipitates death.

3-79. Shock denotes inadequate tissue and organ perfusion, which is left untreated, can culminate in fatality despite the non-lethal nature of the precipitating injury or condition. Preventing shock is a cornerstone of first aid, significantly enhancing casualty survival prospects.

3-80. Effective wound management entails safeguarding injuries against infection, which can impede recovery and exacerbate morbidity. Understanding basic first aid principles include proficiency in wound dressing to avert infection and minimize contamination.

3-81. TCCC serves as the benchmark in prehospital battlefield medicine. Rapid casualty assessment and application of life-saving skills are pivotal in preserving lives. During casualty response, Service members are urged to take decisive action, document provided medical aid for continuity of care and facilitate evacuation to expedite access to advanced medical treatment.

INITIAL ENCOUNTER

3-82. In many cases, the initial care a Service member receives is provided through self-aid, buddy aid, or by CLS. The effectiveness of the actions taken during the initial encounter with a casualty, along with proper movement and transportation, are crucial elements in the successful application of TCCC. Sound decision-making and proficient execution of TCCC tasks have a significant impact on the health outcomes of casualties.

3-83. Upon first encountering a casualty, it is imperative for the responder to assess the situation, understand the nature and severity of the injuries, and determine the appropriate TCCC interventions required quickly and accurately. Accurate assessment is essential not only for the welfare of the casualty but also for the safety of the responder. For example, in the case of electrocution, the responder must avoid direct contact with the casualty to prevent becoming a casualty themselves. This initial assessment forms the foundation for all subsequent care and decisions in the management of battlefield casualties.

TACTICAL COMBAT CASUALTY CARE CLINICAL PRACTICE GUIDELINE

3-84. The JTS is dedicated to establishing the standard of care for all aspects of battlefield medicine, including prehospital care. It operates as a Department of Defense Center of Excellence for Trauma and is responsible for providing the CPGs and performance improvement initiatives for all levels of military trauma care. Under JTS, the CoTCCC is tasked with developing the TCCC CPG.

3-85. The TCCC CPG serves as the benchmark for contemporary battlefield medical care. While the guidelines serve as recommendations, they do not replace clinical judgment. Given that the guidelines undergo periodic review and updates, it is essential for healthcare providers to regularly review them and integrate any changes into their training. For more detailed information on the TCCC CPG, reference the deployed medicine website.

3-86. The content provided in this publication is accurate as of the 23 March 2026; however, standards of care may evolve over time. Continuous learning and training are crucial for Service members. The Deployed Medicine website and its accompanying mobile application, both endorsed by the DOD's JTS, serve as

authoritative resources for TCCC medical training. This application allows users to stay informed about the TCCC guidelines and updates.

TRIAGE OR SORTING OF CASUALTIES

3-87. *Triage* is the process of sorting casualties based on need for treatment, evacuation, and available resources (FM 4-02). It is performed in each role of medical care. Triage is the most effective means of establishing order and the best method for providing the greatest benefit to the greatest number of patients within the limitations of time, distance, and capability. Triage is a constant process as casualties move within and through the roles of medical care.

3-88. During incidents that generate mass casualties, the need to withhold treatment from a severely injured casualty, in-order to redirect lifesaving resources to casualties whose wounds are more survivable, is a difficult decision to make. While such decisions have been infrequent in recent conflicts, they may become more common during large-scale combat operations or situations where evacuation is difficult. The overarching goal of triage is, aiming to provide the greatest good to the greatest number and should guide these difficult decisions. The mission, immediate tactical situation, followed by medical necessity should prioritize resource allocation and should not be influenced by a casualty's nationality or combatant status.

3-89. Triage, or sorting of casualties, is essential to maximize limited resources. It involves evaluating, categorizing, and prioritizing casualties for medical treatment and evacuation. Casualties are sorted into categories based on their treatment urgency: immediate, delayed, minimal, and expectant. The goal is to return the greatest number of Service members back to duty while preserving life, limb, and eyesight.

Immediate Category

3-90. Immediate injuries are those that require attention within minutes to two hours to prevent death or major disability. Examples include airway obstruction, tension pneumothorax, uncontrolled hemorrhage, torso, neck or pelvis injuries with shock, head injuries requiring emergent decompression, threatened limb loss, collection of blood behind the eye, and multiple extremity amputations.

Delayed

3-91. Delayed injuries are those requiring surgery but permitting delay without significant risk to life, limb, or eyesight. Examples include blunt or penetrating torso injuries without shock, fractures, soft tissue injuries, facial fractures without airway compromise, eye injuries, and survivable burns without immediate threat of life (airway, respiratory) or limb.

Minimal

3-92. Minimal injuries are those that can care for themselves or need minimal medical care. Examples include minor lacerations, abrasions, small bone fractures, and minor burns. These casualties may assist with the movement or care of more severely injured individuals.

Expectant

3-93. Expectant injuries are those casualties that have injuries that would overwhelm current medical resources, at the expense of treating salvageable patients. These casualties should receive comfort measures and intermittent reassessment. Examples include casualties without vital signs or signs of life, open pelvic injuries with controlled bleeding and shock, life-threatening burns, high cervical spinal cord injuries affecting respiration, and destructive cardiac injuries.

CASUALTY ASSESSMENT

3-94. When learning, training, or teaching emergency medical care for a medical condition the airway, breathing, and circulation is used to prioritize treatment and may still be applicable in some situations. Service members experiencing a trauma casualty often start with more problems than resources available. In these situations, casualty assessment and management in TCCC follows a deliberate framework to prioritize trauma care known as massive hemorrhaging, airway, respiration, circulation, and hypothermia (MARCH) - pain, wounds, antibiotics, and splinting (PAWS).

3-95. The implementation of TCCC in modern warfare reflects the military's commitment to enhancing the well-being and survival of personnel on the battlefield. TCCC provides a comprehensive approach to immediate medical intervention in combat environments, with a focus on the "MARCH" portion of the MARCH-PAWS treatment protocol. This protocol prioritizes the evaluation and control of:

- Massive Bleeding or hemorrhaging (the top priority) (see Chapter 4).
- Airway (see Chapter 5).
- Respiration (breathing) (see Chapter 6).
- Circulation (see Chapter 7).
- Hypothermia and head injuries (see Chapter 8).

3-96. The second portion of the MARCH-PAWS casualty assessment is PAWS, which is addressed after the MARCH portion. For instance, while treating a casualty, a first responder or CLS might identify a fractured arm during the circulation and second injury portion of the MARCH. However, splinting the fracture should occur during the PAWS portion of the assessment.

3-97. The primary focus of the casualty assessment is to swiftly identify and address life-threatening injuries before treating secondary injuries. For example, if a first responder or CLS were to treat a simple wound or splint a fracture before completing the full MARCH assessment, they might overlook life-threatening bleeding to a lower extremity. Although the simple wound or fracture would be addressed, the casualty would still be at risk of death due to the untreated life-threatening bleed. After addressing life-threatening injuries, the PAWS portion of the framework should be completed.

3-98. Following the MARCH-PAWS sequence guides Service members through the rapid casualty assessment and CLS through the trauma assessment. This involves either a rapid casualty assessment or a full tactical trauma assessment; both aimed at identifying and treating all life-threatening conditions and serious wounds. Rapid and accurate evaluation of casualties is crucial for providing effective first aid.

Rapid Casualty Assessment

3-99. When a casualty has transitioned from CUF into TFC, the Service member should treat all immediate and potential life threats discovered during the rapid casualty assessment. The rapid casualty assessment, performed by all Service members, is very similar to the tactical trauma assessment, which is performed by the CLS. Once the Service member has treated all immediate and potential life threats, the PAWS portion of the rapid casualty assessment should be performed, and all other injuries treated. These may include head injuries, eye trauma, fractures, and burns, among others.

Tactical Trauma Assessment

3-100. The CLS utilizes the tactical trauma assessment (a more in-depth assessment compared to the rapid casualty assessment) to reassess any hemorrhage control measures (hasty TQs) initiated in the CUF phase by first responders and identify any injuries that have not yet been treated. The tactical trauma assessment follows the MARCH-PAWS sequence, allowing the CLS to systematically evaluate and address any remaining life-threatening injuries and secondary injuries.

MEDICAL EQUIPMENT AND DOCUMENTATION FOR TACTICAL COMBAT CASUALTY CARE

3-101. Service members carry a large array of equipment, and survival on the battlefield depends heavily on the ability to properly use that equipment. One piece of equipment that is critically important is the joint first aid kit (JFAK), which is a standardized medical assembling.

Note. The Army uses the individual first aid kit (IFAK) instead of the JFAK.

Medical Equipment

3-102. The JFAK contains both medical and documentation materials. Essential medical supplies for TCCC are typically found in the Service member's first aid kit or Service-specific first aid kit. It's vital for all Service members to be familiar with the contents and location of these kits, as well as any maintenance procedures for trauma supplies required by their unit. See Appendix A for further discussion on first aid kits.

3-103. CoTCCC-identified equipment required to treat injuries and conditions to execute specific aspects of the MARCH-PAWS sequence and can be found in the following chapters—

- Chapter 4 Massive Bleeding.
- Chapter 5 Airway Control.
- Chapter 6 Respiration and Breathing control.
- Chapter 7 Circulation Control.
- Chapter 8 Hypothermia Prevention.
- Chapter 9 Secondary Assessment.

Medical Documentation

3-104. The JFAK includes documentation material, primarily the DD Form 1380. This form is the standard for documenting all assessment and treatment provided to casualties before evacuation or handoff to medical personnel.

3-105. Every piece of equipment in the JFAK is evidence-based, drawing from lessons learned in recent overseas contingency operations. The JFAK's contents support the provision of TCCC per current guidelines and enable first responders or CLS to address issues identified in various phases of casualty care. Each Service may have additional items in a Service-specific first aid kit. See Appendix A for Service-specific equipment.

Combat Lifesaver Aid Bag

3-106. The CLS carries a compact aid bag containing supplies for controlling bleeding, relieving tension pneumothorax, and performing other essential procedures. For a current list of supplies in the aid bag, refer to the medical materiel information portal website. CLS bags should be issued for training events and operational deployments.

3-107. It's the responsibility of each CLS to:

- Ensure their aid bag is stocked per the prescribed packing list.
- Verifying all stocked items are serviceable and not expired.
- Allocate aid bag supplies as needed.

Note. All supplies and materials required for CLS training should be obtained through normal supply channels.

3-108. The CLS should prioritize using the casualty's first aid kit before using supplies from the CLS bag if the casualty's kit is depleted.

3-109. Medical units should maintain stocks of medical materiel for resupplying CLS aid bags during operations. Resupply procedures during deployment mimic those during home station operations, involving bag inventories, supply requests, and emergency resupply from nearby medical units capable of helping.

MAINTENANCE AND RESUPPLY

3-110. Conducting precombat inspections or readiness checks is paramount for every piece of equipment carried or used by Service members. The readiness and serviceability of contents in first aid kits, CLS bags, or unit-specific kits can be a matter of life or death. Regular inspections before, during, and after training events and combat missions are crucial. Unit leaders should anticipate Class VIII resupply kits based on casualty estimates, and routine inventories must be conducted to ensure all required material is present.

Seals and wrappers

3-111. Replace items with broken or unsealed wrappers. If an item was vacuum sealed upon issuance but is no longer sealed during inspection, it should be replaced.

Check expiration dates

3-112. Medications and many medical-grade materials, such as hemostatic dressings, have expiration dates and lot numbers. Check all medications and medical-grade items for expiration dates and replace any expired items or those whose expiration dates do not extend beyond the expected deployment period. While items like TQs typically do not have expiration dates, ensure they are CoTCCC-approved, serviceable, reflect the current generation, and have not been replaced with newer models.

Unapproved or Fake Equipment

3-113. Only a few items in the first aid kit require specific recommendations from the CoTCCC. Ensure that kits are stocked only with CoTCCC-recommended or approved items and be wary of unapproved equipment. Beware of fraudulent manufacturers producing fake, misleading, or substandard medical equipment, especially TQs and hemostatic dressings.

CAUTION

Avoid deploying with equipment that is missing, expired, or previously used in training.

Unit-Specific Evacuation Equipment

3-114. Unit-specific evacuation equipment may include ropes, harnesses, packing materials for mountain or extreme cold conditions, life jackets, or flotation devices. It's essential to incorporate precombat inspections of these unit-specific items into the unit's SOPs. After ensuring all specialized gear is accounted for, proceed to inspect litters for proper functioning and serviceability. Additionally, check the litter straps to ensure their locking functions and placements are correct. Inspect special evacuation equipment according to the manufacturers' instructions or unit guidelines and standards.

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Chapter 4

Massive Bleeding Control

OVERVIEW OF MASSIVE BLEEDING CONTROL

4-1. The number one medical priority in TCCC is early control of severe bleeding. Extremity hemorrhage is the most frequent cause of preventable battlefield deaths. Understanding the human body and how it works along with how casualties survive in combat results in efficient and proper TCCC skills application and an increased survivability of combat casualties.

4-2. Injury to a major blood vessel can quickly lead to shock and death. Only life-threatening extremity bleeding warrants intervention during CUF (if the tactical situation allows). Casualties with injuries to large blood vessels in the extremities (such as the femoral artery in the thigh, or the brachial artery in the upper arm) can result in death in as little as three minutes if bleeding is allowed to continue. If life-threatening bleeding is encountered at the junctional areas (neck, axillary, or inguinal) treatment with a hemostatic dressing is not performed until the casualty transitions to TFC.

4-3. Treatment priorities should follow MARCH-PAWS, with massive bleeding (M) always being the most important. The treatment priorities are not a linear process; if at any time massive bleeding is not controlled, go back to M.

4-4. The initial casualty evaluation should be a rapid head-to-toe check for any unrecognized life-threatening bleeding (a blood sweep). This blood sweep is a systematic way to ensure rapid identification of any unrecognized life-threatening bleeding, and should include a visual and hands-on (palpation) inspection by checking the—

- Neck.
- Axillary (armpit).
- Inguinal (groin).
- Legs.
- Arms.
- Back.

4-5. The priority in caring for a casualty, should be to identify and stop life-threatening bleeding and prevent them from bleeding to death. Bleeding is considered life-threatening when—

- There is pulsing or steady bleeding from the wound.
- The overlying clothes are soaked with blood.
- Blood is pooling on the ground.
- There is a traumatic amputation of an extremity.
- Bandages or makeshift bandages used to cover the wound are ineffective and are steadily becoming soaked with blood.
- There was prior bleeding, and the patient is now in shock (unconscious, confused, and pale).

4-6. A TQ is immediately needed when any of the above examples are present on an extremity. Injury to a major vein or artery can result in shock or death from blood loss in minutes. Extremity (arm or leg) hemorrhage is a leading cause of preventable combat death. Using TQ(s) to stop the bleeding is essential to the survival of casualties with these types of injuries.

4-7. During CUF a TQ is the only medical intervention available to stop life-threatening bleeding from an extremity injury. Other wounds (such as neck, armpit, groin, or abdomen) are not treated during CUF. If the casualty is able, direct them to apply pressure to the wound as self-aid. Airway and other issues are not treated until the TFC phase.

4-8. It may be difficult to determine whether hemorrhage is life-threatening until the TFC phase when the wound can be exposed and evaluated. A suspected life-threatening hemorrhage is immediately treated.

4-9. During CUF, if a casualty shows to have massive bleeding, from an extremity, a hasty TQ is applied “high and tight” on the injured extremity and then cover should be acquired as quickly as possible. The

equipment in the casualty's first aid kit should be used and not that of the responder. Do not put a TQ directly over the knee or elbow, over a holster, or cargo pocket that contains bulky items.

4-10. Direct pressure can and should be used as a temporary measure until a TQ or dressing is in place. It is difficult to use direct pressure alone to control significant bleeding or while moving the casualty. Direct pressure can be used if treatment no longer maintains control of the bleeding while a new treatment is started.

4-11. In CUF, consider placing one or more hasty TQs in a "high and tight" manner to an arm or leg to control massive life-threatening bleeding. This should be accomplished in less than one minute. If the casualty is conscious, the responder should direct the casualty to render self-aid and apply a TQ to themselves, if they have one. A TQ applied in the CUF must be re-evaluated in TFC.

4-12. To effectively perform first aid, Service members need a fundamental understanding of the key body systems, including their structures and functions. Knowledge of the respiratory, circulatory, and musculoskeletal systems is crucial as it helps in assessing injuries and providing the appropriate first aid without causing further harm to the casualty. Familiarity with these systems also aids in recognizing when they are damaged or not functioning properly. Injuries to or compromises of these vital systems can lead to severe, permanent injury or even death. Therefore, a solid grasp of these systems not only supports effective first aid intervention but also helps prevent intensifying the casualty's condition.

CIRCULATORY SYSTEM

4-13. The heart and the blood vessels (arteries, veins, and capillaries) circulate blood throughout the body. The heart is divided into two separate halves, each acting as a pump. The left side pumps oxygenated blood (bright red) through the arteries into the capillaries. Nutrients and oxygen pass from the blood through the walls of the capillaries and into the cells. At the same time, waste products and carbon dioxide leave the cells and enter the capillaries. From the capillaries, the blood is carried through the veins to the right side of the heart and then into the lungs where it expels the carbon dioxide and picks up oxygen. Blood in the veins is dark red because of its low oxygen content. Blood does not flow through the veins in spurts as it does through the arteries. The entire system of the heart, blood vessels, and lymphatics is called the circulatory system.

BLOOD VESSELS

4-14. The body has a complex network of blood vessels that are crucial for blood circulation. These vessels are three main types. First, arteries take oxygen-rich blood from the heart and branch into smaller arterioles, eventually leading to tiny capillaries. Next, capillaries play a key role in exchanging oxygen, nutrients, and wastes between blood and tissues. The blood, now lacking oxygen, moves into small veins called venules, which combine to form larger veins. Lastly, veins are responsible for carrying blood back to the heart, completing the circulation process.

4-15. The blood vessels in the body are all connected and work together. They help deliver oxygen and nutrients to every part of the body and get rid of waste materials. This shows how each type of blood vessel depends on the others, highlighting their important roles in a Service member's overall health.

BLOOD

4-16. Blood is a mixture of plasma (liquid component), white blood cells, red blood cells, and platelets. The body contains about 5 to 6 quarts (about 5 liters) of blood. Once blood is pumped out of the heart, it takes 20 to 30 seconds to circulate through the body and return to the heart.

4-17. Blood performs essential functions as it circulates through the body. It delivers oxygen and essential nutrients (such as fats, sugars, minerals, and vitamins) to the body's tissues. It carries carbon dioxide to the lungs and other waste products to the kidneys for elimination from the body. In addition, it carries components that fight infection and stop bleeding.

WARNING

If bleeding is not stopped, excessive blood loss can quickly lead to the death of the casualty. It is imperative that first aid providers quickly determine if a casualty is losing blood. Many injuries can result in blood vessels being torn which result in excessive blood loss.

HEARTBEAT

4-18. The heart functions as a pump to circulate blood continuously through the blood vessels to all parts of the body. It contracts, forcing the blood from its chambers. The heart then relaxes, permitting its chambers to refill with blood.

4-19. The rhythmical cycle of contraction and relaxation is called the heartbeat. The normal heart rate (heartbeats per minute) varies based on the age and physical condition of the individual at rest. For example—

- Adults: 60 to 100 heartbeats per minute.
- Children (1 to 6 years): 70 to 120 heartbeats per minute.
- Infants (6 to 12 months): 80 to 140 heartbeats per minute.
- Infants (0 to 5 months): 90 to 140 heartbeats per minute.

Note. The typical pulse for Service members and athletes is 40 to 60 heart beats per minute and is lower than the average adult population due to physical conditioning.

4-20. It is common for a Service member's pulse to be elevated above the normal range during combat operations as this is both a stressful and high exertional state. Elevated pulse in combination with suspected injuries causing hemorrhage suggests developing shock.

PULSE

4-21. The heartbeat causes a rhythmical expansion and contraction of the arteries as it forces blood through them. This cycle of expansion and contraction can be felt (monitored) at various points in the body and is called the pulse. The common points for checking the pulse are at the—

- Side of the neck (carotid).
- Groin (femoral).
- Wrist (radial).
- Ankle (posterior tibial).

Note. DO NOT use your thumb to check a casualty's pulse because you may confuse the beat of your own pulse with that of the casualty.

WARNING

It is imperative that first aid providers quickly determine if the casualty has a pulse (heartbeat). Absence of a pulse (heartbeat) will lead to the death of the casualty if not quickly restored.

4-22. The literature on combat casualty care raises questions about the reliability of traditional blood pressure measurements as indicators of specific pulse types. However, it underscores a crucial reality: as hemorrhage severity increases, more central pulses are likely to be lost, demanding prompt and decisive intervention. Identifying and responding to these central pulses swiftly is imperative in managing severe hemorrhage and ensuring the best possible outcomes for casualties.

Carotid Pulse

4-23. To ascertain the carotid pulse, locate the pulse on the casualty's neck nearest to you. Gently place the tips of your first two fingers alongside their thyroid cartilage, commonly referred to as the Adam's apple (see figure 4-1). This method allows for an effective assessment of the pulse, rhythm, and strength.

Femoral Pulse

4-24. To assess the femoral (groin) pulse, apply gentle pressure with the tips of your first two fingers to the center of the groin area (see figure 4-2). This method enables you to accurately detect the presence and strength of the femoral pulse.

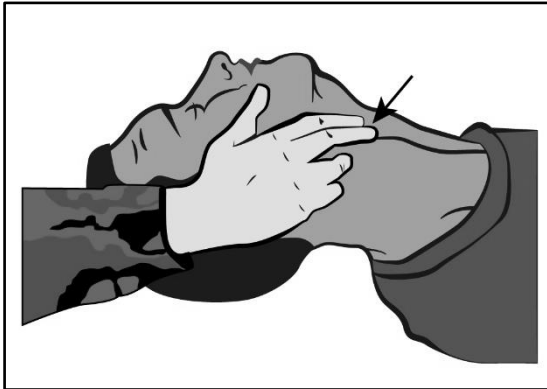


Figure 4-1. Carotid pulse

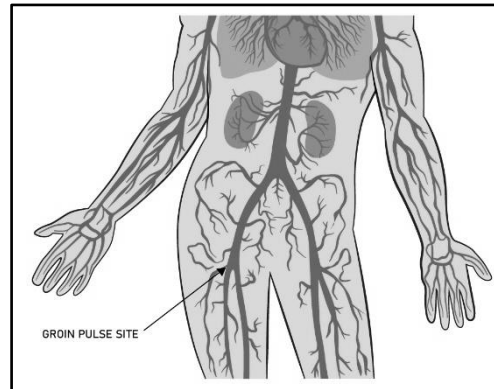


Figure 4-2. Femoral pulse

Radial Pulse

4-25. To ascertain the radial (wrist) pulse, position your first two fingers on the side of the casualty's wrist where the thumb is located (see figure 4-3). This technique allows for a precise assessment of the radial pulse's presence and strength.

Posterior Tibial Pulse

4-26. To assess the posterior tibial (ankle) pulse, position your first two fingers on the inner side of the casualty's ankle (see figure 4-4). This method enables a reliable examination of the posterior tibial pulse's rhythm and strength.

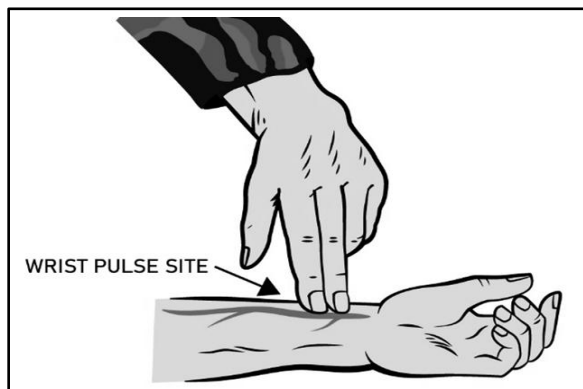


Figure 4-3. Radial pulse

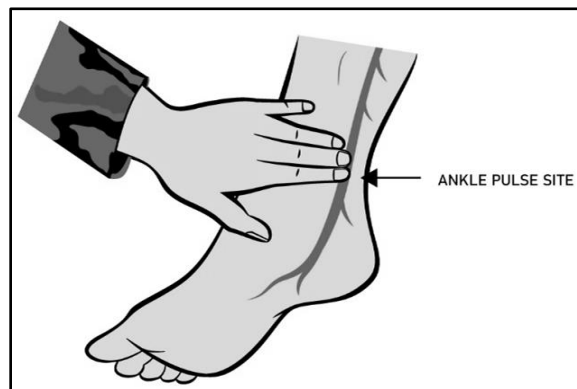


Figure 4-4. Posterior tibial pulse

EQUIPMENT USED FOR MASSIVE BLEEDING CONTROL

4-27. Specialized equipment plays a crucial role in effectively managing massive bleeding, offering immediate and life-saving interventions. TQs, junctional tourniquets, hemostatic agents, hemostatic gauze, and pressure dressings, are among the tools used for this purpose. These items form integral components of the TCCC approach to massive bleeding control, empowering military personnel to address severe hemorrhage swiftly and efficiently in the field and enhancing the likelihood of survival for wounded individuals.

TOURNIQUETS

4-28. TQs serve as critical devices for halting severe bleeding from extremities, such as arms or legs, by effectively restricting blood flow when applied higher to the injury site, thus preventing further blood loss.

4-29. Considered the most vital lifesaving item in the first aid kit, TQs should be kept easily accessible and ready for use during combat missions. The "TQ, Nonpneumatic Combat Application One-Handed," commonly known as the combat application TQ, is the most prevalent type in military inventory, featuring key components such as the windlass rod for twisting and tightening, a windlass clip to secure the rod, a windlass safety strap for added security, and a routing buckle for TQ band management after looping it around an arm or leg.

4-30. It's imperative that all personnel deployed on combat missions have a CoTCCC recommended TQ readily available in a standard location on their battle gear and are trained in its proper usage. Additionally, casualties should be capable of easily and swiftly accessing and applying their own TQs, emphasizing the importance of not storing TQs at the bottom of the pack, and prioritizing the use of the casualty's first aid kit TQ first.

HEMOSTATIC DRESSING AND WOUND PACKING

4-31. Hemostatic dressings, engineered with clotting agents like kaolin or chitosan, are purpose-built to staunch bleeding efficiently. These dressings are either packed directly into deep or narrow wounds or applied directly onto a bleeding wound, accelerating blood clotting far more effectively than traditional dressings.

4-32. When severe bleeding is noted from a junctional area, such as the neck, axillary or inguinal locations of the body (areas where TQs cannot be applied) hemostatic dressings should be used for massive bleeding control. It's essential to note that they should not be used inside the abdomen or chest, where emergency bandages or other trauma dressings are more appropriate.

4-33. Each first aid kit is typically equipped with one hemostatic dressing and one dry sterile gauze. The hemostatic dressings recommended by the CoTCCC are known to be safe and include active ingredients that help blood to clot at the wound site. Hemostatic dressings are available in various types, one type of hemostatic dressing is impregnated with kaolin, another type contains celox granules, or the third type contains chitosan. Chitosan works by binding red blood cells and platelets to the surface of the dressing, enhancing clot formation. Importantly, despite being derived from shellfish, chitosan does not trigger allergic reactions in individuals with shellfish allergies.

PRESSURE DRESSINGS

4-34. Pressure dressings play a crucial role in stemming bleeding by applying direct pressure on wounds. Typically comprised of an elastic bandage and a sterile pad, they are placed over the wound and securely fastened to maintain pressure on the injured area. These dressings not only aid in blood clotting but also compress the open blood vessel, assisting in controlling bleeding effectively. After bleeding has been stemmed with a hemostatic dressing or packing material, it's imperative to maintain pressure on the wound. This is achieved by applying a pressure dressing directly on top of the hemostatic dressing or packing material and wrapping it around the limb, ensuring comprehensive coverage of the wound and gauze applied previously.

4-35. The CoTCCC-recommended emergency bandage or trauma dressing serves as a versatile solution for most wounds, particularly those not severe enough to require a TQ. These dressings can be used alone or in conjunction with other hemorrhage control methods, such as hemostatic gauze and normal gauze, to bolster their effectiveness in controlling bleeding by providing consistent pressure to the wound.

TIER 1 SKILLS FOR MASSIVE BLEEDING CONTROL

4-36. The urgency of promptly stopping massive external bleeding cannot be overstated, as the longer a Service member bleeds from a major wound, the lower their chances of survival become. Severe injuries have the potential to cause death in as little as three minutes due to blood loss. During TFC, when evaluating the casualty for the location, type, and size of the wound or injury, it is essential to cut or tear their clothing and carefully expose the entire area of the wound. This procedure ensures proper visualization of the injury while minimizing the risk of further contamination. It's crucial not to touch the wound directly and to keep it as clean as possible to prevent complications. It's important to note that in a chemical environment, protective clothing should not be removed, and dressings should be applied over the protective clothing. The skills used by all Service members to control massive bleeding include the application of TQs, hemostatic dressings, and pressure bandages.

APPLICATION OF TOURNIQUETS

4-37. In combat scenarios, especially while under enemy fire, a TQ stands as the primary means to control massive extremity bleeding, enabling individuals, battle buddies, or combat medics to swiftly manage life-threatening hemorrhages until the casualty can be moved away from the firefight. Always prioritize treating life-threatening hemorrhage while seeking cover. The application of a TQ to a massive bleed is time-sensitive, with the TQ aiming to halt bleeding within one minute and be fully secured within three minutes, casualties with uncontrolled massive bleeding can succumb within these initial minutes. Expecting pain during application is crucial as it signifies proper TQ application.

4-38. The casualty may need to apply a one-handed TQ (see figure 4-5) to an upper extremity when applying self-aid. One-handed TQs are used to apply self-aid for bleeding from an injury to the upper arm or forearm. (One-handed windlass and one-handed ratchet TQs are the TQs of choice; they are effective and can be applied quickly).



Figure 4-5. Windlass and Ratchet Tourniquets

4-39. A hasty TQ is implemented as "high and tight" on the wounded extremity or when the bleeding source is ambiguous during the CUF phase. However, during TFC, all TQs should be deliberate, placed two to three inches above the wound directly on the skin, not over clothing. Once the casualty is in a safer position, the bleeding source should be reassessed, and the effectiveness of the TQs verified. If bleeding remains uncontrolled, a second TQ may be necessary, particularly for severe bleeding wounds of the thigh.

4-40. It is crucial to recognize that TQs should be used cautiously, with the optimal safe duration being less than two hours. Beyond this timeframe, tissues may start to die, potentially leading to limb loss after six hours. Regular reassessment of TQ effectiveness is necessary, including checking for stopped bleeding, absent distal pulse, and ensuring no pulse below the TQ on the limb. Common errors when applying TQs include not using one, when necessary, not tightening it adequately, and using it for minimal bleeding. Additionally, TQs should never be removed except by medical personnel at a higher role of care. If the source of massive bleeding is evident, such as from an amputation or major wound, immediate bleeding control measures like applying a TQ or wound packing should be initiated, especially in areas where a TQ cannot be

used, such as the groin, armpit, or neck. The following steps address tourniquet application in CUF for all Service Members, CLS, and medical personnel.

4-41. One-Handed (Windlass) TQ Application CUF:

- Remove TQ from JFAK and/or carrying pouch.
- Insert the wounded extremity through the loop of the self-adhering band.
- Position the TQ above the bleeding site, high on the extremity over the clothing/uniform.
- Ensure that all the slack in the self-adhering band is pulled through the routing buckle before the band is fastened back on itself and the windlass is twisted.
- Twist the rod until bleeding has stopped (complete steps 1-5 in under 1 min).
- Lock the windlass rod in place with the windlass clip.
- Route the self-adhering band around the rod and between the clips.
- Secure the windlass safety strap.
- If no other major bleeding is present, move to cover.

Note. Do not document the TQ application time until the TFC phase.

4-42. One-Handed (Ratchet) TQ Application CUF:

- Remove TQ from the JFAK and/or carrying pouch.
- Insert the wounded extremity through the loop of the TQ band.
- Position the TQ above the bleeding site, high on the extremity over the clothing/uniform.
- Grasp the TQ loop with your teeth (or if able lean against a hard surface) to prevent slipping when tightening.
- Tighten the TQ strap as much as possible.
- Lift the lever arm of the ratcheting buckle and tighten by ratcheting the TQ until bleeding has stopped (complete steps 1-5 in under 1 min).
- Lock the ratchet on itself (it will click into place).
- If no other major bleeding is present, move to cover.

Note. If bleeding is not controlled, continue to ratchet the maneuver device until bleeding has stopped. Do not document TQ application time until the TFC phase.

4-43. Two-Handed (Windlass) TQ Application CUF:

- Remove TQ from JFAK and/or carrying pouch.
- Insert the wounded extremity through the loop of the self-adhering band or route the band around the limb, passing the tip through the slit of the routing buckle.
- Position the TQ above the bleeding site, high on the extremity over the clothing/uniform.
- Ensure that all the slack in the self-adhering band is pulled through the routing buckle before the band is fastened back on itself and the windlass is twisted.
- Twist the rod until bleeding has stopped (complete steps 1-5 in under 1 min).
- Lock the windlass rod in place with the windlass clip.
- Route the self-adhering band around the rod and between the clips.
- Secure the windlass safety strap.
- If no other major bleeding is present, move to cover.

Note. Do not document TQ application time until the TFC phase.

4-44. Two-Handed (Ratchet) TQ Application CUF:

- Remove the TQ from the casualty's JFAK and/or carrying pouch.
- Insert the wounded limb through the TQ strap loop or route the strap around the limb; pass the tip through the buckle.
- Position the TQ above the bleeding site, high on the extremity over the clothing/uniform.
- Pull the strap as tightly as possible, removing all excess slack.

- Lift the lever arm of the ratcheting buckle and tighten by ratcheting the TQ until bleeding has stopped (complete steps 1-5 in under 1 min).
- Lock the ratchet on itself (it will click into place).
- If no other major bleeding is present, move the casualty to cover.

Note. If bleeding is not controlled, continue to ratchet the maneuver device until bleeding has stopped. Do not document TQ application time until the TFC phase.

APPLICATION OF HEMOSTATIC DRESSING AND WOUND PACKING

4-45. Hemostatic dressings serve as effective tools for controlling bleeding in areas where a TQ cannot be applied effectively, such as the groin, armpit, or neck. These dressings should be applied with direct and firm pressure and held for at least three minutes.

CAUTION

Do not pack hemostatic dressings into the abdomen or chest.

4-46. After the bleeding has been controlled, it's crucial to check for a distal pulse to ensure the dressing hasn't been applied too tightly. Adjust the dressing if a pulse is not felt to restore circulation.

WARNING

The emergency bandage must be loosened if the skin distal to the injury becomes cool, blue, numb, or pulseless.

4-47. Before applying the dressing, carefully examine the casualty for multiple wounds, as a projectile may have entered at one point and exited at another, with the exit wound typically being larger than the entrance wound.

4-48. When applying a bandage, prioritize using the casualty's emergency bandage. Apply it by opening the plastic dressing package, placing the sterile pad directly over the wound, wrapping the elastic tail around the extremity, running it through the plastic pressure bar, and securing the closure bar to prevent slipping without causing excessive pressure.

4-49. While hemostatic dressings should be packed into limb wounds, they should not be used in the abdomen or chest. For compressible hemorrhage in areas where a TQ cannot be applied effectively, such as the neck, armpit, and groin, use a CoTCCC-recommended hemostatic dressing.

4-50. Despite containing active hemostatic agents, hemostatic dressings require direct pressure for at least three minutes to be effective. Non-CoTCCC-recommended supplies may not be as effective in controlling massive hemorrhage.

WARNING

Continuously monitor the casualty for conditions necessitating basic lifesaving measures. Additionally, all open wounds should be checked for entry and exit points, with first aid administered accordingly.

WARNING

If a projectile remains lodged in the body, do not attempt to remove it. Apply a dressing. Similarly, if an object is impaled in the wound, do not remove it; instead, apply a dressing around the object and stabilize it with additional bulky materials to prevent further injury.

APPLICATION OF PRESSURE DRESSINGS

4-51. Depending on the nature of the injuries, the first responder may need to pack wounds and apply a pressure bandage. Hemostatic dressings, containing special clotting agents, are ideal for wound packing to promote blood clotting. If hemostatic dressings are unavailable, gauze bandage can be used to pack wounds.

WARNING

Avoid packing wounds in the chest or abdomen.

4-52. Pressure dressings are crucial for maintaining the effectiveness of hemostatic dressings, ensuring sustained pressure on the bleeding sources. In situations where pressure dressings are not available, maintaining firm, direct pressure is essential until medical assistance arrives.

4-53. When applying a pressure bandage over a packed wound, use short tugs while wrapping and continue with tension. Care should be taken to avoid wrapping the bandage too tightly, ensuring the skin below remains pink and warm, with a present pulse.

Note. Service members assess tightness based on skin coolness and paleness rather than “capillary refill.”

4-54. Hemostatic dressings are only applied during the TFC phase. The steps for wound packing and pressure bandage are as follows:

- Expose the wound, if not previously exposed.
- Remove clothing and equipment as required.
- Locate the source of the most active bleeding and apply direct pressure.
- Remove the hemostatic dressing from its sterile package.
- If a hemostatic dressing is not available, use gauze or clean, dry cloth material.
- Pack the hemostatic dressing tightly into the wound directly over the site that has the most active bleeding.

Note. Fill and pack the whole wound cavity tightly while keeping firm pressure on the wound. More than one hemostatic dressing or gauze may be required to stop the blood flow.

CAUTION:

If a penetrating object is lodged in the casualty's body, bandage it in place. Do not remove the object.

- Ensure the hemostatic dressing or gauze extends 1-2 inches above the skin.

Note. If the hemostatic dressing or gauze does not extend 1 to 2 inches above the skin, place additional hemostatic dressing or gauze.

- After packing, continue to apply firm, manual pressure until the bleeding stops. Hold continuous direct pressure for a minimum of 3 minutes.
 - Reassess to ensure bleeding has been controlled while maintaining pressure.
-

Note: Ensure blood is not seeping through or around the hemostatic dressing. If bleeding has stopped, leave the hemostatic dressing in place while still maintaining pressure.

- If bleeding has not been controlled and packed initially with a hemostatic dressing, remove the ineffective hemostatic dressing and repack starting at step three of this skill.
 - If the wound was previously packed with gauze or other materials, apply additional gauze or materials and pressure (for another 3 minutes) until bleeding has stopped.
 - Remove the pressure bandage from its package.
 - Place the pad (bandage) directly over the wound on the previously applied hemostatic dressing while continuing to apply direct pressure.
 - Wrap the pressure/elastic bandage tightly around the extremity, focusing pressure over the wound and ensuring that the edges of the pad are covered.
-

Note: If the bandage has a pressure bar, insert the elastic wrap completely into the pressure bar, pull the bandage tight, and reverse it back over the top of the pressure bar forcing it down onto the pad.

- Secure the hooking ends of the hood and loop straps or closure bar onto the last wrap of the bandage.
 - Check for circulation below the pressure bandage by feeling for a distal pulse.
 - Document all findings and treatments on a DD Form 1380 and attach it to the casualty.
-

Note. If the skin below the pressure bandage becomes cool to touch, bluish, or numb, or if the distal pulse is no longer present, the pressure bandage may be too tight and should be loosened and reapplied. If bleeding reoccurs you may need to apply a limb or junctional tourniquet.

4-55. Secure the pressure dressing tail with hooks or a knot, taping them down to prevent loosening. Pressure bandages should not create a TQ effect; if the pulse is absent below the bandage, it's too tight and needs to be loosened and retied.

4-56. Signs of excessive tightness include bluish skin tint, coolness to touch, or numbness in the extremity below the bandage. All bandages should be reassessed after the casualty is relocated.

TIER 2 SKILLS FOR MASSIVE BLEEDING CONTROL

4-57. Tier 2 skills expand upon Tier 1 skills, which Service members apply to themselves or others in urgent situations. In the CUF phase, the tactical situation often limits the initiation of Tier 2 skills for hemorrhage control, typically extending only to the application of a hasty TQ. Tier 2 treatments such as the deliberate TQ and hemostatic pressure dressings are not applied during CUF due to the length of time to apply and the immediate threat to life from enemy fire.

4-58. Although Tier 1 and Tier 2 provide the same hemorrhage control treatment of hasty TQ during CUF, during TFC the Tier 2 CLS treats any massive bleeding, from an extremity, with deliberate TQs. Tier 2 personnel are also trained in junctional wound hemostatic pressure dressings.

APPLICATION OF TOURNIQUETS

4-59. The CLS must possess proficiency in executing all Tier 1 massive bleeding control skills and be capable of directing or assisting Service members in their application. Emphasizing TQs as the primary method for bleeding control during CUF.

4-60. During TFC, where time allows for thorough assessment, exposing the wound becomes feasible, aiding in and pinpointing the precise site of bleeding. Deliberate TQs applied in this phase, positioned two to three inches above the wound directly on the skin, optimize effectiveness while minimizing unnecessary impact on healthy tissue.

4-61. The CLS conducts reassessment of TQs applied during CUF to ensure their efficacy. TQs hastily applied "high and tight" may need adjustment as medical personnel intervene, addressing the placement for optimal effectiveness.

4-62. If bleeding persists despite a previously applied TQ, the CLS should promptly apply a second deliberate TQ adjacent to the original, positioned higher on the limb. Bleeding cessation is expected within one minute, with full TQ securing achieved within three minutes. Any life-threatening bleeding not identified during CUF warrants immediate intervention with a TQ, hemostatic, or pressure dressing.

4-63. Documenting the time of TQ placement on both the TQ itself and the DD Form 1380 during TFC, rather than CUF, is crucial for subsequent medical personnel. Care should be taken to avoid placing TQs over joints or pockets containing bulky items, ensuring the TQs effectiveness.

APPLICATION OF HEMOSTATIC DRESSING

4-64. While hemostatic dressings should be packed into wounds of the limbs, when the source of bleeding is too high on the limb for TQ application (such as the neck, armpit, and groin) they should NOT be packed into the abdomen or chest. For compressible (external) hemorrhage, places where a TQ cannot be effectively applied, use a CoTCCC-recommended hemostatic dressing.

4-65. Although the hemostatic dressing contains the active hemostatic agents, direct pressure must be applied for at least three minutes. Non-CoTCCC-recommended supplies may be available; however, they have not been shown to be effective in controlling massive hemorrhaging.

APPLICATION OF DIRECT PRESSURE

4-66. After packing the wound with a hemostatic dressing or gauze, hold direct pressure for three minutes. Direct pressure can and should be used as a temporary measure until a TQ or dressing is in place. It is difficult to use direct pressure alone to control significant bleeding, or while moving the casualty. While packing a wound, maintain constant, direct pressure, at the source of the bleeding to be effective.

APPLICATION OF WOUND PACKING

4-67. Wounds should not be blindly packed. Effort must be taken to locate the source of bleeding and apply direct pressure immediately while preparing the gauze.

4-68. Wound packing:

- Identify and expose the wound.
- Locate the source of the most active bleeding and apply direct pressure, while preparing the hemostatic dressing.
- Remove the hemostatic dressing from its sterile package and pack it into the wound tightly, directly over the site of the most active bleeding.
- Fill and pack the whole wound cavity tightly while keeping firm pressure on the wound.
- Ensure the hemostatic dressing extends one to two inches above the skin.
- Hold continuous direct pressure for a minimum of three minutes.
- Reassess the wound to ensure that the bleeding has stopped.
- Apply a pressure bandage over the hemostatic dressing to secure it in place.
- Reassess the wound to ensure that the bleeding has stopped. If blood is seeping through or around the hemostatic dressing, the bleeding is not controlled, and the following steps should be applied:

- If extra hemostatic dressings are available, remove the initial, ineffective hemostatic dressing and replace them with new hemostatic dressings.
- If extra hemostatic dressings are unavailable, leave initial, ineffective hemostatic dressing in place and add extra gauze over the wound tract.
- If roller or compressed gauze was used initially, due to unavailability of hemostatic dressings, leave initial, ineffective roller or compressed gauze in place and add extra gauze over the wound tract.
- Once bleeding has been controlled, apply a pressure bandage over the wound packing material to secure it in place.

Note. More than one hemostatic dressing may be required to stop the blood flow.

APPLICATION OF PRESSURE DRESSINGS

4-69. Once bleeding has been controlled with a hemostatic dressing or packing material, pressure must be maintained on the wound. All hemostatic dressings for significant bleeding should be secured with pressure bandages.

4-70. The pressure dressing is placed directly on top of the hemostatic dressing or packing material. While maintaining direct pressure on the hemostatic dressing or packing material, the CLS will wrap the pressure or elastic bandage tightly focusing pressure directly over the wound. Secure the pressure dressing tail either by hooks, or with a knot, and tape both down to ensure they do not come loose.

4-71. Pressure bandages should not have a TQ effect. If there is no pulse below the pressure bandage, it has been applied too tightly and will need to be loosened and retied. Another sign that the pressure bandage has been applied too tightly is the color of the skin. The skin will have a bluish tint and may be cool to the touch. The casualty will complain of numbness in the extremity below the bandage. All bandages should be reassessed by—

- Checking for circulation **BELOW** the pressure bandage by feeling for a distal pulse (a pulse below the bandage).
- Checking the skin **BELOW** the pressure bandage ensures the skin is not cool to the touch, has a bluish tint, or is numb.
- Checking every time a casualty is moved.

4-72. The pressure dressing is too tight if the pulse below the pressure dressing is no longer present, or the skin is cool. If circulation is blocked or stopped, loosen, and re-tie the bandage. Dressings and bandages should be reassessed and checked routinely.

APPLICATION OF JUNCTIONAL PRESSURE

4-73. Junctional pressure is an alternative method to control bleeding. This method uses pressure to press at the site or point where a main artery supplying the wounded area lies near the skin surface or over bone (see figure 4-6 on page 55). These areas are known as junctional areas and are difficult to stop the bleeding even though the vessels are larger than in the extremities. Junctional areas are located at the junction of the extremities and neck with the torso.

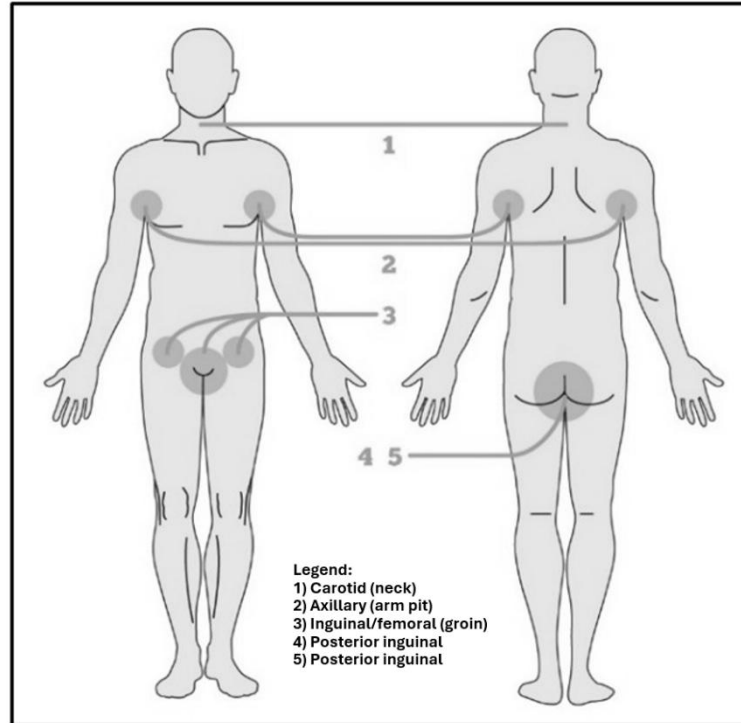


Figure 4-6. Junctional Anatomy

4-74. Junctional hemorrhage can also occur on the extremities if the injury is too close to the torso for a TQ to be applied. Blood vessels at junctional areas are larger than in the limbs but can still be compressed with direct pressure. Junctional pressure may help in instances where bleeding is not easily controlled by TQs, where a hemostatic dressing has not yet been applied, or where pressure dressing may assist in clotting. Junctional pressure is a temporary measure that may buy time for more definitive treatment.

Neck Junctional Hemorrhage Control

4-75. A neck wound should be packed with a hemostatic dressing using your fingers to ensure that the gauze is packed in all parts of the wound. The wound should be packed sufficiently to ensure that the gauze extends one to two inches above the skin surface. Direct pressure must be maintained continuously for three minutes and maintained throughout the application of the dressing. A pressure dressing is applied on top of the gauze.

4-76. Neck junctional hemorrhage control:

- Consider body substance isolation
- Expose the injury and assess the bleeding source.
- Apply direct pressure to the bleeding source, use dressing from the casualty's IFAK or JFAK.
- Pack wound tightly with the hemostatic gauze until the wound cavity is filled.

Note. These steps should be performed within 90 seconds.

- Ensure that the gauze extends 1 to 2 inches above the skin.
- Hold pressure for a minimum of three minutes.
- Reassess to ensure bleeding has been controlled while maintaining pressure.
- If bleeding has not been controlled.
- If packed with hemostatic dressing, and extra hemostatic dressings are available, remove and repack.
- If packed with gauze or initially packed with hemostatic dressing and no more hemostatic dressing is available, apply additional gauze and pressure (for three minutes) until the bleeding stops.

- Place an elastic bandage over the dressing leaving enough tail to tie it into a knot.
- While maintaining pressure, wrap diagonally across the chest under the opposite arm (armpit) around the back to the neck and back over the wound.
- Wrap the bandage (no less than 1 ½ times) over the packing material covering it completely.
- Wrap around the neck and under the arm (on alternating sides of the tail, while maintaining tension/pressure), pulling elastic bandage tightly for pressure, covering the packing material.

Note. Do not use the pressure bar on the neck.

- Secure the dressing by tying a non-slip knot with the end of the elastic bandage and its tail.
- Secure the elastic bandage tails with tape, wrapping a minimum of 1 ½ times around the knot.
- Swath the upper arm (of the injured side) to the chest using a bandage.
- Continue to assess the wound for further bleeding.
- Document all findings and treatments on DD Form 1380.

Axillary Junctional Hemorrhage Control

4-77. The axillary (upper armpit) area is an area where the arm attaches to the torso. The axillary artery runs from the torso through the armpit and feeds oxygenated blood to both arms. Axillary junctional hemorrhage is very difficult to treat if attention to detail is not observed. The steps for controlling Axillary Junctional Hemorrhage are:

- Consider body substance isolation.
- Cut away any clothing (to expose injury) and lift the arm to assess the bleeding source.
- Apply direct pressure to the most active bleed. Prepare hemostatic dressing.
- Extend the casualty's arm at a 90-degree angle by placing it on your shoulder.
- Pack the wound tightly with the hemostatic gauze until the wound cavity is filled.

Note. These steps should be performed within 90 seconds.

- Ensure that the gauze extends 1 to 2 inches above the skin.
- Hold pressure for a minimum of three minutes.
- If the bleeding has not been controlled.
- If packed with hemostatic dressing, and extra hemostatic dressings are available, remove and repack starting at step 5.
- If packed with gauze or initially packed with a hemostatic dressing and no more hemostatic dressing is available, apply additional gauze and pressure (for three minutes) until bleeding stops.

Note. Clothing may need to be cut away to properly expose the injury. The best position to treat the casualty is the seated position. If the casualty cannot be treated in the seated position, you will need to sit the casualty up as much as possible to apply the elastic bandage.

- While maintaining pressure on the dressing/gauze, wrap the pressure (or elastic) bandage around the injured shoulder twice to ensure the gauze underneath is completely covered.
- Wrap the elastic bandage across, back and under the opposite armpit, anchoring around the opposite shoulder in a "figure 8" pattern.
- Depending on the bandage used, secure with the closure bar of tie tails of elastic bandage with a nonslip knot.
- Secure the pressure (elastic) bandage tails and knot using tape, wrapping a minimum of 1 ½ times around the knot.
- Swath the upper arm to the side of the chest using a cravat.
- Continue to assess the wound for further bleeding.

EXPEDIENT TECHNIQUES FOR HEMORRHAGE CONTROL

4-78. Expedient techniques for hemorrhage control are of paramount importance in TCCC due to their ability to rapidly address life-threatening bleeding in high-stress and time-critical combat situations. Expedient techniques enable military personnel to intervene quickly and effectively control hemorrhage and prevent further blood loss. These techniques, such as the use of TQs, pressure dressings, and hemostatic agents, are specifically designed to be simple, easily applied, and accessible to all Service members, regardless of medical training. By swiftly initiating hemorrhage control, expedient techniques significantly increase the chances of survival, allowing injured individuals to receive timely care, and ultimately improving their overall mission success, and preserving the fighting strength of the military forces.

IMPROVISED TOURNIQUET

4-79. Using one of the CoTCCC-recommended TQs is the safe and preferred procedure for hemorrhage control. Improvised TQs are much less effective than commercially available TQs discussed in paragraph 4-37 on page 48. Improvised TQs are difficult to make and apply without extensive practice, however they may be needed in situations where commercially produced TQs are unavailable, or supplies have been exhausted.

CAUTION

The risks associated with using improvised TQs include:

- Damage may occur to the skin if the band is too narrow.
- Bleeding may worsen.
- Bleeding may not be completely controlled.
- Likely to loosen overtime from not being properly secured.

4-80. If a TQ is not available, pack the wound and hold direct pressure over the source of the heaviest bleed, FIRST. If this fails to control the bleeding in an arm or leg, only then consider an alternate improvised TQ.

4-81. An alternate improvised TQ uses other materials on hand. In constructing one, it is recommended that you fold a cravat (or other pliable material) to a width of approximately 1 ½ to 2 inches and tighten it with some sort of windlass rod (a straight rod that can be used to twist the material tighter). Then secure the rod to ensure it does not loosen.

Note. The cravat is included in the JFAK, however it is not part of the Army IFAK Generation 1 or 2. To resource the cravat for the Army IFAK, the nomenclature is: Sterile Burn Dressing Cravat, NSN: 6510-01-587-6579; Muslin Compressed Bandage NSN: 6510-00-201-1755.

4-82. An improvised TQ should not be used except as an absolute last resort when there is no other option to control life-threatening bleeding. If no TQ is available, pack the wound and hold direct pressure over the main source of bleeding.

4-83. Continue to check the TQ, as improvised TQs are prone to loosening due to (but not limited to) casualty movement, shifting fluid, and quality of improvised material.

INGUINAL HEMORRHAGE CONTROL WITH AN IMPROVISED PRESSURE DELIVERY DEVICE

4-84. A pressure delivery device (PDD) is made using improvised materials, such as a shoe or boot, full water bottle, or canteen which applies additional pressure to the wound after it has been packed. A PDD may be needed to apply additional and targeted pressure to control inguinal or groin hemorrhage.

4-85. For groin injuries packed with hemostatic dressings, use an improvised junctional PDD to secure the dressing. The PDD is placed in the groin while continuously maintaining pressure to the dressing.

4-86. The PDD is then secured with a TQ, which is wrapped over the hips with the windlass or ratchet placed directly over the improvised device and tightened to add additional pressure. Two TQs placed together may be needed when improvising a PDD.

Note. Junctional hemorrhages involve larger blood vessels requiring more pressure (and targeted pressure) than can be applied with a pressure dressing alone.

4-87. Inguinal (Groin) Hemorrhage Control with Improvised Junctional Pressure Device:

- Consider body substance isolation.
- Expose the injury and assess the bleeding source.
- Apply direct pressure to the wound using a hand, fist, or elbow and open the hemostatic gauze package.
- Tightly pack the wound with hemostatic gauze until the wound cavity is filled (finishing the packing within 90 seconds).
- Hold pressure for a minimum of 3 minutes.
- Reassess to ensure that the bleeding has been controlled while maintaining pressure.
- If packed with hemostatic dressing, and extra hemostatic dressings are available, remove and repack.
- If packed with gauze or initially packed with hemostatic dressing and no more hemostatic dressing is available, apply additional gauze and pressure (for three minutes) until bleeding stops.
- Select and position a pressure delivery device, such as a canteen, into the inguinal gutter while maintaining pressure.
- Select a TQ that can wrap around the casualty's waist/hip area or connect the two tourniquets together.
- Place the windlass or ratchet TQ directly over the middle of the pressure delivery device; ensure that the routing buckle is located towards the middle of the body.
- Remove all slack from the band or strap using a pushing motion across the casualty's body before tightening the TQ.
- Tighten the TQ and ensure that the bleeding is still controlled, and the distal pulse is absent.

Note. If the bleeding is not controlled and the distal pulse (a pulse located below the TQ) is present, remove any remaining slack in the strapping (if possible) and twist or ratchet the TQ device until bleeding is controlled and/or a distal pulse is absent.

- Visually inspect the placement of the equipment, ensuring that the pressure delivery device is in place and the windlass or ratchet is properly positioned over the device.
- Document all findings and treatments on DD Form 1380.

Chapter 5

Airway Control

OVERVIEW OF AIRWAY CONTROL

5-1. After massive bleeding is controlled, the responder can move on to airway control, the first “A” in the MARCH-PAWS sequence. The casualty must have an open (unobstructed) airway. If the casualty is awake and talking, you can assume the casualty’s airway is open. However, the casualty may still have difficulty breathing. If the casualty is unconscious, their airway must be inspected for any obvious blockages.

5-2. Humans must have oxygen to live. Through the breathing process, the lungs draw oxygen from the air and put it into the blood. The heart pumps the blood through the body to be used by the cells that require a constant supply of oxygen. Some cells are more dependent on a constant supply of oxygen than others. For example, cells of the brain may die within five minutes without oxygen. The brain, heart and lungs are very intolerant to lack of oxygen. The brain is considered one of the organs that is least tolerant to oxygen deprivation. Once these cells die, they are lost forever and do not regenerate. This could result in permanent brain damage, paralysis, or death.

5-3. Airway obstruction on the battlefield is often due to facial trauma. If the casualty is breathing on their own but unconscious or semi-conscious, and there is no airway obstruction, further airway management is achieved through a nasopharyngeal airway (NPA). First responders must assess casualties with facial trauma when contraindications exist to NPA use (such as basilar skull fractures, disruption of the midface, nasopharynx, or roof of the mouth).

5-4. Unconscious casualties can also lose their airway as the muscles of their tongue may have relaxed. This causes the tongue to block the airway by sliding to the back of the mouth and covering the opening to the windpipe. The signs and symptoms of airway obstruction include:

- Casualty is in distress and indicating that they cannot breathe properly.
- Casualty making snoring or gurgling sounds.
- Visible blood or foreign objects are present in the airway.
- Severe trauma to the face is observed.

RESPIRATION

5-5. Respiration is the process where we breathe in oxygen and breathe out carbon dioxide. The efficiency of this process can vary depending on someone's age and physical health. Generally, adults breathe 12 to 20 times a minute. But children and infants have different rates. For instance, children aged one to 10 years take 15 to 30 breaths per minute. Infants who are six to 12 months old breathe faster, at a rate of 25 to 50 times per minute. Meanwhile, the youngest infants, those aged zero to five months, have a rate of 25 to 40 breaths per minute.

ANATOMY AND PHYSIOLOGY OF THE RESPIRATORY SYSTEM

5-6. The respiratory system consists of the airway, lungs, and rib cage. The airway consists of the nose, mouth, throat, voice box, and windpipe. Air passes to and from the lungs through the canal. The lungs are two elastic organs made up of thousands of tiny air spaces or sacs and covered by an airtight membrane. The bronchial tree is a part of the lungs. The rib cage is the structure that joins the spine in the back and the breastbone in front, which protects the heart and lungs. The top part of the rib cage is closed by the structure of the neck, and the bottom part is separated from the abdominal cavity by a large dome-shaped muscle called the diaphragm (figure 5-1 on page 60).

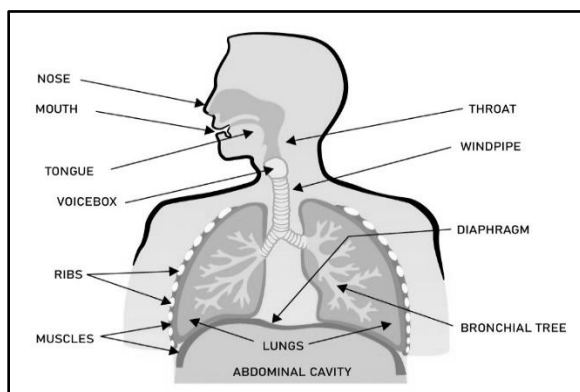


Figure 5-1. Airway, lungs, and the rib cage

5-7. The diaphragm and rib muscles, which are under the control of the respiratory center in the brain, automatically contract and relax. Inspiration increases and expiration decreases the size of the rib cage. When the rib cage increases and then decreases, the air pressure in the lungs is first less and then more than the atmospheric pressure, thus causing the air to rush into and out of the lungs to equalize the pressure.

EQUIPMENT USED FOR AIRWAY CONTROL

5-8. The CoTCCC-recommended NPA with lubricant is a nonsterile, rubber tube-shaped device that is inserted into the nostril of a casualty. It acts as a wedge to keep the airway open by keeping the tongue from falling back into the space behind the mouth leading to the windpipe. Only an NPA, size 32 French, is found within the standard issued JFAKs. Sizing the NPA before insertion is not necessary but is important to minimize nasopharyngeal trauma or elicit the casualty's gag reflex. The lubricant is a water-based substance that assists in inserting the device into the nose. Use of an NPA is a Tier 2 skill.

CAUTION

Do NOT use blood or other body fluids to lubricate the NPA.

TIER 1 SKILLS FOR AIRWAY CONTROL

5-9. Tier 1 TCCC skills for airway control encompass techniques and procedures designed to address various challenges, including airway obstructions and respiratory distress. Proficiency in these skills is imperative for Service members operating in high-stakes, dynamic situations, and where swift and precise intervention can mean the difference between life, and death on the battlefield.

SAFELY POSITION THE CASUALTY TO OPEN THE AIRWAY

5-10. Check for responsiveness (figure 5-2 step A on page 61) and establish whether the casualty is conscious by gently shaking them and asking, "Are you OK?" Call for help (figure 5-2 step B on page 61) and then position the unconscious casualty so that they are lying in the recovery position, head tilted back; chin away from chest (figure 5-2 step C on page 61) by —

- Straightening the casualty's legs.
- Moving the casualty's arm that is nearest to you. Move it so that it is straight and above their head. Repeat the procedure for the other arm placing it by their side.
- Kneeling beside the casualty with your knees near their shoulders (leave space to roll their body).
- Placing one hand behind their head and neck for support. On your other hand, grasp the casualty under their arm (figure 5-2 step C on page 61).
- Rolling the casualty towards you using a steady, even pull. The casualty's head and neck should stay in line with their back.

5-11. The recovery position allows blood and mucus to drain out of the casualty's nose and mouth and not drain back into the airway. However, if a neck injury is suspected and the jaw-thrust technique will be used, kneel at the casualty's head, looking towards their feet.

WARNING

If the casualty is lying on their chest (prone position), cautiously roll the casualty as a unit so that their body does not twist. The twisting may further complicate a back, neck, or spinal injury.

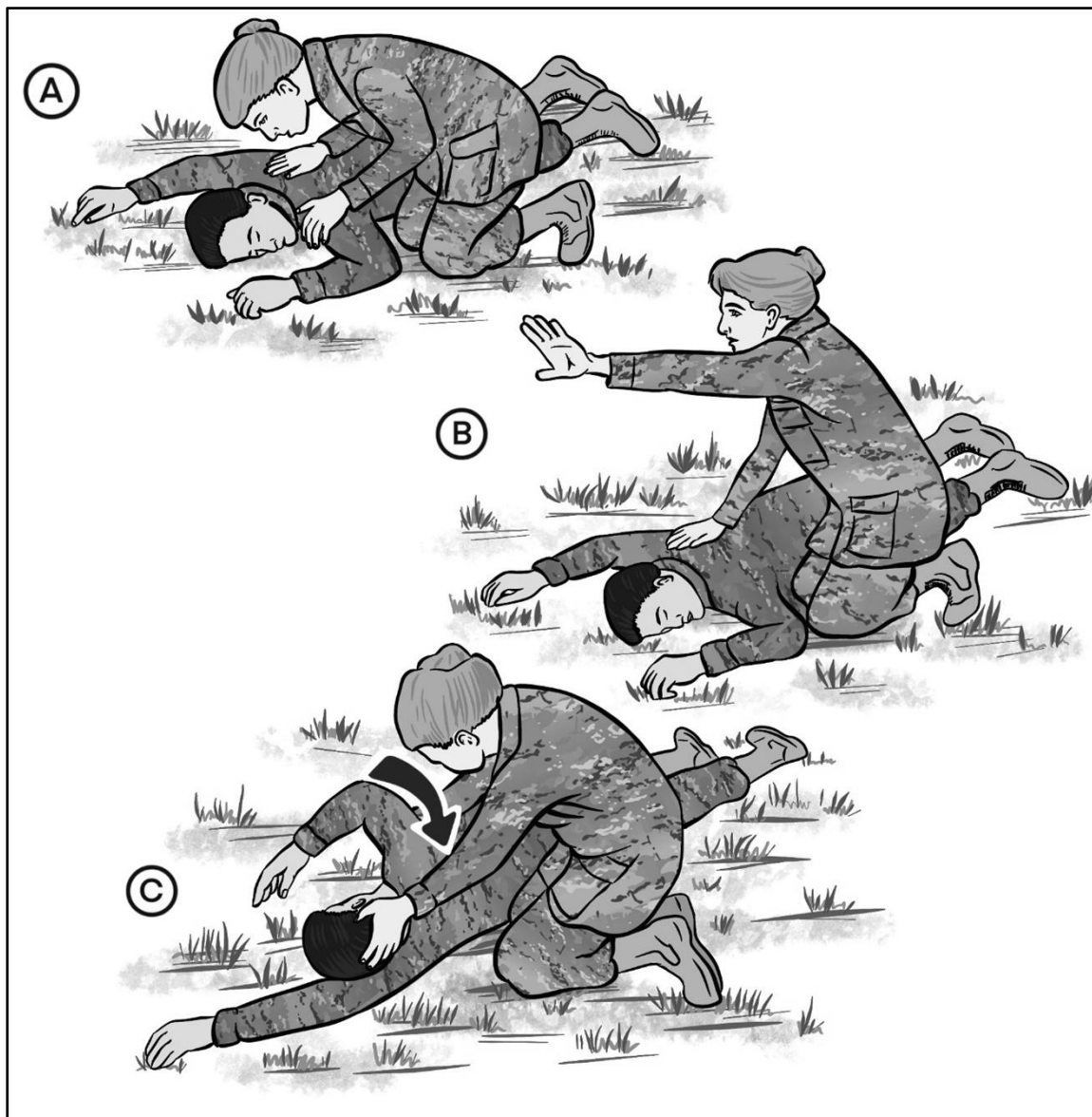


Figure 5-2. Casualty positioning

AIRWAY OBSTRUCTIONS

5-12. Oxygen flows to and from the lungs requires an unobstructed airway, yet obstructions can arise due to various factors:

- Unconsciousness may lead to the tongue falling back into the throat, obstructing the airway if not properly managed.
- Foreign objects, particularly food, may become lodged in the throat during eating, especially when large pieces are swallowed without proper chewing, exacerbated by factors like alcohol consumption or slipping dentures.
- Regurgitation of stomach contents or the formation of blood clots due to head and facial injuries can further contribute to airway blockage.
- Ensure correct positioning and maintenance of the airway in injured or unconscious casualties to mitigate these risks.

5-13. Upper airway obstruction, a potentially life-threatening emergency, can occur as either a partial or complete blockage. The severity and nature of the obstruction underscore the importance of rapid recognition and immediate intervention to prevent respiratory compromise. Identifying the signs early and responding effectively are crucial in maintaining the airway and supporting breathing, which are vital for the casualty's survival. This situation demands prompt and decisive action to alleviate the obstruction and ensure an open airway.

Partial Airway Obstruction

5-14. The casualty may still have an air exchange. A good air exchange means that the casualty can cough forcefully, though they may be wheezing between coughs. The rescuer should not interfere and should encourage the casualty to cough up the object obstructing their airway on their own. Weak coughing with a high-pitched noise between coughs may indicate a poor air exchange. You should assist the casualty and treat them as though they had a complete obstruction.

Complete Airway Obstruction

5-15. A complete obstruction (no air exchange) is indicated if the casualty cannot speak, breathe, or cough at all. They may be clutching their neck and moving erratically. In an unconscious casualty, a complete obstruction is also indicated if after opening their airway you cannot ventilate them. A casualty, experiencing a partial foreign body airway obstruction, with poor air exchange, or a complete airway obstruction will not survive without medical intervention. Both conditions are treated the same.

OPENING THE AIRWAY OF AN UNCONSCIOUS OR NOT BREATHING CASUALTY

5-16. The tongue is the single most common cause of an airway obstruction (see figure 5-3). In most cases, simply using the head-tilt/chin-lift or jaw thrust technique can open the airway. This action pulls the tongue away from the air passage in the throat (see figure 5-4).



Figure 5-3. Airway, blocked

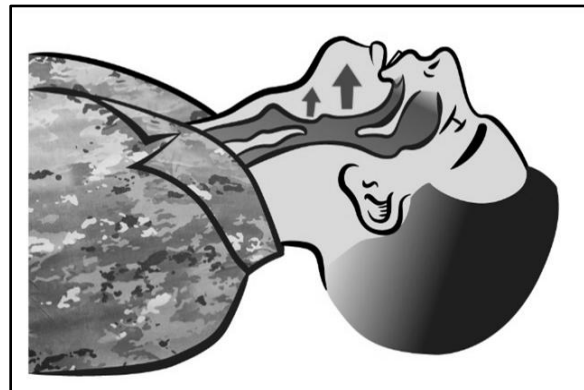


Figure 5-4. Airway, opened

5-17. Call for help and then position the casualty. Move (roll) the casualty onto their back as described in paragraph 5-10 on page 60.

5-18. If a foreign body is seen in the airway, an attempt to remove it should be made. Do not put your fingers in the mouth to try to find or remove any unseen objects that might be causing an obstruction as this may push the object further into the airway. Open the airway using the head-tilt/chin-lift method.

Note. If foreign material or vomitus is visible in the mouth, it should be removed as quickly as possible.

IMPORTANT!

Remove any visible objects, but do not perform a blind finger sweep.

CAUTION

The head-tilt/chin-lift technique is an important procedure in opening the airway; however, use extreme care because excess force in performing this maneuver may cause further spinal injury.

In a casualty with a suspected neck injury or severe head trauma, the safest approach to opening the airway is the jaw-thrust technique because in most cases, it can be accomplished without extending the neck.

Head-Tilt/Chin-Lift Method

5-19. Place one hand on the casualty's forehead and apply firm, backward pressure with the palm to tilt the head back. Place the fingertips of the other hand under the bony part of the lower jaw and lift, bringing the chin forward. The thumb should not be used to lift the chin (see figure 5-5).

Note. The fingers should not press deeply into the soft tissue under the chin because the airway may be obstructed.

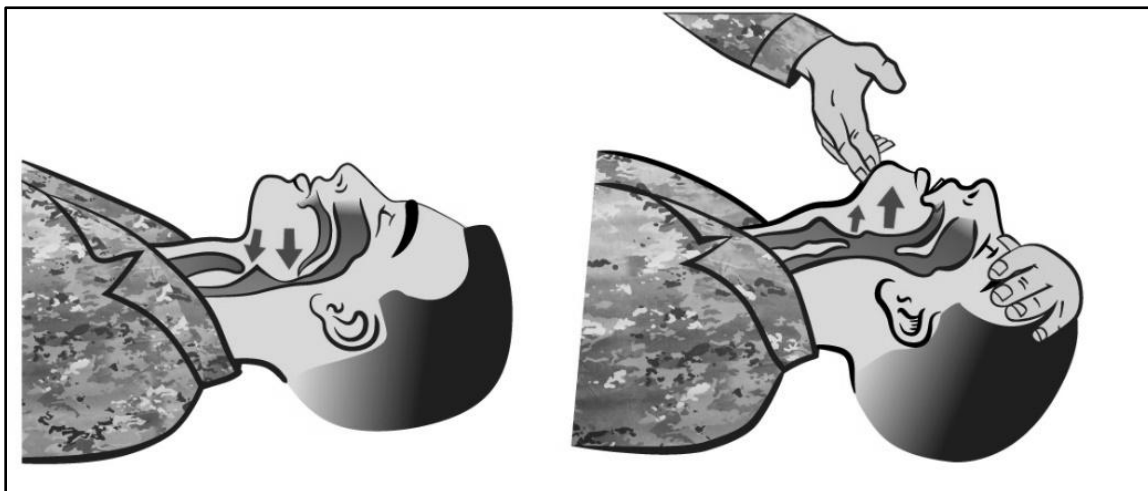


Figure 5-5. Head-tilt/chin-lift technique of opening the airway

Jaw-Thrust Method

5-20. The jaw-thrust method may be accomplished by the rescuer grasping the angles of the casualty's lower jaw and lifting with both hands, one on each side, displacing the jaw forward and up (see figure 5-6). The rescuer's elbows should rest on the surface on which the casualty is lying. If the lips close, the lower lip can be retracted with the thumb. If mouth-to-mouth breathing is necessary, close the nostrils by placing your cheek tightly against them. The head should be carefully supported without tilting it backwards or turning it from side-to-side. If this is unsuccessful, the head should be tilted back slightly. The jaw-thrust is the safest first approach to opening the airway of a casualty who has a suspected neck injury because in most cases it can be accomplished without extending the neck.

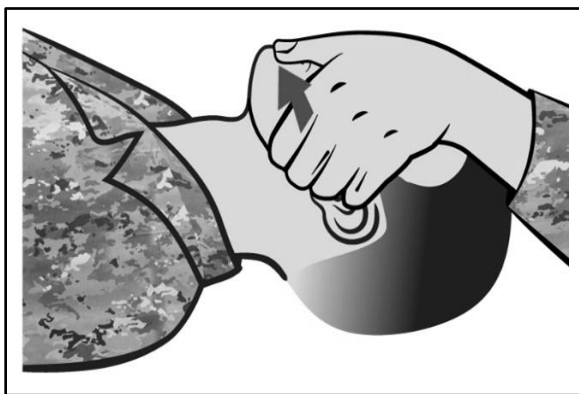


Figure 5-6. Jaw-thrust technique of opening the airway

Note. If the casualty's lips are still closed after the jaw has been moved forward, use your thumbs to retract the lower lip, and allow air to enter the casualty's mouth.

5-21. The jaw-thrust method, commonly emphasized in training, can sometimes serve as a distraction in the field. Neck injuries in trauma cases are typically associated with dramatic incidents such as falls and motor vehicle accidents. Most neck injuries occur upon impact, and the extent of injury is usually immediate and unchangeable at the scene. If neck trauma is present, it is often necessary to perform a jaw-thrust maneuver to secure an airway and maintain that position until additional personnel arrive. This intervention allows the responder to proceed with the MARCH sequence.

MANAGING THE AIRWAY

5-22. If the casualty is breathing independently yet is unconscious or semiconscious, and there is no airway obstruction, managing the airway is best accomplished using an NPA. An NPA is suitable for both conscious and unconscious individuals and is effective in helping to open or maintain an open airway. This method ensures that the airway remains clear, facilitating easier breathing and potentially preventing further complications.

5-23. Casualties with severe facial trauma can often protect their own airways by sitting up and leaning forward. Assisting a conscious casualty by helping them assume any comfortable sitting-up position that allows them to breathe easily. If the casualty is unconscious but breathing, the casualty should be placed in the recovery position (figure 5-7 on page 65).

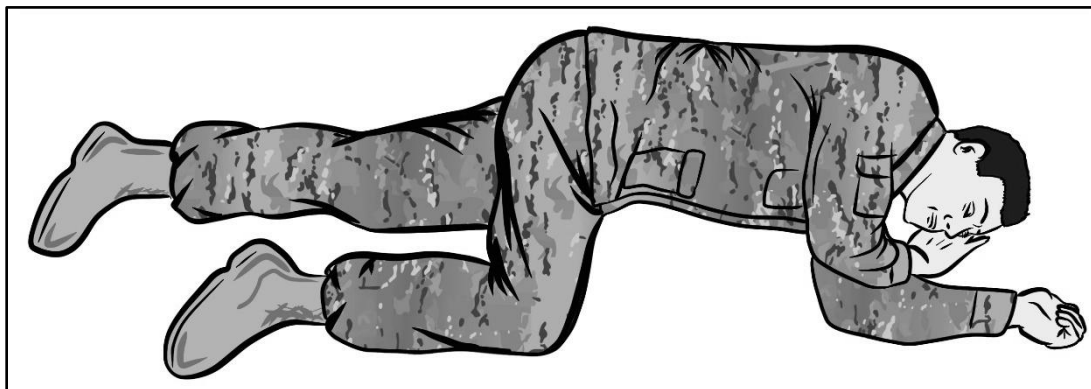


Figure 5-7. Unconscious Casualty Recovery Position

5-24. The recovery position allows blood and mucus to drain out of the casualty's nose and mouth and not drain back into the airway. The recovery position also helps to protect against inhaling vomit if the casualty throws up. If a casualty can breathe on their own, let them assume the best position (position of comfort) that allows them to breathe, including sitting up.

TIER 2 SKILLS FOR AIRWAY CONTROL

5-25. The primary objective of Tier 2 airway control skills is to confirm the initiation of Tier 1 skills, with the addition of Tier 2 insertion of a nasopharyngeal airway and monitor the casualty for any signs of airway compromise, as outlined in paragraph 5-3. If the casualty shows signs of not breathing spontaneously, it's imperative to promptly notify medical personnel. Medical personnel may then require assistance from the CLS in administering breathing assistance using a bag valve mask (BVM) device, as depicted in figure 5-8. In instances where respirations are observed to be diminished, providing ventilation support with a BVM becomes necessary, and the CLS may be requested to assist in its usage by medical personnel.

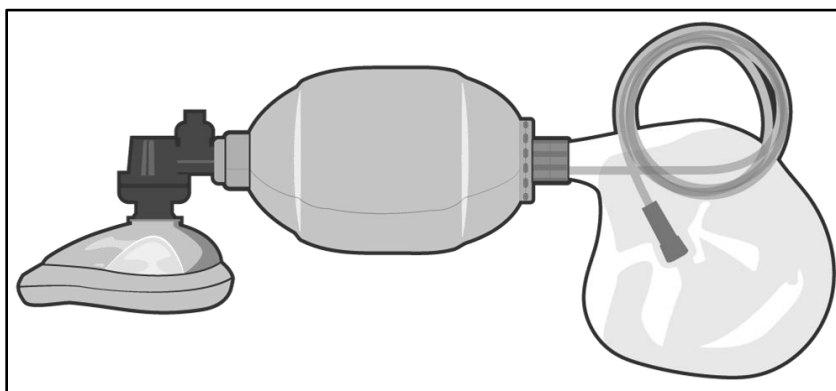


Figure 5-8. Bag valve mask

5-26. Casualties requiring ongoing BVM support due to lack of spontaneous breathing effort, if not just administered a sedating pain medication, might indicate an injury or condition warranting earlier or immediate evacuation (such as severe traumatic brain injury or TBI). If definitive treatment or evacuation is not available, consider triaging the casualty to expectant if resources are limited or better applied to more likely survivable casualties.

5-27. Ventilations can be performed alone or with two people working together. The mask is held using the "E-C" holding technique (figure 5-9 on page 66). The mask is sealed over the casualty's mouth so that air does not escape (figure 5-10 on page 66). The mask is then squeezed firmly for one to two seconds and five to six seconds apart. The Deployed Medicine website has an instruction video on the proper application of the one and two-person BVM.

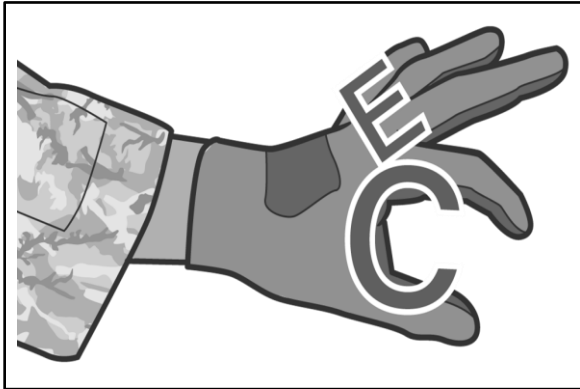


Figure 5-9. Bag valve mask holding technique

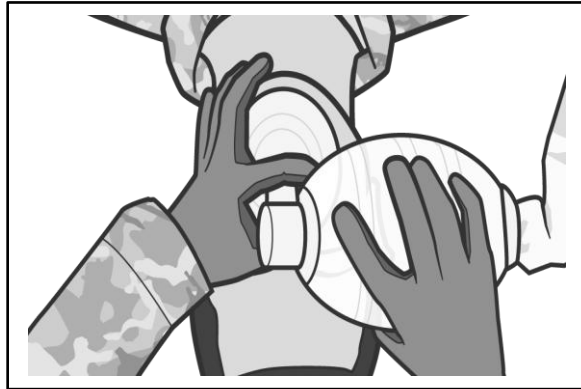


Figure 5-10. Bag valve mask applied to casualty

EXPEDIENT TECHNIQUES FOR AIRWAY CONTROL

5-28. Expedient techniques for airway control are critically important in TCCC due to the immediate threat posed by compromised airways in combat situations. Injured Service members may experience airway obstructions, respiratory distress, or loss of consciousness, which can rapidly progress to life-threatening situations. Expedient airway control techniques, such as the head tilt/chin lift, jaw thrust maneuver and use of NPAs may enable prompt restoration of adequate breathing and oxygenation. These techniques can be quickly implemented by trained personnel, ensuring timely intervention even in chaotic and high-stress environments. By swiftly addressing airway compromises, through the use of expedient techniques in TCCC play a vital role in preserving life, preventing further complications, and maximizing the chances of survival for injured military personnel on the battlefield.

OPENING THE OBSTRUCTED AIRWAY – CONSCIOUS CASUALTY

5-29. Clearing an airway obstruction in a conscious casualty can be managed while they are either standing or sitting. However, it's critical to recognize that an obstructed airway can lead to oxygen deficiency in the brain, resulting in respiratory arrest, unconsciousness and, if not promptly corrected, death.

5-30. To assess and manage airway obstruction, start by asking the casualty if they can speak or if they are choking. Additionally, check for the universal choking sign (see figure 5-11 on page 67). If the casualty can speak or cough effectively, encourage them to continue coughing. This indicates that they still have good air exchange, and you should not interfere with their efforts to expel the obstruction.



Figure 5-11. The universal sign for choking

5-31. However, if you notice high-pitched sounds when the casualty breathes or coughs, indicating poor air exchange, or if the casualty is unable to breathe, immediate action is required. Call for help and prepare to perform manual thrusts to clear the obstruction. Use abdominal thrusts, also known as the Heimlich maneuver, by positioning your hands between the waist and the rib cage and delivering quick upward thrusts. For individuals in advanced stages of pregnancy, those who are markedly obese, or if there is a significant abdominal wound, use chest thrust instead. This involves placing your hands in the center of the breastbone and pushing sharply inward and upward.

5-32. Quick action is essential in these situations to restore breathing and prevent further complications, including death. Always ensure to assess the situation carefully and apply the appropriate technique based on the casualty's condition

Apply Abdominal Thrusts

5-33. Prior to applying abdominal thrust, first responders should administer five back blows. Abdominal thrust can be accomplished by using the following procedures:

- Stand behind the casualty and wrap your arms around their waist.
- Make a fist with one hand and grasp it with the other. The thumb side of your fist should be against the casualty's abdomen, in the midline and slightly above the casualty's navel, but well below the tip of the breastbone (figure 5-12 on page 68).
- Press the fists into the abdomen with a quick backward and upward thrust (figure 5-13 on page 68).

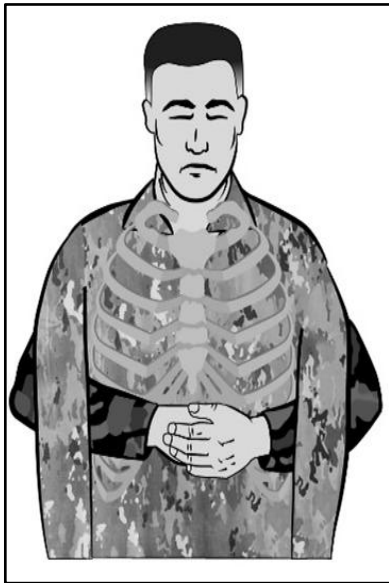


Figure 5-12. Anatomical view of the abdominal thrust procedure



Figure 5-13. Profile view of the abdominal thrust

- Each thrust should be a separate and distinct movement.
- Continue performing abdominal thrusts until the obstruction is expelled or the casualty becomes unresponsive.
- If the casualty becomes unresponsive, call for help as you proceed with the steps to open the airway.

Apply Chest Thrusts

5-34. An alternate technique to the abdominal thrust is the chest thrust. This technique is useful when the casualty has an abdominal wound, is pregnant, or when the casualty is so large that you cannot wrap your arms around the abdomen. To apply chest thrusts with casualty sitting or standing:

- Stand behind the casualty and wrap your arms around their chest with your arms under their armpits.
- Make a fist with one hand and place the thumb side of the fist in the middle of the breastbone (take care to avoid the tip of the breastbone and the margins of the ribs).
- Grasp the fist with the other hand and exert thrusts (figure 5-14 on page 69).



Figure 5-14. Profile view of the chest thrust

- Each thrust should be delivered slowly, distinctly, and with the intent of relieving the obstruction.
- Perform chest thrusts until the obstruction is expelled or the casualty becomes unresponsive.
- If the casualty becomes unresponsive, call for help as you proceed with steps to open the airway, and perform rescue breathing.

OPENING THE OBSTRUCTED AIRWAY – CASUALTY LYING DOWN OR UNRESPONSIVE

5-35. When an individual who is choking becomes unresponsive while lying down, or if an unconscious person is discovered and the cause of their condition is unknown, specific emergency protocols must be followed to address potential airway obstruction.

5-36. If a conscious individual begins choking and subsequently loses consciousness, immediate action is required:

- Summon help right away to ensure assistance is on the way.
- Open the casualty's airway to assess and manage the obstruction.
- Perform a finger sweep only if there is visible obstruction in the mouth to avoid pushing the obstruction deeper.
- Attempt to rescue breathing, which involves assessing whether the air can flow into the lungs. Refer to detailed instructions on mouth-to-mouth resuscitation provided in paragraph 6-49 on page 81. If breathing efforts fail due to a blockage, swiftly move to clear the airway using procedures outlined in paragraph 5-39.

5-37. In scenarios where a casualty is found already unresponsive, the procedure slightly differs:

- Quickly assess the situation to understand the context and severity.
- Call for immediate assistance.
- Position the casualty on their back to manage and secure the airway effectively.
- Check for a pulse to determine the urgency of cardiovascular support.
- Open the airway using recommended techniques.
- Confirm if the casualty is not breathing and, if so, proceed to attempt rescue breathing as per the guidelines detailed in paragraph 6-49 on page 81.

5-38. These critical steps are designed to maximize the chances of recovery by ensuring that the airway is cleared, and that breathing is restored at the earliest possible opportunity. Quick and effective intervention is crucial in such emergencies, underscoring the importance of thorough training and readiness.

5-39. If you are still unable to ventilate the casualty, perform 6 to 10 manual (abdominal or chest) thrusts. To perform the abdominal thrusts:

- Kneel astride the casualty's thighs (see below figure 5-15).

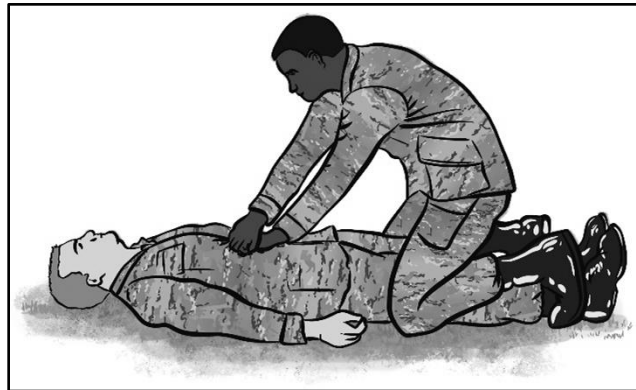


Figure 5-15. Abdominal thrust on an unresponsive casualty

- Place the heel of one hand against the casualty's abdomen (in the midline slightly above the navel but well below the tip of the breastbone). Place your other hand on top of the first one. Point your fingers toward the casualty's head.
- Press into the casualty's abdomen with a quick, forward, and upward thrust. You can use your body weight to perform the maneuver. Deliver each thrust quickly and distinctly.
- Repeat the sequence of abdominal thrusts, finger sweeping, and rescue breathing (attempt to ventilate) as long as necessary to remove the object from the obstructed airway.
- If the casualty's chest rises, proceed to feel for their pulse.

5-40. To perform chest thrusts the unresponsive casualty is placed on their back, face up, and open their mouth. The first responder kneels close to the side of the casualty's body and—

- Locates the lower edge of the casualty's ribs with their fingers by running their fingers up along the rib cage to the notch (figure 5-16 step A on page 71).
- Places the middle finger on the notch in the casualty's chest and the index finger next to the middle finger on the lower edge of the casualty's breastbone.
- Place the heel of the other hand on the lower half of the breastbone next to the two fingers (figure 5-16 step B on page 71).
- Remove the fingers from the notch and place that hand on top of the positioned hand on the breastbone, extending or interlocking the fingers (figure 5-16 step C on page 71).
- Straighten and lock elbows with their shoulders directly above the hands without bending the elbows, rocking, or allowing the shoulders to sag.
- Apply enough pressure to depress the breastbone 1½ to 2 inches and then release the pressure completely (figure 5-16 step D on page 71). This press and release process is delivered quickly and distinctly, six to 10 times. Figure 5-17 on page 71 shows another view of the breastbone being depressed.

CAUTION

Rescuers may feel ribs fracture, and this is OK since it's a life-threatening situation.

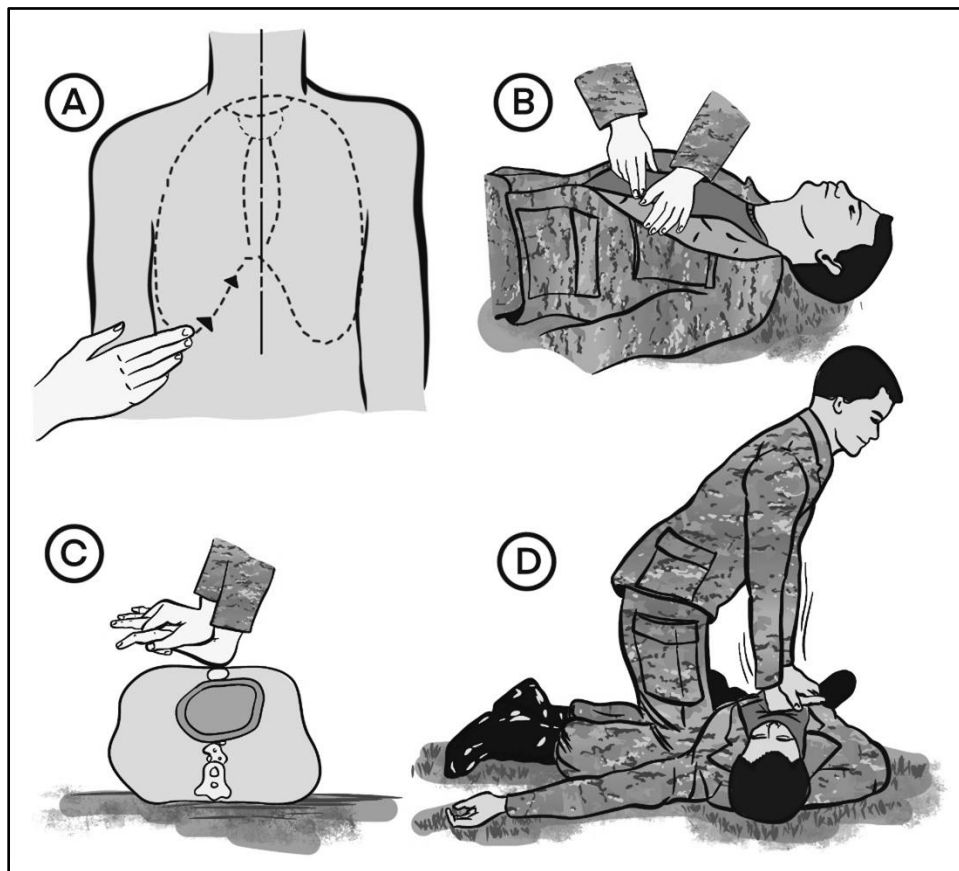


Figure 5-16. Hand placement for the chest thrust

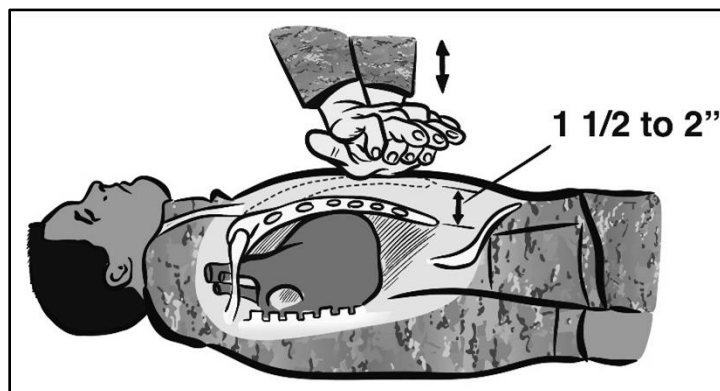


Figure 5-17. Breastbone depressed 1 1/2 to 2 inches

5-41. The first responder repeats the sequence of the chest thrust, finger sweep, and rescue breathing as long as necessary to clear the object from the obstructed airway. If the casualty's chest rises, proceed to feeling for their pulse.

5-42. If rescue breathing cannot be administered due to airway obstruction, remove the airway obstruction by—

- Placing the casualty on their back, face up, turn the unresponsive casualty as a unit, and call out for help.
- Performing a finger sweep.

- Using the tongue-jaw lift by opening the casualty's mouth by grasping both their tongue and lower jaw between your thumb and fingers and lifting (tongue-jaw lift) (see figure 5-18). If you are unable to open their mouth, cross your fingers and thumb (crossed-finger method) and push their teeth apart (see figure 5-19) by pressing your thumb against their upper teeth and pressing your finger against their lower teeth.



Figure 5-18. Opening the casualty's mouth (tongue-jaw lift)



Figure 5-19. Opening the casualty's mouth (crossed-finger method)

- Insert the index finger of the other hand down along the inside of their cheek to the base of the tongue. Use a hooking motion from the side of the mouth toward the center to dislodge the foreign body (see figure 5-20).

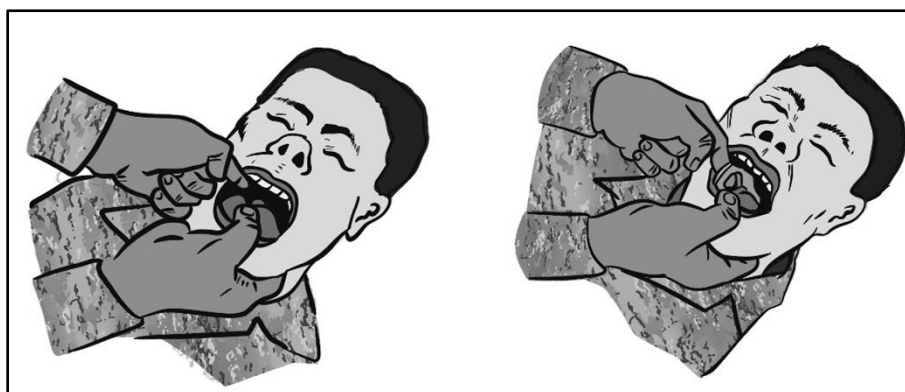


Figure 5-20. Using a finger to dislodge a foreign body

WARNING

Take care not to force the object deeper into the airway by pushing it with the finger.

Chapter 6

Respiration and Breathing Control

OVERVIEW OF RESPIRATION AND BREATHING

6-1. After establishing an open airway, it is important to evaluate the casualty's respirations and maintain that airway in an open position. This is the “R” of MARCH-PAWS. Respiratory distress means difficulty breathing (rapid or abnormally slow breathing), in other words, it is difficult for the casualty to get air in or out. Airway and respiration are not addressed in CUF; however, they must be addressed in TFC.

6-2. The pleural space between the lungs and chest wall naturally has negative pressure, which helps the lungs to collapse (exhale) and expand (inhale). With either a blunt or penetrating injury to the chest wall or lungs, air may counteract the lung's natural tendency to expand and collapse. This is due to positive pressure replacing negative pressure. It results in air being trapped in the pleural space, putting pressure on the affected lung. This forces the lungs to collapse and reduces the ability to get oxygen to the body.

6-3. While it may be easy to see a gunshot wound or shrapnel wound to the chest, it is important to evaluate casualties for additional injuries from blunt force trauma which may have been caused from an explosive device or from the chest hitting the steering wheel during a high-impact vehicle accident.

6-4. Signs of future life-threatening respiration issues include bruising, contusions (swelling around the chest, back, or rib cage), crepitus (crackling, popping, grating) that are felt or heard, or other deformities of the chest not normally seen without removing the clothing. Signs of respiratory distress include:

- Difficulty in breathing.
- Struggling to get air in and out.
- Breathing is too weak to be effective (less than six times per minute).
- Rapid breathing (greater than 20 times per minute).

Note. The above signs may also indicate a penetrating chest wound injury. Report findings of respiratory distress to medical personnel.

EQUIPMENT USED FOR RESPIRATION AND BREATHING CONTROL

6-5. In TCCC, various specialized equipment is used to address respiration and breathing control in combat situations. These tools are designed to promptly manage compromised airways and provide respiratory support. The primary equipment used includes:

- NPA.
- BVM.
- Chest Seal.
- 10–14-gauge catheter over needle device.

6-6. An NPA is a flexible tube inserted through the nose into the oropharynx to maintain an open airway and facilitate adequate breathing. It is particularly useful when there is a risk of tongue obstruction or in cases where the patient is conscious and able to maintain their own airway.

6-7. A BVM device consists of a self-inflating bag and a mask that is manually squeezed to deliver positive pressure ventilation. It is used when there is a need for artificial respiration to support or control the patient's breathing.

6-8. A chest seal is an airtight dressing that is applied over a penetrating chest wound to prevent the entry of air into the pleural cavity, reducing the risk of a tension pneumothorax. It allows for the restoration of normal respiratory function by maintaining appropriate chest pressure. The CoTCCC-recommended chest seal is a vented (preferred) self-adhering chest seal (see figure 6-1 on page 74) is used for treating and sucking chest wounds or open pneumothorax.

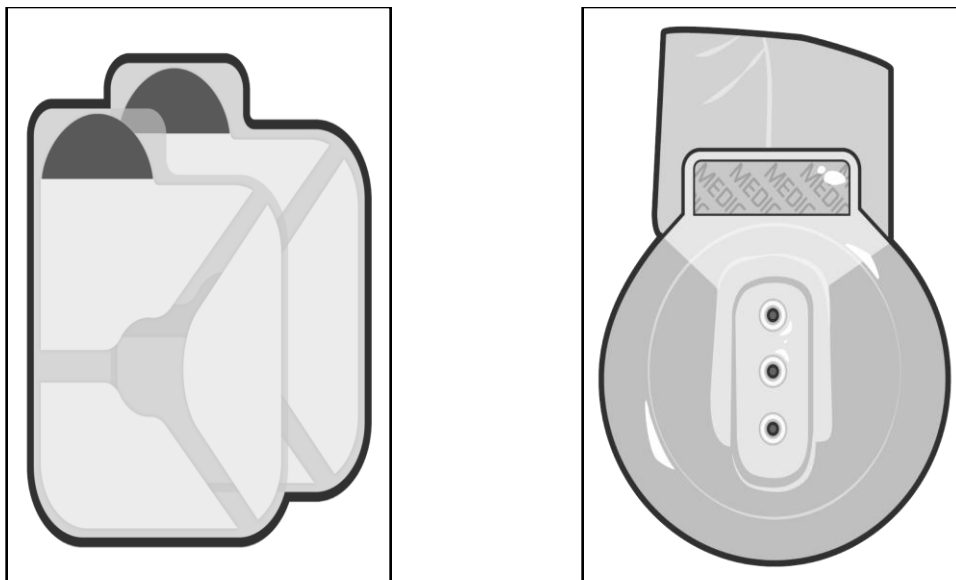


Figure 6-1. Vented chest seal examples

6-9. The equipment is essential components of TCCC's approach to respiration and breathing control. Utilizing this equipment enables military personnel to manage airway obstructions, provide respiratory support, and prevent potentially life-threatening complications associated with compromised breathing in combat scenarios.

6-10. The CoTCCC-recommended catheter-over-needle device is used to treat tension pneumothorax. The recommended catheter-over-needle device is either a 10-gauge, 14-gauge, or 3/4 inch long device (see figure 6-2).

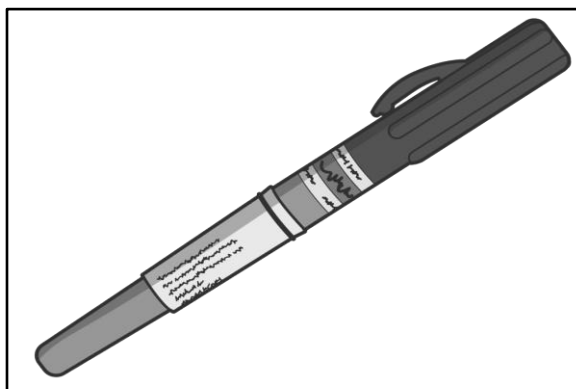


Figure 6-2. Catheter-over-needle device

TIER 1 SKILLS FOR RESPIRATION AND BREATHING CONTROL

6-11. Often the act of just opening and maintaining the airway will allow the casualty to breathe properly. The first responder opens the casualty's airway maintaining the head position to keep the airway open. Failure to maintain the open airway will prevent the casualty from receiving an adequate supply of oxygen. While maintaining an open airway the first responder checks for breathing, respiration rate (breaths per minute) and respiration quality (shallow and labored) by placing the casualty on their back and assessing their breathing by observing the casualty's chest by—

- Looking for the chest to rise and fall.
- Listening for air escaping during exhalation by placing your ear near the casualty's mouth.
- Feeling for the flow of air on your cheek (see figure 6-3).



Figure 6-3. Check for breathing

6-12. If air is not flowing freely in and out, the first responder opens the airway maneuvers as described in paragraphs 5-16 through 5-21.

Note. If the casualty has a suspected neck or spine injury, and the situation allows, use, and maintain the jaw-thrust maneuver.

6-13. If the casualty is unconscious, and the respiratory rate is less than two in 15 seconds, or the casualty is making snoring or gurgling sounds, the first responder inserts an NPA by—

- Keeping the casualty in a face-up position.
- Lubricating the tube of the NPA with water.
- Pushing the tip of the casualty's nose upward gently.
- Positioning the tube of the NPA so that the bevel (pointed end) of the NPA faces toward the septum (the partition inside the nose that separates the nostrils).
- Inserting the NPA into the nostril and advance it until the flange rests against the nostril. Most NPAs are designed to be placed in the right nostril.

CAUTION

Never force the NPA into the casualty's nostril. If resistance is met, pull the tube out and attempt to insert it into the other nostril. If neither nostril will accommodate the NPA, place the casualty in the recovery position.

6-14. The first responder places the casualty in the recovery position by rolling the casualty as a single unit onto their side, placing the casualty's hand of the upper arm under the casualty's chin, and flexing the casualty's upper leg. The first responder then watches the casualty closely for life-threatening conditions and checks for other injuries.

6-15. If the casualty is not breathing, immediately start rescuing breathing and seek medical aid. If the casualty resumes breathing at any time during this procedure, the airway is kept open, and the casualty is continued to be watched. If the casualty continues to breathe, they should be transported to medical aid, or to an MTF IAW the tactical situation.

6-16. Exposure to smoke or toxic inhalants can cause difficulty breathing, but other injuries can also lead to respiratory distress. Life-threatening chest injuries can also result in respiratory distress. There are two types of chest injuries that can potentially be life-threatening, penetrating injuries, like those seen in gunshot or shrapnel wounds, or blunt trauma from blasts or vehicular accidents. It is important to roll the casualty to examine the front and back for signification torso trauma.

6-17. At this point in the MARCH-PAWS protocol, the responder's role is to carefully roll the casualty to examine both the front and back of the torso for any signs of chest trauma. It is crucial for the responder to identify and acknowledge if the casualty has a potentially life-threatening chest injury that is causing respiratory distress. The responder must then effectively communicate these findings to the medical personnel arriving on the scene, ensuring that the casualty receives the appropriate medical attention promptly.

IMPORTANT!

DO NOT pack chest wounds with a hemostatic (or other) dressing.

REPORT a severe chest injury to medical personnel immediately.

TIER 2 SKILLS FOR RESPIRATION AND BREATHING CONTROL

6-18. The Tier 2 skills do not replace all Service member skills for respiration and breathing control. The CLS must reassess respiration and breathing control in TFC and start more advanced respiration and breathing control interventions. The CLS must be ready to treat an open pneumothorax, a tension pneumothorax, and identify if the treatment for those injuries was successful.

OPEN PNEUMOTHORAX

6-19. An open pneumothorax is also known as a sucking chest wound. An open pneumothorax occurs when air enters the pleural space. The pleural space is the thin area between the lungs and chest wall.

6-20. When a casualty experiences penetrating torso trauma the elastic lung deflates and pulls away from the chest wall. If the hole, through the chest wall, is at least the size of a nickel or 2/3 the diameter of the trachea, on inspiration, the air now enters the chest through the hole instead of into the lungs through the normal airways of the mouth and nose. As a result, the affected lung cannot be fully re-inflated by inhalation.

6-21. An open pneumothorax usually occurs with a hole in the chest the size of a nickel or bigger. Not all chest wounds are sucking chest wounds; some do not penetrate as deeply as the lung cavity.

6-22. The first responders check the casualty's respiration and ability to breathe. All open or sucking chest wounds are treated by immediately applying a vented chest seal to cover the defect. If a determination cannot be made if the wound has penetrated the chest wall completely, treat the wound as though it were an open chest wound. A casualty with an open chest wound will exhibit one or more of the following signs and symptoms:

- A “sucking” or “hissing” sound when the casualty inhales.
- Difficulty breathing.
- A puncture wound of the chest.
- Froth or bubbles around the injury.
- Coughing up blood.
- Blood-tinged sputum (spit).

6-23. For respiratory distress not resolved by a chest seal or in a casualty with known or suspected chest or back trauma without an open sucking chest wound, consider a tension pneumothorax, and perform a needle decompression of the chest (NDC). Injuries to the chest are very serious and can be life-threatening. The casualty's condition can change quickly with a chest injury.

6-24. An open chest wound should be treated with a CoTCCC- recommended commercial chest seal found within the first aid kit, which maybe vented or non-vented. Vented chest seals are for treating penetrating wounds to the chest. The vented chest seals allow air to escape out of the chest while non-vented chest seals do not. The injured lung will remain partially collapsed, but the mechanics of respiration will be better.

6-25. Recommended treatment for open or sucking chest wounds is prompt application of a vented chest seal. If a vented chest seal is NOT available, a non-vented chest seal should be used. When the casualty inhales,

the plastic is sucked against the wound, preventing the entry of air. When the casualty exhales, trapped air can escape from the wound and out the valve.

WARNING

Suspect a tension pneumothorax if the casualty's condition worsens.

Treat the tension pneumothorax by burping the wound or temporarily removing the dressing.

6-26. The wound should be uncovered unless clothing is stuck to the wound or in a CBRN environment. Once an open wound is discovered, place a gloved hand over the wound to provide a temporary seal.

Note. If unsure that the wound has penetrated the chest wall completely, treat the wound as though it were an open chest wound. If multiple wounds are found, treat as soon as they are discovered. First wound found; first wound treated.

6-27. Fully open the outer wrapper of the casualty's commercial chest seal located in the casualty's first aid kit. Wipe any dirt, blood, or sweat from the skin with gauze supplied in the commercial chest seal. Instruct the casualty to fully exhale and then immediately place the vent of the chest seal directly over the first wound encountered. Do not touch the inner adhesive surface of the commercial chest seal. Ensure that the edges of the chest seal extend two inches beyond the wound. Firmly press the chest seal on the skin to ensure a good airtight seal. If the chest seal does not provide a good airtight seal, apply a two-inch tape (from the casualty's first aid kit) to all four sides of the chest seal, ensuring the tape does not cover any vents, and secure it firmly to the casualty's chest.

6-28. Roll the casualty on their side or have the conscious casualty sit up and examine their back for any wounds. If any wounds are present on the back, treat as a chest injury with a commercial chest seal. Encourage the casualty to breathe normally. Place the casualty in a position of comfort (usually sitting) if conscious, or in the recovery position (on the casualty's side), with the injured side down if unconscious. Continue to monitor the casualty for signs of respiratory distress (increased difficulty breathing). If the casualty has increased difficulty breathing, lift the chest seal, release any trapped air by wiping blood or debris from the wound, and reseal or replace the chest seal upon exhalation. Seek medical aid and continue to monitor the casualty until relieved by medical personnel.

TENSION PNEUMOTHORAX

6-29. Tension pneumothorax is caused by significant torso (chest) trauma, or a blast injury resulting in severe and progressive respiratory distress. Progressive difficulty breathing is labored, and rapid breathing worsens overtime.

6-30. A tension pneumothorax is the second-leading cause of preventable deaths on the battlefield. As a tension pneumothorax develops, air enters the chest cavity through the wound with every inspiration but does not leave with expiration and is trapped. Every breath adds more air to the air space inside the rib cage and outside the lung, and the pressure inside the chest builds up (figure 6-4).

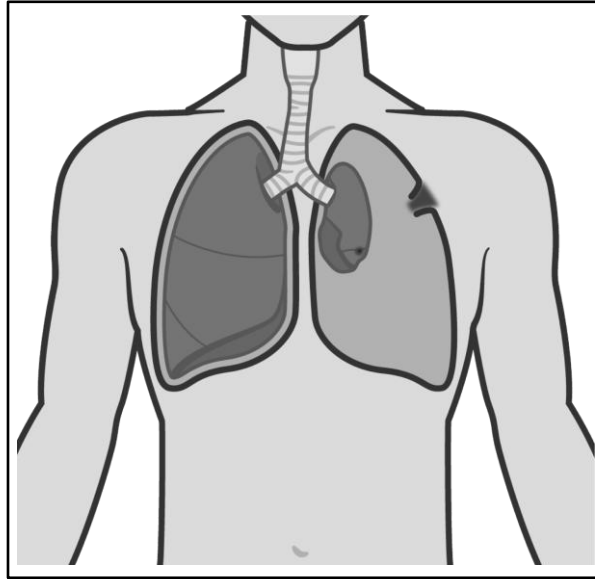


Figure 6-4. Tension Pneumothorax

6-31. When a lung is injured, it can act like a one-way valve, trapping air between the lung and chest wall. This trapped air builds pressure, potentially compressing both lungs and the heart, impairing their function and leading to respiratory distress and shock. In tension pneumothorax, this elevated pressure stops the lungs and heart from expanding normally, which can be fatal. Signs of tension pneumothorax vary from early to late stages:

- Early signs include:
 - Increasing difficulty breathing over time.
 - Shortness of breath.
 - Confusion, lightheadedness, or agitation from low oxygen.
 - Bluish discoloration around the mouth and lips.
 - Rapid pulse.
- Late signs, which may appear as the condition worsens, include:
 - Swollen neck veins.
 - Tracheal deviation (the windpipe shifting to one side).
 - Severe difficulty breathing, often with chest pain or anxiety.

6-32. These signs might be hard to detect in a combat situation. Responders must be vigilant for tension pneumothorax if there is any chest wound accompanied by increasing breathing difficulties.

6-33. It's crucial to monitor these signs during TFC. Like controlling bleeding, treating a tension pneumothorax is crucial; if left untreated, it can lead to severe respiratory issues, shock, and death. The initial treatment for a suspected tension pneumothorax is NDC. After this, any chest injury, whether an open pneumothorax or tension pneumothorax, needs further evaluation by medical personnel and likely evacuation.

UNSUCCESSFUL TREATMENT OR RECURRENCE OF TENSION PNEUMOTHORAX

6-34. Casualties with a tension pneumothorax must be continuously monitored for reoccurring symptoms. The condition may initially be treated unsuccessfully and then spontaneously recur. The immediate remedy for the unsuccessful treatment or recurrence of a tension pneumothorax is to burp the chest seal, if one is in place.

6-35. If initial treatment of tension pneumothorax with NDC does not result in improvement, a second NDC should be attempted at the alternate recommended site. If tension pneumothorax initially responds to NDC, but symptoms later recur, then repeat NDC at the same site right beside the original NDC. If no improvement is noted after the second NDC, proceed with circulation assessment and treatment following the MARCH-PAWS protocol.

WARNING

DO NOT put an NDC through a chest seal! Use an alternate site instead.

6-36. The treatment is to let the air trapped under pressure escape by inserting an NDC into the chest. The NDC site selection is based on the mechanism of injury and the physical findings during the tactical trauma assessment. The NDC (see figure 6-5 below) is placed either at the—

- Second, intercostal space in the mid-clavicular line.
- If the needle is used at the second intercostal space, ensure the site selection is outside the nipple line.
- Anterior axillary line in the fifth intercostal space.

CAUTION

Watch needle placement to avoid the heart and arteries.

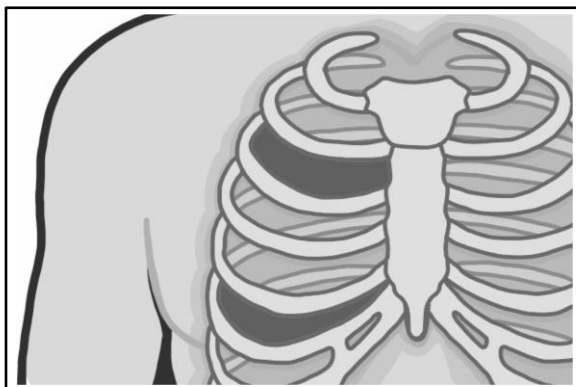


Figure 6-5. Needle Decompression of the Chest Site Selection

6-37. The NDC should be placed perpendicular to the body. After the NDC is placed, the needle should be held in place for 5-10 seconds before removing the needle and leaving the catheter. As with all other treatment given, it is critical to document all interventions on DD Form 1380.

6-38. After the casualty has been treated with an NDC, the casualty should be placed in the position of comfort. If the casualty is unconscious, place the casualty in the recovery position. If the casualty is conscious, allow the casualty to adopt the sitting position, if breathing is more comfortable.

EXPEDIENT TECHNIQUES FOR RESPIRATION AND BREATHING CONTROL

6-39. Expedient techniques for respiration and breathing control in TCCC are crucial for immediate and effective management of a compromised casualty in combat situations. These techniques are designed to be quickly implemented by military personnel with minimal equipment, providing life-saving interventions in high-stress and time-critical environments. Expedient techniques may include maneuvers such as the head-tilt/chin-lift or jaw-thrust to open the airway, clearing obstructing objects, or repositioning the casualty to optimize breathing. Additionally, rescue breaths or mouth-to-mouth ventilation can be employed to support respiration. These expedient techniques are crucial time until advanced medical care can be provided, ensuring that the injured Service members receive immediate attention to maintain adequate respiration and oxygenation, thereby increasing their chances of survival in the challenging conditions of the battlefield.

PENETRATING CHEST INJURIES

6-40. Penetrating chest injuries involve the intrusion of foreign objects such as bullets, knives, or projectiles through the chest wall, penetrating into the thoracic cavity. This breach can lead to severe damage to vital structures including the lungs, heart, major blood vessels, or other organs within the thoracic cavity.

6-41. Penetrating chest injuries can result in significant internal bleeding, pneumothorax (collapsed lung), hemothorax (accumulation of blood in the chest cavity), tension pneumothorax (air buildup under pressure in the chest), or cardiac tamponade (compression of the heart due to blood accumulation). They are considered severe and life-threatening, requiring immediate intervention to stabilize the casualty and prevent further complications. Penetrating chest injuries can also be referred to as an open chest wound.

6-42. A CoTCCC vented commercial chest seal is the preferred type of chest seal. However, if one is unavailable, a non-vented commercial chest seal may be applied in the same manner as a vented commercial chest seal. If there is not a commercial chest seal available, the following examples can be used as occlusive (airtight) materials:

- Plastics such as the outer wrapper of the emergency dressing from the first aid kit.
- The combat ration packing.
- Cellophane.

6-43. An improvised chest seal may be used to treat an open chest wound by—

- Wiping any dirt, blood, or sweat from the skin with gauze or any available clothing.
- Instructing the casualty to fully exhale and then place the inner surface of the outer wrapper or other occlusive material directly over the wound.

Note. Do not touch the inner surface of the occlusive material. Ensure that the edges of the occlusive material extend two inches beyond the wound.

- Applying two-inch tape from the casualty's first aid kit to three sides of the occlusive material and secure it firmly to the casualty's chest. Leave the side nearest to the ground un-taped.
- Rolling the casualty on their side or have the conscious casualty sit up and examine the back for any wounds.
- Checking the casualty for any wounds on the back and treating it as a chest injury with a large improvised occlusive chest seal (such as the plastic wrapper from the emergency dressing in the first aid kit).
- Placing the casualty in a position of comfort (usually sitting) if conscious, or in the recovery position (on the casualty's side) with the injured side down if unconscious.
- Continuing to monitor the casualty for signs of respiratory distress (increased difficulty breathing).

Note. If the casualty has increased difficulty breathing, lift the chest seal, release any trapped air by wiping blood or debris from the wound, and reseal or replace the chest seal upon exhalation.

EXTERNAL CHEST COMPRESSIONS

6-44. External chest compressions, also known as Cardiopulmonary Resuscitation (CPR), in combat casualties is nearly always the result of hemorrhagic or obstructive shock, neither of which will respond to standard CPR. Conducting CPR may delay definitive treatment in nearly all cases. The CoTCCC guidelines recommend performing bilateral needle decompression and expedite blood product administration. Mission variables or the credentialed provider may guide exceptions to this rule in support of CPR on combat casualties. The administration of CPR on a combat casualty should be a rare circumstance and not the standard of care in this trauma population.

6-45. CPR is conducted using a specific technique to provide manual circulation support during cardiac arrest or when the heart has stopped beating effectively. A general outline of the steps involved in conducting external chest compressions is:

- Position: Ensure the person is lying on a firm, flat surface, such as the ground or a CPR board. Kneel beside the person's chest.

- **Hand Placement:** Place the heel of one hand on the center of the person's chest, between the nipples. For infants or children, use two or three fingers instead of the entire hand.
- **Compression Depth:** Press down firmly and smoothly. For adults, compress the chest at least two inches (five centimeters) deep. For infants or children, compress the chest about one-third to one-half of its depth.
- **Compression Rate:** Perform compressions at a rate of 100 to 120 compressions per minute. This equates to about two compressions per second.
- **Complete Chest Recoil:** Allow the chest to fully recoil between each compression, without lifting your hands off the chest. This allows the heart to refill with blood.
- **Compression to Ventilation Ratio:** If you are performing CPR alone, do cycles of 30 chest compressions to two rescue breaths (mouth-to-mouth or mouth-to-mask). If there are two rescuers, the ratio can be 30:2 or continuous chest compressions with ventilation being delivered by the other rescuer.
- **Continue until Help Arrives:** Keep performing chest compressions until professional medical help arrives, or until the person shows signs of movement or starts breathing.

6-46. It is important to note that these steps provide a general overview of external chest compressions. Proper CPR training and certification are needed to gain a thorough understanding of the technique and to ensure accurate and effective application during emergency situations.

6-47. The duration of CPR can vary depending on the circumstances and response of the individual being treated. In general, CPR should be continued until one of the following conditions is met:

- **Signs of Life:** If the person shows obvious signs of life, such as breathing normally, coughing, or moving, CPR can be stopped.
- **Professional Medical Help Arrives:** CPR should be continued until professional medical personnel take over the resuscitation efforts.
- **Provider Fatigue:** If the person performing CPR becomes too exhausted to continue effectively, they can switch with another trained individual to maintain the quality of compression.

6-48. In many cases, CPR is performed continuously without interruptions until one of the above conditions is met. However, it is important to note that high-quality chest compressions are crucial for improving the chances of survival. Therefore, proper training in CPR techniques and regular updates on the latest guidelines are recommended to ensure effective and efficient resuscitation efforts.

RESCUE BREATHING

6-49. Rescue breathing, also known as artificial ventilation, is a technique used in CPR to provide oxygen to a person who is not breathing or has inadequate breathing. It involves delivering breaths into the person's lungs to support their respiratory function. Rescue breathing is conducted executing the following:

- **Position:** Ensure the casualty is lying on a flat surface, preferably with their head tilted back slightly to open the airway.
- **Clear the Airway:** Check for any visible obstructions in the person's mouth or throat, and if present, gently remove them if possible.
- **Seal the Mouth:** If the casualty's mouth is open, use your hand to gently lift the chin upward to maintain an open airway. With your other hand, pinch the person's nose closed to prevent air from escaping.
- **Breathing Technique:** Place your mouth over the casualty's mouth, forming a seal, and blow into their mouth while watching for the chest to rise. Deliver the breath over one second and make sure it is sufficient to visibly inflate the chest.
- **Breathing Rate:** Provide rescue breaths at a rate of about one breath every five to six seconds (10 to 12 breaths per minute).
- **Chest Recoil:** Allow the chest to fully recoil between breaths. This enables the chest to return to its resting position, which helps in the exchange of gases during respiration.
- **Continue Until Help Arrives:** Perform rescue breathing until professional medical help arrives or until the person starts breathing on their own.

6-50. There are several methods for rescue breathing, each suitable for different situations and age groups. The main methods include:

- Mouth-to-Mouth: This common method involves the rescuer forming a seal over the casualty's mouth with their own and delivering breaths into their lungs.
- Mouth-to-Nose: Used when the casualty's mouth is injured, obstructed, or inaccessible, the rescuer closes the casualty's mouth and breathes into their nose to create a seal.
- Mouth-to-Mask: In situations like professional CPR or with specialized equipment, a pocket mask or BVM device is used. The mask covers the casualty's mouth and nose, and the rescuer delivers breaths by squeezing the bag or using the valve.
- Mouth-to-Stoma: For individuals with a tracheostomy or a surgical stoma in the neck for breathing, rescue breaths can be delivered directly into the stoma using the rescuer's mouth or a specialized device.

6-51. It is important to note that the technique and ratio of rescue breaths may vary depending on the age of the person receiving CPR (such as an adult, child, or infant) and the specific guidelines in the area or operation. Proper training in CPR techniques, including rescue breathing, is crucial to ensure accurate and effective administration of artificial ventilation during emergency situations.

6-52. It is important to follow the specific guidelines and protocols for rescue breathing based on the person's age, condition, and the resources available. Proper training and certification in CPR techniques are essential to gain proficiency in rescue breathing and to ensure the best possible outcomes during resuscitation efforts.

Chapter 7

Circulation Control

OVERVIEW OF CIRCULATION CONTROL

7-1. Hemorrhage control assessment and management in TFC falls under the “C,” for Circulation, in the MARCH-PAWS sequence. The body requires good blood flow throughout, and a lack of good blood flow may cause the casualty to go into hypovolemic shock.

7-2. Hemorrhage control in the TFC phase takes place after massive bleeding control that occurs in the CUF phase. It is possible that it did not occur, and this may be the first opportunity to address massive bleeding as well, so there is significant urgency. The first step is to reassess all prior hemorrhage control interventions for effectiveness.

7-3. The casualty’s hemorrhage control interventions must be frequently reassessed to ensure continued hemorrhage control. This includes reassessments at frequent time intervals and any time the casualty is moved or the casualty’s status changes. The casualty must be continually watched for signs of shock.

7-4. Controlling shock is the primary method of circulation control. Shock is the inadequate blood flow to body tissues. Inadequate blood volume inside the circulatory system results in inadequate oxygen delivery to the body’s cells. As cells, tissues, and organs cease to function, eventually the whole body will fail, and death follows.

Note. Hypovolemic shock can also result from blood loss due to internal bleeding into the abdominal or chest cavity. The first responder or CLS will not be able to control internal bleeding. This condition requires rapid evacuation of the casualty.

7-5. Shock is caused by a decrease in the amount of blood volume circulating in the casualty's blood circulatory system. Shocks can have many causes such as:

- Low blood volume or hypovolemia (such as dehydration or blood loss).
- Low blood pressure.
- Heart failure (pump failure).
- Neurologic damage (infection, allergic reaction).
- Obstructive causes (such as tension pneumothorax or pericardial tamponade causes obstruction of the heart).

INDICATORS OF SHOCK

7-6. Shock is usually caused by severe bleeding, but it can also be caused by severe burns classified as partial thickness burns (2nd degree) or full thickness burns (3rd degree) on 20 percent or more of the body surface. On the battlefield, assume shock is from severe blood loss (also called hemorrhagic shock). The signs and symptoms of shock include:

- Mental confusion.
- Weak or absent radial pulse.
- Rapid breathing.
- Nausea.
- Sweaty, cool, or clammy skin.
- Excessive thirst.
- Pale or gray skin.
- Previous severe bleeding

IMPORTANT!

Alert medical personnel if the casualty is presenting signs of shock.

7-7. If the casualty has both indicators of mental confusion and weak or absent radial pulse of shock, the casualty has lost a significant amount of blood. Shock will lead to the casualty's death if not quickly recognized and treated. Table 7-1 provides an overview of the effects of blood loss and highlights why it is important to quickly apply a TQ during CUF, reassess, and evaluate for additional bleeding sources during TFC.

Table 7-1. General Indicators of Shock

Blood Volume Remaining	Blood Volume Loss	Signs and Symptoms	Effects and Outcomes
4,500cc	500cc	Possible increased HR.	Usually, no effects.
4,000cc	1,000cc	Radial pulse >100. Breathing is probably normal.	Unlikely to die from this amount of loss.
3,500cc	1,500cc	Change in mental status. Weak radial pulse >100. Increased respirations.	If there is no further blood loss, still unlikely to die.
3,000cc	2,000cc	Confusion and lethargy. Very weak radial pulse >120. High respiratory rate >35.	Very possibly fatal if not managed.
2,500cc	2,500cc	Unconscious. No radial pulse, carotid pulse, HR >140. Respirations >35.	Fatal without immediate and rapid interventions.
Legend: cc cubic centimeter HR heart rate > greater than			

PREVENTION OF SHOCK

7-8. It is more effective to prevent shock through hemorrhage control than to treat it once it has developed. Even in cases where shock is already present, the most critical first action is to control any bleeding. Continuously reassess all bleeding control measures, such as tourniquets and pressure dressings, to ensure they remain effective and are adjusted tightly as soon as possible. This is a crucial step in treating shock.

7-9. Internal bleeding caused by blunt or penetrating trauma to the chest or abdomen may not be immediately controlled and can lead to shock later on. Continuous assessment of the casualty is essential. Even if a casualty is not currently in shock but has sustained serious injuries, they should be carefully monitored for signs of developing shock. It's important not to wait until signs and symptoms of shock appear. While medical personnel will provide comprehensive treatments, controlling any external bleeding early can save valuable time.

EQUIPMENT USED FOR CIRCULATION CONTROL

7-10. At present, no specialized equipment has been designated exclusively for managing circulation control. The equipment mentioned in the earlier section of this publication serves to support and enhance circulation control measures.

TIER 1 SKILLS FOR CIRCULATION CONTROL

7-11. The best way to prevent shock is to control bleeding. Responders should continuously monitor and reassess the effectiveness of bleeding control measures to ensure they are working, TQ application, wound packing with hemostatic dressing, a pressure dressing, or a combination of these.

7-12. After reassessing TQs and hemostatic dressings, the TCCC guideline on shock states: “Assess for hemorrhagic shock” (altered mental status in the absence of brain injury or weak and absent radial pulse). Reassess all bleeding control measures are still effective and TQs remain tight.

BASIC MEASURES TO CONTROL CIRCULATION

7-13. Control bleeding and then take steps to prevent shock by elevating the casualty’s legs higher than the casualty’s heart if their airway tolerates lying flat on their back.

7-14. Conduct a bleeding re-check by:

- Reassess every TQ and dressing that may have been applied earlier.
- Expose the area and determine if bleeding is controlled.
- Expose and clearly mark all TQ sites with the time of TQ application if not previously done.
- When time and the tactical situation permit, check for further bleeding, and for pulses further out on the limb than the TQ.

POSITION THE CASUALTY FOR SHOCK

7-15. Move the casualty under a permanent or improvised shelter to shade them from direct sunlight. If the casualty is conscious, place the casualty in the shock recovery position (see figure 7-1), which is on their back with their feet elevated slightly above the level of their heart. This will help their blood circulation. A log, field pack, box, rolled field jacket, or other stable objects can be used to elevate their feet. If a sitting position makes it easier for the casualty to breathe, that is fine.



Figure 7-1. Shock recovery position

7-16. The following steps will also significantly help in the prevention of shock:

- Loosen clothing at the neck, waist, or anywhere is binding.

WARNING

Do not loosen clothing if in a chemically contaminated area.

- Prevent the casualty from getting chilled or overheated. Using a blanket or clothing, cover the casualty to avoid loss of body heat by wrapping the blanket completely around the casualty.

Note. Ensure no part of the casualty is touching the ground, as this increases loss of body heat.

- Calm and reassure the casualty by—
 - Taking charge and showing self-confidence.
 - Assuring the casualty that they are being taken care of.
- Watch the casualty closely for life-threatening conditions and check for other injuries, if necessary.
- Seek medical aid.
- Record the treatment on DD Form 1380.

WARNING

If the casualty has a fractured lower extremity, do not place the casualty in the shock position until the fracture has been splinted.

FOOD OR DRINK

7-17. When providing first aid for shock, DO NOT give the casualty any food or drink. If you must leave the casualty or if they are unconscious, turn their head to the side (see figure 7-2) to prevent them from choking, if they vomit.



Figure 7-2. Casualty's head turned to the side

TIER 2 SKILLS FOR CIRCULATION CONTROL

7-18. Tier 2 skills for this section specifically address the crucial aspect of circulation control. Emphasizing the need for rapid and appropriate interventions to prevent further harm and optimize the chances of survival for wounded Service members. Circulation control involves techniques and procedures aimed at managing bleeding and vascular injuries. These are often encountered in high-stakes and dynamic combat environments.

PELVIC FRACTURES

7-19. Pelvic fractures can be a cause of massive internal bleeding and impact circulation, which is the “C” in the MARCH-PAWS sequence. A pelvic fracture may be suspected if the casualty's injuries are a result of blunt force or blast trauma with one or more of the following:

- Pelvic pain.
- Major lower-limb amputation or lower-limb near amputations.
- Pelvic deformities, penetrating injuries, or bruising near the pelvis.
- Pelvic instability or crepitus, which is a crinkly, grating feeling, or sound under the skin.
- Unconsciousness or shock.

7-20. If a pelvic fracture is suspected, the casualty will require advanced evaluation by medical personnel, and you should notify medical personnel of the potential for a pelvic fracture as soon as possible.

PRESSURE BANDAGE REASSESSMENT

7-21. The regular reassessment of pressure bandages is crucial in managing wounds effectively. The CLS are tasked with ensuring that there is no ongoing bleeding from the wound site. Additionally, it's important to check for circulation below the pressure bandage by feeling for a distal pulse—that is, a pulse found below the level of the bandage.

7-22. If the skin below the bandage is cool, bluish, numb, or if the distal pulse is absent, these may be signs that the bandage is applied too tightly, potentially hindering blood flow. In such cases, the bandage should be loosened and then securely retied to ensure proper circulation is restored.

7-23. All dressings and bandages must be checked routinely, especially every time a casualty is moved. This practice helps to monitor the effectiveness of the initial treatment and to make timely adjustments as needed.

7-24. In situations where a pressure bandage or hemostatic dressing fails to control bleeding, a TQ should be applied approximately two to three inches above the site of bleeding, if feasible. If placing a TQ is not possible and the existing bandage is either ineffective or saturated with blood, replace it with a fresh hemostatic dressing. Ensure to pack the wound thoroughly with the hemostatic material, applying constant, direct pressure at the bleeding source for optimal effectiveness.

EXPEDIENT TECHNIQUES FOR CIRCULATION CONTROL

7-25. Medical personnel may not always be readily available. Nonmedical personnel must rely on themselves, other Service members' knowledge, fundamentals of first aid for circulation control, until medical assistance arrives. This section will provide additional first aid, non-TCCC, procedures that nonmedical personnel can use for circulation control.

TECHNIQUES FOR NON-MEDICAL PERSONNEL

7-26. When medical personnel are not immediately available, non-medical personnel must depend on themselves and the knowledge of other Service members. The fundamentals of first aid for controlling circulation are crucial until professional medical help arrives. This section aims to provide essential, non-TCCC first aid procedures that non-medical personnel can use effectively to maintain circulation control.

7-27. Non-medical personnel are often the first responders in critical situations, making it essential for them to be familiar with basic first aid techniques. Understanding and applying these methods can be lifesaving when delays in professional medical treatment occur. Training in these expedient techniques empowers individuals to act confidently and efficiently, providing critical care in emergencies.

7-28. The guidance provided here includes several key first aid actions that non-medical personnel can take to manage circulation issues effectively. These include methods to minimize bleeding, stabilize the affected individual, and prepare for the arrival of medical professionals. Each technique has been chosen for its simplicity and effectiveness, ensuring that those with minimal medical training can apply them under pressure.

7-29. It is vital for all personnel, especially those in high-risk environments, to be trained in these techniques. Such training ensures that every member of a team can provide essential care in emergencies, enhancing overall safety and increasing the likelihood of positive outcomes in critical situations.

TREATMENT FOR ANAPHYLACTIC SHOCK

7-30. Anaphylactic shock is a severe and life-threatening allergic reaction that significantly impacts the circulatory system. This medical emergency is marked by a swift release of histamines and other chemicals into the body, leading to vasodilation and increased permeability in blood vessels. Such changes can disrupt the adequate perfusion of vital organs and tissues, putting the casualty's life at great risk.

7-31. Vasodilation, a key feature of anaphylactic shock, involves the relaxation of smooth muscle cells within blood vessel walls, causing these vessels to widen and blood pressure to drop significantly. This physiological process can rapidly deteriorate the casualty's condition.

7-32. The treatment of anaphylactic shock is urgent and time sensitive. Immediate actions include calling for medical help and, if available, using the casualty's epinephrine auto-injector. Anaphylactic reactions can

manifest within seconds or minutes after exposure to an allergen, affecting the skin, respiratory, and circulatory systems.

7-33. Prompt recognition and response to anaphylactic shock are critical. Assessing both the signs and symptoms early on is essential for effective intervention. Quick and accurate identification of the condition allows for the necessary steps to be taken to mitigate its life-threatening effects.

7-34. To effectively check for anaphylactic shock, Service members should follow a systematic approach:

- Assess Symptoms:
 - Observe for sudden and severe symptoms like difficulty breathing, wheezing, or shortness of breath.
 - Check for a rapid or weak pulse.
 - Look for swelling of the face, lips, or tongue.
 - Note any hives or widespread skin redness.
 - Be alert to signs of confusion, dizziness, or loss of consciousness.
- Ask Questions:
 - Inquire about any known allergies or recent exposures to potential allergens.
 - Ask if they have ingested or come into contact with a known allergen recently.
- Observe Breathing:
 - Check for signs of respiratory distress, such as rapid or labored breathing.
 - Listen for wheezing or other abnormal breathing sounds.
- Check Circulation:
 - Assess the pulse to determine its rate and strength.
 - Look for signs of poor circulation, such as pale or bluish skin.
- Look for Other Symptoms:
 - Be aware of nausea, vomiting, abdominal pain, or a feeling of impending doom.

7-35. If anaphylactic shock is suspected, immediate medical assistance is crucial. The primary treatment involves administering epinephrine (adrenaline). If an epinephrine auto-injector is available, assist the affected individual in using it, or administer it yourself if you are allowed and trained. Dosage guidelines are typically provided with the injector—0.15 milligrams (mg) for children and 0.3 mg for adults weighing more than 66 pounds.

7-36. Further doses of epinephrine may be necessary, administered every 5 to 10 minutes if prescribed and symptoms persist. Additionally, provide supportive measures for treating shock, respiratory failure, circulatory collapse, or cardiac arrest:

- Position the casualty supine with legs elevated if injuries permit.
- Perform rescue breathing and external chest compressions if required.
- Monitor vital signs every three to five minutes until the casualty is stable.

7-37. Record all first aid actions on DD Form 1380 and request MEDEVAC immediately. Use CASEVAC only if MEDEVAC is unavailable and transport the casualty to the nearest MTF as soon as possible. Anaphylaxis is life-threatening, and immediate action is crucial. Seek professional medical help promptly and provide supportive care while awaiting arrival of medical personnel.

Chapter 8

Hypothermia Control

OVERVIEW OF HYPOTHERMIA CONTROL

8-1. The “H” in MARCH-PAWS stands for hypothermia prevention. Hypothermia begins when the body’s core temperature drops below 97° Fahrenheit (F). It can be mild (93-96°F), moderate (89-93°F), or severe (less than 89°F), depending on the temperature. To accurately diagnose hypothermia, the core temperature must be measured rectally with a special low-range thermometer, as mouth and ear measurements won't work well in cold environments.

8-2. Hypothermia can be caused by being in a cold environment or by losing a lot of blood, a condition known as hypovolemia. This is when there’s a significant drop in the volume of blood circulating in the body. Both environmental exposure and loss of blood can lead to dangerously low body temperatures, which can be life-threatening if not treated.

8-3. Environmental hypothermia happens when the body loses heat faster than it can produce it, often from being in cold weather for too long. Hypothermia from hypovolemic shock happens due to a large loss of blood or fluids, making it hard for the body to keep warm. Both causes can lead to serious health issues, including death, if not managed quickly.

8-4. Hypothermia can happen suddenly, like from falling into cold water, or slowly, if someone is exposed to cold conditions for a long time. It can even happen in mild temperatures if a person is wet, which makes it easier to lose body heat. Factors like wet clothing, not enough food, exhaustion, and drinking alcohol or using drugs can make hypothermia worse.

8-5. Massive bleeding can cause hypothermia, which in turn can cause more bleeding because cold temperatures can prevent blood from clotting properly. This is a problem even in warm weather, and especially for people in shock who can’t generate enough body heat. It’s crucial to stop this cycle to save the person’s life.

SIGNS AND SYMPTOMS OF HYPOTHERMIA

8-6. In the TFC phase of care, the focus has shifted from immediate life-threatening hemorrhage control during CUF phase, to the reassessment of all previous interventions. This is followed by the prevention and treatment of other injuries and complications such as hypothermia. It is important to identify the signs of hypothermia. Some easily recognizable signs of hypothermia are:

- Vigorous shivering is typically present.
- Shivering may decrease or cease as the core temperature continues to fall.
- Conscious, but usually apathetic or lethargic.
- Confusion.
- Sleepiness.
- Slurred speech or mumbling.
- Slow breathing and drowsiness.
- Weak pulse.
- Low or unattainable blood pressure.
- Change in behavior with or without poor control over body movements, and with or without slow reactions.

8-7. The signs and symptoms of hypothermia may vary depending on the severity and can be categorized into mild, moderate, and severe stages:

- Mild Hypothermia:
 - Shivering.
 - Cold and pale skin.
 - Numbness or tingling in extremities.
 - Fatigue and drowsiness.

- Slurred speech.
- Impaired coordination and fine motor skills.
- Moderate Hypothermia:
 - Intense shivering or shivering stops.
 - Difficulty in speaking and thinking clearly.
 - Clumsiness and lack of coordination.
 - Cold, pale, or blue-gray skin.
 - Slow breathing and heart rate.
 - Weak pulse.
- Severe Hypothermia:
 - Shivering may stop completely.
 - Confusion and disorientation.
 - Loss of consciousness.
 - Very cold, pale, or bluish skin.
 - Slow or irregular breathing.
 - Weak or absent pulse.

FIRST AID FOR HYPOTHERMIA

8-8. The goals for first aid for hypothermia are to rescue, examine, insulate, and rapidly transport. If untreated, hypothermia is a true medical emergency and requires evacuation. Evacuate as soon as possible with the casualty lying down.

CAUTION

Do not allow the casualty to use tobacco, consume alcohol, or caffeinated drinks.

8-9. Re-warming techniques include—

- Remove the casualty from the cold environment.
- Replace wet clothing with dry clothing.
- Cover the casualty with insulating material or blanket.
- Wrap the casualty from head to toe.
- Avoid unnecessary movement from the casualty.
- If the casualty is conscious, slowly give high caloric sweet warm fluids.
- Seek medical aid.

8-10. Hypothermia is a serious condition that requires professional evaluation and treatment. Attempting to rewarm a severely hypothermic person should be done by medical professionals using appropriate techniques. While waiting for help to arrive, focus on preventing further heat loss and providing support to the person.

PREVENTION OF HYPOTHERMIA

8-11. Preventing hypothermia is critical in reducing the effects of heat loss and preventing deaths from uncontrolled bleeding. Immediate action to prevent hypothermia should commence right after an injury occurs. Without preventative measures, there is a real risk that a casualty may succumb to hemorrhage from a wound that could otherwise be survivable. As blood loss itself can lead to hypothermia, it is vital for first responders and CLS to control bleeding and actively engage in both passive and active measures to prevent the casualty's body temperature from dropping.

8-12. The quicker hypothermia prevention begins, the less severe its impact will be on the casualty's bleeding and potential shock. Minimizing exposure to the elements is essential. Actions such as replacing wet clothing with dry ones and placing the casualty on an insulated surface should be prioritized. Environmental factors such as daily temperature fluctuations, nightfall, changes in altitude, and wind exposure must also be considered as they can significantly affect body heat retention.

8-13. Prevention of hypothermia is far more manageable than treating it once it has set in. Starting hypothermia prevention measures early is crucial because a decreased body temperature can adversely affect blood clotting processes, heightening the risk of continued bleeding—even in warm conditions.

8-14. CLS personnel must be proactive in assuming that every casualty is at risk of developing hypothermia, utilizing all known prevention techniques. If a casualty is already hypothermic, the response should be even more aggressive, treating the condition as a direct threat to the casualty's life.

MANAGEMENT OF HYPOTHERMIA

8-15. To prevent hypothermia, it is essential to keep the casualty warm, even in very hot environments. This is particularly crucial when a casualty is suffering from hemorrhagic shock due to blood loss, as they are at an extreme risk for developing hypothermia. Both active and passive methods should be employed to ensure the casualty's body temperature is maintained. This involves proactive measures such as providing thermal blankets, using heat packs, and minimizing exposure to cold or windy conditions. Maintaining warmth not only aids in preventing hypothermia but also helps stabilize the casualty's overall condition.

EQUIPMENT USED FOR HYPOTHERMIA CONTROL

8-16. The hypothermia control equipment falls into either active warming (external source of heat) or passive warming blankets, or a poncho liner to treat or prevent hypothermia. The most common equipment seen is the hypothermia prevention and management kit, blankets, or ponchos.

ACTIVE HYPOTHERMIA MANAGEMENT AND MATERIAL

8-17. Medical personnel will distribute active hypothermia blankets based on the unit mission and load. Active hypothermia blankets are activated when their heating elements are exposed to air and can produce temperatures reaching 104° F for up to eight hours. Active hypothermia blankets are applied to a casualty who cannot generate their own heat, but not directly on their skin, because the activated blankets can cause burns.

8-18. Apply the active warming blanket from the active hypothermia materials to the casualty's torso, not directly on the skin, and cover the casualty with the passive hypothermia shell. Continue to reassess the casualty to determine if additional methods are needed to prevent or treat hypothermia. The heat reflective shell will help to retain the heat produced by the ready-heat blanket. It has an incorporated hood and Velcro closures down each side to allow exposure of an arm or a leg. Such exposure allows the medic to attend to IVs and TQs.

8-19. If an active hypothermia device is not available, a combination of the passive warming blanket and active warming blanket may be used. Active hypothermia treatment uses heating sources such as the ready-heat blanket to warm the casualty. This requires a chemical reaction with oxygen, so at higher altitudes there may not be enough oxygen to sustain the chemical reaction required to generate heat.

8-20. You must determine based on the environmental conditions whether the casualty needs both the outer shell and the heating blanket. It is sufficient to just keep them covered with the shell, because it is already hot outside.

8-21. If no active rewarming equipment is available, then use passive warming devices that will retain heat and keep the casualty dry. The casualty should be assessed for hemorrhagic shock and ensure bleeding is controlled.

WARNING:

Active warming is contraindicated in a casualty who is showing signs and symptoms of hyperthermia from heat related illness.

PASSIVE HYPOTHERMIA MANAGEMENT AND MATERIAL

8-22. Passive hypothermia blankets are designed to retain the casualty's body heat by enclosing them within a thermal barrier, keeping them off the cold ground. While passive strategies prevent further cooling, they do not actively warm the casualty. Therefore, it's critical to avoid placing the casualty directly onto cold, wet, or damp surfaces; instead, a blanket or poncho should be positioned underneath them to provide insulation. In the absence of specialized re-warming equipment, dry blankets, poncho liners, sleeping bags, or similar items should be used to retain heat and keep the casualty dry and elevated from the ground.

8-23. Key considerations include recognizing that blood loss can significantly reduce body temperature, even in warmer climates. When using a blanket or any passive heating material, it is crucial to wrap the casualty completely, including their head, but ensure the face remains uncovered to prevent breathing obstructions. Leaving any part of the body exposed can result in heat loss and worsen hypothermia. These measures are vital in managing a hypothermic casualty, especially in the absence of active warming tools.

TIER 1 SKILLS USED FOR HYPOTHERMIA CONTROL

8-24. The first responder must prevent, and address hypothermia compounded by massive blood loss. This is not hypothermia due to cold weather, but can be prevented by—

- Keeping clothing or uniform on the casualty.
- Replacing extremely wet clothing if possible and keeping the casualty dry.
- Keeping the casualty off the ground, by placing a barrier between the casualty and the ground or lifting them off the ground, if possible.

Note. Contact with the ground increases loss of body heat.

- Using dry blankets, poncho liners, sleeping bags, or anything that will retain heat.
- Minimizing the casualty's exposure to the environment.

TIER 2 SKILLS USED FOR HYPOTHERMIA CONTROL

8-25. The Tier 2 TCCC skills emphasize advanced yet practical approaches to managing battlefield casualties under hypothermic conditions. Personnel trained in Tier 2 TCCC are equipped with the knowledge and skills to effectively use available medical equipment to prevent and treat hypothermia. This includes the application of specialized gear such as hypothermia prevention and management kits, which are designed to maintain body temperature and prevent further heat loss in injured soldiers. In addition, Tier 2 personnel are trained to monitor vital signs closely, ensuring that the core body temperature is maintained within a safe range, thereby reducing the risk of hypothermia-related complications.

EXPEDIENT SKILLS USED FOR HYPOTHERMIA CONTROL

8-26. In situations where standard medical supplies are not available, field expedient techniques for hypothermia control become crucial. Soldiers are taught to use their environment and available resources to prevent and manage hypothermia. For instance, using space blankets or even regular blankets to wrap the victim can significantly reduce heat loss. If these are not available, utilizing materials such as leaves, grass, or even newspapers as insulation can be a life-saving improvisation. Creating a barrier between the victim and the cold ground, such as with a sleeping pad, backpack, or any available material, can also prevent further body heat loss. These improvised methods are essential skills in TCCC, enabling soldiers to provide effective hypothermia care even in resource-limited settings.

Chapter 9

Secondary Injury Assessment Using Pain, Antibiotics, Wounds, Splinting and Mechanism of Injury, Injury Type, Signs and Symptoms, and Treatments

OVERVIEW OF PAIN, ANTIBIOTICS, WOUNDS, AND SPLINTING AND MECHANISM OF INJURY, INJURY TYPE, SIGNS AND SYMPTOMS, AND TREATMENTS

9-1. As discussed in paragraph 3-96 on page 39, PAWS is the second part of the MARCH-PAWS casualty assessment and is conducted after the MARCH portion. The MIST reporting was introduced as a standard component of the MEDEVAC request during Operation Enduring Freedom in Afghanistan.

9-2. While not officially part of the NATO and U.S. standard MEDEVAC request, MIST reporting has become customary in some operations. Often referred to as the 10th line in the 9-line MEDEVAC request, the completed DD Form 1380 serves as the MIST report. However, the high volume of casualties expected during large scale combat operations may make it challenging to complete MIST reporting for each casualty.

9-3. The MIST report facilitates the transfer of medical information to the receiving health care provider by providing a structured approach to the casualty handoff. A MIST report delivers additional evacuation information that might be required by theater commanders and assists the receiving MTF in preparing for specific inbound casualties. However, transmitting a MIST report for each casualty during operations and mass casualty incidents may not always be timely or feasible.

MANAGEMENT OF PAIN AND ANTIBIOTICS

9-4. The combat wound medication pack (CWMP) is both the “P” (pain) and the “A” (antibiotics) in the MARCH-PAWS sequence. Pain is common with battlefield injuries. Some injuries and levels of pain can be treated safely by using pain medications (also known as analgesics) in the CWMP. These include fractures, burns, and eye injuries.

9-5. The CWMP is a prepackaged medication packet with drugs and dosages specifically chosen for use in combat casualty care. The CWMP contains drugs for mild to moderate pain (Meloxicam and Acetaminophen) and an antibiotic specific for penetrating wounds (Moxifloxacin). The CWMP should be used only for combat related injuries and is not for routine use. It may be procured through medical logistics systems as a pre-packaged blister pack or, may be locally packaged by medical personnel.

9-6. For pain relief on the battlefield of mild to moderate pain that will not keep the casualty out of the fight, ensure they take their CWMP. The casualty should take all three medications in the CWMP. This can give significant pain relief and will not alter the casualty’s mental status. This is a good option when the casualty’s pain and wounds are not severe enough to keep them out of the fight.

Note. If the casualty has wounds or pain severe enough to render them unable to fight, then medical personnel have other options for more effective pain relief. The medications given by medical personnel will generally require that the casualty be disarmed because the medications can alter the casualty’s mental status.

9-7. CWMP is a prepackaged pill pack containing CoTCCC-recommended medications for use in combat casualty care. The CWMP is found in the casualty’s first aid kit. It contains medications for pain and antibiotics that can be taken by mouth.

9-8. The medication in the CWMP should be taken as soon as possible after life-threatening conditions have been addressed. All medications administered should be documented along with the time given on the DD Form 1380.

9-9. The CWMP contains the following components:

- Two 500mg caplets of acetaminophen (total 1,000mg) in extended-release form.
- One 400mg tablet of moxifloxacin.
- One 15mg tablet of meloxicam.

9-10. Each of the three medications in these dosages is contained in a blister pack.

Note. The popular nonsteroidal anti-inflammatory drugs such as ibuprofen, naproxen, and aspirin interfere with platelet function and blood clotting and can significantly increase the risk of bleeding in combat casualties.

9-11. Acetaminophen and meloxicam do not alter platelet function and are safe and effective for use in combat casualties. Meloxicam does not alter platelet function and is the preferred Non-Steroidal Anti-Inflammatory Drug for personnel who may see combat in the next 7-10 days.

9-12. The CWMP should be given when the casualty:

- Is conscious and able to swallow.
- It has mild to moderate pain.
- Is still able to fight if needed.
- Has any penetrating wounds or break of the skin.

9-13. The CWMP should NOT be given if the casualty:

- Is unable to swallow or take oral meds, such as when the casualty is unconscious, or has severe facial trauma, or burns.
- Has known allergies to the CWMP medications.
- If the casualty is unconscious, refer them to medical personnel as soon as possible.

Note. If the casualty has a break in the skin in a traumatic injury, they should take the CWMP. Otherwise, consult with medical personnel before giving it.

9-14. Battlefield wounds are often dirty and susceptible to infection. Early administration of antibiotics from the CWMP may reduce the chance of later infections. Wound infections can kill the casualty or delay their recovery. The CWMP should be given as soon as possible for wounds after life-threatening issues have been addressed.

MANAGEMENT OF WOUNDS

9-15. Beyond MARCH, there are additional or secondary injuries that should be identified and may need treatment. The secondary injuries identified in TCCC are organized by eye trauma, head injury, burns, and fractures. These secondary injuries are treated in TFC.

9-16. Wound management is the “W” in the MARCH-PAWS sequence. Once the casualty's hemorrhage has been controlled, earlier control interventions are completed, and wound management must be rechecked. All wounds must be frequently reassessed to ensure continued hemorrhage control.

9-17. The first responder or CLS should confirm all wounds have been addressed and reassess for re-bleeding under gauze or bandages to ensure bleeding has been controlled. The first responder or CLS must look for blood flowing around or under TQs, bandages, and dressings. If the bleeding has not been controlled, tighten the TQ or pressure bandage for that wound if possible, and redress any wounds as necessary. Also, check other wounds to make sure that:

- Eye injuries are shielded.
- Open chest wounds are accounted for.
- All wounds are wrapped.
- Fractures are splinted.

9-18. Wounds should always be monitored and frequently reassessed to ensure hemorrhage has been controlled. Any wounds that continue to bleed must be packed with new hemostatic dressing. Once the dressing has been applied with pressure for three minutes, carefully observe for blood continuing to flow

from under the gauze to determine if bleeding has been controlled. When the bleeding has stopped, a new pressure bandage should be applied over the hemostatic dressing.

9-19. Any previously untreated wounds should be dressed by applying (or packing) gauze with direct pressure. Non-life-threatening bleeding usually does not need hemostatic dressing. If dressings or gauze are not available, use clean, dry cloth such as torn clothing, or a cravat if available.

9-20. Minor wounds include minor lacerations and abrasions, such as road rashes. Other wounds that may need to be dressed include major wounds that are no longer bleeding, such as:

- Amputation stumps.
- Gunshot wounds that required a TQ.
- Major lacerations.
- Shrapnel wounds, possibly with shrapnel still in place.
- Impaled objects.

9-21. Applied pressure bandages must be continuously reassessed for:

- Increased pain.
- Skin color.
- Loss of the pulse.

9-22. If any of the above develops, it might indicate an emergency. Ensure that the applied bandage isn't too tight and loosen it as needed while keeping the bleeding controlled.

Note. Pulses are reassessed after dressings are placed to ensure that they are not too tight. Do not ever apply a dressing and forget it!

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Chapter 10

Eye Trauma

OVERVIEW OF EYE TRAUMA

10-1. The eye is a vital sensory organ, and blindness is a severe physical handicap. Timely first aid for eye injuries may relieve pain and may help prevent shock, permanent eye injury, and possible loss of vision. The eye is very sensitive: any injury can be easily aggravated if it is improperly handled. Injuries in the eye may be quite severe. Cuts of the eyelids can appear to be very serious, but if the eyeball is not involved, a person's vision usually will not be damaged. However, lacerations (cuts) of the eyeball can cause permanent damage or loss of sight.

10-2. Exposure to intense light can burn an individual. Infrared rays, eclipse light (if the casualty has looked directly at the sun), or laser burns cause injuries of the exposed eyeball. Ultraviolet (UV) rays from arc welding can cause a superficial burn to the surface of the eye. These injuries are generally not painful but may cause permanent damage to the eyes. Immediate first aid is usually not required. Loosely bandaging the eyes may make the casualty more comfortable and protect their eyes from further injury caused by exposure to other bright lights or sunlight. Eyesight is one of our most important senses. Damage to an eye can be irreversible if not treated correctly and as early as possible. The following signs should make you suspect a penetrating eye injury:

- Bleeding surrounding the eye, inside the eyeball, or coming from the eyeball.
- Obvious penetration of shrapnel or debris into the eyeball or eye socket.
- Protruding objects from the globe of the eyeball.
- Swelling or lacerations of the globe of the eyeball.
- Protrusion of the globe of the eyeball from the eye socket.
- Reduced vision and swelling of the eye area.
- Misshapen or distorted parts of the eye from normal.

EQUIPMENT USED FOR THE TREATMENT OF EYE TRAUMA

10-3. The rigid eye shield is a barrier to environmental hazards and to protect the eye by providing a safe enclosure, until additional medical attention is available. The rigid eye shield is normally plastic or metallic, with a domed shape to enhance the protection of eye injuries, without applying pressure. It may be self-adhering or require tape. If a rigid eye shield is not available, or already used, the following may be used as an improvised or expedient rigid eye shield:

- Protective glasses.
- Clean cup.
- Dip can.

10-4. The Military Acute Concussion Evaluation 2 Card is an acute assessment tool for all medically trained personnel who treat Service members involved in a potentially concussive event. The Military Acute Concussion Evaluation 2 incorporates current state-of-the-science traumatic brain injury information, which looks at the systems in charge of integrating balance, vision, and movement.

TIER 1 SKILLS USED TO TREAT EYE TRAUMA

10-5. If a penetrating eye injury is seen or suspected, the first responder must shield the injured eye. The casualties injured eye must be covered with a rigid eye shield, not a pressure patch. In typical situations, a concave shield is placed over the injured eye, (not both eyes), and taped in place. If there is a protruding object in the eye, bandage in place — do not remove the object. Do not waste critical time by conducting further treatment. The casualty should be evacuated as soon as possible.

WARNING

DO NOT apply pressure to the injured eye.

10-6. The rigid eye shield is secured over the injured eye with tape at a 45-degree angle across the forehead and cheek. Rigid eye shields should be placed over both eyes only when you are sure and at least suspect that both eyes have been injured. If a rigid eye shield is not available, tactical eyewear may be used to protect the eyes from further trauma.

10-7. When only one eye has been injured, do not place a shield over the uninjured eye to prevent eye movement. Movement has not been shown to worsen the outcome for the injured eye. Blindness, resulting from placing eye shields over both eyes unnecessarily, makes an otherwise ambulatory casualty a litter casualty, and is psychologically stressful. The first responder should be prepared to render first aid for lacerations of the eyelid, foreign bodies in the eye, extrusions of the eye, and chemical burns of the eye.

INITIATE FIRST AID FOR LACERATIONS OF THE EYELID OR EYEBALL WITH EYE-SHIELD

10-8. Lacerated or torn eyelids may bleed heavily, but bleeding usually stops quickly. Cover the injured eye with a sterile dressing. **DO NOT** put pressure on the wound because you may injure the eyeball. Handle torn eyelids very carefully to prevent further injury. Place any detached pieces of the eyelid on a clean bandage or dressing and immediately send them with the casualty to the medical facility.

10-9. Lacerations or cuts to the eyeball may cause serious and permanent eye damage. Cover the injury with a loose sterile dressing. **DO NOT** put pressure on the eyeball because additional damage may occur.

CAUTION

DO NOT apply pressure when there is a possible laceration of the eyeball. The eyeball contains fluid. Pressure applied over the eye will force the fluid out, resulting in permanent injury. **APPLY A PROTECTIVE DRESSING WITHOUT ADDED PRESSURE.**

10-10. Survey the situation; if the injury is from chemical exposure, immediately wash the eye. Otherwise, **DO NOT** put anything on the eye **ITSELF**, especially if you think the eye is cut.

10-11. If necessary, remove the casualty's headgear. Position and treat the casualty as follows:

- The conscious casualty will be placed in a seated position.
- The unconscious casualty will be placed in a supine position with their head slightly elevated.
- Gently clean any dirt or blood from the affected area around the eye.

WARNING

DO NOT put pressure on the eye or remove anything from the eye surface. This could cause additional harm, such as loss of eyesight or loss of the eye.

10-12. Examine the eyes for the following:

- Objects protruding from the eyes.
- Look for foreign objects in or on the eyes and damage to the eyes.
- Swelling or lacerations of the eyes.
- Bloodshot appearance of the white of the eyes.
- Bleeding—
 - Surrounding the eye.

- From inside the eyeball.
- Coming from the eyeball.

10-13. Ask the casualty if they are wearing contact lenses, but DO NOT force the eyelids open. Record and communicate to medical personnel (first responders, emergency medical technicians, or combat medics) that contact lenses are being worn.

10-14. Categorize the injury as follows:

- Injury to the tissue surrounding the eye (lacerations and contusions).
- Injury to the eyeball.
- Extrusion (the eye is protruding out of the eye socket).
- Foreign bodies.
- Protruding (impaled) objects.

Note. Torn eyelids should be handled carefully. Wrap any fragments in a separate moist dressing and evacuate with the casualty.

10-15. Provide first aid for the injury as follows:

- Control bleeding with light pressure from a dressing.

WARNING

Pressure is used for orbital injuries, DO NOT use any pressure if injury to the eyeball itself is suspected.

Injuries to the eye itself require an immediate covering with an eye shield.

- Gently clean any dirt or blood from the affected area around the eye.
- Cover the eyelid with the rigid eye-shield provided in the first aid kit, or anything that provides a hard cover over the eye such as—
 - Eye protective gear, ballistic glasses, or goggles.
 - The bottom of a disposable cup for example, one made of extruded polystyrene (foam), plastic, or paper.

Note. This helps prevent any undue pressure or further damage to the eye.

- Secure the shield by applying tape from the cheek to the forehead (figure 10-1 on page 100). If no tape is available, use gauze, cloth dressing, shoelace, or string to secure in place.

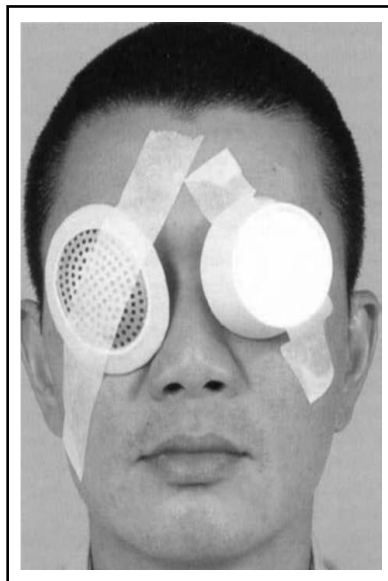


Figure 10-1. Rigid eye-shield or cup properly secured over the injury

CAUTION

ALWAYS use a rigid eye-shield to provide a barrier to prevent further damage. NEVER place anything under the rigid eye-shield or directly on the eye.

10-16. Preserve any avulsed (detached) skin or tissue in a clean sterile gauze soaked in saline (if available) and transport it with the casualty for possible grafting.

10-17. While seeking CASEVAC or MEDEVAC:

- Do not cause further injury.
- Limit movements that could cause further injury to the eye.
- Do not administer medication to prevent coughing, sneezing, or vomiting.

10-18. Consider giving the entire combat pill pack if medical treatment takes more than three hours from the time of injury and if giving the medication to the casualty does not cause nausea.

10-19. Record first aid administered on the DD Form 1380. Evacuate the casualty:

- On their back, with the head elevated and immobilized.
- With their corrective eyeglasses, even if they are broken.
- Ensure the casualty's personal belongings, such as identification and any critical medications, accompany them during evacuation.

10-20. Do not cause further injury to the eye or casualty while waiting for evacuation. Provide assurance to the casualty and stay with the casualty until relieved by medical personnel or competent authority.

INITIATE FIRST AID FOR FOREIGN BODIES IN THE EYE WITH AN EYE-SHIELD

10-21. With impaled objects or significant sized foreign bodies, both eyes are usually bandaged to help secure the foreign body in the injured eye. In a battlefield environment, leave the uninjured eye uncovered so that the casualty can see.

10-22. Survey the situation, if the injury is from chemical exposure, immediately wash the eye. Otherwise, DO NOT put anything on the eye ITSELF, especially if you think the eye is cut. Examine the eyes as follows:

- Look at the eyes — are the pupils equal and round?

- Look for bleeding or foreign bodies in the eye.

WARNING

DO NOT put pressure on the eye or remove anything from the eye surface. This could cause additional harm, such as loss of an eye.

- Foreign bodies in the eye are treated by appropriate medical personnel or by the casualty if it can easily be removed (for example, a surface non-embedded material such as an eyelash or speck of dirt).

10-23. Use the rigid eye-shield provided in the first aid kit, or anything that provides a hard cover over the eye, such as—

- Eye protective gear, ballistic glasses, or goggles.
- The bottom of a disposable cup, for example one made of extruded polystyrene (foam), plastic, or paper.

Note. This helps prevent any undue pressure or further damage to the eye. DO NOT put any pressure on the eye.

- Secure the rigid eye-shield by applying tape from the cheek to the forehead. If tape is unavailable, use gauze, cloth dressing, shoelace, or string to secure in place.

CAUTION

Removing foreign bodies from the eye is NOT a first aid procedure. Seek medical treatment.

Note. In hazardous conditions (such as combat or unsure footing), leave the good eye uncovered long enough to ensure the casualty's safety.

CAUTION

DO NOT attempt to remove a foreign object stuck to or sticking into the eyeball. A medical officer (physician) must remove such objects.

10-24. Apply dry sterile dressings to build around and support the object. This will help prevent further contamination and minimize movement of the object. Cover the injured eye with a paper cup or cardboard cone.

10-25. Seek further medical aid immediately.

10-26. Obtain details about the injury—

- Source and type of foreign bodies if known.
- Whether the foreign bodies were windblown, or by what type of high-velocity mechanism.
- Time of onset and length of discomfort.
- Any previous injuries to the eye.

10-27. Do not cause additional injury to the eye—

- Do not probe for foreign bodies.
- Do not put pressure on the eyeball.
- Do not remove the impaled object.

- Limit movements that could cause further eye injury.
- Do not administer medication to prevent coughing, sneezing, or vomiting.

10-28. Consider giving the entire combat pill pack if medical treatment takes more than three hours from the time of injury, and if giving the medication to the casualty does not cause nausea.

10-29. Record the first aid administered and other information as required on the DD Form 1380 (provided in the first aid kit).

10-30. Do not cause further injury to the eye or casualty while waiting for evacuation. Provide assurance and stay with the casualty until relieved by medical personnel or competent authority.

INITIATE FIRST AID FOR EXTRUSIONS OF THE EYE WITH AN EYE-SHIELD

10-31. Service members may encounter casualties with severe eye injuries that include an extruded eyeball (eyeball out-of-socket). In such instances, you should gently cover the extruded eye with a loose moistened dressing and cover the unaffected eye. **DO NOT** bind or exert pressure on the injured eye while applying the dressing. Keep the casualty quiet, place them on their back, treat for shock, and evacuate them immediately.

10-32. Survey the situation and remove or avoid hazards.

10-33. Remove the casualty's headgear, if necessary.

10-34. Position a—

- Conscious casualty in a seated position.
- Reassure and inform the conscious casualty of the first aid steps that you are providing.
- Unconscious casualty lying on their back with their head slightly elevated.

WARNING

DO NOT put pressure on the eye or remove anything from the eye surface. This could cause additional harm, such as loss of eyesight or loss of the eye.

10-35. Examine the eyes for—

- Objects protruding from the eye or eyes.
- Swelling or lacerations of the eyes.
- Bloodshot appearance of the eyes.
- Bleeding—
 - Surrounding the eye.
 - From inside the eyeball.
 - Coming from the eyeball.
- Contact lenses. Ask the casualty if they are wearing contact lenses, however, do not force the eyelids open.
- Extrusion (the eye is protruding from the socket).

CAUTION

DO NOT attempt to reposition the eyeball or replace it in the eye socket.

10-36. Provide appropriate first aid for the injury as follows:

- Position the casualty face up.
- Cut a hole into several layers of dressing material and then moisten it. Use sterile or the cleanest water if available, preferably potable.

- Gently place the dressing so the injured eyeball protrudes through the hole, with minimal contact. Surround and protect the eyeball, minimizing the movement of the dressing. The dressing should be built up higher than the eyeball.

Note. If available, place a paper cup or cone-shaped piece of cardboard over the eye. This material should not touch the eye. Do not apply pressure to the injury site. Apply roller gauze to hold the cup in place.

- The provided rigid eye-shield will not be deep enough to properly protect the eye extrusion injury.

10-37. Call for medical aid immediately and arrange for MEDEVAC as soon as possible.

10-38. While calling for CASEVAC or MEDEVAC —

- Limit movements that could cause further eye injury.
- Do not administer medication to prevent coughing, sneezing, or vomiting.
- Consider giving the entire combat pill pack if medical treatment takes more than three hours from the time of injury, and if giving the medication to the casualty does not cause nausea.

10-39. Record first aid administered on the DD Form 1380. Evacuate the casualty:

- On their back, with the head elevated and immobilized.
- With their corrective eyeglasses, even if they are broken.
- Ensure the casualty's personal belongings, such as identification and any critical medications, accompany them during evacuation.

10-40. Do not cause further injury to the eye or casualty while waiting for evacuation. Provide assurance and stay with the casualty until relieved by medical personnel or competent authority.

INITIATE FIRST AID FOR CHEMICAL BURNS OF THE EYE WITH AN EYE-SHIELD

10-41. Injuries from chemical burns require immediate first aid. Mainly acids or alkalis cause chemical burns. The first aid measures consist of flushing the eyes immediately with large amounts of water for at least 5 to 20 minutes, or as long as necessary to flush out the chemical and, once flushed, bandaging the eyes. If the burn is an acid burn, you should flush the eye for at least 5 to 10 minutes. If the burn is an alkali burn, you should flush the eye for at least 20 minutes. After the eye has been flushed, evacuate the casualty immediately.

10-42. When an individual suffers burns of the face from a fire, the eyes will close quickly due to extreme heat. This reaction is a natural reflex to protect the eyeballs; however, the eyelids remain exposed to possibly being burned. If a casualty receives burns of the eyelids or face—

- DO NOT apply dressing.
- DO NOT touch.
- SEEK medical assistance immediately.

WARNING

DO NOT attempt to force open the eyes with thermal burns.

10-43. Reassure the casualty and quickly survey the situation.

10-44. Quickly identify the substance that the casualty was exposed to such as—

Note. Do not delay first aid to perform this step. Time is critical and the eyes must be irrigated as soon as possible.

- Alkali substances, which are the most dangerous of all substances are due to the penetrating factor. Common substances that contain the hydroxides of ammonia, lye, potassium, magnesium, and lime are—

- Fertilizers.
- Cleaning products such as drain cleaners, and oven cleaners.
- Plaster and cement.
- Acid substances are usually less severe than alkali burns. Common acids contain sulfuric, hydrochloric, nitric, acetic, chromic, and hydrofluoric acids are—
 - Glass polish, vinegar, and nail polish remover.
 - Automobile battery acid.
- Irritants and substances that have a neutral power of hydrogen tend to cause more discomfort to the eye. Examples of irritants that may cause this damage are—
 - Most household detergents.
 - Pepper spray.
 - Smoke from burning chemicals or rubber.

WARNING

DO NOT put pressure on the eye or remove anything from the eye surface. This could cause additional harm, such as loss of eyesight or loss of the eye.

10-45. Check for the following signs and symptoms:

- Irritation.
- Pain and redness.
- Watering and tearing.
- Possible erosion of the corneal surface.
- Inability to keep the eye open.
- Swelling of the eyelid.
- Blurred vision.

10-46. Initiate first aid for the chemical burn as follows:

- IMMEDIATELY flood the eye with water.
- Keep irrigating the eye with running water from a faucet, low-pressure hose, bottle, cup, or intravenous setup (if available).
- Hold the irrigating tip 1 to 1½ inches away from the casualty's eye or eyes, direct the irrigating solution gently from the inner eye (nearest the nose) to the outer eye (towards the ear) from the casualty's unaffected eye.
- Start the transport and continue washing out the eye or eyes for at least 20 minutes or until the casualty's arrival at the MTF.
- Dry the area around the eye or eyes by gently patting with gauze sponges or other appropriate clean material. Do not touch the casualty's eye.
- Cover the injured eye or eyes with the rigid eye-shield provided in the first aid kit, or an appropriate device that provides a hard cover over the eye such as:
 - Eye protective gear, ballistic glasses, or goggles.
 - The bottom of a disposable cup made of extruded polystyrene (foam), plastic, or paper.
- Secure the rigid eye-shield by applying tape from the cheek to the forehead. If tape is unavailable, use gauze, cloth dressing, shoelace, or string to secure in place.

Note 1. In hostile environments (such as combat or unsure footing), the eyes may have to remain uncovered so the casualty can see getting away from danger.

Note 2. Burned eyelids swell to protect the underlying eyes. The eyes should be protected with the rigid eye-shield to prevent further injury.

CAUTION

ALWAYS use a rigid eye-shield to provide a barrier to prevent further damage. NEVER place anything under the shield or directly on the eye.

10-47. Seeking MEDEVAC for the casualty.

10-48. Do not cause further injury to the eye by limiting movement that could cause further eye injury. Do not administer medication that prevents coughing, sneezing, or vomiting.

10-49. Consider giving the entire combat pill pack if medical treatment takes more than three hours from the time of injury, and if giving the medication to the casualty does not cause nausea.

10-50. Record the first aid given on the DD Form 1380 (provided in the first aid kit).

10-51. Evacuate the casualty without causing further harm.

TIER 2 SKILLS USED TO TREAT EYE TRAUMA

10-52. If a penetrating eye injury is noted or suspected, four steps must be taken:

- Perform a rapid field test of visual acuity. Rapid visual acuity testing includes:
 - Read print.
 - Count fingers.
 - Identity hand motion.
 - Sense light perception.
- Cover the affected eye with a rigid eye shield, NOT a pressure patch. A pressure dressing could result in permanent loss of vision. Place a rigid eye shield on only one eye, unless both eyes are injured. Covering both eyes turns an otherwise ambulatory casualty into a litter casualty. Tactical eyewear is always a good way to prevent eye issues and can be used for protection if no eye shield is available.
- Administer the casualty's complete CWMP, including all the medications contained in it. One of the medications is 400mg of moxifloxacin, which can be administered to prevent infection.
- Document all findings and treatments on the casualty's DD Form 1380.

10-53. Pressure on the eye could force the interior contents of the eye out of the eyeball through a cut or laceration. For protruding or impaled objects extending past the eye shield, cut a hole in the eye shield to allow the object to fit through, and secure it in place. If the eye shield cannot be applied around the impaled object, then use an improvised eye shield to avoid pressure on the eye.

EXPEDIENT SKILLS USED TO TREAT EYE TRAUMA

10-54. There are several expedient skills that can be employed to provide immediate care for eye trauma. Eye trauma should always be evaluated and treated by a qualified medical professional as soon as possible. The following are some skills or techniques that may be used in the initial management of eye trauma:

- Protecting the eye: If there is a foreign object, such as debris or a penetrating object, near or embedded in the eye, it's crucial to prevent further injury. Avoid touching or rubbing the eye, as this can exacerbate the damage. Place a protective shield or cup over the eye, if available, to prevent accidental contact.
- Flushing the eye: If there are chemicals or irritants in the eye, use immediate and gentle irrigation with clean water, or saline solution can help flush out the substance. Tilt the head to the side and pour a gentle stream of fluid over the eye from the inner corner to the outer corner. Use a clean container or a commercially available eye wash bottle.
- Covering injuries: If there is a cut or laceration around the eye, apply a sterile dressing or clean cloth to cover the wound. Avoid applying pressure directly on the eye. Secure the dressing in place with a bandage or adhesive tape.

- Immobilizing the eye: If there is a suspected or evident fracture of the eye socket or orbit, it may be necessary to immobilize the eye to prevent further injury. Place a protective shield or a cup over the eye and use a bandage or tape to secure it in position. This can help stabilize the injured area until medical attention is available.
- Applying cold compresses: In cases of eye trauma accompanied by swelling or bruising, a cold compress can be applied gently over the closed eyelid. This can help reduce swelling and alleviate pain. Use a clean cloth or ice pack wrapped in a towel. Avoid placing ice directly on the skin.

10-55. The above expedient skills are meant to provide initial first aid and should not replace professional medical care. The casualty should be evacuated as soon as possible to supporting MTF to ensure a thorough evaluation and appropriate treatment for eye trauma.

Chapter 11

Head Injury

OVERVIEW OF HEAD INJURIES

11-1. Head injuries are a common concern that can occur due to accidents, falls, or sports-related incidents. When the head experiences a forceful impact, it can result in different types of injuries. Understanding head injuries can help Service members take the necessary precautions to protect themselves or others from potential harm. It is critically important to report findings of a head injury to medical personnel at the scene since a lack of rapid, and appropriate medical care can worsen the head injury, and the eventual outcome. Head trauma can be either blunt or penetrating. Head injuries can be broadly classified as open or closed head injuries.

OPEN HEAD INJURY

11-2. An open head injury, also known as an open skull injury, is a type of injury that occurs when there is a break or penetration of the skull, exposing the brain to the external environment. It typically involves visible damage to the scalp, skull, or meninges (protective coverings of the brain). The signs and symptoms of an open head injury can vary depending on the severity and specific location of the injury. Open head injuries are considered medical emergencies. If an open head injury is suspected, it is crucial to seek immediate medical attention for evaluation, diagnosis, and appropriate treatment.

11-3. The common signs and symptoms associated with an open head injury are:

- Visible head wound. An open head injury often involves a visible wound or laceration on the scalp or skull. There may be bleeding, or an open fracture where the bone is exposed.
- Profuse bleeding. Open head injuries can cause significant bleeding due to damage to blood vessels. Blood may be visible on the scalp, face, or in severe cases, there may be profuse bleeding from the head wound.
- Skull deformity. In some cases, an open head injury may cause a visible deformity or depression in the skull, indicating a skull fracture or bone displacement.
- Loss of consciousness. Depending on the severity of the injury, loss of consciousness can occur, ranging from a brief period to an extended duration.
- Clear fluid or blood drainage. Leakage of clear fluid (cerebrospinal fluid) or blood from the nose, ears, or wound can indicate a more severe injury and should be treated as an emergency.
- Severe headache. An intense and persistent headache is common following an open head injury.
- Nausea and vomiting. Feelings of nausea and vomiting may be present after an open head injury.
- Confusion and disorientation. The Service member may exhibit confusion, memory loss, difficulty concentrating, or may appear disoriented.
- Seizures. Seizures can occur after an open head injury, especially if the brain is directly affected.
- Changes in speech or vision. The person may have difficulties with speech, vision problems, double vision, or other sensory disturbances.
- Weaknesses or numbness. Weakness or numbness in certain body parts can be a sign of nerve damage resulting from the head injury.

CLOSED HEAD INJURY

11-4. A closed head injury refers to trauma to the head that does not involve an open wound or penetration of the skull. Instead, the injury occurs internally, within the skull. Closed head injuries are often the result of blunt force impacts, falls, or accidents, where the brain is subjected to sudden acceleration, deceleration, or rotational forces. Concussions are the most common type of a closed head injury. Closed head injuries may not be obvious or detectable.

11-5. Closed head injuries can range in severity from mild concussions to more severe forms, such as contusions, diffuse axonal injuries, or intracranial hemorrhages. It is essential to seek medical attention if a

closed head injury is suspected, as the impact on the brain can have serious consequences and requires appropriate evaluation and management.

11-6. The signs and symptoms of a closed head injury can vary depending on the severity of the injury. Mild cases may be present with subtle symptoms, while more severe injuries can have immediate and pronounced effects. The common signs and symptoms associated with a closed head injury are:

- Headache or pressure in the head.
- Nausea and vomiting.
- Dizziness and balance problems.
- Confusion and disorientation.
- Memory loss or difficulty concentrating.
- Mood and behavior changes.
- Sensory changes, such as blurred vision, or ringing in the ears.
- Sensitivity to light and sound.
- Loss of consciousness (in severe cases).

11-7. It is important to seek medical attention if any of these signs are present after a head injury, as they can indicate a range of severity from a mild concussion to more serious brain trauma. The above symptoms can occur immediately after the injury or may develop gradually over time. Symptoms may vary from person to person.

CONCUSSIONS

11-8. A concussion is a type of traumatic brain injury that occurs when the brain experiences a sudden and forceful movement within the skull. It is typically caused by a blow, jolt, or bump to the head or body, which can disrupt the normal functioning of the brain.

11-9. Concussions can occur in various situations, such as sports-related injuries, falls, motor vehicle accidents, or any other incident involving a significant impact to the head. Concussions can happen even without a direct impact to the head. The forceful movement or shaking of the body can still cause the brain to move inside the skull, leading to a concussion.

11-10. The primary characteristic of a concussion is a temporary disruption of brain function. Concussions may not always involve a loss of consciousness. In fact, most concussions occur without a person losing consciousness. Common symptoms of a concussion can include:

- Headache or pressure in the head.
- Dizziness or feeling lightheaded.
- Confusion or feeling mentally foggy.
- Memory problems or difficulty concentrating.
- Fatigue or feeling tired.
- Nausea or vomiting.
- Sensitivity to light or noise.
- Changes in sleep patterns.
- Irritability or changes in mood.
- Blurred vision or difficulty with balance.

11-11. It's important to seek medical attention if a concussion is suspected, as health care professionals can evaluate the severity of the injury and provide appropriate guidance for recovery. The initial management of a concussion involves rest and avoiding activities that could further injure the brain. Physical and cognitive rest are recommended to allow the brain to heal properly. In some cases, individuals may need to temporarily limit activities like school or work to promote recovery. A gradual return to normal activities is typically recommended under the guidance of a health care professional.

11-12. Every concussion is unique, and the symptoms and recovery time can vary from person to person. With proper rest, most individuals recover fully from a concussion. However, in some cases, symptoms may persist or worsen, and a condition known as post-concussion syndrome may develop.

CONTUSIONS RELATED TO HEAD INJURIES

11-13. A contusion refers to a type of injury characterized by damage to the tissue or blood vessels beneath the skin, usually resulting from a blunt force impact. It is commonly known as a bruise. When a contusion occurs, blood vessels are damaged, leading to bleeding and the subsequent discoloration of the skin.

11-14. In relation to head injuries, contusions can occur because of a direct blow or trauma to the head. When the head experiences a significant impact, the brain can collide with the inner surface of the skull, leading to contusions. These types of injuries are known as cerebral contusions.

11-15. Cerebral contusions can be classified as coup-contrecoup injuries, which means that contusions can occur both at the site of impact (coup) and on the opposite side of the brain due to the brain rebounding and hitting the skull (contrecoup). These injuries are concerning because they involve damage to the brain tissue itself.

11-16. The severity of a cerebral contusion can vary, ranging from mild to severe. In more severe cases, contusions can lead to increased pressure within the skull, swelling of the brain, and potentially life-threatening complications.

11-17. It is crucial to seek immediate medical attention if a head injury is suspected, as prompt evaluation and treatment are essential to prevent further complications. Symptoms of a cerebral contusion may include:

- Changes in vision.
- Confusion.
- Difficulty speaking.
- Dizziness.
- Headache.
- Loss of consciousness.
- Nausea.
- Seizures.
- Vomiting.
- Weaknesses.

11-18. Treatment for cerebral contusions focuses on managing the underlying injury and preventing further damage. It may involve measures such as rest, pain management, observation, close monitoring of neurological status, and, in more severe cases, surgical intervention to relieve pressure and remove any blood clots or damaged tissue.

EQUIPMENT USED FOR THE TREATMENT OF A HEAD INJURY

11-19. Treating a head wound requires few essential pieces of equipment. Service members will need disposable gloves to protect themselves and the injured person from any potential infection. Sterile gauze pads or clean clothes are vital for covering and applying pressure to the wound to stop the bleeding. Adhesive bandages or medical tape are used to secure the gauze in place. Additionally, a mild antiseptic solution or wipes are important for cleaning the wound to prevent infection. Lastly, a cravat or gauze bandage can be used to create a makeshift dressing or sling to support the head if there is a suspected skull or neck injury.

TIER 1 SKILLS FOR THE TREATMENT OF A HEAD INJURY

11-20. Service members may not have the time or expertise to conduct extensive treatment for a head injury. Service members experiencing any of the symptoms explained in paragraph 11-17 should be evacuated as soon as tactically feasible.

TIER 2 SKILLS FOR THE TREATMENT OF A HEAD INJURY

11-21. Service members experiencing any of the symptoms explained in paragraph 11-17 should be evacuated as soon as tactically feasible. However, the CLS should complete the injury/evaluation/distance (commonly known as I.E.D.) checklist that is outlined in DODI 6490.11. This checklist is used to gather information, and to identify the signs and symptoms of a head injury.

11-22. Commanders or their representatives are required to assess all Service members involved in potentially concussive events, including those without apparent injuries, as soon as possible using the injury/evaluation/distance checklist (figure 11-1).

<i>Injury</i>	<i>Physical damage to the body or body part of a Service member?</i>	<i>(Yes/No)</i>
Evaluation	H – Headaches or vomiting?	(Yes/No)
	E – Ear ringing?	(Yes/No)
	A – Amnesia, altered consciousness, or loss of consciousness?	(Yes/No)
	D – Double vision or dizziness?	(Yes/No)
	S – Something feels wrong or is not right?	(Yes/No)
Distance	Was the Service member within 50 meters of the blast? Record the approximate distance from the blast.	(Yes/No) Not Applicable

Figure 11-1. The Injury/Evaluation/Distance Checklist

11-23. Service members who are involved in a potentially concussive event, as specified in DODI 6490.11, must be referred for a medical evaluation. This referral is mandatory if they answer "Yes" on the injury/evaluation/distance checklist or exhibit any symptoms associated with concussion at any point following the incident. Once the injury/evaluation/distance assessment is concluded, it is crucial to document the results for all individuals involved in the event. These records should then be submitted as part of the significant activities report, which is required for all blast-related incidents or other events outlined in DODI 6490.11. This procedure ensures that all potential concussions are promptly and effectively addressed, safeguarding the health and well-being of the Service members.

11-24. When dealing with trauma casualties suspected of having a head injury, it is crucial to report specific critical observations to medical personnel in accordance with the Military Acute Concussive Evaluation 2 (known as MACE 2). Immediate evaluation following the injury significantly enhances the effectiveness of the assessments. A traumatic brain injury should be suspected if the casualty exhibits any of the following signs:

- A deteriorating level of consciousness.
- Experiencing double vision.
- Exhibiting increased restlessness, combative, or agitated behavior.
- Repeated vomiting.
- Positive results from a structural brain injury detection device, if available.
- Seizures.
- Weakness or tingling sensations in the arms or legs.
- Severe or worsening headache.

11-25. If any of these signs or symptoms are present, they must be reported immediately to the arriving medical personnel. For comprehensive guidelines on reporting and treating head injuries, refer to DODI 6490.11. This protocol ensures that potential traumatic brain injuries are promptly identified and appropriately managed.

EXPEDIENT SKILLS USED FOR THE TREATMENT OF HEAD, NECK, AND FACIAL INJURIES

11-26. Some wounds and burns will require special precautions and procedures when applying these measures. This section discusses specific first aid procedures for wounds of the head, neck, and facial injuries. It also discusses the techniques for applying dressings and bandages to specific parts of the body.

11-27. Head injuries range from minor abrasions or cuts on the scalp, to severe brain injuries that may result in unconsciousness and sometimes death. Head injuries are classified as open or closed wounds. An open wound is one that is visible, has a break in the skin, and usually has evidence of bleeding. A closed wound may be visible (such as a depression in the skull) or the first aid provider may not be able to see an apparent injury (such as internal bleeding). Some head injuries result in unconsciousness; however, a Service member may have a serious head wound and still be conscious. Casualties with blunt trauma should be treated as though they also have a spinal injury. The casualty should not be moved until the head and neck are stabilized unless they are in immediate danger (such as close to a burning vehicle).

11-28. Prompt first aid measures should be initiated for casualties with suspected head and neck injuries. The conscious casualty may be able to provide information on the extent of their injuries. However, because of the head injury, they may be confused and unable to provide accurate information. The signs and symptoms that a first aid provider might observe are:

- Nausea and vomiting.
- Convulsions or twitches.
- Slurred speech.
- Confusion and loss of memory. (Do they know who they are? Do they know where they are? Do they know what day it is?).
- Recent unconsciousness.
- Dizziness.
- Drowsiness.
- Blurred vision, unequal pupils, or bruising (black eyes).
- Paralysis (partial or full).
- Complaint of headache.
- Bleeding or other fluid discharge from the scalp, nose, or ears.
- Deformity of the head (depression or swelling).
- Staggering while walking.

UNCONSCIOUS CASUALTY

11-29. When encountering an unconscious casualty, it is essential to act swiftly and methodically, as their condition may be critical. Unconsciousness can result from various causes, such as trauma, medical emergencies, or underlying health conditions. The primary objective of first aid for an unconscious patient is to ensure their immediate safety, preserve their airway, and seek professional medical help. By following specific guidelines and employing basic life support techniques, first responders can play a vital role in providing initial assistance until medical professionals arrive. An unconscious casualty does not have control of all their functions and may choke on their tongue, blood, vomitus, or other substances resulting in:

- Loss of Breathing. The brain requires a constant supply of oxygen. A bluish (or an individual with dark skin-grayish) color of skin around the lips and nail beds indicates that the casualty is not receiving enough oxygen. Immediate action must be taken to clear the airway, to position the casualty on their side, or to initiate rescue breathing.
- Bleeding. Bleeding from a head injury usually comes from blood vessels within the scalp. Bleeding can also develop inside the skull or within the brain. In most instances visible bleeding from the head can be controlled by application of the field first aid dressing.

CAUTION

DO NOT attempt to put unnecessary pressure on the wound or attempt to push any brain matter back into the head (skull). DO NOT apply pressure dressing.

- Convulsions. Convulsions (seizures or involuntary jerking) may occur even after a mild head injury. When a casualty is convulsing, protect them from hurting themselves. Take the following measures:
 - Ease them to the ground if they are standing or sitting.
 - Support their head and neck.
 - Maintain their airway.
 - Protect them from further injury (such as hitting close-by objects).

Note. DO NOT forcefully hold their arms and legs if they are jerking because this can lead to broken bones. DO NOT force anything between the casualties' teeth-especially if they are tightly clenched because this may obstruct the casualty's airway. Maintain the casualty's airway if necessary.

- **Brain Damage.** In severe head injuries where brain tissue is protruding, leave the wound alone. Carefully place a loose moistened dressing (moistened with sterile normal saline if available) and a first aid dressing over the tissue to protect it from further contamination. **DO NOT** remove or disturb any foreign matter that may be in the wound. Position the casualty so that their head is higher than their body. Keep them warm and seek medical assistance immediately.

Note 1. If there is an object extending from the wound, **DO NOT** remove the object. Improvise bulky dressings from the cleanest material available and place this material around the protruding object for support, then apply the field dressing.

Note 2. Unconscious casualties with open head injuries are unlikely to survive, regardless of how quickly they are evacuated.

FIRST AID FOR HEAD INJURIES

11-30. Provide direct manual stabilization of the head by placing your hands on each side of the casualty's head, maintaining the position of the head in line with the body. Assess the casualty's level of consciousness utilizing the alert, verbal, pain, and unresponsive (AVPU) method. AVPU is determined by asking—

- Is the casualty **alert**? A.
- Does the casualty respond to **verbal** stimuli or commands? V.
- Does the casualty respond to **painful** stimulus? P.
- Is the casualty **unconscious**? U.

11-31. The casualty with a head injury (or suspected head injury) should be continually monitored for the development of conditions that may require basic lifesaving measures. After initiating first aid measures, request medical assistance and evacuation. If dedicated MEDEVAC assets are not available, transport the casualty to an MTF as soon as the situation permits. The first aid provider should not attempt to remove a protruding object from the head or give the casualty anything to eat or drink. Furthermore, the first aid provider should be prepared to—

- Clear the airway.
- Control bleeding (external).
- Administer first aid measures for shock.
- Keep the casualty warm.
- Protect the wound.

Initiate Treatment for a Superficial Head Injury

11-32. Provide first aid for a superficial head injury by—

- Applying a dressing.
- Observing for abnormal behavior or evidence of complications.
- Documenting on the TCCC card.

Head Injury Involving Trauma

11-33. Provide first aid for a head injury involving trauma as follows:

- Maintain a patent airway using the jaw-thrust maneuver.
- Dress the head wounds.
- Control bleeding.
- Treat for shock.
- Seek medical aid.
- Monitor the casualty for convulsions or seizures.

11-34. Monitor unstable casualties for the following—

- Level of consciousness by using the AVPU indicators.
- Pupils respond to light and are equal in size.
- Motor functions, casualty's strength, mobility, coordination, and sensation.

- Seek medical aid.
- Evacuate the casualty as soon as possible.

NECK INJURIES

11-35. Neck injuries may result in heavy bleeding. Apply pressure above and below the injury. Do not interfere with the breathing process and attempt to control the bleeding. Apply a dressing. When blunt trauma occurs, always evaluate the casualty for a possible neck fracture or spinal cord injury; if suspected, seek medical treatment immediately.

Note. Establish and maintain the airway in cases of facial or neck injuries. If a neck fracture or spinal cord injury is suspected, immobilize the injury and, if necessary, perform basic life support measures.

FACIAL INJURIES

11-36. Soft tissue injuries of the face and scalp are common. Abrasions (scrapes) of the skin do not cause serious problems. Contusions (injury without a break in the skin) usually cause swelling. A contusion of the scalp looks and feels like a lump. Laceration (cut) and avulsion (torn away tissue) injuries are common. Avulsions are frequently caused when a sharp blow separates the scalp from the skull beneath it. The face and scalp are richly supplied with blood vessels (arteries and veins) and the wounds of these areas usually bleed heavily.

Ear Injuries

11-37. Lacerated (cut) or avulsed (torn) ear tissue may not be a serious injury. Bleeding, or the drainage of fluids from the ear canal, however, may be a sign of a head injury, such as a skull fracture. DO NOT attempt to stop the flow from the inner ear canal or put anything into the ear canal to block it. Instead, you should cover the ear lightly with a dressing. For minor cuts or wounds to the external ear, apply a cravat or gauze bandage as follows:

- Place the middle of the bandage over the ear (figure 11-2 step A).
- Cross the ends, wrap them in opposite directions around the head, and tie them (figure 11-2 step B and figure 11-2 step C).

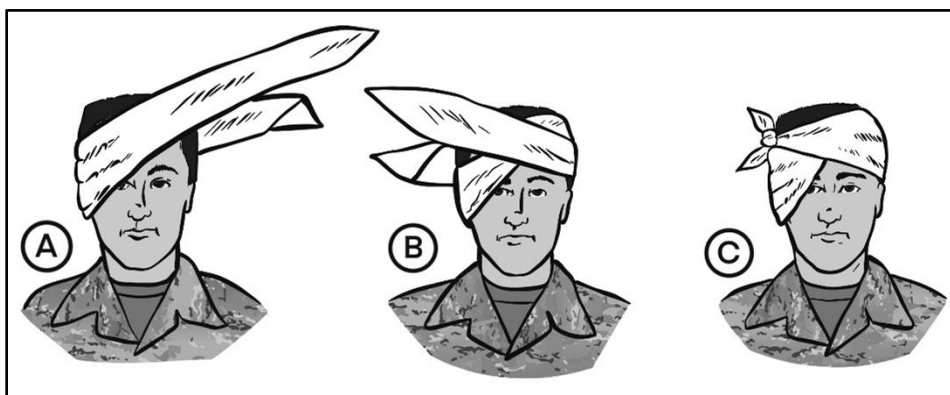


Figure 11-2. Applying a bandage to the ear

- If possible, place some dressing material between the back of the ear and the side of the head to avoid crushing the ear against the head with the bandage.

Nose Injuries

11-38. Nose injuries generally produce bleeding. The bleeding may be controlled by placing an ice pack (if available) over the nose or pinching the nostrils together. The bleeding may also be controlled by placing torn gauze (rolled) between the upper teeth and the lip.

CAUTION

DO NOT attempt to remove objects inhaled into the nose. An untrained person who removes such an object could worsen the casualty's condition and cause permanent injury.

Jaw Injuries

11-39. Before applying a bandage to a casualty's jaw, remove all loose or free-floating foreign material from the casualty's mouth. If the casualty is unconscious, check for obstructions in the airway and remove if possible. If there is profuse bleeding in the oral cavity, the cavity may require loose packing with soft bandaging material prior to applying a bandage. Care should be taken to avoid occluding the airway.

11-40. When applying the bandage, allow the jaw enough freedom to permit passage of air and drainage from the mouth. Dress the wound, and apply the bandages by—

- Placing the bandage under the chin and pull its ends upward. Adjust the bandage to make one end longer than the other (figure 11-3 step A).
- Taking the longer end over the top of the head to meet the short end at the temple and cross the ends over (figure 11-3 step B).
- Wrapping the ends in opposite directions to the other side of the head and tying them over the part of the bandage that was applied first (figure 11-3 step C).

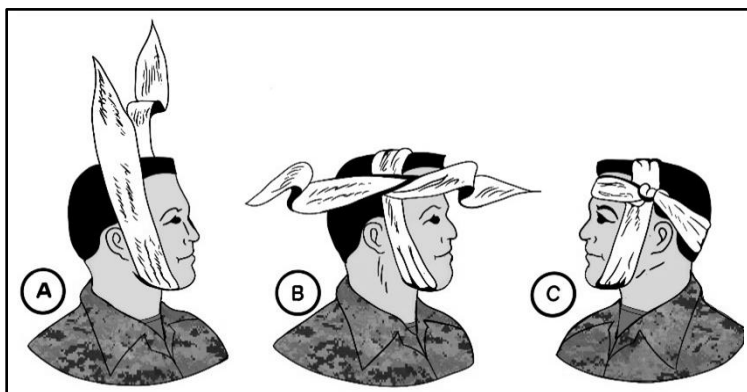


Figure 11-3. Applying a bandage to the jaw

Note. The bandaging procedure outlined is the cravat technique used to immobilize a fractured jaw

Chapter 12

Burns

OVERVIEW OF BURNS

12-1. A burned trauma casualty should be treated for all other life-threatening injuries first, following the MARCH-PAWS sequence, before addressing burns. Even if the skin is burned, trauma treatments can still be performed on or through it.

12-2. Burns can result from various causes:

- Friction Burns. These are common in situations where skin is rubbed off against a hard surface, often seen in road accidents where motorcycle riders, cyclists, or skateboarders slide across pavement, getting what's known as "road rash." Carpet burns occur from skin scraping against carpet, often during indoor sports or play. Rope burns are common during activities like climbing or tug-of-war, and sliding burns frequently happen in sports like baseball when sliding into bases.
- Cold Burns. Also known as frostbite, these burns damage the skin by freezing it. Frostbite can occur from prolonged exposure to freezing temperatures or direct contact with extremely cold objects.
- Thermal Burns. These occur when the skin touches something very hot, like hot metals, scalding liquids, flames, or steam, raising the skin temperature high enough to kill skin cells.
- Radiation Burns. Sunburn is a typical example. Other sources include X-rays or radiation therapy for cancer.
- Chemical Burns. These happen when strong acids, solvents, or detergents come into contact with the skin.
- Electrical Burns. These burns are the result of contact with electrical currents.

12-3. Burns are classified based on how deeply they damage the skin:

- Superficial or First Degree Burns (figure 12-1 on page 116): Superficial burns affect only the outer layer of the skin, causing redness, pain, and mild swelling. They are like mild sunburn and usually heal within a few days without leaving scars.
- Partial thickness or Second Degree Burns (figure 12-2 on page 116): Partial thickness burns reach the second layer of the skin and can cause blisters, severe pain, redness, and swelling. Healing times vary, and these burns may leave scars.
- Full thickness or Third Degree Burns (figure 12-3 on page 116): Full thickness burns are the most severe, these burns go through the entire thickness of the skin and can damage muscles or bones underneath. The skin may look white, blackened, or charred, and there might be no feeling in the area because of nerve damage. These burns require immediate medical attention and often need skin grafts to heal.

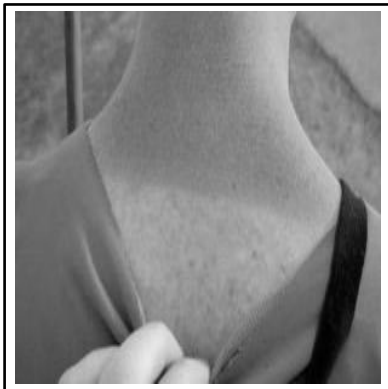


Figure 12-1. First degree or Superficial burn



Figure 12-2. Second degree or Partial thickness burn



Figure 12-3. Third degree or Full thickness burn

12-4. The size of or percentage of the burn area will be documented on DD Form 1380. The “Rule of Nines” is used to determine or estimate the burn percentage. Figure 12-4 shows the approximation for each area of the body. The body is divided into areas with each having a percentage of the total body surface area as follows:

- Head – 9 percent.
- Torso (front) – 18 percent.
- Torso (back) – 18 percent.
- Arms – 9 percent each.
- Legs – 18 percent each.
- Pubic area – 1 percent.
- Buttocks – 1 percent.

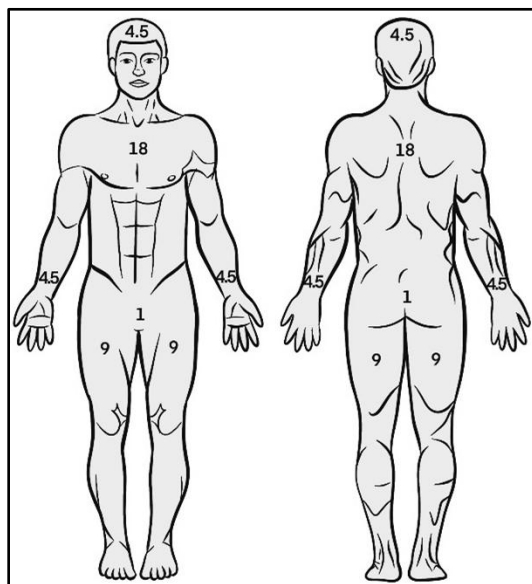


Figure 12-4. Burn area percentage “Rule of Nines”

Note. First degree or superficial burns are not part of the burn percentage calculation and are rarely of medical significance.

12-5. General guidelines are that the size of the casualty's palm of the hand represents approximately 1 percent of the burned area. When estimating, it is easiest to round up to the nearest 10 percent. If half of the front or rear area is burned, the area would be half of the area value. For example, if half of the front upper leg or front lower leg were burned, it would be half of 9 percent, or 4.5 percent. If half of the front torso were burned, say either the upper or the lower part of the front torso, then it would be half of 18 percent or 9 percent. Remember, the higher the percentage burned, the higher the chance is for hypothermia.

EQUIPMENT USED FOR THE TREATMENT OF BURNS

12-6. The CoTCCC-recommended compressed gauze and or gauze rolls are 6-ply sterile cotton that has been compressed. They are used to stop minor bleeding, cover wounds or burns, pack wounds, act as bulky material for pressure dressings, or pad pressure points in splinting.

12-7. The CoTCCC-recommended elastic bandage is a stretchable bandage. They are used to hold dressings or splints in place or can be applied more tightly to apply localized pressure on a wound. Elastic or restrictive bandages are normally not applied to a burn casualty. The application of these types of bandages are very painful and swelling secondary to the injury may restrict circulation.

12-8. The cravat is a sterile triangular shaped bandage. It can be used as both a bandage and a sling. It is commonly used to secure a splint, create a sling, or swath for an injured limb.

TIER 1 SKILLS FOR THE TREATMENT OF BURNS

12-9. Burn care in TCCC is very basic. Eliminating the source of the burn should be the initial priority. Providing any treatment prior to removing the casualty from the source of the burn or extinguishing the cause of the burn will be ineffective and harmful to the casualty and casualty response personnel.

12-10. Electrical burn care consists of—

- Eliminating the source of electrical burns.
- Turning off any sources of electricity that have been exposed to the casualty (if applicable).
- Using nonconductive material to drag the casualty away from the electrical source (if it cannot be turned off).

Note. Types of nonconductive material include rope, clothing, and dry wood.

- Moving the casualty to a safe place.

WARNING

Do not touch the casualty or the electrical source with your bare hands. You will be injured too! High voltage electrical burns from an electrical source or lightning may cause temporary unconsciousness, difficulties in breathing, or difficulties with the heart (irregular heartbeat).

12-11. Thermal burn care consists of—

- Stopping the source of the thermal burn and remove the casualty from the source of the burn.
- Extinguishing any flames on the casualty, cut the clothing around the burned area, and gently lift away.

CAUTION

If clothing is stuck to the burn, ensure you cut around the clothing and leave the stuck clothing in place.

- Avoid grabbing the burned area while moving or picking up the casualty.

12-12. Chemical burn care consists of—

- Eliminating the source of chemical burns.
- Brushing off dry chemicals (carefully) with a clean, dry cloth.
- Flush the burned area with copious amounts of flowing water to remove any liquid chemicals.
- Applying a wet barrier (water-soaked gauze, clothing, or mud) with an occlusive dressing.
- Advising medical personnel immediately.

WARNING

Do not submerge a chemical burn casualty in water. This will greatly contribute to hypothermia and spread the irritation.

12-13. All burn care consists of—

- Monitoring the casualty for life-threatening conditions.
- Checking for additional injuries.
- Treating for shock (if applicable).
- Seeking medical aid.
- Annotating all treatment and care on DD Form 1380.

12-14. Close attention to the casualty's airway, breathing is required and always assess for secondary injuries and shock.

TIER 2 SKILLS FOR THE TREATMENT OF BURNS

12-15. Along with the treatment measures outlined in paragraph 13-9 through 13-14 extra emphasis should be placed on heat loss prevention methods. Burn patients are particularly susceptible to hypothermia. Casualties should be kept off the ground and placed onto an insulated surface as soon as possible.

12-16. For extensive burns, those with greater than 20 percent of the area burned place the casualty in the Heat Reflective Shield to cover the burned areas and prevent hypothermia. Preventing and managing hypothermia for burn patients is important and occurs regardless of the ambient temperature in the environment.

12-17. Facial burns, especially those that occur in closed spaces, may be associated with an inhalation injury. These casualties should be monitored closely for potential airway issues. Do not place an NPA in a casualty with signs of inhalation burns. Notify medical personnel as soon as possible if an inhalation injury is suspected.

12-18. Analgesia may be administered to treat burn pain. Antibiotic therapy is not solely indicated for burns; it should also be given to prevent infection in penetrating wounds. Consider giving the entire combat pill pack if medical treatment takes more than three hours from the time of injury, and if giving the medication to the casualty does not cause nausea.

12-19. Interventions for warm and cool weather should be considered. The addition of blood loss can cause the body's temperature to drop even when it is hot outside. The TQs should not be covered; keep them visible so medical personnel can easily see it.

EXPEDIENT SKILLS USED FOR THE TREATMENT OF BURNS

12-20. In emergency scenarios, especially in remote or combat zones, effectively managing burn injuries with limited resources is essential. Basic materials such as gauze bandage, clean clothing, and cravats can be used to cover burns and control bleeding. Cooling and cleaning burns should be done with available clean water, avoiding ice to prevent further skin damage. For thermal burns, it's crucial to remove the casualty from the heat source and cut away clothing gently around the burn area. In cases of electrical burns, disconnecting the power and using non-conductive materials to move the casualty is vital, while chemical burns require immediate decontamination by brushing off dry chemicals and flushing the area thoroughly with water.

12-21. Preventing hypothermia is critical in burn care due to the high risk posed by skin damage. Using emergency blankets, reflective heat shields, or makeshift insulation from natural materials can help maintain body temperature. Pain management, when possible, should include safe administration of over-the-counter pain relievers, and continuous monitoring for shock is essential. These field expedient techniques are pivotal in stabilizing burn victims and preventing further injury until more advanced medical help can be accessed, underscoring the importance of training and preparedness in these emergency interventions.

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Chapter 13

Fractures and Splinting

OVERVIEW OF FRACTURES

13-1. Prompt and appropriate management of fractures, or breaks in bones, is essential to minimize further harm and optimize the chances of survival for injured individuals. TCCC emphasizes a systematic approach to fracture care, incorporating principles such as rapid assessment, stabilization, and pain management. The goal is to provide efficient and effective treatment, considering the challenges of the tactical environment where resources may be limited, and evacuation to definitive care may be delayed. Mastery of fracture management in TCCC equips military personnel and first responders with the skills necessary to stabilize fractures, prevent complications, and facilitate the evacuation process, ultimately improving the overall outcomes for casualties in the field.

MUSCULOSKELETAL SYSTEM

13-2. The musculoskeletal system provides a structural framework for the body and, because bones are rigid, provides support and protection for vital organs and softer tissues. Skeletal muscles and bones work together to make body movement possible.

13-3. The musculoskeletal system is composed of—

- Bones.
- Joints.
- Muscles.
- Cartilage.
- Ligaments.
- Tendons.

Bones

13-4. Bones are the substance that forms the skeleton of the body. It is composed chiefly of calcium phosphate and calcium carbonate. The human body has 206 bones. There are 80 axial (head and trunk) bones and 126 appendicular (upper and lower extremity) bones. Bones provide structural support for the body. Individual and groups of bones provide structure for the attachment of soft tissues and protect organs.

Joints

13-5. The human body's mobility hinges on its joints, where bones connect and facilitate movement. These pivotal junctures come in various forms, each serving distinct functions in enabling motion.

Muscles

13-6. The body's muscles serve as its powerhouse, providing the force needed for movement and various physiological functions. Within the body, there exist three main types of muscles, each with its unique roles and characteristics. These include skeletal muscles, which control movement in the extremities and outer body areas, cardiac muscles, found exclusively in the heart, and smooth muscles, lining the walls of arteries and the gastrointestinal tract.

Cartilage

13-7. Cartilage serves as a resilient and pliable tissue that provides cushioning and support between bones at the joints, facilitating smooth movement. In addition to its role in joint function, cartilage also forms essential structures elsewhere in the body, such as the larynx and the external portions of the ears, contributing to their flexibility and shape.

Ligaments

13-8. Ligaments play a crucial role in the body's structural integrity, serving as sturdy bands of connective tissue that link bones together at joints. They provide stability and support, helping to prevent excessive movement and potential joint injuries.

Tendons

13-9. Tendons serve as soft tissue connectors that attach muscles to bones, enabling movement and transmitting the force generated by muscles to the bones they are connected to. Despite their flexibility, tendons possess considerable toughness, providing durability to withstand the stresses of movement and physical activity.

TYPES OF FRACTURES

13-10. A fracture can be a closed or open fracture. A closed fracture (see figure 13-1) means that there is no external wound associated with the fracture. An open fracture (see figure 13-2) means that there is a break in the skin in conjunction with the fracture, often with protruding bone. The warning signs that an arm or leg might be fractured include:

- Significant pain along with marked swelling.
- An audible or perceived “snap.”
- Different length or shape of limb.
- Loss of pulse or sensation in the injured arm or leg.
- Crepitus (a crackling or popping sound under the skin).



Figure 13-1. Closed Fracture



Figure 13-2. Open Fracture

EQUIPMENT USED FOR THE TREATMENT OF FRACTURES

13-11. The CoTCCC endorses the use of the semi-rigid splint, a 36-inch-long malleable device comprised of moldable aluminum, encased in foam padding. This type of splint is designed for versatility in the field and is most commonly employed for stabilizing fractures. Its flexible yet sturdy structure allows it to be shaped according to the specific contours of the injury site, providing effective support and immobilization needed to prevent further injury during transport or until more definitive care can be administered.

TIER 1 SKILLS FOR THE TREATMENT OF FRACTURES

13-12. During the TFC phase, it is highly advisable to splint all fractures if circumstances permit. Splinting is crucial to immobilize the affected area, preventing further damage, and minimizing pain for the casualty. Proper splinting techniques ensure that fractures are adequately supported until the injured individual receives comprehensive medical treatment. For detailed guidance on how to correctly apply splints, refer to paragraph 14-13, which provides specific instructions and considerations for effective splinting in a tactical environment.

TIER 2 SKILLS FOR THE TREATMENT OF FRACTURES

13-13. The "S" in MARCH-PAWS protocol denotes splinting, a crucial step in managing both open and closed fractures. For more information on fractures see paragraph 14-10 on page 122. Splinting involves immobilizing the injured limb to prevent further movement, typically using semi-rigid splints or improvised materials like boards or tree limbs. It's essential to incorporate the joints above and below the fracture site for stability and secure the splint firmly with wraps, belts, or tape. Additionally, securing arm fractures to the shirt using the sleeve as a sling can provide added support. However, caution must be exercised to avoid excessive tightness, which can compromise circulation to the fingers or toes.

13-14. Caution must be taken when splinting to ensure that it is not wrapped too tightly. If wrapped too tightly, the splint may cut off circulation below the splint. If there is numbness, tingling or increased pain, or if the fingers or toes turn pale or bluish, loosen the bandage. Increased swelling may occur; continued monitoring of the splinted fracture is required.

13-15. When splinting, first responders or CLS should avoid—

- Manipulating the fracture site too much, resulting in pain, additional damage to blood vessels, and nerves.
- Securing too tightly and cutting off blood flow.
- Failing to immobilize the joint above and below the fracture when possible.
- Causing further injury.
- Making the casualty uncomfortable during transport or evacuation.
- Splinting near or over a wound that has not been properly treated.

SPLINTING LEG FRACTURES

13-16. Care should be taken on cravat placement. First responders or CLS should not apply a cravat on the fracture site, as the pressure from the cravat could cause additional injury to the fracture site.

13-17. The ends of the splint should not be placed against the groin, as this could interfere with blood circulation. Extra padding is used for joints or sensitive areas, such as the groin.

13-18. Before splinting a suspected fracture, the first responder or CLS should—

- Identify the location of the fracture.
- Check the pulse below the fracture.
- Watch for signs of poor circulation, including coolness, numbness, or lack of a pulse.
- Ensure the ends of the rigid objects are not interfering with blood circulation.
- If there are any signs of poor circulation, loosen the splint, reposition, and reapply.

13-19. The first responder or CLS should follow the below when applying a leg splint:

- Apply the splint to the injured extremity with the limb in the position of function or a normal resting position if possible.
- Secure the splint in place with appropriate materials.
- Ensure the joints above and below the fracture are immobilized in the splint whenever possible.
- Recheck the pulse after applying the splint.
- Watch for signs of poor circulation, including coolness, numbness, or lack of a pulse. Ensure the ends of the rigid objects are not interfering with blood circulation. If there are any signs of poor circulation, loosen the splint, reposition, and reapply.
- Administering the CWMP as needed.
- Document all treatment on a DD Form 1380 and attach it to the casualty.

SPLINTING ARM FRACTURES

13-20. Movement or manipulation of a fractured extremity may result in increased pain. The following will help to reduce the casualty's pain:

- If possible, have the casualty support their injured arm while preparing the equipment.
- Mold the padded splint using the casualty's unaffected limb to avoid unnecessary manipulation.
- Use cravats or gauze bandages to immobilize the fracture and secure the limb to the body to minimize unnecessary movement during transport.

13-21. The first responder or CLS should follow the following when applying an arm splint:

- Place rigid objects so that one is on each side of the injured arm or forearm.
- Positioning the rigid objects so that the joint above the fracture and the joint below the fracture will be immobilized.
- Apply padding between the arm and the splint.
- Secure the splint with cravats, gauze bandages, strips of cloth, or other securing materials.
- If possible, place two cravats above and two below the fracture site.
- Use square knots to secure bandages tied away from the body on the splint.
- Check pulses lower to the affected extremity before and after the application of the splint.
- Administering the CWMP as needed.
- Document all treatment on a DD Form 1380 and attach it to the casualty.

EXPEDIENT TECHNIQUES FOR SPLINTS, PADDING, BANDAGES, SLINGS, AND SWATHES

13-22. Splints may be improvised from such items such as boards, poles, sticks, tree limbs, or cardboard. If nothing is available for a splint, the chest wall can be used to immobilize a fractured arm, and the uninjured leg can be used to immobilize (to some extent) the fractured leg.

13-23. Padding may be improvised from such items as a jacket, blanket, poncho, shelter half, or leafy vegetation. Padding when splinting a fracture is essential for the well-being of the casualty. It promotes comfort, even pressure distribution, stability, protection against skin issues, and helps manage swelling, all of which contributes to an effective and safer management of fractures, especially in situations where immediate medical care may not be readily available.

13-24. Bandages may be improvised from belts, rifle slings, handkerchiefs, or strips torn from clothing, or blankets. Narrow materials such as wire, or cord should not be used to secure a splint in place. The application of wire or narrow material to an extremity could cause tissue damage and a TQ effect.

13-25. A sling is a bandage suspended from the neck to support an upper extremity. If a bandage is not available, a sling can be improvised by using the tail of a coat, shirt, or pieces of cloth torn from such items as clothing and blankets. The cravat is ideal for this purpose. Remember that the casualty's hand should be higher than their elbow, and their fingers should be always showing. The sling should be applied so that the supporting pressure is on the uninjured side.

13-26. Swathes are any bands (pieces of cloth or straps) that are used to further immobilize a splinted fracture. Cravats are often used and are called swathe bandages. The purpose of the swathe is to immobilize; therefore, the swathe bandage is placed above or below the fracture, not on top of the fracture.

PROCEDURES FOR SPLINTING SUSPECTED FRACTURE

13-27. Before beginning first aid procedures for a fracture, gather whatever splinting materials are available. Ensure that splints are long enough to immobilize the joint above and below the suspected fracture. If possible, use at least four ties (two above and two below the fracture) to secure the splints. The ties should be square knots and should be tied away from the body on the splint. Distal pulses of the affected extremity should be checked before and after the application of the splint.

13-28. Evaluate the casualty and be prepared to perform any necessary lifesaving measures. Monitor the casualty for development of conditions that may require you to perform necessary lifesaving measures.

WARNING

Unless there is immediate life-threatening danger, such as a fire or an explosion, DO NOT move the casualty with a suspected back or neck injury. Improper movement may cause permanent paralysis or death.

13-29. Locate the site of the suspected fracture by:

- Asking the casualty—
 - Where are you injured?
 - Do they have any pain?
 - Where is it tender?
 - Can you move the extremity?

Note. With the presence of an obvious deformity, do not make the casualty move the extremity.

- Look for an unnatural position of the extremity.
- Look for a bone sticking out (protruding).

13-30. Prepare the casualty for splinting the suspected fracture.

- Reassure the casualty. Tell them that you will be providing first aid for them, and that medical help is on the way.
- Loosen any tight or binding clothing.
- Remove all jewelry from the injured part and place it in the casualty's pocket. Tell the casualty you are doing this because if the jewelry is not removed and swelling occurs later, they may not be able to remove it and further bodily injury could result.
- Boots should not be removed from the casualty unless they are needed to stabilize a neck injury, or there is actual bleeding from the foot.

13-31. Gather splinting materials. If splinting materials (splints, padding, and cravats) are not available, gather improvised materials. If splinting materials are not available and the suspected fracture cannot be splinted, then swathes or a combination of swathes and slings can be used to immobilize the extremity.

13-32. Pad the Splints. Pad the splints where they touch any bony part of the body, such as the elbow, wrist, knee, ankle, crotch, or armpit areas. Padding prevents excessive pressure on the area, which could lead to circulation problems.

13-33. Check the circulation below the site of the injury.

- Note any pale, white, or bluish-gray color of the skin, which may indicate impaired circulation.
- Check the temperature of the injured extremity. Use your hand to compare the temperature of the injured side with the uninjured side. The body area below the injury may be colder to the touch indicating poor circulation.
- Question the casualty about the presence of numbness, tightness, cold, or tingling sensations.

WARNING

Casualties with fractures of the extremities may show impaired circulation, such as numbness, tingling, cold or pale to bluish skin tone. These casualties should be evacuated to medical personnel and treated as soon as possible. Prompt medical treatment may prevent possible loss of the limb.

WARNING

If it is an open fracture and the bone is protruding from the skin, DO NOT ATTEMPT TO PUSH THE BONE BACK UNDER THE SKIN. Apply a field dressing over the wound to protect the area.

13-34. Apply the splint in place by—

- Splinting the fracture in the position found. DO NOT attempt to reposition or straighten the injury. If it is an open fracture, stop the bleeding and protect the wound.
- Covering all wounds with field dressings before applying a splint. Remember to use the casualty's field dressing, before using your own.
- Placing one splint on each side of the fracture. Make sure that the splints reach, if possible, beyond the joints above and below the fracture.
- Tying the splints by—
 - Securing each splint in place above and below the fracture site with cravats, or improvised material. Improvised materials, such as gauze bandage, strips of cloth, belts, or whatever other materials you have, may be used.
 - With minimal motion to the injured areas, place and tie the splints with the bandages.
 - Push splint material through and under the natural body curvatures, and then gently position splint material, and tie in place.
 - Using square knots. Tie all knots on the splint away from the casualty (figure 13-3). DO NOT tie knots directly over the suspected fracture site.

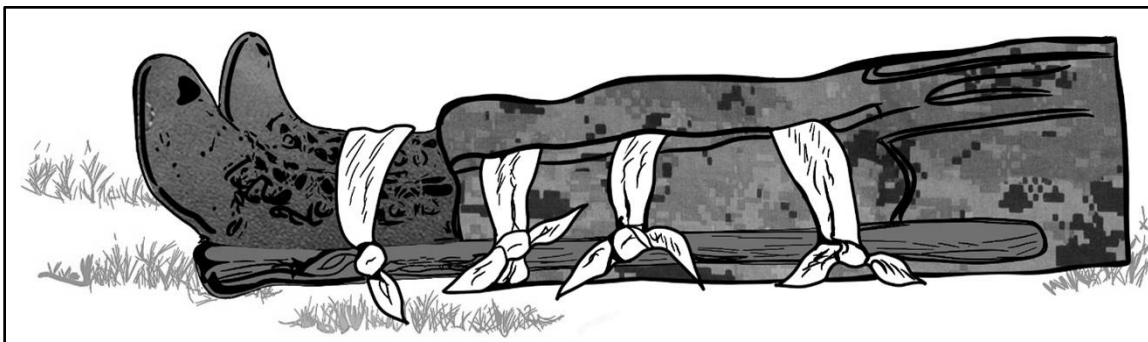


Figure 13-3. Square knots tied away from the casualty

13-35. Check the splint for tightness—

- CHECK to ensure that the bandages are tight enough to securely hold splinting materials in place, but not so tight that circulation is impaired.
- RECHECK the circulation after application of the splint. Check the skin color and temperature. This is to ensure that the bandages holding the splint in place have not been tied too tightly. A fingertip check can be made by inserting the tip of the finger between the bandaged knot and the skin.
- MAKE any necessary adjustments without allowing the splint to become ineffective.

APPLY AN IMPROVISED SLING

13-36. An improvised sling may be made from any available non-stretching piece of cloth, such as a utility uniform shirt or trousers, poncho, or shelter half. Slings may also be improvised using the tail of a coat (figure 13-4), belt (figure 13-5), or a piece of cloth may also be used for support. A sling is used to place the supporting pressure on the casualty's uninjured side (figure 13-6). The hand on the supported arm is positioned slightly higher than the elbow, showing the fingers.



Figure 13-4. Shirttail used for support



Figure 13-5. Belt used for support



Figure 13-6. Improvised sling

13-37. The sling is secured by bringing the ends of the sling up and tie them at the side (or hollow) of the neck on the uninjured side (figure 13-7). The end or corner of the sling is then twisted and tucked at the elbow (figure 13-8).



Figure 13-7. Ends of improvised sling tied to the side of neck



Figure 13-8. Corner of improvised sling twisted and tucked at the elbow

13-38. Apply swathes to the injured arm by wrapping the swathe over the injured arm, around the casualty's back, and under the arm on the uninjured side. Tie the ends on the uninjured side (figure 13-9 on page 128). You may use any large piece of cloth, Service member's belt, or pistol belt, to improvise a swathe. A swathe is applied to an injured leg by wrapping the swathe around both legs and securing it on the uninjured side.



Figure 13-9. An arm immobilized with strips of clothing

WARNING

The swathe should not be placed directly on top of the injury but positioned either above or below the fracture site.

13-39. Seek medical assistance. Notify medical personnel, watch closely for development of life-threatening conditions and, or impaired circulation to the injured extremity.

IMMOBILIZING A FRACTURE

13-40. Splinting or immobilizing a fracture is essential for providing immediate and effective care to casualties in a tactical environment. It contributes to stabilizing the injury, managing pain, preventing complications, and facilitating safe evacuation, ultimately improving the chances of a positive outcome for the injured individual.

Immobilizing an Upper Extremity Fracture

13-41. To splint an upper extremity fracture—

- Have the casualty sit up.
- Have someone support the fractured extremity.
- Remove all jewelry from the fractured extremity.
- Expose the fracture site.

13-42. Figure 13-10 on page 129 through figure 13-18 on page 131 shows you how to apply slings, splints, and cravats (swathes) to immobilize and support fractures of the upper extremities. Although the padding is not visible in some of the illustrations, it is always preferable to apply padding along the injured part for the length of the splint, especially where it touches any bony parts of the body.



Figure 13-10. Application of a cravat to form a sling (two methods)

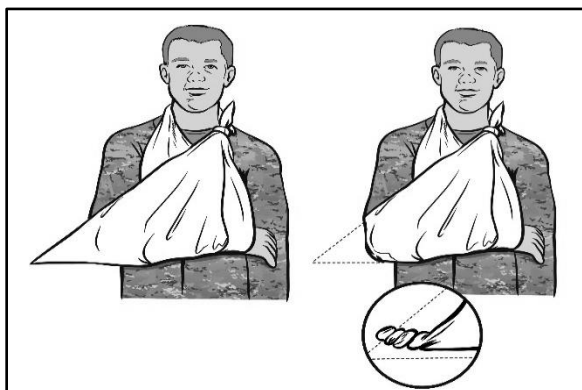


Figure 13-11. Completing the sling sequence by twisting and tucking the corner of the sling at the elbow

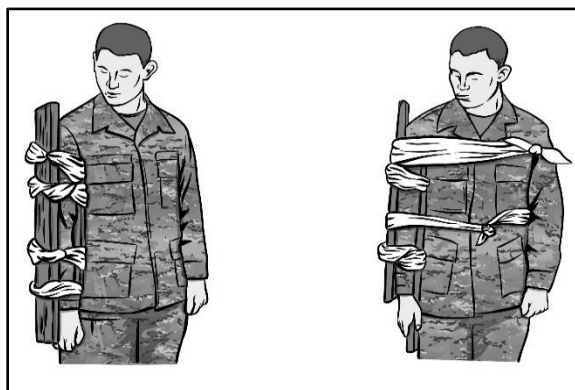


Figure 13-12. Board splints applied to the fractured elbow when the elbow is not bent (two methods)

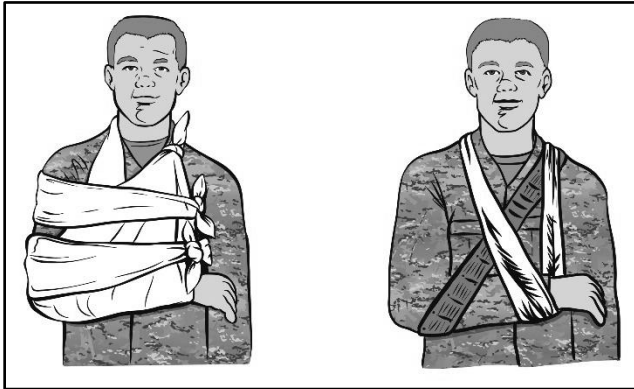


Figure 13-13. Chest wall used as a splint for the upper arm fracture when no splint is available



Figure 13-14. Chest wall, sling, and cravat used to immobilize the fractured elbow when the elbow is bent



Figure 13-15. Board splint applied to the fractured forearm



Figure 13-16. Fractured forearm or wrist splinted with sticks, and supported with a tail of the shirt and strips of material

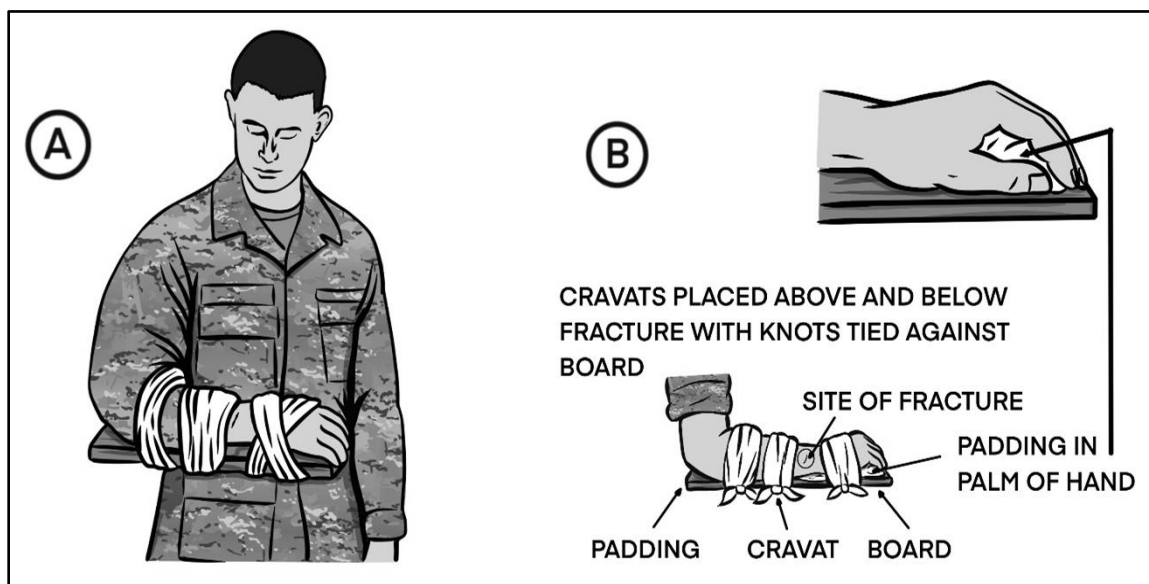


Figure 13-17. Board splint applied to a fractured wrist and hand

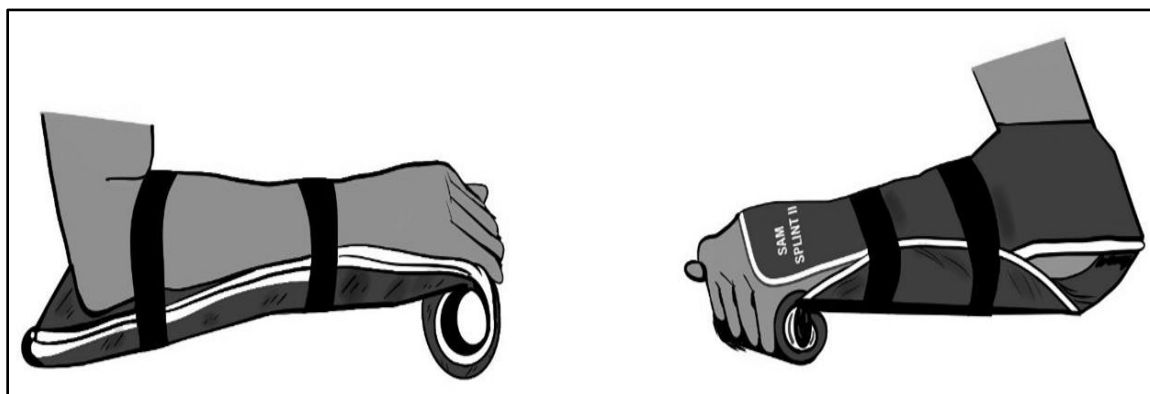


Figure 13-18. Semi-rigid splint applied to a fractured wrist or forearm

Immobilizing a Lower Extremity Fracture

13-43. To splint a lower extremity fracture—

- Have the casualty sit or lie down.
- Have another person manually immobilize the fractured extremity.
- Remove the footgear and expose the fracture site.
- Check distal (at the wrist) pulse and capillary refill on the injured extremity. Prepare the splint for application.

Note. Measuring and forming of splints should be done on the unbroken extremity prior to placing on the broken one.

13-44. Figure 13-19 on page 132 through figure 13-25 on page 135 show how to apply splints to immobilize fractures of the lower extremities. Although padding is not visible in some of the illustrations, it is always preferable to apply padding along the injured part for the length of the splint, especially where it touches any bony parts of the body.

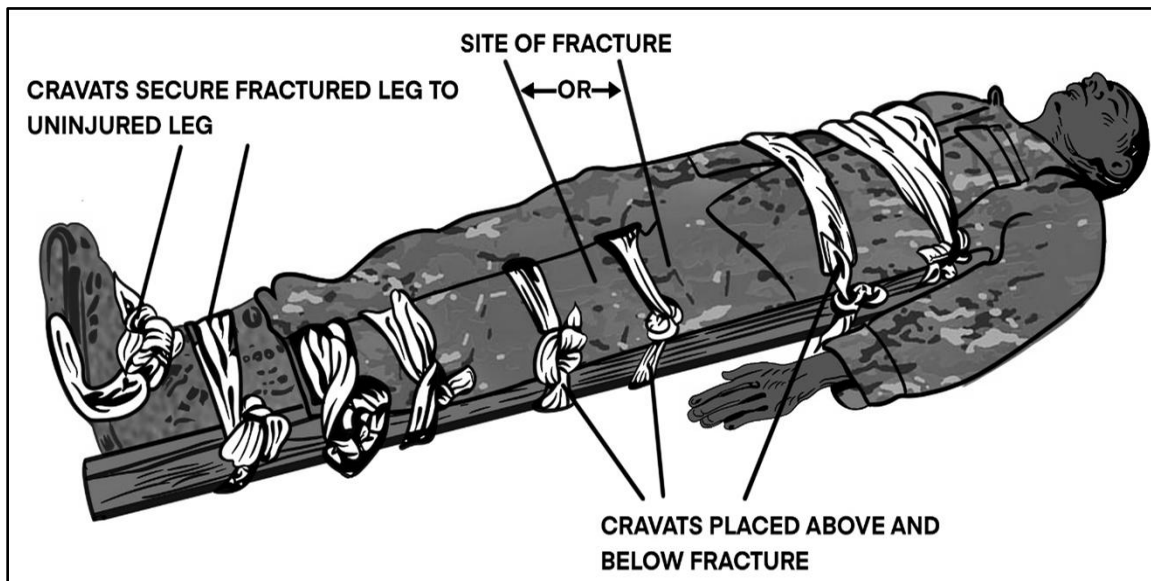


Figure 13-19. Board splints applied to the fractured hip or thigh

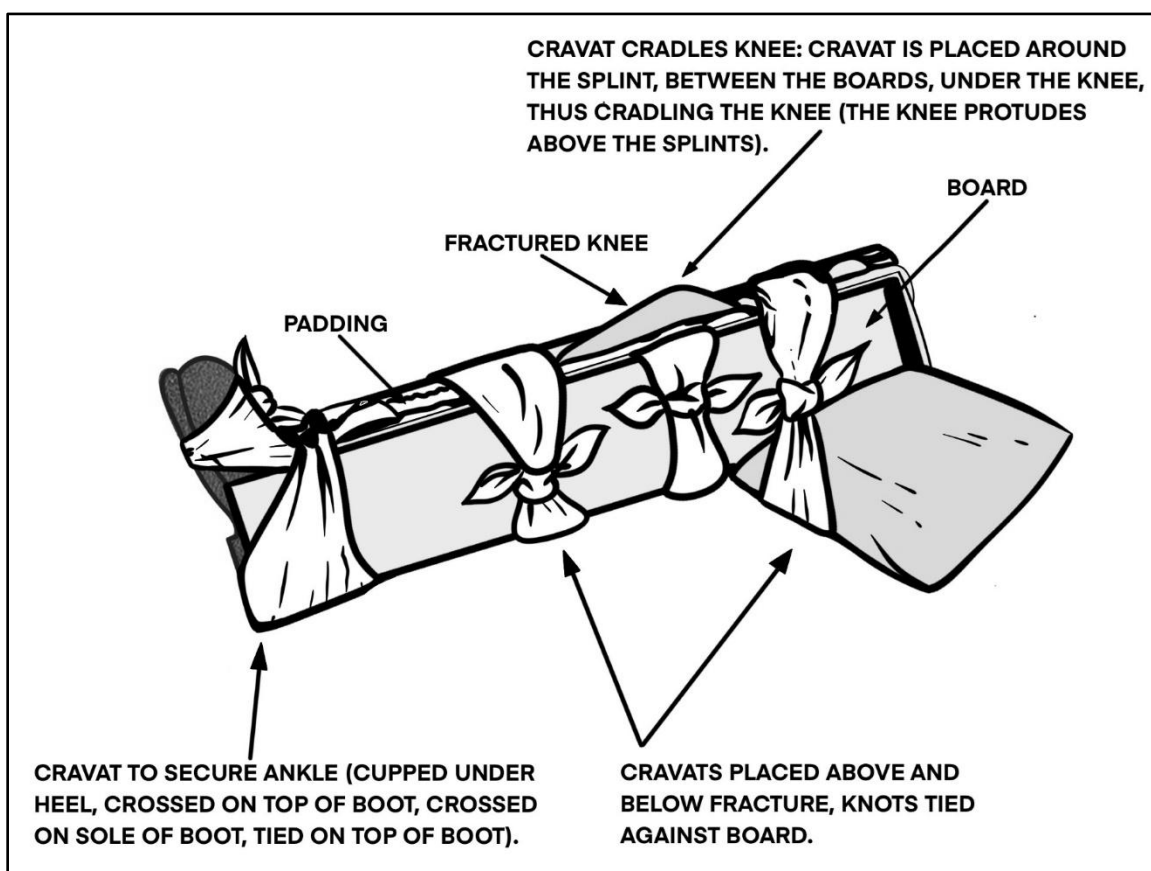


Figure 13-20. Board splint applied to the fractured or dislocated knee

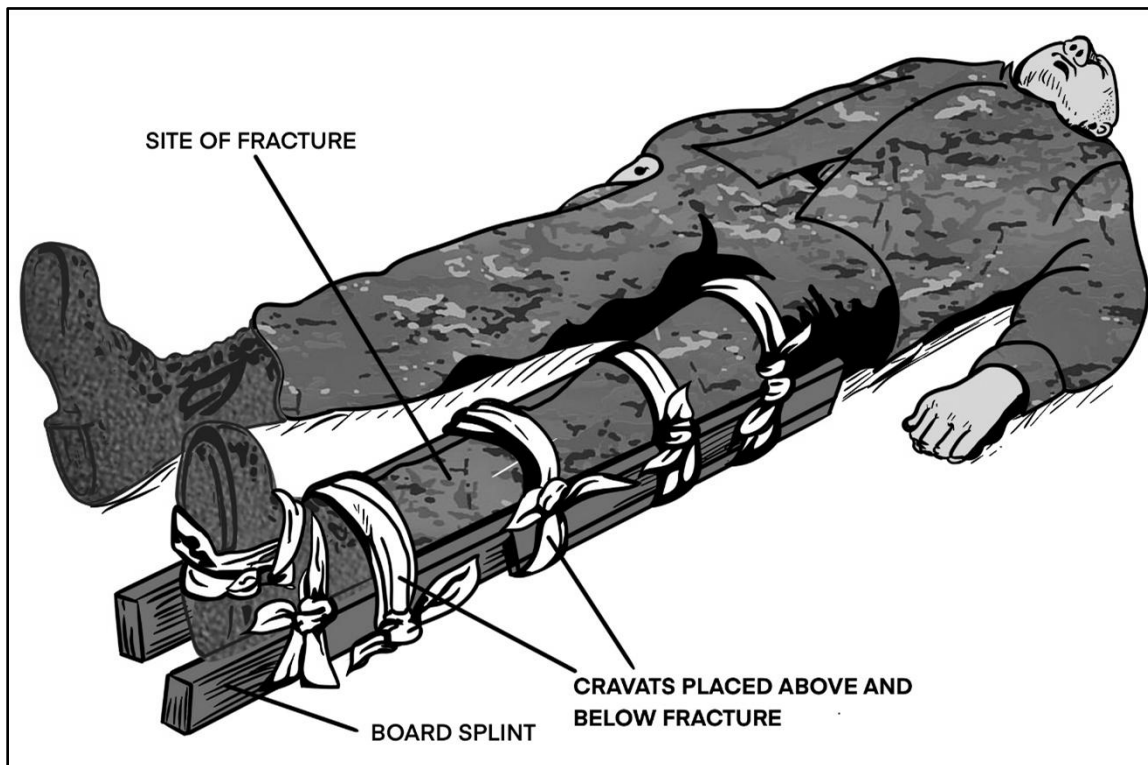


Figure 13-21. Board splints applied to the fractured lower leg or ankle

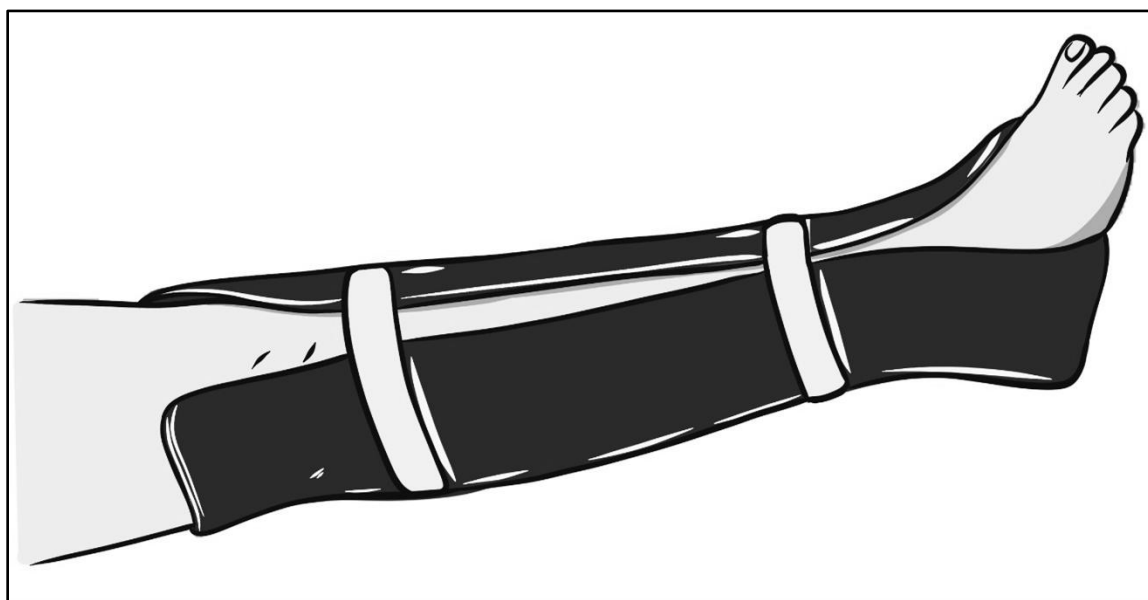


Figure 13-22. Moldable splint applied to the fractured lower leg or ankle

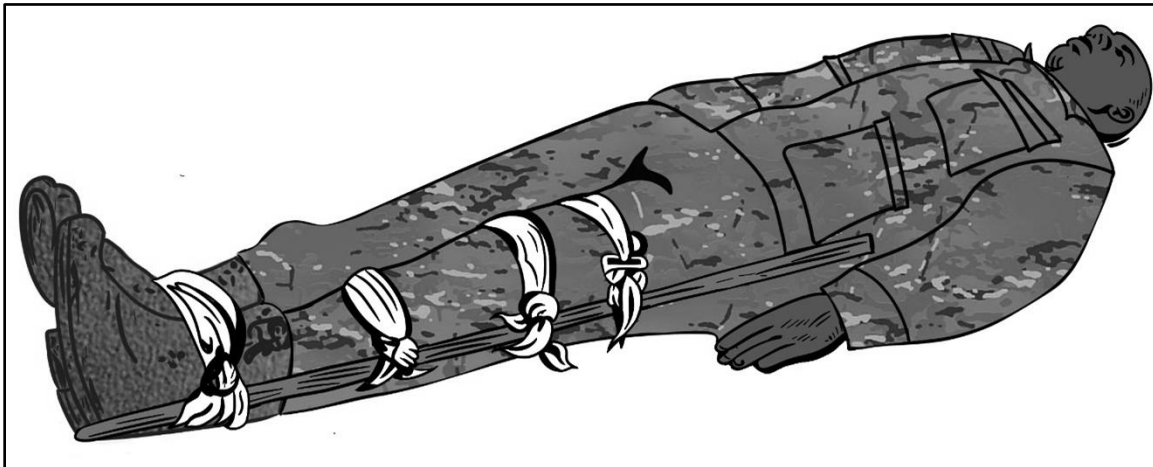


Figure 13-23. Improvised splint applied to the fractured lower leg or ankle

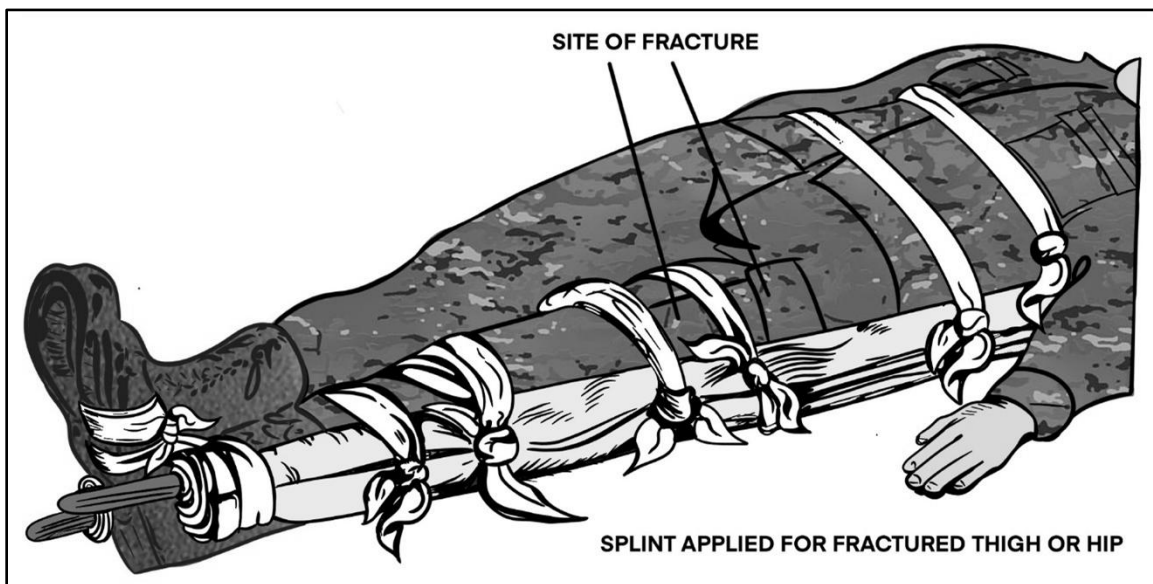


Figure 13-24. Poles rolled in a blanket and used as splints that are applied to the fractured lower extremity

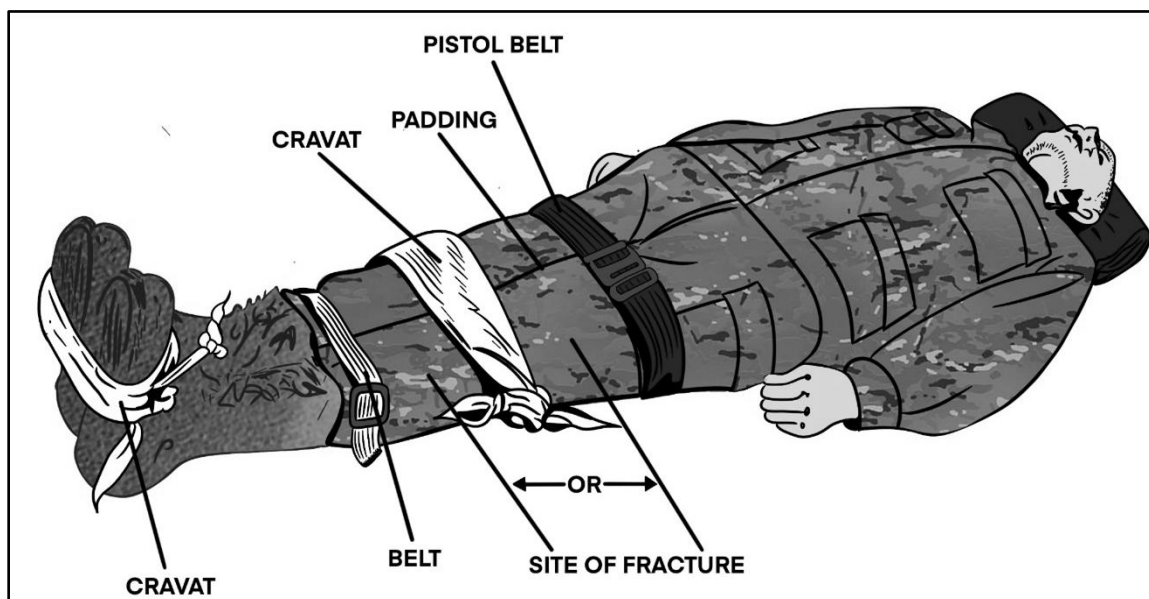


Figure 13-25. Uninjured leg used as a splint for the fractured leg (anatomical splint)

Immobilizing a Jaw, Collarbone, or Shoulder Fracture

13-45. To immobilize a fractured jaw, apply a cravat or gauze bandage following the guidelines outlined in paragraph 11-39 on page 114. Ensure that all bandaging support is directed to the top of the casualty's head rather than the back of their neck. Placing the bandage incorrectly may result in pulling the casualty's jaw backward, potentially obstructing their breathing and exacerbating the injury.

WARNING

Casualties with lower jaw (mandible) fractures cannot be laid flat on their backs because facial muscles will relax and may cause an airway obstruction.

13-46. A fractured collarbone can be immobilized by applying two belts, a sling, and a swathe. Figure 13-26 illustrates how to immobilize a fractured collarbone.



Figure 13-26. The application of belts, sling, and swathe to immobilize a fractured collarbone

13-47. A suspected dislocated or fractured shoulder can be immobilized by applying a sling and a swathe. Figure 13-27 on page 136 illustrated how to properly immobilize a dislocated or fractured shoulder.

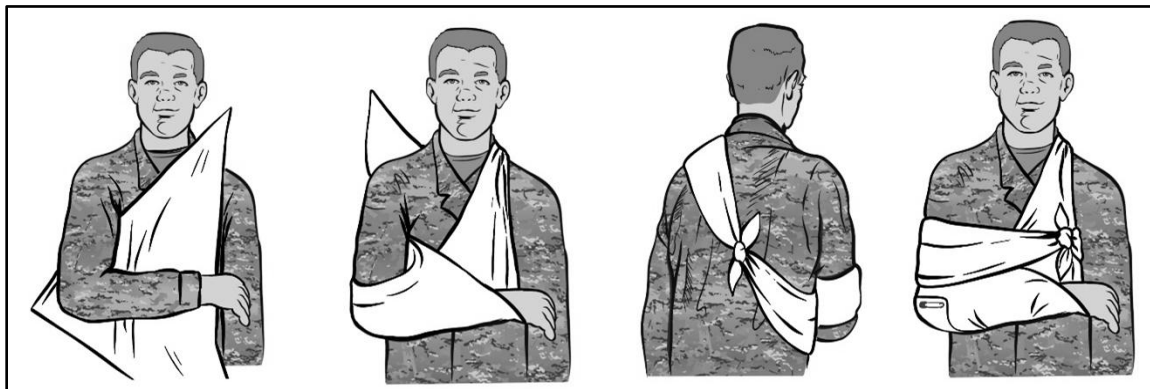


Figure 13-27. The application of the sling and swathe to immobilize a fractured or dislocated shoulder

Immobilizing a Spinal Column Fracture

13-48. It is often impossible to be sure if a casualty has a fractured spinal column. Be suspicious of any back injury, especially if the casualty has fallen or if their back has been sharply struck or bent. If a casualty has received such an injury and does not have feeling in their legs or cannot move them, you can be reasonably sure that they have a severe back injury which is managed as a fracture. A neck fracture or injury to the back should always be suspected, and it is often impossible to be sure if a casualty has a fractured spinal column. If the spine is fractured (figure 13-28 step A on page 137), bending it can cause the sharp bone fragments to bruise or cut the spinal cord and result in permanent paralysis or death. The spinal column must maintain normal spinal position at the lower back (lumbar region) to help remove pressure from the spinal cord.

13-49. If the casualty is not to be transported until medical personnel arrive—

- Caution them not to move. Ask if they are in pain or, if they are unable to move any part of their body.
- Leave them in the position in which they were found. DO NOT move any part of their body unless they are in imminent danger.
- If the casualty is lying face up, slip a blanket or other supporting material under the arch of their lower back to help support the spine in a normal position (figure 13-28 step B on page 137). Take care not to place so much bulky padding under your casualty, as this will cause potential damage and cause undue pressure on the spine.
- If the casualty is lying face down, DO NOT put anything under any part of their body.

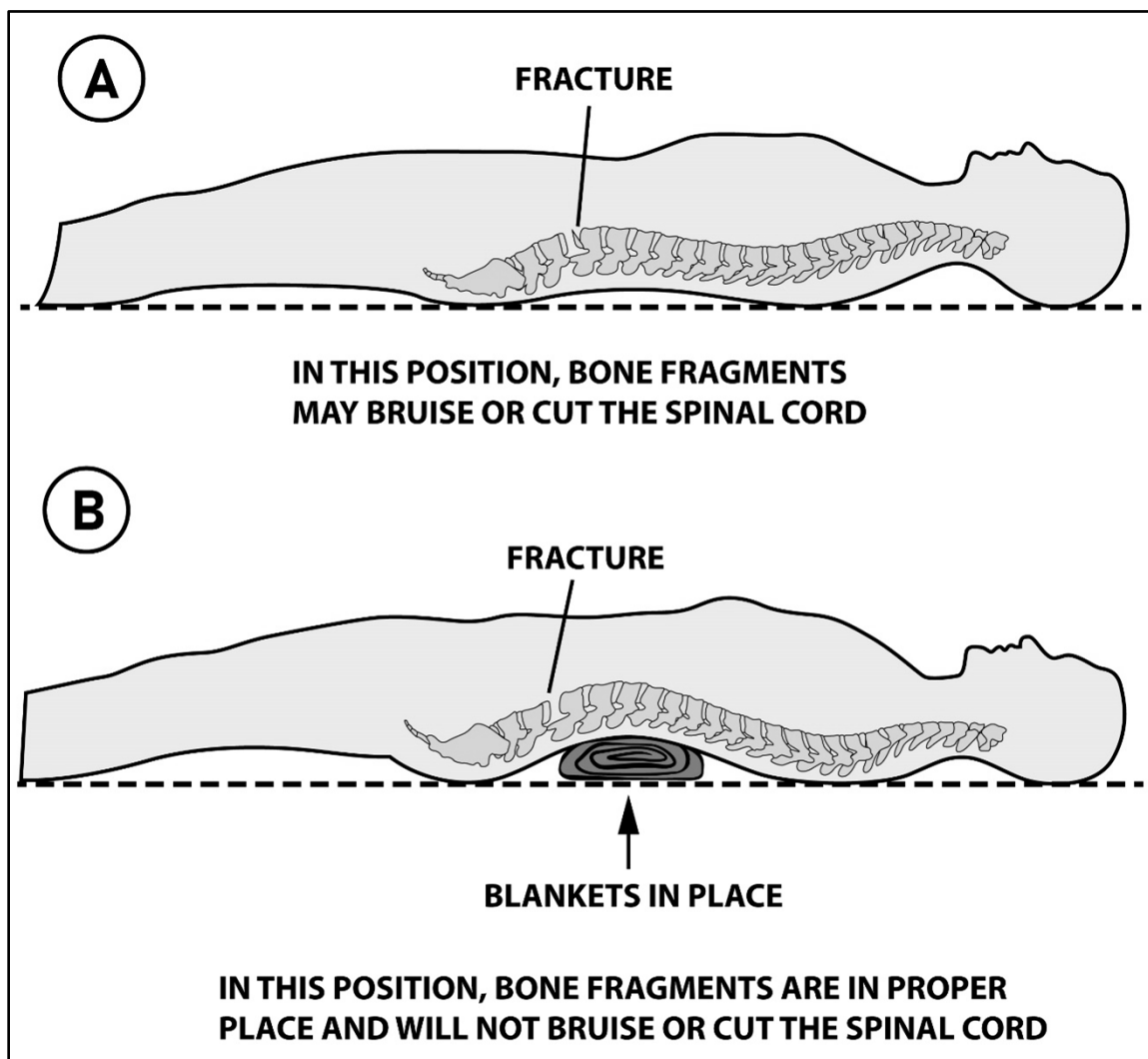


Figure 13-28. The spinal column must maintain a normal spine position

13-50. If the casualty must be transported to a safe location before medical personnel arrive and if the casualty is in a—

- Face-up position, transport them by litter or use a firm substitute, such as a wide board or a door longer than their height. Loosely tie the casualty's wrists together over their waistline, using a cravat, gauze bandage, or a strip of cloth. Tie their feet together to prevent the accidental dropping or shifting of their legs. Lay a folded blanket across the litter where the arch of their back is to be placed. Using a four-person team (figure 13-29 on page 138), place the casualty on the litter without bending their spinal column or their neck. Avoid prolonged use, temporary if possible.



Figure 13-29. Placing the casualty face-up with a fractured back onto the litter

- The number two Service member positions themselves at the casualty's head. Their responsibility is to provide manual in-line (neutral) stabilization of the head and neck. The number three, and four Service members position themselves on one side of the casualty; all kneel on one knee along the uninjured side of the casualty.
- The number one Service member positions themselves preferably to the opposite side of the casualty as the number three and four members. The number one member can be on the same side as number three and four, if needed. Numbers two, three, and four Service members gently place their hands under the casualty. The number one Service member on the opposite side places their hands under the injured part to assist.
- When all four Service members are in position to lift, the number two Service member commands, "PREPARE TO LIFT" and then, "LIFT." On the command of "LIFT" all Service members, in unison, gently lift the casualty about eight inches off the ground.
- With the casualty lifted, the number one Service member recovers and slides the litter under the casualty, ensuring that the blanket is in proper position. The number one Service member then returns to their original lift position.
- When the number two Service member commands, "PREPARE TO LOWER" and then "LOWER." On the command of "LOWER" all Service members, in unison, gently lower the casualty onto the litter.

13-51. Casualties in a facedown position must be transported in the facedown position. The four-person team lifts the casualty onto a regular or improvised litter, keeping the spinal column in a normal spinal position. If a regular litter is used, first place a folded blanket on the litter at the point where the chest will be placed.

Immobilizing a Neck Fracture

13-52. A fractured neck is extremely dangerous. Bone fragments may bruise or cut the spinal cord just as they might in a fractured back. If the casualty is not to be transported until medical personnel arrive—

- Caution them not to move. Moving may cause permanent injury or death.
- Leave the casualty in the position in which they are found. If their neck and head (cervical spine) are in an abnormal position, immediately immobilize the neck and head.
- Keep their head still, if the casualty is lying face up, raise their shoulders slightly, and slip a roll of cloth that has the bulk of a bath towel under their neck (figure 13-30). The roll should be thick enough to arch their neck only slightly, leaving the back of their head on the ground. DO NOT bend their neck or head forward. DO NOT raise or twist their head.
- Immobilize the casualty's head (figure 13-31). Do this by padding heavy objects (such as rocks or their boots filled with dirt, sand, gravel, or rock) and placing them on each side of their head. If it is necessary to use boots, after filling them, tie the top tightly or stuff with pieces of cloth to secure the contents.



Figure 13-30. Casualty with a roll of cloth (bulk) under neck



Figure 13-31. Immobilization of the fractured neck with boots

13-53. DO NOT move the casualty if they are lying face down. Immobilize the head and neck by padding heavy objects and placing them on each side of their head. DO NOT put a roll of cloth under the neck.

13-54. DO NOT bend the neck or head, nor roll the casualty onto their back. If the casualty must be prepared for transportation before medical personnel arrive at least two persons are needed to move the casualty's head and trunk in unison.

13-55. Care is taken should a casualty have to be moved prior to medical personnel arrival. The casualty is prepared by—

- A wide board is placed lengthwise beside the casualty. It should extend at least 4 inches beyond the casualty's head and feet (figure 13-32 step A on page 140). Avoid prolonged use, if possible.
- If the casualty is lying face up, the number one Service member steadies the casualty's head and neck between their hands. At the same time, the number two Service member positions one foot and one knee against the board to prevent it from slipping. They then grasp the casualty underneath their shoulder and hip and gently slide the casualty onto the board (figure 13-32 step B on page 140).
- If the casualty is lying face down, the number one member steadies the casualty's head and neck between their hands, while the number two Service member gently rolls the casualty over onto the board (figure 13-32 step C on page 140).
- The number one Service member continues to steady the casualty's head and neck. The number two Service member simultaneously raises the casualty's shoulders slightly, places padding under their neck, and immobilizes the casualty's head. Any improvised supports are secured in position with a cravat, gauze bandage, or strip of cloth extended across the casualty's forehead and under the board (figure 13-32 step D on page 140).
- The board is lifted onto a litter or blanket in order to transport the casualty (figure 13-32 step E).

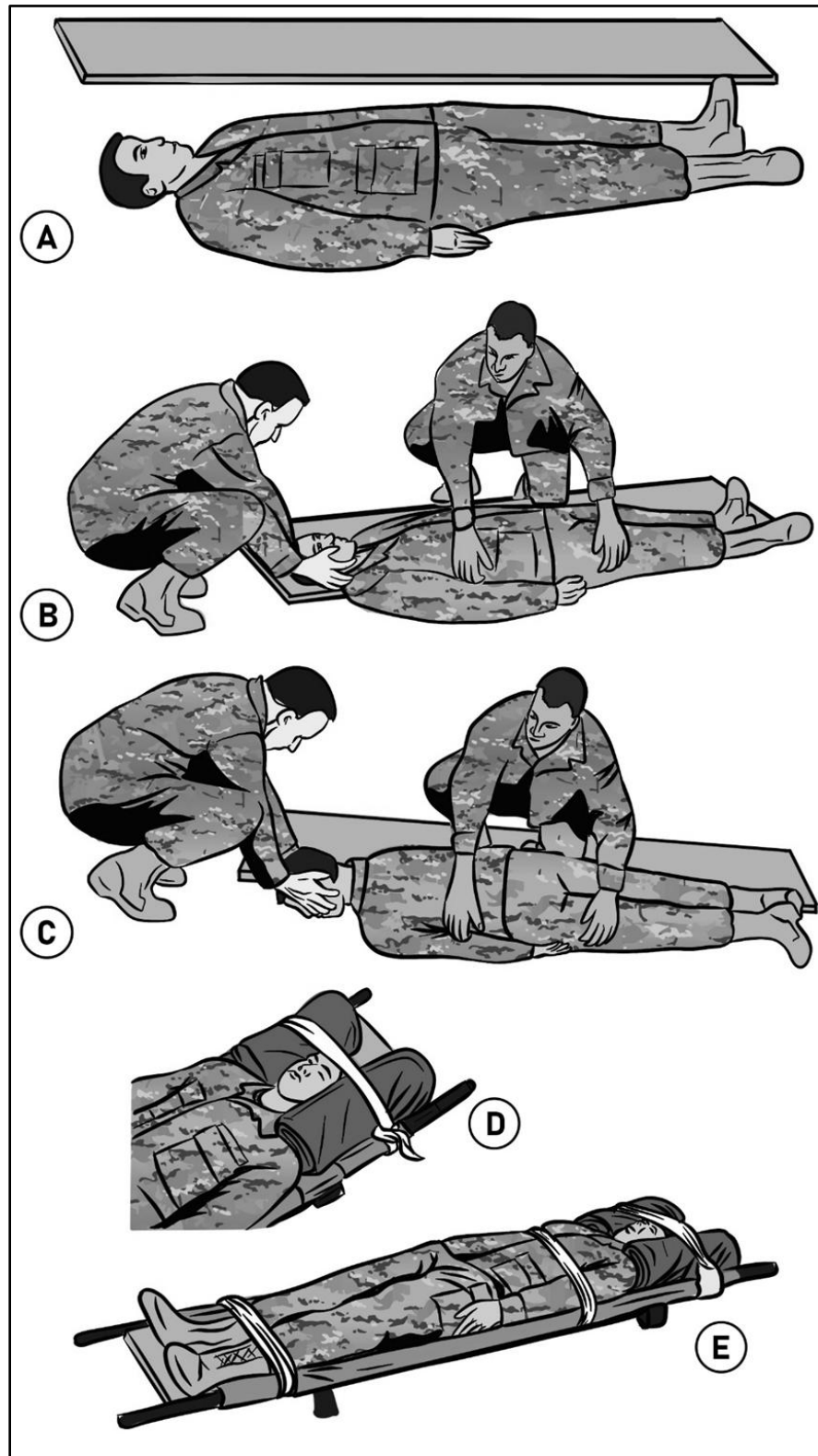


Figure 13-32. Preparing the casualty with a fractured neck for transportation

Chapter 14

Expedient Techniques for other Secondary Injuries

OVERVIEW OF SECONDARY INJURIES

14-1. Because medical personnel will not always be readily available, nonmedical personnel must rely on themselves and other Service members' knowledge and fundamentals of first aid for secondary injuries until medical assistance is gained and initiated. This section will provide additional first aid, non-TCCC, procedures that nonmedical personnel can use in a garrison environment, or in combat situations.

ABDOMINAL WOUNDS

14-2. The most serious abdominal wound is one in which an object penetrates the abdominal wall and pierces internal organs or large blood vessels. In these instances, bleeding may be severe, and death can occur rapidly.

14-3. The first step in treating an abdominal wound is to position the casualty. Place and maintain the casualty on their back with their knees in an upright (flexed) position (figure 14-1). The knees-up position helps relieve pain, assists in the treatment of shock, prevents further exposure of the bowel (intestines) or abdominal organs, and helps relieve abdominal pressure by allowing the abdominal muscles to relax.

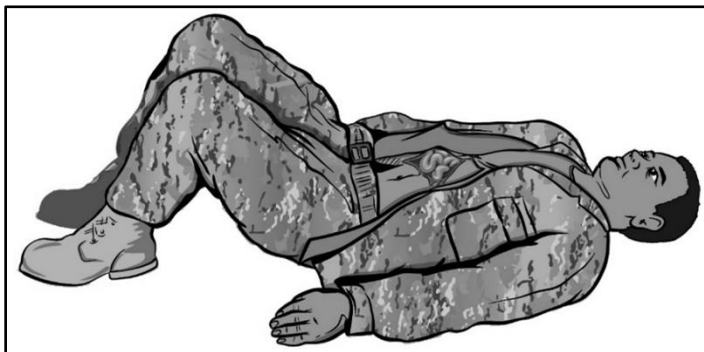


Figure 14-1. The casualty positioned (lying) on their back with their knees (flexed) up

14-4. Expose the wound by—

- Removing the casualty's loose clothing to expose the wound. However, DO NOT attempt to remove clothing that is stuck to the wound; removing it may cause further injury.
- Gently pick up any organs that may be on the ground. Do this with a clean, dry dressing or with the cleanest available material. Place the organs on top of the casualty's abdomen (figure 14-2).



Figure 14-2. Protruding organs placed near the wound

Note. DO NOT probe, clean, or try to remove any foreign object from the abdomen. DO NOT touch with bare hands any exposed organs. DO NOT push organs back inside the body.

14-5. It may be necessary to wear improvised dressing (for example strips of cloth, a t-shirt, or the cleanest material available). Stabilize any protruding or impaled objects and apply a sterile abdominal dressing as follows:

Note. Protruding abdominal organs should be kept moist to prevent the tissue from drying out. A moist sterile dressing should be applied if available.

- Using the sterile side of the dressing, or other clean material, place any protruding organs near the wound. (DO NOT try to push organs back inside the body.)
- Ensure that the dressing is large enough to cover the entire mass of protruding organs or area of the wound.
- If large enough to cover the affected area, place the sterile side of the plastic wrapper directly over the wound.
- Moisten the dressing with clean water prior to placing a dressing over the exposed organs.
- Place the dressing directly on top of the wound or plastic wrapper, if used.
- Tie the dressings loosely at the casualty's side.

14-6. If two dressings are needed to cover a large wound, repeat steps in paragraph 15-5. If necessary, loosely cover the dressing with a cravat, elastic bandage, or gauze bandage. Tie them on the side of the casualty opposite to that of the dressing ties.

14-7. To avoid causing further injury to the casualty—

- Do not touch any exposed organs.
- Do not try to push any exposed organs back into the body.
- Do not tie the dressing tails tightly or directly over the wound.
- Do not give the casualty anything by mouth.

Note. Continue to evaluate the casualty and provide necessary first aid.

14-8. Prepare the casualty for evacuation as follows:

- Place the casualty on their back (face up) with the knees flexed.
- Keep the patient warm, cover with a blanket as needed.
- If evacuation is delayed, check the casualty for shock every five minutes.
- Seek medical aid.
- Record the procedures applied on DD Form 1380.

14-9. The first responder continues to evaluate the casualties and is prepared to perform basic first aid measures. Always check for both entry and exit wounds. If there are two wounds (entry and exit), treat the wound that appears more serious first (for example, the heavier bleeding, protruding organs, larger wound, and so forth). It may be necessary to improvise dressings for the second wound by using strips of cloth, a t-shirt, or the cleanest material available.

14-10. Use the casualty's dressing from the first aid kit, not your own. If the dressing is not large enough to cover the entire wound, the plastic wrapper from the dressing may be used to cover the wound first (placing the dressing on top). Open the plastic wrapper carefully without touching the inner surface. If necessary, other improvised dressings may be made from clothing, blankets, or the cleanest materials available.

14-11. Hypothermia prevention measures are taken with abdominal injuries due to the risk of rapid heat loss from the exposed abdominal cavity or organs. Moisten the dressing with clean water prior to placing a dressing over the exposed organs and seeking medical assistance.

WARNING

If there is an object extending from the wound, **DO NOT** remove it. Place as much of the wrapper over the wound as possible without dislodging or moving the object. **DO NOT** place the wrapper over the object.

WARNING

When the dressing is applied, **DO NOT** put pressure on the wound or exposed internal parts, because pressure could cause further injury (such as vomiting or ruptured intestines). Therefore, tie the dressing ties (tails) loosely at the casualty's side, not directly over the dressing.

CAUTION

DO NOT give casualties with abdominal wounds food or water (moistening the lips is allowed).

BANDAGING

14-12. Expedient bandaging techniques are of paramount importance for the U.S. military due to their ability to provide immediate and effective medical care in combat situations. In the fast-paced and unpredictable nature of military operations, Service members often face severe injuries that require immediate attention to stabilize the wounded and prevent further harm. Expedient bandaging techniques enable military personnel to apply life-saving measures to control bleeding and increase the chances of survival for injured personnel. These techniques not only buy valuable time until medical help arrives but also contribute to the overall mission readiness by ensuring that Service members can continue their duties with confidence and knowing that essential medical support is readily available when needed.

SHOULDER BANDAGE

14-13. To apply shoulder bandages attached to the field first aid dressing—

- Take one bandage across the chest, the other across the back, and under the arm opposite the injured shoulder.
- Tie the ends with a square knot (figure 14-3 on page 144).
- Ensure that the bandages are snug but not too tight, allowing for proper circulation and comfort for the casualty.



Figure 14-3. The shoulder bandage

14-14. To apply a bandage to the shoulder or armpit—

- Make an extended bandage by using two cravats or an elastic bandage (figure 14-4 step A on page 145); place the end of the first cravat along the base of the second one (figure 14-4 step B on page 145).
- Fold the two bandages into a single extended bandage (figure 14-4 step C on page 145).
- Fold the extended bandage into a single cravat (figure 14-4 step D on page 145). After folding, secure the thicker part (overlap) with tape or two or more safety pins (figure 14-4 step E on page 145).
- Place the middle of the cravat under the armpit so that the front end is longer than the back end and safety pins are on the outside (figure 14-4 step F on page 145).
- Cross the ends on top of the shoulder (figure 14-4 step G on page 145).
- Take one of the bandages, which ends across the back and under the arm on the opposite side and the other end across the chest. Tie the ends (figure 14-4 step H on page 145).

14-15. Be sure to place sufficient wadding in the armpit. DO NOT tie the bandage too tightly. Avoid compressing the major blood vessels in the armpit.

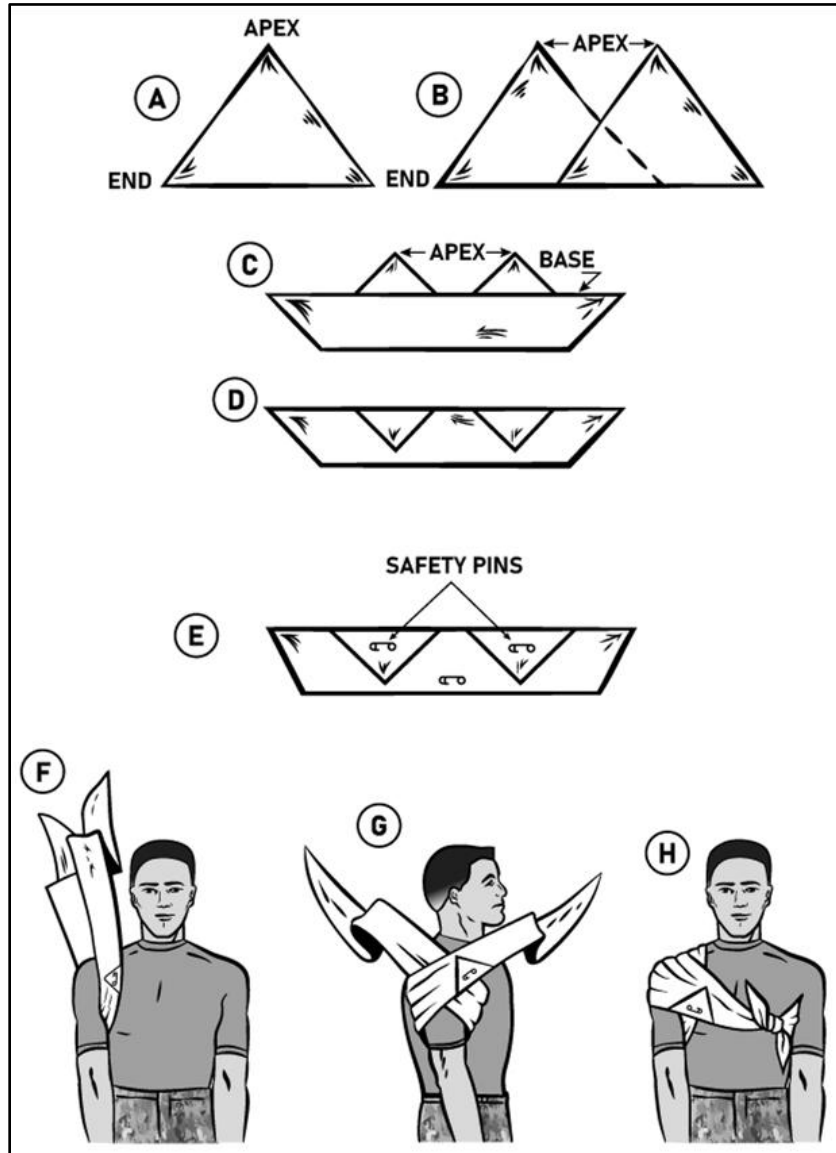


Figure 14-4. The extended cravat applied to the shoulder or armpit

ELBOW BANDAGE

14-16. To apply a cravat to the elbow—

- Bend the arm at the elbow and place the middle of the cravat at the point of the elbow bringing the ends upward (figure 14-5 step A on page 146).
- Bring the ends across, extending both downward (figure 14-5 step B on page 146).
- Take both ends around the arm and tie them with a square knot at the front of the elbow (figure 14-5 step C on page 146).

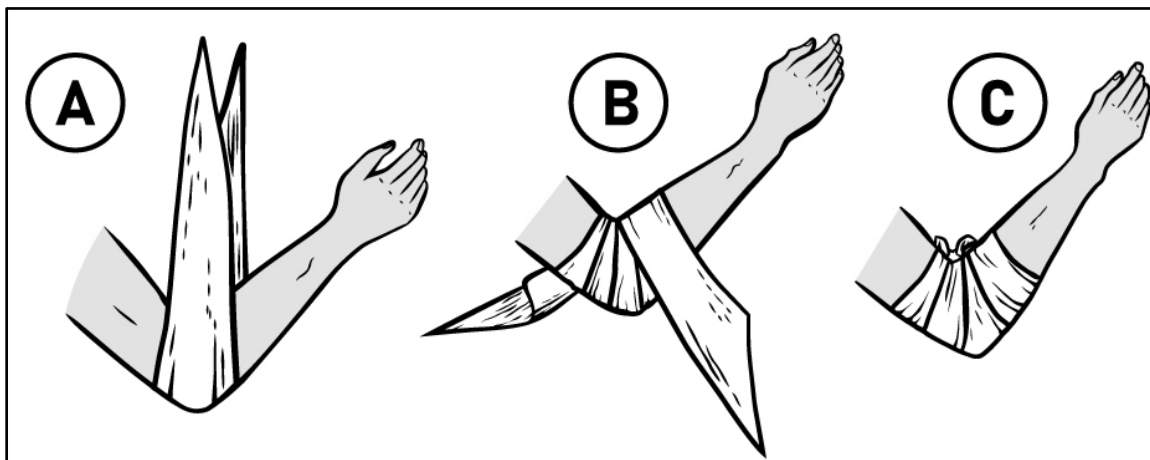


Figure 14-5. The elbow bandage

CAUTION

If an elbow fracture is suspected, DO NOT bend the elbow; bandage it in the position found.

HAND BANDAGE

14-17. To apply a bandage to the hand—

- Place the hand in the middle of the cravat or gauze bandage with the wrist at the base of the bandage (figure 14-6 step A). Ensure that the fingers are separated with absorbent material to prevent chafing and irritation of the skin.
- Place the apex over the fingers and tuck any excess material into the pleats on each side of the hand (figure 14-6 step B).
- Cross the ends on top of the hand, take them around the wrist, and tie them (figure 14-6 step C through step E) with a square knot.

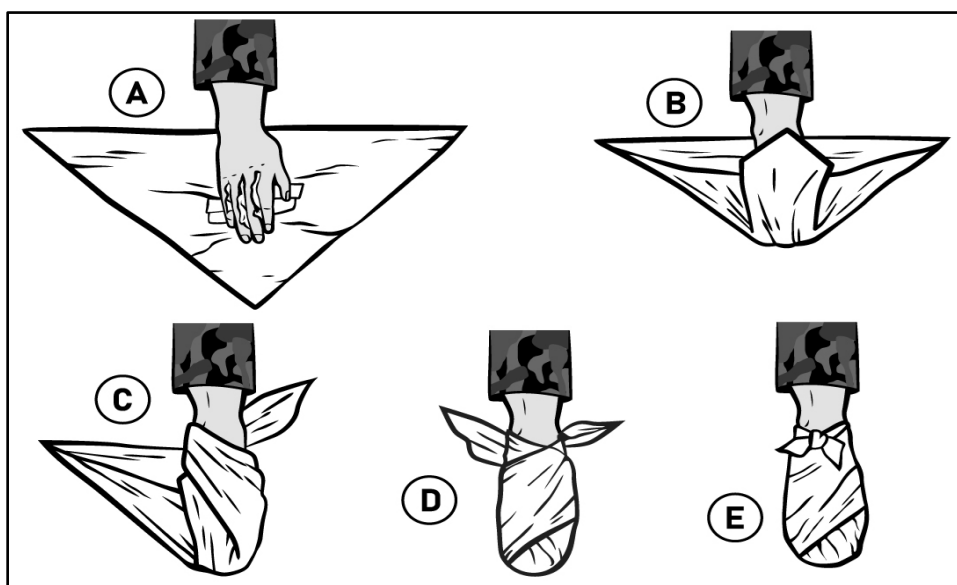


Figure 14-6. Bandage applied to the hand

14-18. To apply a bandage to the palm of the hand—

- Lay the middle of the cravat or gauze bandage over the palm of the hand with the ends hanging down on each side (figure 14-7 step A).
- Take the end of the cravat or gauze bandage at the little finger across the back of the hand, extending this upward over the base of the thumb; then bring it downward across the palm (figure 14-7 step B).
- Take the thumb end across the back of the hand, over the palm, and through the hollow between the thumb and palm (figure 14-7 step C).
- Take the ends to the back of the hand and cross them; then bring them up over the wrist and cross them again (figure 14-7 step D).
- Bring both ends down and tie them with a square knot on top of the wrist (figure 14-7 step E through step F).

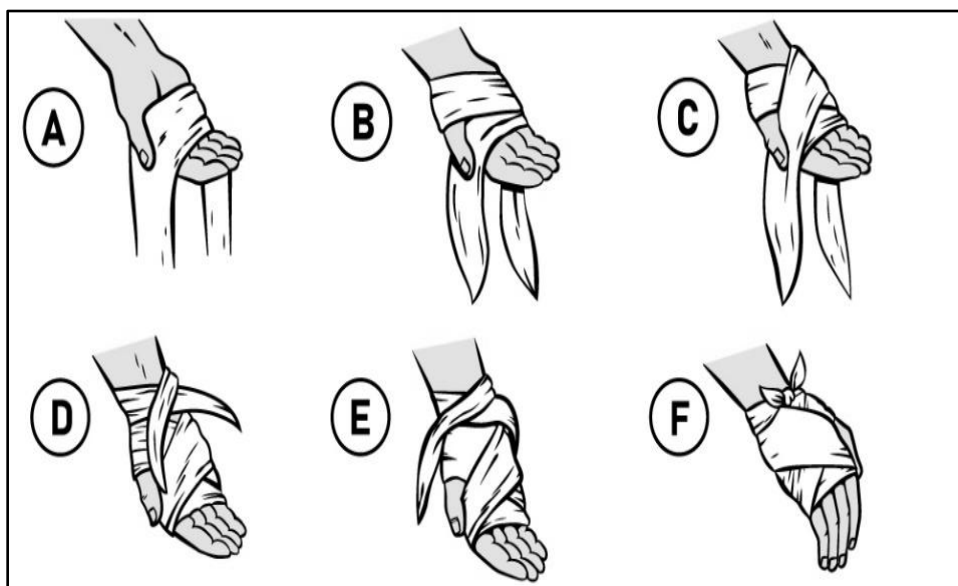


Figure 14-7. Bandage applied to the palm of the hand

LEG (UPPER AND LOWER) BANDAGE

14-19. To apply a bandage to the leg—

- Place the center of the cravat or gauze bandage over the dressing (figure 14-8 step A on page 148).
- Take one end around and up the leg in a spiral motion and the other end around and down the leg in a spiral motion, overlapping part of each preceding turn (figure 14-8 step B on page 148).
- Bring both ends together and tie them (figure 14-8 step C on page 148) with a square knot.

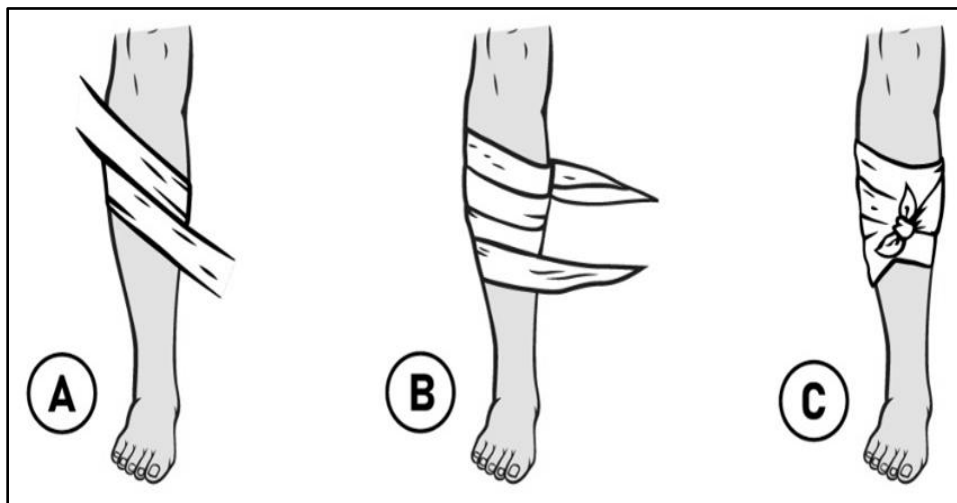


Figure 14-8. Bandage applied to the leg

KNEE BANDAGE

14-20. To apply a bandage to the knee as illustrated in figure 14-9, use the same technique applied in bandaging the elbow described in paragraph 14-16 on page 145.

CAUTION

If a fracture of the kneecap is suspected, DO NOT bend the knee; bandage it in the position found.

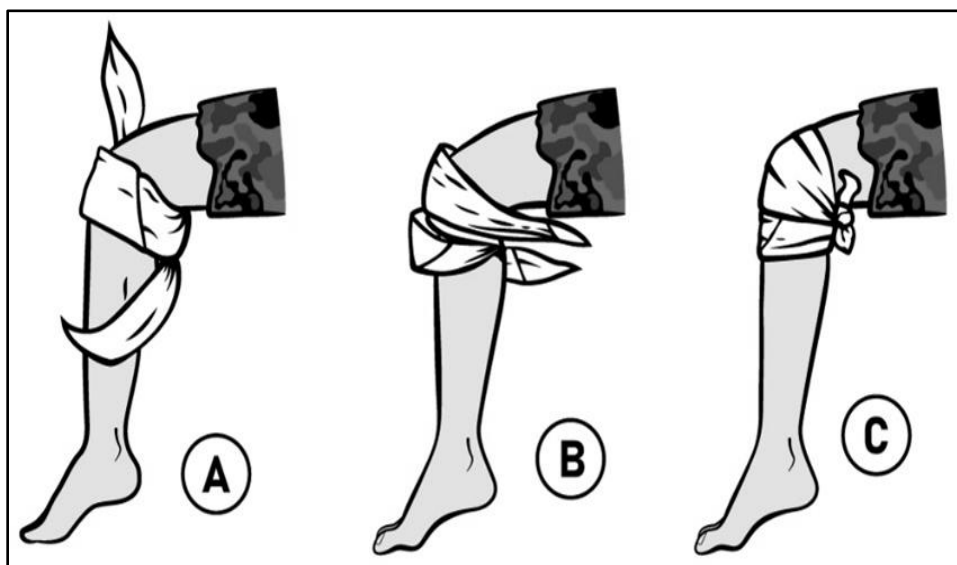


Figure 14-9. Bandage applied to the knee

FOOT BANDAGE

14-21. To apply a bandage to the foot—

- Place the foot in the middle of the cravat, elastic bandage, or gauze bandage with the heel well forward of the base (figure 14-10 step A). Ensure that the toes are separated by absorbent material to prevent chafing and irritation of the skin.
- Place the apex over the top of the foot and tuck any excess material into the pleats on each side of the foot (figure 14-10 step B).
- Cross the ends on top of the foot, take them around the ankle, and tie them at the front of the ankle (figure 14-10 step C through step E).

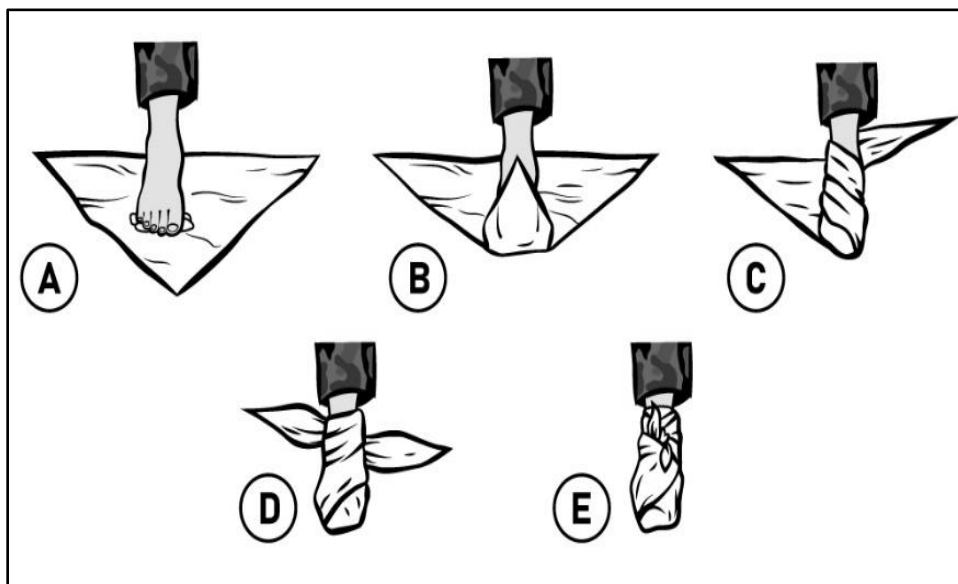


Figure 14-10. Bandage applied to the foot

IMPALEMENT INJURIES

14-22. Depending upon where the impaled object is located on the body, extreme care must be exercised. Improper methods may cause severe injury and possess the potential for further disability and death.

14-23. Apply first aid for an impaled object as follows:

- Tell the casualty to remain still and not to move the impaled object.
- Expose the injury by cutting away or removing the clothing or equipment around the wound site.
- If the impalement is on an extremity, check the pulse distal (situated away from the center of the body or injury) from the injury site.

Note. If an assistant is available, one person should immobilize the object while the other applies the dressings and bandages.

- If necessary, apply direct pressure using gloved hands on either side of the object.
- Place several layers of bulky dressings around the injury site so that the dressings surround the object.
- Use additional bulky materials or dressings to build up the area around the object.
- Apply the support bandages.
- Apply the bandage over the bulky support material to hold it in place.
- Apply the bandage tightly but not so tightly as to impair circulation or breathing.
- Check circulation after applying the support bandages.

Note. If a pulse was felt on an extremity and cannot be felt after the bandage has been applied, the bandage must be loosened until a pulse can be felt.

- Immobilize the affected area with a splint or sling, if applicable.
- Check for a pulse distal (far side of the wound) to the injury site.
- Treat for shock, if necessary.
- Seek medical assistance.

14-24. If the impalement is found in the cheek and both ends of the object can be seen, perform the following:

- Remove the object in the direction it entered the cheek.
- Position the casualty to allow for drainage.
- If both ends of the object in the cheek cannot be seen, immobilize the impaled object.

Chapter 15

Casualty Monitoring and Evacuation Preparation

CASUALTY MONITORING

15-1. After the first responder and CLS have treated the casualty in CUF and TFC phase of TCCC they may be required to continue to monitor the casualty and prepare the casualty for evacuation. The first responder and CLS cannot treat a casualty and then forget about them. The casualty must be continually monitored and reassessed prior to the casualty's evacuation. The continued monitoring utilizing the MARCH-PAWS sequence will allow the first responder and CLS to identify changes in the casualty's condition and initiate those interventions necessary to prevent unnecessary death.

15-2. Casualty monitoring follows the same MARCH-PAWS process and will guide the first responder and CLS through their reassessments. After the initial casualty assessment and any treatments that are performed, the casualty must be reassessed every 15 minutes, or if seriously wounded every five to 10 minutes. The reassessment should start by reexamining any massive bleeding issues or interventions, should proceed through the rest of the MARCH-PAWS sequence, and be repeated until that casualty is transferred to the care of medical personnel.

15-3. The first responder or CLS should next evaluate any changes in respiratory status, look for signs or symptoms of shock, hypothermia, or head injuries. Also, monitor the casualty's respiratory rate, pulse, and level of consciousness. Assessing the casualty's respiratory status involves checking the rate and the quality of the respirations. By looking, listening, and feeling for breaths on the cheek, the respiratory rate can be determined (documented in the number of breaths per minute) and the respiratory effort (shallow breaths, difficulties moving air in and out, and loss of air movement on one side of the chest).

15-4. If a casualty becomes unconscious or breathing rate drops below eight respirations within one minute, insert an NPA. The casualty must be reassessed for tension pneumothorax and treated as necessary by performing an NDC. The NDC must be reassessed to confirm that it was successful. Document the rate, respiratory effort, and time assessed on DD Form 1380. If the casualty's respiratory status begins to change, reassess their status using the same approach used in the tactical trauma assessment.

15-5. The level of consciousness is best expressed by addressing the casualty's response using the AVPU method. Document any AVPU assessments or any changes in the casualty's status on the DD Form 1380. This documentation helps medical personnel better understand the casualty's situation.

Note. If the casualty is not ALERT, indicating decreased mental status, the casualty should not have weapons or communications equipment.

15-6. Assessing the casualty's circulation status is done by checking for pulses. Depending on the casualty and their injuries, the first responder and CLS can check the casualty's pulse at the carotid artery (neck) or radial artery (wrist). The first responder and CLS should use their index and middle fingers, NOT their thumb, to check pulses (figure 15-1 on page 152). Document pulse rates, locations, and time taken on the DD Form 1380.

Note. The absence of a radial pulse is an indication that the casualty is in shock.

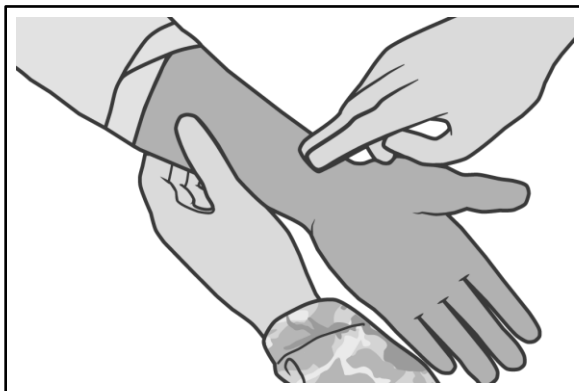


Figure 15-1. Technique to Check the Pulse

15-7. When requesting MEDEVAC, several preparatory actions are crucial for ensuring the casualty's safety and stability during transport. These steps include:

- **Securing the Casualty's Weapons and Equipment:** It is vital to secure all weapons and equipment according to the SOPs or mission requirements. Due to the potential inability of casualties to manage their own equipment owing to injuries or altered mental status, it is essential to secure these items to prevent accidents. If possible, weapons should not accompany the casualty during evacuation. If securing the weapon is not feasible, ensure it is cleared and rendered safe.
- **Selecting Appropriate Litter:** Choose a litter that aligns with both the casualty's specific needs and the mission profile to facilitate a safe and efficient evacuation.
- **Preparing Equipment for Evacuation:** Organize and prepare any equipment that is needed to accompany the casualty. This includes medical supplies or specialized gear necessary for their condition.
- **Anticipating Environmental Factors:** Consider environmental challenges that might impact the casualty's stability during evacuation. This includes extreme noises, vibrations from the transport vehicle, high winds from propeller wash, and exposure to cold temperatures due to altitude or the environment.
- **Developing a Mitigation Plan:** Formulate a plan to address each potential environmental threat to maintain the casualty's stability throughout the evacuation process.

15-8. It is important to note that medical personnel receiving the casualty may not be familiar with the equipment or have the means to secure it. Thus, preparing adequately and ensuring everything is in order prior to evacuation is critical for the safety and care of the casualty.

PRE-EVACUATION PROCEDURES, COMMUNICATION, AND DOCUMENTATION

15-9. Pre-evacuation procedures bridge both TFC and Tactical Evacuation Care. Immediate life-threatening hemorrhage control, followed by the prevention and treatment of other injuries and complications have all been completed before most pre-evacuation procedures are initiated. Although, some of the communication and documentation may be ongoing during the TFC phase.

COMMUNICATING WITH THE CASUALTY

15-10. When providing care, the first responder or CLS should always talk to the injured person. Getting hurt can cause a lot of worry and fear, on top of the stress from combat. It helps to speak openly with them about what injuries they have and what you're doing to help. Let them know that you're doing everything possible for them and assure them that they're in good hands. This kind of support can help lessen their anxiety.

15-11. It's important to be truthful about the injuries but keep a hopeful tone when talking about rescue and treatment. Chatting with the injured person is not only comforting but also lets you check how they are feeling mentally. Explaining what you're doing as you provide care can boost both your own confidence and the injured person's trust in you.

COMMUNICATING WITH LEADERS

15-12. Communicate with leaders as soon as possible and throughout casualty treatment. Leaders need to understand the impact of casualties on the mission. Communicate with medical providers about details of the casualty injuries. This includes 9-line communication and ongoing MIST reports. Communication is also important, as the injured casualty may affect the success of the mission or change the tactical landscape.

MEDICAL DOCUMENTATION

15-13. Medical documentation may be difficult to accomplish in tactical prehospital settings, but it is important to the casualty's subsequent care that every effort should be made to document the care provided by first responders and medics throughout the trauma care continuum from point of wounding and injury to definitive care at the hospital. A DD Form 1380 is provided in each first aid kit.

15-14. Based on the principles of TCCC, this card provides an easy way to document initial lifesaving care provided at the point of wounding. The card also serves as a prompt to remind first responders and CLS of the assessment and treatment steps of the MARCH-PAWS sequence. DD Form 1380 is relatively self-explanatory once personnel have been trained on using the form.

15-15. The MIST information about the casualty informs the MEDEVAC request and can be collected simultaneously with the other required information. The MIST report transmits medical information to the receiving combat medic, evacuation platform, and MTF to better prepare for specific inbound casualties. A MIST report conveys additional evacuation information that may be required by commanders.

15-16. Transitioning casualty care to another medical team is best accomplished with an oral discussion of the casualty's status, along with the written documentation on the DD Form 1380. When an oral hand-off is not a viable option, the written information may be the only way receiving medical personnel will know what the provider has done to help the casualty and what the next steps should be to provide the best care going forward.

15-17. The DD Form 1380 (figure 15-2 on page 154) should be completed on every casualty before handoff to another provider or evacuation to an MTF. DD Form 1380 will follow the patient and will be used until the patient arrives at a role of care where a DD Form 3019 (*Resuscitation Record*), can be used.

Note 1. This document should remain with the casualty throughout the prehospital evacuation system and become part of the casualty's medical record upon reaching an MTF. The data from the DD Form 1380 will also be uploaded into the DOD Trauma Registry maintained by the JTS.

Note 2. Many commercially available casualty cards do not replicate all information requirements of the DD Form 1380. While the DD Form 1380 may be reproduced locally, it should be on waterproof paper or laminated with a means of attaching it to the casualty. It is best to have an indelible or permanent marker available for writing on the TCCC Card.

TACTICAL COMBAT CASUALTY CARE (TCCC) CARD

BATTLE ROSTER #: **BL6553**

EVAC: ☒ Urgent ☐ Priority ☐ Routine

NAME (Last, First): **JOE, BRIAN** LAST 4: **6553**

SEX: ☒ M ☐ F DATE (DD-MMM-YY): **17-JAN-24** TIME: **1345**

SERVICE: **ARMY** UNIT: **1-9 INF BN** ALLERGIES: **PENICILLIN**

Mechanism of Injury: (X all that apply)
☐ Artillery ☐ Blunt ☐ Burn ☐ Fall ☐ Grenade ☒ GSW ☐ IED
☐ Landmine ☐ MVC ☐ RPG ☐ Other: _____

Injury: (Mark injuries with an X)

TQ: R Arm TYPE: **IMPROV** TIME: **1350**

TQ: L Arm TYPE: _____ TIME: _____

TQ: R Leg TYPE: _____ TIME: _____

TQ: L Leg TYPE: _____ TIME: _____

Signs & Symptoms: (Fill in the blank)

	Time	1400	1415
Pulse (Rate & Location)		116	198
Blood Pressure		112 / 76	/ /
Respiratory Rate		20	20
Pulse Ox % O2 Sat		98	98
AVPU		ALERT	ALERT
Pain Scale (0-10)		6	5

DD Form 1380, JUN 2014 TCCC CARD

BATTLE ROSTER #: BL6553

EVAC: ☒ Urgent ☐ Priority ☐ Routine

Treatments: (X all that apply, and fill in the blank) Type

C: TQ: ☒ Extremity ☐ Junctional ☐ Truncal

Dressing: ☐ Hemostatic ☒ Pressure ☐ Other

A: ☒ Intact ☐ NPA ☐ CRIC ☐ ET-Tube ☐ SGA

B: ☐ O2 ☐ Needle-D ☐ Chest-Tube ☐ Chest-Seal **NONE**

C:

	Name	Volume	Route	Time
Fluid				
Blood Product				

MEDS:

	Name	Dose	Route	Time
Analgesic (e.g., Ketamine, Fentanyl, Morphine)	MELOXICAM	15MG	PO	1420
	ACETAMINOPHEN	500MG	PO	1420
Antibiotic (e.g., Moxifloxacin, Ertapenem)	MOXIFLOXACIN	400MG	PO	1420
Other (e.g., TXA)	ONDANSETRON	4MG	PO	1425

OTHER: ☐ Combat-Pill-Pack ☐ Eye-Shield (☐ R ☐ L) ☐ Splint
☒ Hypothermia-Prevention Type: _____

NOTES:

SINGLE GSW TO RIGHT UPPER ARM, BLEEDING CONTROLLED WITH TOURNIQUET AND PRESSURE DRESSING

FIRST RESPONDER

NAME (Last, First): **DOE, JANE** LAST 4: **9501**

DD Form 1380, JUN 2014 (Back) TCCC CARD

Legend:

AVPU	alert, verbal, pain, and unresponsive	M	male
BN	battalion	MG	milligram
CRIC	cricothyrotomy	MVC	motor vehicle collision
DD Form	Department of Defense form	NPA	nasopharyngeal
DD-MMM-YY	day-month-year	O2	oxygen
ET-Tube	endotracheal tube	OX	oximeter
EVAC	evacuation	PO	by mouth
F	female	R	right
GSW	gunshot wound	RPG	rocket propelled grenade
IED	improvised explosive device	Sat	saturation
INF	infantry	SGA	supraglottic airway
IV	intravenous	TCCC	tactical combat casualty care
L	left	TQ	tourniquet

Figure 15-2. Sample DD Form 1380, The Tactical Combat Casualty Care (TCCC) Card

15-18. All Service members should have at least one or more DD Form 1380s within their first aid kit. For additional information and instructions on completing the DD Form 1380 refer to AR 40-66. Use the following steps to complete a DD Form 1380:

- Remove the DD Form 1380 from the casualty's first aid kit.
- Complete all entries as fully as possible.

Note. As the DD Form 1380 is the first, and sometimes the only record of treatment of combat casualties, accuracy and thoroughness of information provided is of the utmost importance.

- Front of DD Form 1380 (figure 15-2):
 - Battle Roster # - Write first letter of the casualty's first name, then write first letter of the casualty's last name, write the last four numbers of the casualty's identification number. In figure 15-2, the Service member's battle roster number is "BL6553."

- Evacuation (EVAC) - Mark an “X” on the casualty's evacuation precedence (Urgent; Priority; or Routine).
- Name - Write casualty's name (Last, First).
- Last 4 - Write last four numbers of casualty's identification number.
- Sex - Mark an “X” on the casualty's sex Male (M) or Female (F).
- Date - Write date of injury in DD-MMM-YY format. For example, “29-JUN-13.”
- Time - Write 24-hour time of injury and indicate whether Local (L) or Zulu (Z) time. For example, “1300Z.”
- Service - Write casualty's branch of service (United States Army [USA], United States Air Force [USAF], United States Coast Guard [USCG], United States Navy [USN], United States Marine Corps [USMC], and United States Space Force [USSF]. For U.S. civilians write “U.S. CIV”. For non-U.S. personnel, write “NON-U.S.” or a standard abbreviation for the casualty's nationality.
- Unit - Write casualty's unit name.
- Allergies - Write casualty's known drug allergies. If no drug allergies, write no known drug allergies (NKDA).
- Mechanism of Injury - Mark an “X” on the mechanism or cause of injury (mark all that apply) (artillery, blunt, burn, fall, grenade, gunshot wound [GSW], improvised explosive device [IED], landmine, motor vehicle crash or collision [MVC], rocket-propelled grenade -propelled grenade [RPG], and other (specify).
- Injury - Mark an “X” at the site of the injury(s) on the body picture. For burn injuries, circle the burn percentage(s) on the figure. If multiple mechanisms of injury and multiple injuries, draw a line between the mechanism of injury and the anatomical site of the injury.
- TQ: R Arm - If a TQ is applied to the right arm, write the type of TQ (hasty or deliberate) used, and the time of TQ application.
- TQ: L Arm - If a TQ is applied to the left arm, write the type of TQ (hasty or deliberate) used, and the time of TQ application.
- TQ: R Leg - If a TQ is applied to the right leg, write the type of TQ (hasty or deliberate) used, and the time of TQ application.
- TQ: L Leg - If a TQ is applied to the left leg, write type of TQ (hasty or deliberate) used, and the time of TQ application.
- Time - Write the time that vital signs are taken.
- Pulse (rate and location) - Write casualty's pulse rate.
- Blood Pressure - Write casualty's blood pressure.
- Respiratory Rate - Write casualty's respiratory rate.
- AVPU - Write casualty's level of consciousness.
- Pain Scale (0-10) - Write casualty's level of pain in numeric rating scale of 0 to 10, with 0 being no pain and 10 being the worst pain.
- Back of Card (figure 15-2 on page 154):
 - Battle Roster # - Write first letter of casualty's first name, then first letter of casualty's last name, write the last four numbers of the casualty's identification number. In figure 15-2, the Service member's battle roster number is “LD2270.”
 - Evacuation (EVAC) - Mark an “X” on the casualty's evacuation precedence (Urgent, Priority, or Routine).
 - C - Mark an “X” for all Circulation hemorrhage control interventions. For TQ, mark category (Extremity, Junctional, and, or Truncal) and write name of TQ(s) used. For dressings, mark category (Hemostatic, Pressure, and, or Other) and write type of dressing(s) used.
 - A - Mark an “X” for all airway interventions (Intact, NPA, cricothyroidotomy (known as CRIC), endotracheal (known as ET) tube, supraglottic airway (SGA) and write the type of device(s) used.
 - B - Mark an “X” for all Breathing interventions oxygen (O2), needle decompression (Needle-D), Chest-Tube, (Chest-Seal) and write the type of device(s) used.
 - C: Fluid - Circulation resuscitation interventions. Write name, volume, route, and time of any fluids given.
 - C: Blood Product - Circulation resuscitation interventions. Write name, volume, route, and time of any blood products given.

Note. When more space is needed for documentation, attach another DD Form 1380 to the original by safety pin or other means. The second form will be labeled DD Form 1380 #2, showing the Service member's name and unit.

- Analgesic Medications - Write name, dose, route, and time that any analgesics are given.
 - Antibiotic Medications - Write name, dose, route, and time that any antibiotics are given.
 - Other Medications - Write name, dose, route, and time of any other administered medications.
 - Other - Mark an "X" for other treatments administered (combat pill pack, eye shield (mark right (R) or left (L)), splint, hypothermia prevention) and the type of device(s) used.
 - Notes - Use this space to record any other pertinent information or clarifications.
 - First Responder Name - Print the first responder's name (Last, First).
 - First Responder Last 4 - Write the last four numbers of the first responder's social security number or identification number.
- Attach the completed DD Form 1380 to the casualty in a visible area where follow on medical staff will see it.

Note. Do not attach DD Form 1380 to the casualty's body armor, as this equipment most likely will be separated from the casualty once they arrive at the MTF.

15-19. First responders and CLS personnel are advised to fill out the DD Form 1380 to the best of their ability. DD Form 1380 is crucial as it records evidence of initial treatment and provides subsequent levels of medical care with essential information. This documentation is vital for medical personnel to assess and determine the appropriate continuation of care for the casualty. This process ensures that each step of medical intervention is informed and based on prior actions taken, facilitating a cohesive and effective treatment pathway.

MEDICAL EVACUATION REQUEST

15-20. Every Service member should be able to initiate a 9-line MEDEVAC request. The standard MEDEVAC request has nine lines. However, lines one through five are required before a MEDEVAC can be launched. Using a phonetic alphabet and following the unit's normal communications procedures, call in:

- Grid location.
- Radio frequency and call sign.
- Number of casualties by precedence.
- Equipment that needs to be brought by the evacuation team.
- Number of litter and ambulatory casualties.

15-21. Before initiating a 9-line MEDEVAC request, collect all the information needed. When calling in, be sure to follow all appropriate communication protocols and guidance.

15-22. Depending on operational, mission variables, and availability, evacuation options may involve MEDEVAC, CASEVAC, or a combination of the two. During CASEVAC, casualties are moved without medical regulating and do not possess a protected status under the Geneva Conventions. As such, CASEVAC platforms may be engaged by enemy forces. With MEDEVAC, patients may be regulated or unregulated and the ambulance is properly marked, MEDEVAC cannot be used for nonmedical missions without losing its protected status. The 9-line MEDEVAC request is communicated through the designated evacuation system or Patient Evacuation Coordination Cell, depending on orders and the tactical SOP.

15-23. When a Service member requests a MEDEVAC, they are not directly coordinating with a medical provider but are explaining the requestor's evacuation requirements with someone who coordinates the movement assets. Although they still require some general information about the status of the casualty, much of the information that they need to coordinate evacuation is not clinical and relates to logistical and operational issues.

15-24. Communicate with medical providers on the evacuation asset if possible and relay the MIST report. Provide additional information as appropriate. For additional information on MEDEVAC request refer to ATP 4-02.2.

EVACUATION EQUIPMENT AND CASUALTY PACKAGING

15-25. Essential evacuation equipment (for example, litters, packaging materials, and BVMs) should be prepared by other unit personnel while medical treatment continues and coordinating with the casualty treatment team to save time. For example, other unit members can prepare litters while treatment is being provided. Do not delay getting casualties onto litters. The first responder and CLS can better prevent hypothermia by getting casualties off the ground.

LITTERS

15-26. Casualty movement in TFC may be better accomplished using litter due to the tactical situation and the need to move casualties rapidly. The litter exists only to facilitate casualty movement, and the casualty can be placed in the best position that facilitates their care and comfort. Casualties do not have to be placed on their backs on a litter. It is easier to move them if they are already on the litter; however, they must be secured to the litter before movement. Litter is also usually better for moving a casualty long distance.

15-27. All unit members should know how to open and set up litters and rehearse their use during pre-mission training. All unit members should know who will carry litter or where litter is located on vehicles.

15-28. Selecting the proper litter is dependent on several different factors:

- The evacuation platform.
- The terrain at the location of pick-up.
- The casualty's injuries.
- The availability of different types of litters.

15-29. Two of the more common litters often used are the quad-folding litter and the lightweight flexible stretchers. Each of them has advantages and disadvantages and maybe the proper litter depends on the situation. For example, flexible litter can be moved using a one-person drag (terrain permitting) but are more difficult to carry longer distances than a quad-folding litter.

15-30. Each unit will have litters that have been proven to support their mission profile, and it is important to be aware of the limitations and advantages of each litter at your disposal before being deployed in a situation where you will require them.

CASUALTY PACKAGING

15-31. When preparing a casualty for evacuation, the first responder or CLS need to anticipate the environmental factors that could prevent the casualty from remaining stable. Examples include extreme noises, vibrations, high winds from propeller washing, and exposure to cold environments from either altitude or low ambient temperatures. A plan should be developed to mitigate each of those potential threats when preparing for your casualty.

15-32. All loose ends of bandages, medical equipment, and hypothermia prevention materials should be secured. The first responder or CLS should prevent dressings and other medical items from being blown around by the rotor wash or becoming entangled with other equipment.

15-33. Loose materials can catch on anything from tree limbs to body-armor of litter bearers, to parts of the aircraft or vehicles. Any snag like this can cause delays in evacuation or even further injury to casualties or providers. Blankets and foil-based hypothermia materials are especially susceptible to being caught in the wind.

15-34. Hypothermia prevention equipment should be tucked and secured beneath the casualty and litter straps. Loose edges can be caught up in wind or rotor wash or snagged on objects in the helicopter as the casualty is loaded aboard.

15-35. The TQs should remain visible and not be hidden. The DD Form 1380 should be attached to the casualty.

Evacuation Considerations for Suspected Spinal Injuries

15-36. A spinal injury should be expected in certain situations, like fast-roping falls or being near a significant blast. In those cases, when tactically feasible, ensure the C-spine is immobilized and the casualty is kept straight during evacuation. Keep the litter type and evacuation platform in mind and ensure the evacuation ground or air vehicle can accommodate the selected litter.

Walking Wounded

15-37. Not all casualties require a litter for evacuation. For those casualties who are still ambulatory, provide instructions or assistance as needed. Depending on the nature of their injuries, they may be able to assist with carrying litters or providing security.

15-38. It is best to guide disoriented or visually impaired casualties by having them place their hand on the shoulder of the casualty in front of them, or a nonmedical attendant as they move to the evacuation platform. Instruct them on repeatedly checking their own wounds and dressings to ensure that bleeding remains controlled throughout the evacuation process.

CASUALTY STAGING

15-39. A particular importance when moving more than one casualty is the process of staging the casualties. Based on the guidance from the MEDEVAC personnel, certain casualties have priority and may need to be loaded last, so that they are the first ones offloaded at the destination.

15-40. If not given specific guidance, place the most serious casualty in a position where they are loaded last, and have casualties who may be less severely injured loaded before them. The first responder or CLS may also need to assist in marking the LZ, providing security, or assisting in marshalling the aircraft or litter-bearer teams.

EVACUATION CARE AFTER-ACTION REVIEW

15-41. It is critical that documentation of personnel, injury, and treatment be annotated using DD Form 1380 and should be completed prior to any handoff of the casualty. If care cannot be documented using the DD Form 1380 at the point of care by the first responder or CLS, it should be documented using the TCCC after action review, found on the JTS website, as soon as possible (within 72 hours) after treatment. The nonmedical first responder or CLS should follow up with the supporting unit medical personnel to complete and submit the TCCC after action review.

PART THREE

FIRST AID PROCEDURES FOR NON-TACTICAL COMBAT CASUALTY CARE INJURIES

In the operational environment, Service members are exposed to risks and receive injuries beyond those caused by an operational event. First aid skills play a vital role in ensuring the safety and well-being of Service members. As personnel participate in outdoor activities and training, the potential for accidents or injuries increases. Having a solid foundation in first aid equips Service members with the necessary knowledge and abilities to respond swiftly and effectively in times of emergencies. Whether it's treating minor cuts and bruises, providing initial care for more serious injuries, or even saving lives, the importance of first aid skills cannot be overstated. By acquiring these skills, Service members not only become more responsible and capable individuals, but they also contribute to creating a safer environment for themselves and others.

Chapter 16

Bites and Stings

OVERVIEW OF BITES AND STINGS

16-1. Bites and stings are common occurrences that can happen in almost every environment. From pesky mosquito bites to venomous snakebites, these encounters with creatures big and small can lead to a range of discomfort, pain, and even potential danger. Understanding the various types of bites and stings, their causes, and proper treatment is crucial for minimizing the associated risks and ensuring a swift recovery.

16-2. Poison and venom are terms often used interchangeably, but they have distinct differences in their mode of delivery and the way they affect living organisms. A poison is a substance that is harmful or lethal when ingested, inhaled, or absorbed through the skin. It is typically produced by plants, animals, or microorganisms as a defensive mechanism. Poisonous substances can be found in various forms, such as toxins in certain plants, chemicals in household products, or even certain medications when taken in excessive amounts. The key characteristic of poison is that it needs to be ingested or absorbed into the body to cause harm.

16-3. Venom is a specialized secretion that is actively injected into another organism. It is typically produced by certain animals, such as snakes, spiders, scorpions, and some insects. Venom is delivered through specialized organs, such as fangs, stingers, or spines, and is primarily used for hunting or self-defense. Unlike poison, venom is actively injected into the victim's bloodstream, tissues, or body cavities, often leading to localized or systemic effects, which can range from pain and swelling to paralysis or even death. The main distinction between poison and venom lies in their delivery method. Poison is harmful when ingested or absorbed, while venom is specifically injected into another organism through specialized organs.

16-4. If bites and stings are not treated promptly and correctly, they may cause serious illness or death. The severity of a snakebite depends upon whether the snake is venomous or nonvenomous, the type of snake, the location of the bite, and the amount of venom injected. Bites from humans and other animals, such as dogs, cats, bats, raccoons, and rats, can cause severe bruises and infection and tears or lacerations of tissue. Awareness of the potential sources of injuries can reduce or prevent them from occurring. Knowledge and prompt application of first aid measures can lessen the severity of injuries from bites and stings and keep the Service member from becoming a serious casualty.

HUMAN OR ANIMAL BITES

16-5. First aid for human or animal bites aims to prevent infection and provide initial care for the wound. To perform first aid for a human or animal bite:

- Remove yourself from further harm. If dealing with an animal bite, ensure that you are no longer at risk of being bitten again.
- Control bleeding. If the bite wound is bleeding, apply direct pressure with a clean cloth or sterile gauze to stop or minimize the bleeding. Elevating the wound, if possible, can also help reduce bleeding.
- Clean the wound. Rinse the bite wound thoroughly with clean water or saline solution to flush out any dirt, bacteria, or saliva that may have entered the wound. Gently remove any debris from the wound using mild soap and water.
- Apply an antiseptic. After cleaning, apply an antiseptic solution or ointment to the wound to help reduce the risk of infection. Follow the instructions on the product label and ensure it is suitable for use on open wounds.
- Cover the wound. Use a sterile dressing or clean bandage to cover the wound. This will protect it from further contamination and promote healing. Change dressing regularly to maintain cleanliness.
- Seek medical attention. It is important to seek medical evaluation for all animal bites, especially those from unfamiliar or wild animals, deep wounds, or those that become red, swollen, or increasingly painful. A health care professional can assess the severity of the bite, prescribe antibiotics if necessary, and administer tetanus prophylaxis if required.

16-6. If someone gets bitten by an animal, whether known, unknown, wild, or domestic, the primary concern is rabies. Unless the animal is someone's pet and verified to be up to date on rabies, the individual who has been bitten must get the rabies post-exposure vaccine. Seemingly minor bites can lead to infection, so it is essential to consult a health care professional to determine the appropriate course of action and receive any necessary medical treatment.

Note. If unable to capture or kill the animal, provide medical personnel with any information that will help identify it.

MARINE (SEA) ANIMALS

16-7. Except for sharks and barracuda, most marine animals will not deliberately attack. The most frequent injuries from marine animals are wounds by biting, stinging, or puncturing. Wounds inflicted by marine animals can be very painful but are rarely fatal.

16-8. Wounds from sharks, barracuda, and crocodilian can involve major trauma because of bites and lacerations. Bites from large marine animals are potentially the most life-threatening of all injuries from marine animals. Major wounds from these animals can be treated by controlling the bleeding, preventing shock, giving basic life support, splinting the injury, and by securing prompt medical aid.

16-9. Turtles, moray eels, and corals normally inflict minor wounds. Treat by cleansing the wound(s) thoroughly and by splinting if necessary.

16-10. Jellyfish, anemones, and others marine animals inflict injury by means of stinging cells in their tentacles. Contact with the tentacles produces burning pain with a rash and small hemorrhages on the skin. Shock, muscular cramping, nausea, vomiting, and respiratory distress may also occur. Gently remove the clinging tentacles with a towel and wash or treat the area. Use diluted ammonia or alcohol, a meat tenderizer, and talcum powder. If symptoms become severe or persist, seek medical assistance.

16-11. Spiny fish, urchins, stingrays, and cone shells inject their venom by puncturing the skin with their spines. General signs and symptoms include swelling, nausea, vomiting, generalized cramps, diarrhea, muscular paralysis, and shock. Deaths are rare. Treatment consists of soaking the wound(s) in hot water (when available) for 30 to 60 minutes. This inactivates the heat sensitive toxin. In addition, further first aid measures should be taken such as controlling bleeding or applying a dressing. This should be carried out as necessary.

CAUTION

Be careful not to scald the casualty with water that is too hot because the pain of the wound will mask the normal reaction to heat.

SNAKE BITES

16-12. Snakebites can occur unexpectedly during outdoor activities or encounters with snakes in their natural habitats. Understanding how to identify snakes, knowing the proper steps to take after a snakebite, and having a grasp of basic first aid techniques can significantly increase the chances of a positive outcome in such situations. Snakes are categorized between nonvenomous and venomous snakes.

16-13. Snakes can be categorized into various groups based on different characteristics. One common classification system categorizes snakes into four main categories of colubrid snakes, constrictor snakes, non-venomous snakes, and venomous snakes. These categories may overlap, and some snakes may possess traits from multiple categories. Understanding the different categories of snakes can aid in snake identification, assessing potential risks, and knowing how to respond appropriately in various situations.

Colubrid Snakes

16-14. Colubrid snakes represent the largest snake family and include a wide range of non-venomous species. These snakes have diverse habitats, sizes, and feeding habits. Common colubrid snakes include rat, king, gopher, and water snakes.

Constrictor Snakes

16-15. Constrictor snakes are non-venomous snakes that subdue their prey by coiling around them and squeezing until they can no longer breathe. Constrictors then swallow their prey whole. Well-known constrictor snakes include boas, pythons, and anacondas.

Non-venomous Snakes

16-16. Non-venomous snakes include snakes that lack venom glands or fangs and pose no direct threat to humans. Non-venomous snakes play an essential role in controlling rodent populations and maintaining ecological balance. Examples of non-venomous snakes are boas, pythons, corn, and garter snakes.

16-17. There are approximately 130 different varieties of nonvenomous snakes in the U.S. They have oval-shaped heads and round eyes. Unlike venomous snakes, discussed below, nonvenomous snakes do not have fangs with which to inject venom. Figure 16-1 depicts the characteristics of a nonvenomous snake.

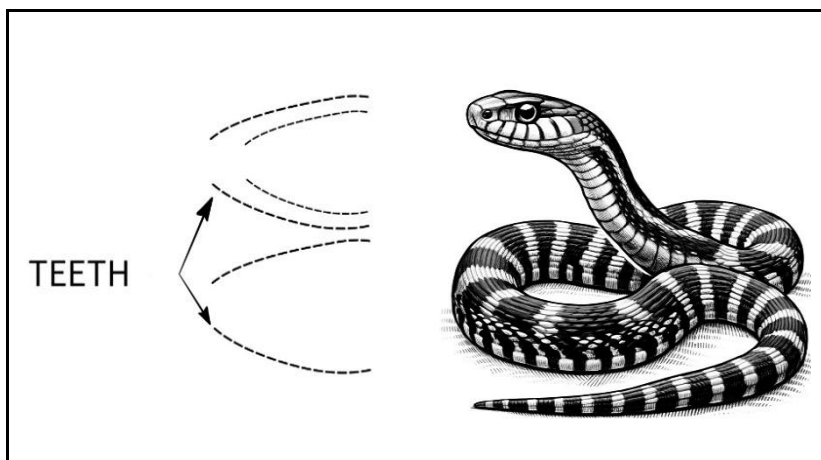


Figure 16-1. Characteristics of the nonvenomous snake

Venomous Snakes

16-18. Venomous snakes possess venom glands and specialized fangs to inject venom into their prey or as a defense mechanism. It is important to exercise caution and seek immediate medical attention in the event of a venomous snakebite.

16-19. It is important to note that the distribution of venomous snakes may vary within states, and some species may be specific to certain regions. When spending time in areas known to have venomous snakes, it is crucial to be aware of the local species, understand their habitats and behavior, and take appropriate precautions to avoid encounters and bites.

16-20. Venomous snakes in other parts of the world include sea snakes, the Fer-De-Lance, the Bushmaster, tropical rattlesnake in tropical Central America; the Malayan pit viper in the tropical Far East; the cobra in Africa and Asia; the mamba (or black mamba) in central and southern Africa; and the krait in India and Southeast Asia.

Categories of Venomous Snakes

16-21. Venomous snakes can be categorized into several groups based on their characteristics and taxonomic classification. The following are some common categories of venomous snakes:

- Vipers.
- Elapids.
- Rear-Fanged Snakes.
- Burrowing Asps.
- Sea Snakes.

16-22. These categories provide a general overview of the different types of venomous snakes based on their taxonomic families and key characteristics. It's important to note that there may be variations within each category, and some venomous snakes may exhibit characteristics from multiple families.

16-23. This family of snakes includes some of the most well-known venomous snakes. Vipers are characterized by their large, triangular heads, heat-sensing pits between their eyes and nostrils, and hinged fangs that can fold back when not in use. Examples include rattlesnakes, copperheads, and cottonmouths or water moccasins. See figure 16-2 for characteristics of a venomous pit viper and figure 16-3 on page 163 provides examples of viper snakes.

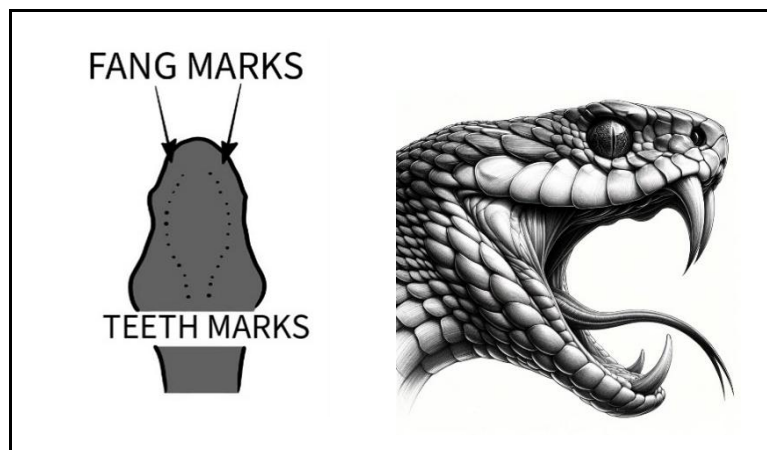


Figure 16-2. Characteristics of the venomous pit viper

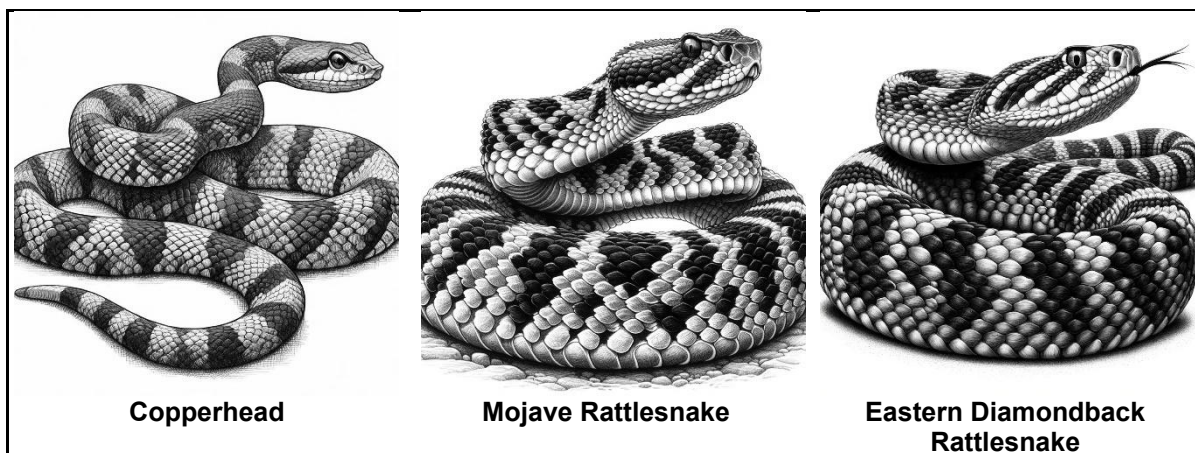


Figure 16-3. The Viper Snake examples

16-24. In the U.S, there are several species of venomous snakes. The following are some of the venomous snakes commonly found in different regions of the country:

- Rattlesnakes: Rattlesnakes are a group of venomous snakes known for the rattling sound they produce by shaking their tails. Species such as the eastern and western diamondback, timber and Mojave rattlesnakes are found in various parts of the U.S.
- Copperheads: Copperheads are venomous snakes with distinctive copper-colored heads. They are commonly found in the eastern and southeastern regions of the U.S.
- Cottonmouths or Water Moccasins: Cottonmouths or Water Moccasins are venomous semi-aquatic snakes primarily found in the southeastern U.S. They are known for their aggressive behavior and are often encountered near bodies of water.
- Coral Snakes: Coral snakes are venomous snakes with colorful bands of red, yellow, and black. The Eastern and Texas Coral Snakes are found in specific regions of the southern U.S.
- Various Pit Vipers: Pit vipers include venomous snakes that possess heat-sensing pits between their eyes and nostrils. Along with rattlesnakes and copperheads, this category includes species such as the eastern and western Massasauga Rattlesnakes, and the Mojave Desert Sidewinder.

16-25. The casualty's condition provides the best information about the seriousness of the situation and how much time has passed since the bite occurred. Pit viper bites are characterized by severe burning pain. Discoloration and swelling around the fang marks usually begins within five to ten minutes after the bite. If only minimal swelling occurs within 30 minutes, the bite will almost certainly have been from a nonvenomous snake or possibly from a venomous snake, which did not inject venom. The venom destroys blood cells, causing a general discoloration of the skin. Blisters and numbness in the affected area follow this reaction. Other signs, which can occur, are weakness, rapid pulse, nausea, shortness of breath, vomiting, and shock.

16-26. The venom of viper bites affects the circulatory system and destroys muscle. Severe pain, swelling and change in skin color usually develop within five to 10 minutes and slowly spreads. After several hours, the following signs and symptoms will occur:

- Further swelling and changes in skin color.
- Bleeding blisters.
- Numbness of the affected area.

16-27. Other signs and symptoms which may occur include:

- Weaknesses.
- Rapid pulse.
- Nausea.
- Vomiting.
- Shortness of breath.
- Sweating.
- Shock.

16-28. Elapids are a family of venomous snakes known for their fixed front fangs and potent neurotoxic venom. They generally have slender bodies and smooth scales. Famous elapid snakes include cobras, coral snakes, kraits, and mambas. Corals (figure 16-4), cobras (figure 16-5), kraits, and mambas all belong to the same group even though they are found in different parts of the world. All four inject their venom through short, grooved fangs, leaving a characteristic bite pattern.

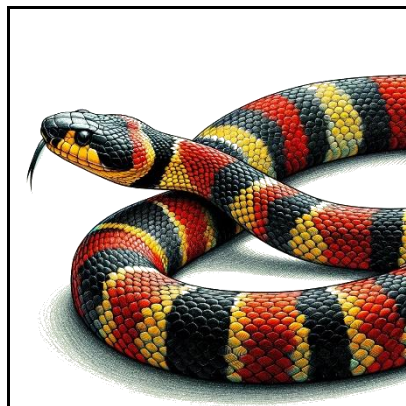


Figure 16-4. The Coral snake example

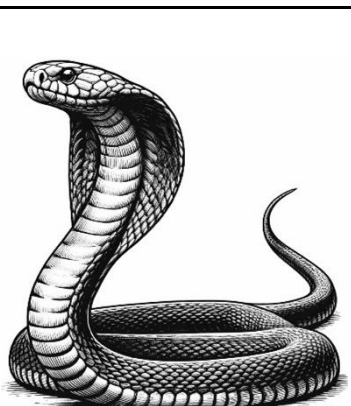


Figure 16-5. The Cobra snake example

16-29. To identify a coral snake, a common phrase used in the U.S. is “Red or yellow, kill a fellow; red on black, venom lack.” This phrase refers to the color pattern on the snake's bands and serves as a mnemonic device to help distinguish coral snakes from non-venomous snakes with similar markings, such as milk or scarlet kingsnakes. Coral snakes have red bands touching yellow bands, while non-venomous snakes have red bands touching black bands. It's important to remember that this phrase is specific to the U.S. and may not apply to coral snake species found in other regions. When encountering any snake, it is advisable to maintain a safe distance and seek professional assistance if unsure of its venomous nature.

16-30. The venom from corals, cobras, kraits, and mambas, which belong to the elapid family, notably affects the central nervous system, producing a distinct set of symptoms that diverge from those caused by pit vipers. Initially, these bites may result in only minimal pain and swelling, leading many individuals to underestimate the severity of the bite. Typically, the onset of neurological symptoms occurs within one to seven hours post-bite, including blurred vision, drooping eyelids, slurred speech, drowsiness, and an increase in salivation and sweating.

16-31. If the bite is not addressed quickly with appropriate medical intervention, more severe symptoms can manifest, such as nausea, vomiting, respiratory difficulties, shock, and eventually more critical conditions like paralysis, convulsions, and coma. This progression underscores the necessity of recognizing the potential danger of elapid bites early and seeking immediate medical treatment to mitigate these serious outcomes.

16-32. While most rear-fanged snakes are non-venomous, some species within this family possess venomous rear fangs. These fangs are located toward the back of the mouth and are used to subdue their prey. Rear-fanged venomous snakes include boomslangs, mangrove, and vine snakes.

16-33. Burrowing asps are a family of venomous snakes primarily found in Africa and parts of the Middle East. They have small, fixed front fangs and are adapted for burrowing in sandy or loose soil. Examples include the stiletto snakes and mole vipers.

16-34. Sea snakes are a group of venomous snakes that have adapted to life in aquatic environments. They have paddle-shaped tails and can be found in the coastal waters of the Indian and Pacific Oceans. Sea snakes have highly potent venom but are generally not aggressive towards humans.

16-35. Sea snakes (figure 16-6 on page 165) are found in the warm water areas of the Pacific and Indian Oceans, along the coasts, and at the mouths of some larger rivers. Their venom is VERY venomous, but their fangs are only $\frac{1}{4}$ inch long.

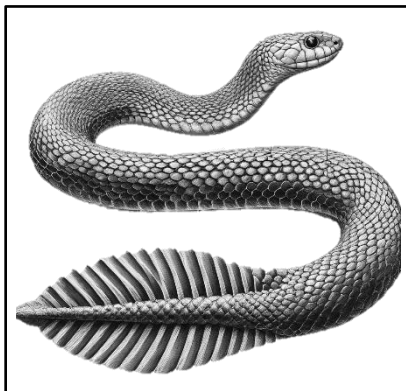


Figure 16-6. The Sea snake example

Venomous Snakebite

16-36. Venomous snakes DO NOT always inject venom when they bite or strike a person. However, all snakes may carry tetanus (lockjaw); anyone bitten by a snake, whether venomous or nonvenomous, should immediately seek medical attention.

16-37. Venom is injected from the venom sacs through grooved or hollowed fangs. Depending on the species, these fangs are either long or short. Pit vipers have long hollow fangs. These fangs are folded against the roof of the mouth and extend when the snake strikes. This allows them to strike quickly and then withdraw. Cobras, coral snakes, kraits, mambas, and sea snakes have short, grooved fangs. These snakes are less effective in their attempts to bite, since they must chew after striking to inject enough venom (poison) to be effective.

16-38. In the event you are bitten, do not attempt to identify and or kill the snake. Freshly killed snakes can continue to bite due to reflexive action. Snake identification is not critical. Medical personnel will treat symptoms by geography and not someone's best guess at a species.

16-39. The venoms of different snakes cause different effects. Pit viper venom (hemotoxin [blood toxin]) destroys tissue and blood cells. Cobras, adders, and coral snakes inject powerful venom (neurotoxin [nerve toxin]) which affects the central nervous system, causing respiratory paralysis. Water moccasins and sea snakes have venom that is both hemotoxic and neurotoxic.

FIRST AID FOR A SNAKEBITE

16-40. In the event of a snakebite, prompt first aid is crucial to minimizing the potential effects of the venom. To perform first aid for a snakebite:

- Stay calm and call for medical assistance immediately. Provide medical personnel with information about the snake, if possible, but do not attempt to capture or kill the snake as it may increase the risk of further bites.
- Keep the bitten Service member as still and calm as possible. Restrict movement and avoid unnecessary exertion, as it can accelerate the spread of venom in the bloodstream.
- Remove any constricting items, such as jewelry or tight clothing, from the affected area, as swelling may occur.
- Position the affected limb slightly below the level of the heart, if possible. This can help slow the spread of venom.
- Cleanse the wound gently with mild soap and water to reduce the risk of infection. Do not apply ice or use a TQ, as they are not effective and can cause additional harm.
- Apply a sterile bandage or dress over the bite, ensuring it is snug but not too tight. This can help immobilize the area and reduce the spread of venom.
- Encourage the person to remain calm and still, while waiting for medical help to arrive. Monitor their vital signs, such as breathing and heart rate, and be prepared to administer CPR if necessary.

16-41. The above steps are general guidelines and may vary depending on the snake species and the region. It is crucial to seek medical attention immediately after a snakebite, as they can provide appropriate anti-venom treatment and further evaluate the situation.

CAUTION

DO NOT attempt to cut open the bite or suck out the venom. If the venom should seep through any damaged or lacerated tissues in your mouth, you could immediately lose consciousness or even die.

16-42. If accompanying the casualty on CASEVAC to an MTF, observe the casualty carefully for signs and symptoms of an allergic reaction. Continue to monitor the airway, breathing, circulation, vital signs (for first aid purposes; pulse, and respiration), and distal pulse of the bitten extremity, treating for shock, and conserving body heat.

INSECT (ARTHROPOD) BITES AND STINGS

16-43. An insect bite or sting can cause great pain, allergic reaction, inflammation, and infection. If not treated correctly, some bites or stings may cause serious illness or even death. When an allergic reaction is not involved, first aid is a simple process. In any case, medical personnel should examine the casualty at the earliest possible time. It is important to identify spiders, bees, or creatures that cause bites or sting, especially in cases of an allergic reaction.

16-44. The insects found throughout the world that can produce a bite or sting are too numerous to mention in detail. Commonly encountered stinging or biting insects include the brown recluse spiders, black widow spiders, tarantulas, scorpions, urticating caterpillars, bees, wasps, centipedes, conenose beetles (kissing bugs), ants, and wheel bugs. Upon being reassigned, especially to overseas areas, take the time to become acquainted with the types of insects to avoid.

16-45. Arthropods are segmented invertebrates with jointed antennae and legs. They include insects, arachnids, centipedes, and millipedes. See figure 16-7 for examples of venomous arthropods.

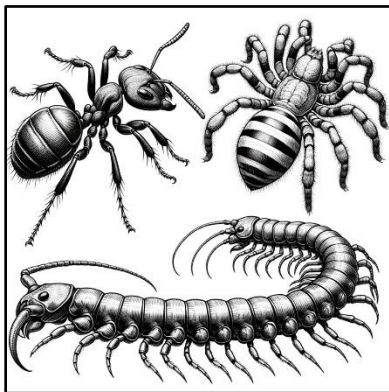


Figure 16-7. Venomous Arthropods

16-46. Of all deaths due to venomous stings or bites in a year, most are caused by arthropods. Most injuries resulting from venomous arthropods can be prevented by doing the following:

- If you are allergic to certain stings or bites, be prepared with the proper medication.
- Ask the local inhabitants about the local venomous and harmful arthropods.
- Avoid touching unfamiliar arthropods or allowing them to touch you.
- Watch where you put your hands and feet.
- Before using it, check places where harmful arthropods may hide.
 - Closets, wall spaces, doorways, and eaves.
 - Outdoor privies and latrine boxes.

- Bed clothing, beds, and sleeping bags.
- Shoes, boots, and clothing.
- Logs, felled trees, and leaves.
- Rocks, boards, and debris.
- Avoid beehives, ant and wasp nests, and spider webs.
- Wear foot protection (shoes, boots, and sandals) always.
- Hang up clothes.
- Practice good sanitation by removing debris, trash, and board piles.
- Avoid wearing deodorants, colognes, perfumes, or cosmetics with fragrances.

INSECTS

16-47. An insect is a small invertebrate animal characterized by having a segmented body, three pairs of legs, and often one or two pairs of wings. Insects are a diverse group, with over a million known species, and they play important roles in ecosystems as pollinators, decomposers, and prey for other organisms.

16-48. First aid treatment for insect bites can vary depending on the type of insect and the individual's reaction. The general first aid steps for treating insect bites are:

- Remove the stinger (if applicable): In the case of stinging insects like bees or wasps, it's important to remove the stinger as quickly as possible to prevent further venom injection. Gently scrape it off the skin using a blunt edge like a credit card or your fingernail. Avoid squeezing the venom sac, as it may release more venom.
- Wash the bite area: Clean the bite site with mild soap and water to reduce the risk of infection. Be gentle and avoid excessive scrubbing.
- Apply a cold compress: To alleviate pain, reduce swelling, and minimize itching, apply a cold compress or an ice pack wrapped in a cloth to the bite area. Do this for 10-15 minutes at a time, several times a day.
- Use over-the-counter remedies: Over-the-counter topical creams, ointments, or sprays containing ingredients such as hydrocortisone, calamine lotion, or antihistamines can help relieve itching and inflammation. Follow the instructions and precautions on the product packaging.
- Take oral antihistamines: If itching or swelling persists, you can take over-the-counter oral antihistamine to help reduce allergic reactions and relieve symptoms. Consult a pharmacist or health care professional for appropriate dosage and recommendations.
- Avoid scratching: Although it may be tempting, refrain from scratching the insect bite to prevent further irritation, infection, or the development of a secondary skin condition.
- Monitor for severe reactions: In rare cases, insect bites can cause severe allergic reactions, known as anaphylaxis. If you experience symptoms such as difficulty breathing, dizziness, swelling of the face or throat, or a widespread rash, seek immediate medical attention.

16-49. If the insect type is unknown or if the Service member is experiencing severe or worsening symptoms, it is best to consult a health care professional for proper evaluation and guidance.

ARACHNIDS

16-50. An arachnid is a class of joint-legged invertebrate animals that include spiders, scorpions, ticks, and mites. Arachnids are part of the phylum Arthropoda, which also includes insects, crustaceans, and millipedes.

16-51. Arachnids are characterized by having two main body segments: the cephalothorax (a fused head and thorax region) and the abdomen. They typically have four pairs of legs attached to the cephalothorax and possess specialized mouthparts called chelicerae, which are used for feeding and defense. Most arachnids also have specialized structures like spinnerets for producing silk (used by spiders for web-building) and a pair of pedipalps that may serve various functions depending on the species.

16-52. While spiders are perhaps the most well-known arachnids, other arachnids exhibit unique features. Scorpions have large pincers and a long tail ending in a venomous stinger. Ticks and mites are tiny arachnids that often parasitize animals, including humans, and can transmit diseases. Some arachnids, such as daddy longlegs and sun spiders, have distinct body shapes and behaviors.

16-53. Arachnids play diverse roles in ecosystems. Spiders, for example, are predators of insects and help control their populations. Arachnids have adapted to various habitats worldwide, from deserts to forests to aquatic environments. Arachnids are a diverse group of invertebrates that exhibit a wide range of adaptations, behaviors, and ecological roles within the animal kingdom.

SPIDERS

16-54. Among the many species of spiders found around the world, a few are known for their potentially dangerous bites. The most well-known dangerous spider species are the:

- Brazilian Wandering Spider.
- Sydney Funnel-web Spider.
- Black Widow Spider.
- Brown Recluse Spider.

16-55. The Brazilian Wandering Spider is found in South and Central America and is known for its highly venomous bite. It is considered one of the most venomous spiders in the world. Their venom can cause severe pain, priapism (prolonged erection), and, in rare cases, can be life-threatening.

16-56. The Sydney Funnel-web Spider is found in Australia, the Sydney Funnel-web Spider is highly venomous and can be aggressive when threatened. Their venom is potent and can cause severe symptoms, including muscle spasm, difficulty breathing, and in rare cases, it can be fatal. Prompt medical attention is crucial if bitten.

16-57. The Black Widow Spider is found in various parts of the world. Females have a distinctive black body with a red hourglass-shaped mark on the abdomen. Their venom contains neurotoxins that can cause intense pain, muscle cramps, nausea, and in severe cases, can affect the nervous system. While bites are rarely fatal, medical attention is still recommended.

16-58. The Brown Recluse Spider is found primarily in the U.S. Their bites can cause tissue necrosis (cell death), leading to an ulcerous wound that may require medical attention. Other symptoms can include pain, itching, and systemic effects.

16-59. While these spiders are considered dangerous, actual bites and severe reactions are relatively rare. Most spiders are not aggressive and will only bite in self-defense. If bitten by a spider and experiencing severe symptoms, seek immediate medical attention.

16-60. Specific risks associated with spider bites can vary depending on a Service member's health, sensitivity, and potential allergies. Therefore, it is always advisable to exercise caution around spiders and seek professional medical advice if a bite occurs.

SCORPION

16-61. The most dangerous scorpions are typically found in regions with warm climates, particularly in arid or desert areas. Some locations known for housing dangerous scorpion species are:

- North Africa.
- Middle East.
- Mexico.
- Brazil.
- India.

16-62. Countries in North Africa, such as Morocco, Algeria, Tunisia, and Egypt, are home to highly venomous scorpion species. These species include the Deathstalker scorpion and the *Androctonus australis*.

16-63. Several countries in the Middle East have venomous scorpions, including Saudi Arabia, Iran, Iraq, Jordan, and Yemen. The Deathstalker scorpion is found in this region, along with other dangerous species such as the Arabian Fat-tailed scorpion and the Palestine Yellow scorpion.

16-64. Mexico is known for its diverse scorpion fauna, and it is home to several highly venomous species. The Bark scorpion is found in various parts of the country, including the notorious Arizona Bark scorpion, which is also found in the southwestern U.S.

16-65. In Brazil, the Brazilian Yellow scorpion is considered one of the most venomous scorpions. It is primarily found in southeastern and southern regions of the country.

16-66. India is home to several dangerous scorpion species. The Indian Red scorpion and the Indian Black scorpion are found to live in this region.

16-67. While these regions have known populations of dangerous scorpions, encounters with venomous scorpions and severe envenomation are relatively rare. However, it is always advisable to exercise caution and take necessary precautions when in scorpion-prone areas. Figure 16-8 provides an example of a scorpion.

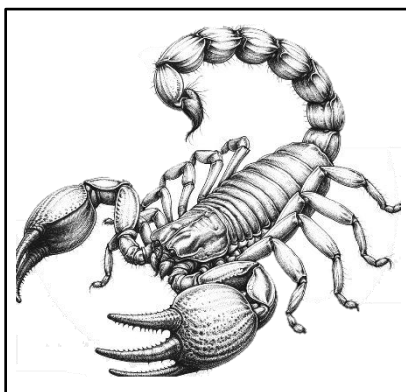


Figure 16-8. Scorpion

CENTIPEDES

16-68. A centipede is a type of arthropod characterized by elongated, multi-segmented creatures with numerous pairs of legs, ranging from tens to hundreds depending on the species. Centipedes are found in various habitats worldwide, including forests, deserts, and tropical regions.

16-69. While most centipedes are not considered dangerous to humans, some larger species possess venomous pincers or fangs located on their first pair of legs. When threatened or provoked, they can deliver a painful sting or bite. The severity of the reaction to a centipede sting can vary depending on the species and the individual's sensitivity to the venom. Centipede bites are painful but not life-threatening. If a Service member is stung by a centipede, first aid steps to take are:

- Stay calm and move away from the centipede to avoid further stings.
- Wash the affected area gently with soap and water to clean the wound and reduce the risk of infection.
- Apply a cold compress or ice pack wrapped in a cloth to the sting site. This can help alleviate pain and reduce swelling.
- Take over-the-counter pain relievers, such as acetaminophen or ibuprofen, to manage pain if necessary. Follow the instructions and recommended dosage.
- Keep the sting site clean and dry and avoid scratching or irritating the area to prevent infection.

16-70. If there are signs of an allergic reaction, such as difficulty breathing, swelling of the face or throat, or widespread hives, seek immediate medical attention.

MILLIPEDES

16-71. Millipedes, belonging to the class Diplopoda, are arthropods distinguished by their elongated, segmented bodies with two pairs of legs per body segment, excluding the first few segments. These creatures are commonly found in moist environments such as forests, gardens, and areas with decaying vegetation.

16-72. Unlike centipedes, millipedes lack venomous pincers or fangs and are generally harmless to humans. However, they possess defensive mechanisms that can release toxins or irritating chemicals, which may cause mild reactions in some individuals, particularly those with sensitive skin.

16-73. If a Service member encounters a millipede and experiences irritation or a sting, the recommended first aid steps include:

- Washing the affected area gently with soap and water to cleanse the site and remove any secretions from the millipede.
- Applying a cold compress or an ice pack wrapped in a cloth to reduce swelling and alleviate discomfort at the sting site.
- Using over-the-counter topical antihistamines or hydrocortisone creams to relieve itching or irritation.
- Avoiding scratching or further irritating the site to prevent infection.

16-74. In cases where there is a severe allergic reaction, characterized by symptoms such as difficulty breathing, swelling of the face or throat, or widespread hives, it is crucial to seek immediate medical attention. While millipede stings are typically mild and self-limiting, if symptoms persist, worsen, or if there are concerns about an individual's reaction, consulting a healthcare professional for further advice and treatment is advisable.

FIRE ANTS

16-75. Fire ants are a type of stinging ant that are native to South America but have spread to various parts of the world, including the southern U.S, Southeast Asia, and Australia. Fire ants are known for their aggressive behavior and painful stings.

16-76. When a fire ant stings, it injects venom into the skin, resulting in a sharp, burning sensation and the formation of a raised, reddish bump. Multiple ants may attack simultaneously, leading to multiple stings. If stung by a fire ant, the first aid steps are to:

- Move away from the area where the ants are present to avoid further stings.
- Remove any clothing or jewelry that may have ants trapped in them, as they can continue to sting.
- Gently wash the affected area with soap and water to clean the site.
- Apply a cold compress or ice pack wrapped in a cloth to the sting site. This can help reduce pain and swelling.
- Over-the-counter pain relievers, such as acetaminophen or ibuprofen, can be taken to help manage pain if necessary. Follow the instructions and recommended dosage.
- Avoid scratching the sting site to prevent infection. If itching is severe, over-the-counter antihistamines or hydrocortisone creams may be used under the guidance of a health care professional.
- Monitor the individual for signs of an allergic reaction, such as difficulty breathing, swelling of the face or throat, or widespread hives. If these symptoms occur, seek immediate medical attention.

16-77. Fire ant stings can be particularly problematic for individuals who are allergic or hypersensitive to ant venom. In such cases, immediate medical attention should be sought. These first aid steps are general guidelines, and the severity of the reaction may vary from person to person. If there are concerns about the Service member's health or if the symptoms worsen or persist, consult a health care professional for appropriate advice and treatment.

TICKS

16-78. A tick is a small arachnid parasite that feeds on the blood of animals, including humans. Ticks are found in various environments, such as forests, grasslands, and shrubby areas. They latch onto their hosts and can transmit diseases through their bites.

16-79. When a tick bites, it burrows its mouth parts into the skin to feed on blood. If you discover a tick attached to your skin, it's important to remove it promptly and take the following first aid steps:

- Prepare for tick removal: Use fine-tipped tweezers to grasp the tick as close to the skin's surface as possible. Avoid squeezing or crushing the tick's body.
- Remove the tick: Gently and steadily pull the tick upward with even pressure. Do not twist or jerk it. The goal is to remove the entire tick, including its mouthparts. Dispose of the tick by placing it in a sealed bag or container or by submerging it in alcohol.
- Clean the bite area: Thoroughly clean the bite area with soap and water or an antiseptic solution to reduce the risk of infection.

- Apply an antiseptic: After cleaning the bite area, you can apply an antiseptic ointment or cream to further prevent infection.
- Monitor for symptoms: Keep an eye on the bite site for any signs of an allergic reaction, infection, or the appearance of a rash. If you notice any concerning symptoms, seek medical attention.

16-80. Certain tick-borne diseases, such as Lyme disease or Rocky Mountain spotted fever, may take time to manifest symptoms. If in an area where these diseases are prevalent or if you develop symptoms such as fever, rash, joint pain, or flu-like symptoms after a tick bite, it is crucial to consult a health care professional for proper evaluation, diagnosis, and treatment.

16-81. Prevention is key when it comes to ticks. To minimize the risk of tick bites, wear protective clothing when in tick-prone areas, use insect repellents containing DEET, perform regular tick checks after outdoor activities, and take appropriate measures to protect pets from ticks as well. DEET (chemical name, N, N-diethyl-meta-toluamide) is the active ingredient in many repellent products. It is widely used to repel biting pests such as mosquitoes and ticks.

WARNING

Lyme disease, transmitted by Blacklegged “Deer” tick, can cause long- term neurological disorders and other complications if not identified and treated early.

16-82. The signs and symptoms of a tick bite can vary depending on factors such as the type of tick, the individual's sensitivity to the tick's saliva, and the presence of any tick-borne diseases. The common signs and symptoms associated with tick bites are:

- Redness and swelling: After a tick bite, it is common to experience localized redness and swelling at the site of the bite. This is a normal reaction to the tick's saliva.
- Itching or irritation: The bite site may be itchy or irritated, like a mosquito bite. Avoid scratching to reduce the risk of infection.
- Tick remnants: If a tick is not promptly and properly removed, parts of the tick's body, such as its mouthparts, may remain embedded in the skin. This can lead to ongoing irritation and a risk of infection.
- Rash: In some cases, a small red spot or rash may develop around the tick bite. This can expand over time and may resemble a target-like pattern. It is important to note that not all tick bites result in a rash.
- Flu-like symptoms: Tick-borne diseases such as Lyme disease, Rocky Mountain spotted fever, or babesiosis can cause flu-like symptoms. These may include fever, fatigue, headache, muscle aches, joint pain, and swollen lymph nodes. However, these symptoms usually develop days or weeks after the tick bite.
- Allergic reaction: In rare cases, individuals may have an allergic reaction to tick bites, resulting in more severe symptoms such as difficulty breathing, swelling, hives, or anaphylaxis. Immediate medical attention is necessary in such cases.

16-83. Not all tick bites lead to the transmission of diseases, and most tick bites are harmless. If a Service member experiences any concerning symptoms or if you live in an area where tick-borne diseases are prevalent, seek medical attention for proper evaluation, diagnosis, and appropriate treatment.

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Chapter 17

Climatic and Environmental Injuries

OVERVIEW OF CLIMATIC INJURIES

17-1. Climatic injuries, also known as weather-related injuries, occur when people's health is negatively affected by extreme weather conditions. These injuries can range from heat-related illnesses to cold-related conditions, and they are often a result of prolonged exposure to harsh environmental elements. Climate plays a significant role in determining the risks individuals face, with hot climates posing dangers of heat exhaustion and heatstroke, while cold climates can lead to frostbite and hypothermia. Understanding the signs, symptoms, and prevention strategies for these climatic injuries is essential in ensuring our well-being and staying safe in different weather conditions.

17-2. Specialized clothing and equipment (such as cold weather gear) for a specific environment should be obtained and used properly. For information on the prevention of heat and cold injuries, refer to TC 4-02.3, *Field Hygiene and Sanitation*.

17-3. Personal factors can significantly influence health and well-being in a climatic environment. Some personal factors that play a role are:

- **Physical Health and Fitness:** A person's physical health and fitness levels affect how they cope with different climatic conditions. Individuals with good overall physical health, including cardiovascular fitness, strength, and endurance, are generally more resilient to extreme temperatures, humidity, and other climate-related challenges.
- **Adaptability:** Personal adaptability and the ability to adjust to changing climatic conditions can impact well-being. Some people are naturally more adaptable to different climates, while others may struggle with the challenges posed by extreme temperatures, high humidity, or other climatic factors.
- **Sensitivity and Allergies:** Some individuals may be more sensitive to specific climate-related factors, such as pollen, air pollution, or mold, which can trigger allergies or respiratory issues. Understanding personal sensitivities and taking necessary precautions can help maintain health and well-being in a climatic environment.
- **Mental Resilience:** Mental resilience plays a vital role in dealing with challenging climatic conditions. People with strong coping mechanisms, stress management skills, and the ability to maintain a positive mindset are better equipped to handle adverse weather conditions, such as extreme heatwaves or cold snaps.
- **Knowledge and Awareness:** Being well-informed about the climatic environment and understanding how it can impact health is crucial. Knowing the risks associated with specific climate-related factors, such as heatstroke or frostbite, and taking appropriate preventive measures can help protect health and well-being.
- **Lifestyle Choices:** Personal lifestyle choices, including clothing preferences, outdoor activities, and exposure to natural elements, can affect health in a climatic environment. Wearing appropriate clothing for protection against extreme temperatures, staying hydrated, and engaging in activities that are suitable for the climate, contribute to well-being.
- **Pre-existing Health Conditions:** Individuals with pre-existing health conditions may be more vulnerable to the effects of certain climatic conditions. For example, individuals with respiratory conditions like asthma may experience worsening symptoms in highly polluted or allergen-rich climates. Managing pre-existing health conditions effectively becomes crucial for maintaining well-being in a climatic environment.
- **Access to Resources:** Personal access to resources, such as reliable shelter, heating or cooling systems, and clean water, can impact health and well-being in different climates. Having access to appropriate resources to adapt to climatic conditions can significantly improve overall well-being.

17-4. Personal factors interact with other environmental and social factors, such as the availability of infrastructure, socioeconomic conditions, and cultural norms. Understanding these personal factors and their

interactions can help individuals make informed choices and take the necessary steps to protect their health and well-being in a climatic environment.

WARNING

Service members should use extreme caution when starting fad diets or taking over-the-counter herbal supplements. Death and severe injury may occur in individuals using dietary or herbal supplements without medical monitoring.

A trained health care provider should supervise weight loss and the use of weight loss supplements.

HEAT INJURIES

17-5. Heat injury prevention is of utmost importance in the military, where Service members face intense physical challenges and operate in diverse climatic conditions. When the body gets overheated, it can lead to serious health risks and impair a Service member's ability to perform their duties effectively. Heat-related injuries, such as heat exhaustion and heat stroke, can be prevented through various measures, including wearing appropriate clothing, taking regular rest breaks, acclimatization and fluid replacement. By prioritizing heat injury prevention, Leaders ensure the well-being and operational readiness of their personnel, allowing them to fulfill their responsibilities and accomplish their missions with strength and resilience.

17-6. Exertional heat illness refers to a spectrum of heat-related conditions that occur when the body's core temperature rises excessively during physical activity or exercise. It is primarily caused by prolonged exertion in hot and humid environments, leading to an imbalance between heat production and heat dissipation. Exertional heat illness encompasses various conditions ranging from mild heat exhaustion to severe heat stroke, which is a medical emergency. Symptoms of heat injuries may include:

- Fatigue.
- Dizziness.
- Headache.
- Nausea.
- Excessive sweating.
- Rapid heartbeat.
- Confusion.
- Loss of consciousness in severe cases.

17-7. Prompt recognition and treatment of exertional heat illness are essential to prevent complications and potentially life-threatening consequences. Effective prevention strategies, such as acclimatization, appropriate clothing, fluid replacement, and regular rest breaks, are crucial in minimizing the risk of exertional heat illness. For more information on heat illness refer to TB MED 507 and TC 4-02.3.

17-8. While there is a range of adverse effects that can result from the body overheating, the two major kinds of heat illnesses that are referred to as heat casualties are heat exhaustion (can be mild or more severe) or heat stroke. Heat stroke is the most severe form of heat illness and possibly fatal.

Note. For more information on heat illness prevention, refer to the Defense Centers for Public Health – Aberdeen website.

HEAT EXHAUSTION

17-9. Heat exhaustion is a heat-related condition that occurs when the body becomes overheated due to prolonged exposure to high temperatures and inadequate fluid replacement. It is considered a milder form of heat illness compared to heat stroke. Signs and symptoms of heat exhaustion may include:

- Heavy sweating.

- Weaknesses and fatigue.
- Dizziness or lightheadedness.
- Headache.
- Nausea or vomiting.
- Pale, cool, or clammy skin.
- Rapid and shallow breathing.
- Elevated heart rate.
- Muscle cramps.
- Feeling faint or fainting.

17-10. If a Service member is suspected to have heat exhaustion, immediate first aid treatment should be provided. The first aid steps to treat heat exhaustion are:

- Move the Service member to a cooler environment: Get them out of the heat and into a shaded or air-conditioned area.
- Encourage rest and rehydration: Have the Service member lie down and elevate their legs slightly. Provide them with cool water and or an electrolyte-containing sports drink to sip on. Avoid alcohol or caffeine.
- Cool the Service member's body: Loosen or remove any tight or unnecessary clothing. Apply a cool, wet cloth or towels to their skin, especially on the neck, face, and armpits. Use a fan or create air movement to aid in evaporation.
- Monitor their condition: Stay with the Service member and observe their symptoms. If they do not improve within 30 minutes or their condition worsens, seek medical assistance immediately.

17-11. Heat exhaustion can progress to heat stroke, a life-threatening condition. If the person's symptoms worsen, they become disoriented, confused, or lose consciousness, immediately call for medical assistance.

17-12. Prevention is key in avoiding heat exhaustion. Practice proper fluid replacement, especially when engaging in physical activity or spending time in hot environments. Wear lightweight and loose-fitting clothing, use sunscreen, and take regular breaks in cool areas to avoid prolonged exposure to heat.

17-13. Heat exhaustion is often preceded by heat cramps, muscle cramps of the arms, legs, or abdomen. Heat cramps and heat exhaustion often act as indicators. These conditions need to be identified and treated before they get to an extreme case of heat stroke. Catch these conditions early, as the casualty needs rest, water, shade, evaluation, and possible medical care.

HEAT STROKE

17-14. Heat stroke is a severe and potentially life-threatening illness characterized by a rapid increase in body temperature. It occurs when the body's cooling mechanisms fail to regulate internal heat, often because of prolonged exposure to high temperatures or physical exertion in hot environments. Heat stroke is a medical emergency that requires immediate intervention. The signs and symptoms of heat stroke may include:

- Altered mental state or confusion.
- Headache, dizziness, or fainting.
- Nausea or vomiting.
- Difficulty breathing.
- Hot, dry skin (lack of sweating).
- High body temperature (typically above 104°Fahrenheit or 40°Celsius).
- Rapid and strong heartbeat.
- Seizures.

17-15. If someone is suspected of having heat stroke, it is crucial to seek emergency medical help immediately. While waiting for medical professionals to arrive, follow these first aid steps:

- Move the person to a cooler location: Get them out of the heat and into a shaded or air-conditioned area as quickly as possible.
- Call for emergency assistance: Dial 911 or the appropriate local emergency number to request immediate medical help to include MEDEVAC.

- Cool the person rapidly: Reduce their body temperature as quickly as possible. You can immerse the person in a tub of cold water, use ice cooling sheets or apply cold water to their body using wet towels or sponges. Fan the person while misting them with cool water to enhance cooling.
- Monitor their vital signs: Keep a close eye on the person's breathing, pulse, and level of consciousness. If they stop breathing and you can no longer find a pulse, begin CPR if you are trained to do so.

17-16. It's important to note that heat stroke is a severe condition, and professional medical treatment is essential. Rapid cooling of the body is crucial in preventing potential organ damage and reducing the risk of complications.

17-17. To prevent heat stroke, it is important to avoid excessive physical exertion in hot environments, wear lightweight and breathable clothing, and take regular breaks in cool areas. Understanding the signs and symptoms of heat stroke and promptly seeking medical help can save lives.

Note. The same person should observe the Service member during cooling and evacuation to spot symptom changes.

HYPONATREMIA (WATER INTOXICATION)

17-18. Hyponatremia (water intoxication) is a condition characterized by abnormally low levels of sodium in the blood. Sodium plays a crucial role in maintaining the body's fluid balance and proper nerve and muscle function. When sodium levels become significantly diluted, usually due to excessive water intake or inadequate sodium intake, hyponatremia can occur. The signs and symptoms of hyponatremia can vary depending on the severity of the condition. Common manifestations include:

- Nausea and vomiting.
- Headache.
- Confusion or altered mental state.
- Fatigue and weakness.
- Seizures.
- Muscle cramps or spasms.
- Loss of appetite.
- Swelling in the hands, feet, or legs.
- Low blood pressure.
- Coma (in severe cases).

17-19. If someone is suspected to have hyponatremia, it is important to seek medical attention. In the meantime, you can provide the following first aid measures:

- Assess the person's condition: Check for any signs of severe symptoms such as seizures, loss of consciousness, or difficulty breathing. If any of these are present, call emergency services immediately.
- Move the person to a cool, shaded area: Ensure they are away from direct sunlight or heat sources.
- Encourage fluid intake: If the person is conscious and able to swallow, give them small sips of a beverage containing electrolytes, such as a sports drink. This can help restore sodium levels.
- Do not give more water. Providing additional water will make the casualty more hyponatremic.
- If awake, allow the Service member to consume salty foods or snacks.
- Monitor the person's vital signs: Keep an eye on their breathing, heart rate, and level of consciousness. If their condition worsens, be prepared to administer CPR if necessary.

17-20. Hyponatremia can be mistaken for heat stroke, and the treatment is very different. Hyponatremia is a medical emergency that requires professional diagnosis and treatment. It is important to address the underlying cause and restore sodium levels under the guidance of a health care professional.

Note. This condition most often occurs during initial entry training; however, it may occur anytime when overhydration is encountered.

COLD WEATHER INJURIES

17-21. Preventing cold weather injuries is a vital aspect of military training and operations in cold climates. When Service members face extreme cold conditions, they are at risk of various cold weather injuries, such as frostbite and hypothermia, which can significantly impact their ability to fulfill their duties effectively. Cold weather injury prevention measures are essential to ensure the well-being and operational readiness of military personnel. By implementing strategies such as proper clothing layering, maintaining adequate hydration, practicing cold weather survival techniques, and promoting awareness of cold weather risks, the military prioritizes the safety and health of its Service members. By emphasizing cold weather injury prevention, the military ensures that its personnel can endure the challenging cold environments and accomplish their missions with strength, resilience, and readiness.

17-22. Cold weather injuries encompass various conditions that occur due to exposure to extreme cold temperatures. The main cold weather injuries are:

- Frostbite.
- Hypothermia.
- Trench Foot.
- Nonfreezing injuries (such as chilblains and snow blindness).

17-23. These injuries are often caused by a combination of factors. Prolonged exposure to cold temperatures, inadequate protection against the cold, wet clothing, wind chill. Other contributing factors include poor nutrition, dehydration, fatigue, and certain medical conditions can increase the susceptibility to cold weather injuries. Proper preparation, adequate clothing, layering, regular breaks to warm up, maintain hydration, and practicing good cold weather safety measures can help reduce the risk of cold weather injuries. For more information on heat illness prevention, refer to the Defense Centers for Public Health–Aberdeen (Cold Weather Casualties and Injuries) website. For additional information on cold weather injuries refer to TC 4-02.3.

FROSTBITE

17-24. Frostbite is a cold weather injury that occurs when tissues freeze due to prolonged exposure to extreme cold temperatures. It typically affects extremities, such as the fingers, toes, ears, and nose, but can also occur in other exposed areas of the body. The signs and symptoms of frostbite may vary depending on the severity, and they can include:

- Cold, pale, or bluish skin.
- Numbness or a loss of sensation in the affected area.
- Tingling or a “pins and needles” sensation.
- Hard or waxy appearance of the skin.
- Swelling or blistering (in severe cases).
- Pain or a burning sensation (as the affected area rewarms).
- Blackened or gangrenous tissue (in severe, untreated cases).

17-25. If someone is suspected of having frostbite, it is important to provide immediate first aid and seek medical attention. Take the following first aid steps:

- Move to a warm environment: Get the person out of the cold and into a warm area as soon as possible to prevent further damage.
- Remove wet or constrictive clothing: Take off any wet clothing or accessories that could contribute to heat loss or restrict blood flow to the affected area.
- Gradually rewarm the affected area: Immerse the frostbit area in warm (not hot) water or use warm compresses. Avoid using direct heat sources like fires or heating pads, as the person may not be able to sense heat properly and could burn the skin.
- Avoid rubbing or massaging the area: Vigorous rubbing can further damage the tissues. Let the affected area rewarm naturally.
- Protect and elevate the affected area: Once the frostbitten area has thawed, cover it with a sterile dressing or clean cloth. Keep the area elevated to reduce swelling, if possible.

- Offer pain relief if needed: Over-the-counter pain relievers like ibuprofen can help alleviate pain and reduce inflammation. However, consult a health care professional before administering any medication.
- Seek medical attention: Frostbite requires medical evaluation and treatment. Even if the frostbite appears mild, professional medical care is essential to assess the extent of the injury and prevent complications.

17-26. Frostbite can cause long-term damage, including tissue loss and nerve damage. Prompt and appropriate treatment is crucial for minimizing complications and maximizing recovery.

17-27. Frostbite accounts for the largest number of cold weather injuries each year and occurs when tissue is exposed to temperatures that are usually below 32°Fahrenheit depending upon wind chill factor, length of exposure, and adequacy of protection.

17-28. Frostbite can occur suddenly due to contact with cold metal or super-cooled liquids such as alcohol, fuel, or antifreeze or can develop over time due to prolonged cold exposure. Frostbite is most common in exposed skin such as the nose, ears, cheeks, but can also occur in the hands and feet due to reduced blood flow.

Note. The onset is signaled by a sudden blanching of the skin of the nose, ears, cheeks, fingers, or toes followed by a momentary tingling sensation. Frostbite is indicated when the face, hands, or feet stop hurting.

HYPOTHERMIA

17-29. Hypothermia is a concern regarding TCCC and the body's ability to form clots is impaired. This condition can cause a tendency toward prolonged or excessive bleeding, which may occur spontaneously or following an injury. Hypothermia is a potentially life-threatening condition that occurs when the body's core temperature drops significantly below normal.

17-30. The risks differ with environmental hypothermia. Victims of moderate to severe environmental hypothermia are known to have a risk of an irregular heartbeat associated with "afterdrop." Afterdrop is the continued fall of deep body temperatures during rewarming after hypothermia. Afterdrop is thought to endanger the heart by further cooling from cold blood presumed to be returning to the body's center or core.

17-31. Environmental hypothermic casualties should be monitored and gently handled as afterdrop is not always observed. The goals for overall casualty treatment (such as hemorrhage control and resuscitation in TCCC versus warming measures along with gentle handling in environmental hypothermia) should be understood, trained, and rehearsed. For further discussion on hypothermia see Chapter 8.

TRENCH FOOT

17-32. Trench foot, also known as immersion foot, is a non-freezing cold injury that occurs when the feet are exposed to prolonged damp and cold conditions. It can lead to tissue damage and infection. Contributing factors include wet and cold environments, inadequate foot hygiene, prolonged periods of immobility, and improper footwear.

17-33. Trench foot is often associated with Soldiers during World War I who endured long periods in wet trenches. Trench foot can occur in any situation where the feet are exposed to excessive moisture and cold temperatures. The signs and symptoms of trench foot include:

- Redness and swelling of the feet.
- Numbness or tingling sensation.
- Itching or a burning sensation.
- Pain or discomfort.
- Blisters or open sores.
- Coldness and paleness of the skin.
- Foul odor.

17-34. If someone is suspected to have trench feet, the following first aid steps can be taken:

- Move them to a warm and dry environment: Get the person out of wet and cold conditions as soon as possible to prevent further damage.
- Remove wet footwear and socks: Take off wet footwear and socks to allow the feet to dry. Avoid rubbing or massaging the feet, as it can further damage the skin.
- Elevate the feet: Raise the feet slightly to help reduce swelling and improve circulation.
- Clean and dry the feet: Gently wash the feet with warm water (not hot) and mild soap. Pat the feet dry with a clean towel, avoiding excessive friction.
- Apply clean, dry socks: Once the feet are dry, put on clean, dry socks to help insulate and protect the feet. Avoid using tight or constrictive socks.
- Avoid direct heat sources: Do not use direct heat sources like fires, heaters, or heating pads to warm the feet. The affected feet may be numb, and exposure to high heat can cause burns.
- Seek medical attention: It is important to seek medical evaluation and treatment for trench foot. Health care professionals can provide appropriate care, clean, and dress any blisters or open sores, and monitor any signs of infection.

17-35. Prevention is key in avoiding trench foot. Keep the feet dry, wear appropriate footwear that is waterproof and well-insulated, change wet socks and shoes promptly, and take breaks to allow the feet to dry and warm up. Proper foot hygiene and regular inspection of the feet can help identify any early signs of trench foot and prevent its progression.

NONFREEZING INJURIES

17-36. Exposure to temperatures from 32°Fahrenheit to 60°Fahrenheit may cause nonfreezing cold injuries of skin, fingers, toes, ears, and facial parts. Exposure of skin to cold metal; super cold fuel, petroleum, oil, and lubricants; wind chill; and the wear of tight, circulation-restricting clothing (particularly boots).

17-37. Riding in open vehicles, exposure to propeller or rotor-generated wind, running or skiing, and altitude exposure where there is little tree cover can all contribute to greater wind chill.

17-38. Nonfreezing cold injuries are chilblain and snow blindness. Trench foot occurs when tissues are exposed to temperatures from 32°Fahrenheit to 60°Fahrenheit for prolonged periods of time (>12 hours), whereas chilblains, which is a more superficial injury, can occur after repeated prolonged exposure of bare skin to low temperatures from 60°Fahrenheit down to 32°Fahrenheit. Snow blindness is caused by unprotected exposure to UV rays. Snow blindness can be prevented using appropriate eye protection (sunglasses). Snow blindness can be painful as it gets closer to three to five hours later.

Chilblains

17-39. Chilblains, also known as pernio, is a nonfreezing cold weather injury that can occur after one to five hours in cold, wet conditions at temperatures from about 50°Fahrenheit down to 32°Fahrenheit. Chilblains are a localized inflammatory response of the skin that occurs due to prolonged exposure to cold and damp conditions.

17-40. The most affected areas are the fingernail-side of the fingers, but the ears, face, and other exposed skin are also areas of occurrence. There are no lasting effects from chilblain. They are more common in colder climates and during the winter season. The signs and symptoms of chilblains include:

- Redness and inflammation of the affected area.
- Itching or a burning sensation.
- Swelling and tenderness.
- Skin discoloration (bluish or purplish appearance).
- Formation of small, fluid-filled blisters.
- Dry or cracked skin.
- Pain or discomfort, especially when exposed to warmth after being cold.

17-41. If someone is experiencing chilblains, the following first aid procedures can be performed:

- Move to a warm environment: Get the person indoors or to a warmer area to prevent further exposure to cold temperatures.

- Gradually rewarm the affected areas: Avoid sudden temperature changes or direct heat sources like heaters or fires, as this can worsen the condition. Instead, gently warm the affected areas by placing them near warm body parts (such as under the armpits) or by using warm water. The water should be comfortably warm, not hot.
- Protect and cover the affected areas: Once the affected areas have been warmed, keep them protected from further exposure to cold. Cover them with warm clothing, gloves, or socks to help insulate and maintain a stable temperature.
- Avoid scratching or rubbing the affected areas: Itching can be intense, but scratching or rubbing can further damage the skin and increase the risk of infection. Encourage the person to resist the urge to scratch.
- Apply soothing lotions or creams: If recommended by a health care professional, apply mild lotions or creams to alleviate itching and provide relief. Avoid using excessive amounts or substances that may irritate the skin.
- Seek medical advice if needed: If symptoms worsen, if blisters or open sores develop, or if signs of infection (such as increased redness, swelling, or discharge) are present, it is important to seek medical attention for proper evaluation and treatment.

17-42. Preventing chilblains is essential. It can be achieved by keeping the extremities warm and dry, wearing appropriate protective clothing and footwear, avoiding prolonged exposure to cold and damp environments, and gradually acclimating the body to temperature changes. Regularly checking the skin for any signs of chilblains can help identify and address the condition early on.

Snow Blindness

17-43. Snow blindness, also known as photokeratitis or UV keratitis, is a condition that occurs when the eyes are exposed to excessive UV radiation, typically from sunlight reflected off snow or ice. It can also happen when exposed to other intense light sources, such as welding arcs or tanning beds. Snow blindness is essentially a temporary sunburn of the cornea, the clear front surface of the eye. The signs and symptoms of snow blindness include:

- Eye pain or a sensation of grittiness.
- Redness and inflammation of the eyes.
- Watery eyes and excessive tearing.
- Blurred vision or sensitivity to light.
- Feeling as if there is something in the eye.
- Headache.
- Temporary vision loss in severe cases.

17-44. If someone is experiencing snow blindness, the following first aid treatment can be administered:

- Move to a dark and sheltered area: Get the person indoors or to a shaded area to protect the eyes from further exposure to bright light.
- Rest the eyes: Keep the eyes closed or minimize eye movements to reduce discomfort and prevent further irritation.
- Avoid rubbing the eyes: Rubbing the eyes can exacerbate the condition and potentially cause additional damage.
- Apply cold compresses: Place a cold, damp cloth or cool pack over closed eyes to help reduce inflammation and alleviate pain. Avoid using ice directly on the skin.
- Use artificial tears: Over-the-counter artificial tear drops can be used to provide relief and moisture to the eyes. Follow the instructions on the packaging or consult a health care professional for appropriate usage.
- Wear protective eyewear: If the person needs to be in a bright environment, ensure they wear sunglasses that offer 100 percent UV protection. Wrap-around sunglasses or goggles with side shields provide the best protection.
- Seek medical attention: It is advisable to seek medical advice if the symptoms persist, worsen, or if vision loss occurs. A health care professional can examine the eyes and provide appropriate treatment and guidance.

17-45. Prevention is crucial in avoiding snow blindness. To protect the eyes from excessive UV radiation, wear sunglasses or goggles that provide 100 percent UV protection, especially in snowy or high-altitude environments. Additionally, using wide-brimmed hats or visors can help shield the eyes from indirect sunlight.

MOUNTAINOUS TERRAIN INJURIES

17-46. Approximately 27 percent of the earth's land surface is considered mountainous terrain. These areas are characterized by rugged landscapes, elevated peaks, and varying degrees of slope. Mountains play a crucial role in shaping the Earth's geography, influencing climate patterns, and providing habitat for diverse ecosystems.

17-47. Service members operating in mountainous terrain should be well-prepared, have the necessary skills, and carry a first aid kit to address potential injuries and emergencies. Additionally, understanding the terrain, weather conditions, and having proper equipment can contribute to safer operations in mountainous terrain.

17-48. Mountainous terrain poses various challenges and risks, and injuries can range from minor to severe. Common injuries in mountainous terrain include:

- Sprains and Strains. Uneven and rocky surfaces can lead to twisted ankles or muscle strains.
- Fractures. Falls or accidents may result in broken bones, particularly in extremities.
- Cuts and Abrasions. Sharp rocks and uneven surfaces can cause cuts and scrapes.
- Altitude Sickness. At higher elevations, individuals may experience symptoms like headache, nausea, and dizziness due to reduced oxygen levels.
- Hypothermia. Cold temperatures, especially at higher altitudes, can lead to a drop in body temperature.
- Dehydration. Strenuous activity, combined with lower humidity and potential difficulty accessing water sources, can lead to dehydration.
- Exhaustion. Hiking or climbing in mountainous terrain can be physically demanding, leading to fatigue and exhaustion.
- Frostbite. Exposure to cold conditions, especially at high altitudes, can result in frostbite, where skin and tissues freeze.
- Avalanche and Rockfall Injuries. In certain mountainous regions, avalanches or rockfalls can pose significant dangers, causing trauma and injuries.
- Wildlife Encounters. Depending on the location, encounters with wildlife can lead to injuries.

17-49. Many of the above injury types and treatments have been discussed throughout this publication, however operating in mountainous terrain can be challenging, but it's crucial to be aware of the potential risks, including altered mental status, high-altitude pulmonary edema (HAPE), and high-altitude cerebral edema (HACE).

ALTITUDE EMERGENCIES

17-50. Changes in consciousness, confusion, or disorientation can occur due to factors like altitude sickness, dehydration, or exhaustion. Recognizing and responding to an altered mental status promptly is essential for ensuring the well-being of Service members. Some general first aid measures for Service members experiencing altered mental status in mountainous terrain include:

- Move to Lower Altitude: If the altered mental status is due to altitude sickness, move the Service member to a lower altitude immediately. Descending to lower elevations allows the body to acclimatize and receive more oxygen, potentially alleviating symptoms.
- Rest and Hydration: Ensure the Service member rests and stays hydrated. Dehydration and fatigue can contribute to an altered mental status, so providing water and allowing for adequate rest are important.
- Monitor Breathing: Keep a close eye on the Service member's breathing. Shallow or irregular breathing may indicate respiratory distress. If breathing difficulties persist, promptly seek medical assistance.
- Check for Signs of Hypoxia: Assess for signs of hypoxia, which occurs when the body doesn't receive enough oxygen. Look for bluish discoloration of the lips or fingertips. If observed, it may indicate a more severe oxygen deficiency, requiring immediate medical attention.

- **Keep Warm:** If the altered mental status is due to cold temperatures, provide warmth by using blankets, jackets, or other insulating materials. Preventing hypothermia is crucial.
- **Monitor for Improvement or Worsening:** Keep a close watch on the Service members' condition. If there is no improvement or if the symptoms worsen, seek medical assistance as soon as possible.
- **Administer Oxygen (if available):** If you have access to supplemental oxygen and the person is experiencing severe symptoms, administering oxygen can be beneficial. However, this may require specific training and equipment.
- **Evacuation if Necessary:** In severe cases or if the person's condition doesn't improve, consider evacuating them to a medical facility at a lower altitude. This may involve calling for professional rescue or using available means of transportation.

17-51. First aid of an altered mental status is a temporary measure, and medical assistance should be sought for any significant or worsening symptoms. If in doubt or if the altered mental status persists, it is essential to prioritize the Service member's safety and seek timely medical help.

HIGH-ALTITUDE PULMONARY EDEMA

17-52. High-Altitude Pulmonary Edema is a serious condition that can occur at higher elevations, typically above 2,500 meters (8,200 feet). It involves the accumulation of fluid fluids in the lungs and can be life-threatening. Some general first aid measures for Service members experiencing HAPE in mountainous terrain include:

- **Immediate Descent.** The most critical first aid measure for HAPE is an immediate descent to lower altitudes. Every effort should be made to get the affected individual to a lower elevation as quickly as possible, as this is the most effective way to alleviate symptoms.
- **Rest and Oxygen.** Ensure the person rests and provide supplemental oxygen if available. Oxygen therapy can help improve oxygen levels in the blood and alleviate breathing difficulties.
- **Keep Warm.** Maintain the person's warmth by using blankets and insulating layers. Cold temperatures can exacerbate symptoms, so preventing hypothermia is important.
- **Stay Hydrated.** Encourage the individual to drink fluids to stay hydrated. Dehydration can worsen the condition, so maintaining adequate fluid intake is essential.
- **Monitor Breathing.** Keep a close watch on the person's breathing. If there are signs of severe respiratory distress, such as blue lips or fingertips, it may indicate a critical situation, and professional medical help should be sought immediately.
- **Administer Medications (if available).** In some cases, medications such as nifedipine or sildenafil may be used to help manage HAPE symptoms. The medical professional should determine the availability and use of these medications.
- **Evacuation.** Seek evacuation to the MTF at a lower altitude as soon as possible. Evacuation may involve contacting medical personnel and requesting evacuation to a lower elevation where more definitive medical care can be provided.

17-53. HAPE is a medical emergency, and prompt action is crucial. While these first aid measures can provide initial assistance, professional medical help should be sought without delay. If you suspect someone has HAPE, it's essential to prioritize their safety and act swiftly to get them to a lower altitude and seek immediate medical attention.

HIGH-ALTITUDE CEREBRAL EDEMA

17-54. High-Altitude Cerebral Edema is a severe and life-threatening condition that can occur at high elevations, typically above 3,000 meters (9,800 feet). It involves swelling of the brain and can lead to neurological symptoms. Some general first aid measures for Service members experiencing HACE in mountainous terrain include:

- **Immediate Descent.** The most critical first aid measure for HACE is an immediate and rapid descent to lower altitudes. Descending to a lower elevation is essential to reduce the impact of high altitude on the brain.
- **Rest and Oxygen.** Ensure the affected person rests and provide supplemental oxygen if available. Oxygen therapy can help improve oxygen levels in the blood and alleviate neurological symptoms.

- **Keep Warm.** Maintain the person's warmth using blankets and insulating layers. Cold temperatures can worsen symptoms, so preventing hypothermia is important.
- **Hydration.** Encourage the individual to drink fluids to stay hydrated. Dehydration can exacerbate the condition, so maintaining adequate fluid intake is crucial.
- **Administer Medications (if available).** In some cases, medications such as dexamethasone may be used to help manage HACE symptoms. Medical professionals should determine the use of medications, and access to such medications may be limited in remote mountainous areas.
- **Monitor Neurological Symptoms.** Keep a close watch on the person's neurological symptoms. Signs of severe neurological impairment, such as confusion, loss of coordination, or changes in consciousness, require urgent medical attention.
- **Evacuation.** Seek evacuation to an MTF at a lower altitude as quickly as possible. HACE is a medical emergency, and professional medical help should be sought without delay.
- **Reassure and calm.** Keep the affected individual calm and reassure them. Stress and anxiety can exacerbate symptoms, so a supportive and calming environment is beneficial.

17-55. HACE is a critical condition, and time is of the essence. While these first aid measures can provide initial assistance, medical assistance should be sought urgently. If HACE is suspected in a Service member, prioritize their safety by facilitating a rapid descent and seeking immediate medical attention.

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Chapter 18

Sickle Cell Trait

OVERVIEW OF SICKLE CELL TRAIT

18-1. Sickle cell trait (SCT) is an inherited genetic condition in which a person carries one copy of the sickle cell gene and one normal hemoglobin gene. It is different from sickle cell disease (SCD), which occurs when a person inherits two copies of the sickle cell gene. The SCT is usually asymptomatic and generally does not cause health problems. However, under certain conditions, such as extreme physical exertion, high altitudes, or dehydration, individuals with SCT may experience complications.

18-2. SCD is a genetic disorder that affects the red blood cells, which are responsible for carrying oxygen throughout the body. It is caused by a mutation in the gene responsible for producing hemoglobin, a protein that enables red blood cells to carry oxygen. In individuals with SCD, the hemoglobin gene mutation leads to the production of abnormal hemoglobin, known as hemoglobin S. This abnormal hemoglobin causes red blood cells to become rigid and take on a sickle or crescent shape instead of their normal disc shape. These sickle-shaped cells can get stuck in small blood vessels, leading to various complications.

18-3. For more information on SCT and SCD, refer to the National Institute of Health website. Service members with SCT can typically participate fully in any physical activity. Under certain circumstances Service members with SCT may suffer negative effects such as muscle weakness, pain, and collapse. This collapse is termed exercise collapse associated with sickle cell trait (ECAST).

18-4. ECAST refers to a specific syndrome that can occur in individuals with SCT during or immediately after strenuous exercise. It is characterized by symptoms such as dizziness, weakness, fatigue, and in rare cases, collapse, or loss of consciousness. The ECAST can occur due to a mismatch between oxygen supply and demand in the muscles during intense physical activity.

18-5. The ECAST is a relatively rare occurrence, and most individuals with SCT can engage in regular physical activities without any issues. However, to minimize the risk of complications, individuals with SCT are advised to take precautions during exercise, such as staying well-hydrated, avoiding excessive heat or cold exposure, and gradually increasing the intensity and duration of physical activity. It is also recommended to consult with a health care professional for personalized guidance and address any concerns related to exercise.

18-6. The ECAST, as a cause of collapse in Service members, is considered because the relative risk of death is significantly higher in Service members with SCT than those without. Early recognition and proper treatment of ECAST can increase survival and decrease complications. For more information on ECAST, refer to the American Medical Society for Sports Medicine website.

RISK FACTORS

18-7. While the exact causes of ECAST are not fully understood, there are several risk factors that can contribute to its occurrence. These include:

- Strenuous exercise: Engaging in high-intensity or prolonged physical activity can increase the risk of ECAST.
- Dehydration: Inadequate fluid intake or excessive fluid loss through sweating can contribute to the risk of ECAST.
- Heat and humidity: Exercising in hot and humid environments can further exacerbate the risk of ECAST.

18-8. Preexisting medical conditions and environmental factors that may increase the likelihood of an ECAST episode include:

- Poor Hydration.
- Exercise-induced asthma.
- Dietary supplements contain stimulants.
- Excessive heat exposure.

- High altitude.
- High humidity.
- Inadequate rest between intense exercise periods (for example, prolonged wind sprints).
- Intense exercise for an extended period.
- Pre-exercise fatigue due to illness.
- Returning from prolonged leave after little physical activity.
- Inadequate acclimatization to environment.

SIGNS AND SYMPTOMS

18-9. Service members experiencing ECAST may have—

- Dizziness or lightheadedness.
- Weakness or fatigue.
- Nausea or vomiting.
- Shortness of breath.
- Muscle pain or cramping.
- Paleness or cyanosis (bluish discoloration) of the skin.
- Collapse or loss of consciousness (rare cases).
- Increased pain and weakness in the muscles, especially the large muscles of the legs, buttock(s), and low back.
- Muscle tenderness.
- Slumping to the ground.
- Rapid breathing.
- Slightly elevated body temperature (usually less than 103° Fahrenheit).

18-10. Some of the differences between the signs and symptoms of ECAST and other causes of exercise related collapse include:

- Service members with ECAST will have soft, weak muscles while Service members with muscle cramps will have firm, tight muscles.
- Service members with cardiac issues experiencing an ECAST event may not be able to speak or think normally.
- Service members with ECAST will have a slightly elevated body temperature while Service members experiencing exertional heat stroke will have a core body temperature above 104° Fahrenheit.
- Service members with ECAST will usually be able to speak and think normally, while Service members suffering from heat illness may appear to be confused.

18-11. If someone experiences symptoms of ECAST, the following first aid measures can be taken:

- Stop the activity: Immediately cease any physical exertion and allow the person to rest.
- Move to a cool and shaded area: Transfer the individual to a cooler environment to minimize heat stress.
- Rehydrate: Encourage the person to drink fluids to rehydrate, preferably water or sports drinks.
- Elevate the legs: If the person is conscious and able to do so, elevate their legs to improve blood flow and circulation.
- Monitor vital signs: Keep an eye on the person's breathing, heart rate, and level of consciousness. Seek medical assistance if the symptoms worsen or persist.

PREVENTIVE MEASURES

18-12. It is crucial to note that ECAST is a relatively rare occurrence, and most individuals with SCT can engage in regular physical activities without complications. However, taking preventive measures, such as staying well-hydrated, avoiding excessive heat exposure, and gradually increasing exercise intensity can help reduce the risk of ECAST. If an individual with SCT experiences recurrent or severe episodes of ECAST, it is recommended to consult with a health care professional who specializes in SCD for further evaluation and personalized guidance.

18-13. Early identification of SCT in Service members is needed to prevent ECAST this is done with a blood test. Defense Health Agency Procedural Instruction (DHA-PI) 6025.14, *Active Duty Service member (ADSM) Erythrocyte Glucose-6-Phosphate Dehydrogenase (G6PD) Deficiency and Sickle Cell Trait Screening*; DODI 6465.01, *Erythrocyte Glucose-6-Phosphate Dehydrogenase Deficiency, (G6PD) and Sickle Cell Trait Screening Programs*; and All Army Activities 094/2020, *Implementation of Army Sickle Cell Trait Screening of All Soldiers* establish the Department of Defense's, Defense Health Agency's, and the Army's procedures to implement erythrocyte G6PD deficiency and SCT screening at—

- Entry to the Military Services (at appointment, enlistment, or induction).
- Pre-deployment.
- Other points as indicated for validation of results.

18-14. Safe exercise practices that decrease the risk of ECAST include:

- Participate in exercise and conditioning activities year-round to maintain physical fitness.
- Gradually increase exercise intensity when starting a training regimen or returning to training after an absence, such as illness, injury, or extended leave.
- Take adequate recovery time between repetitive sprints or drills (for example, wind sprints or intervals).
- Acclimate to the altitude or temperature at which training will occur.
- Exercise for shorter periods and with less intensity during hot or humid conditions.
- Hydrate before, during, and after exercise.
- Understand the signs and symptoms of ECAST. If any symptoms arise, report immediately to the Holistic Health and Fitness staff, unit leaders, or health care personnel.

DUTIES AND RESPONSIBILITIES

18-15. Preventing ECAST is a primary and critical responsibility within the military, given the demanding physical exertion and operational requirements faced by Service members. ECAST can have severe consequences on both the health and operational readiness of military personnel. Recognizing the significance of these risks, the military assumes a proactive stance in fulfilling its responsibilities. This commitment involves the identification of command, personal, medical and first responder responsibilities.

COMMAND RESPONSIBILITIES

18-16. Commanders ensure that—

- Every Service member in their command has been screened for SCT.
- Every Service member with a positive SCT screening is issued and wears a red medical warning tag and receives a formal counseling from a unit or MTF, and health care provider to discuss their SCT-positive result.
- Service members receive educational information and counseling that includes:
 - Erythrocyte G6PD and SCT deficiency diagnosis, causes, complications, treatment, and strategies to mitigate symptoms.
 - Materials and resources that enable the Service member to self-identify symptoms experienced during exercise and know when to seek immediate medical attention.
- Service members without record of an SCT screening is referred to a lab for screening unless the Service member can provide documentation of a prior screening. The date and results of the SCT screening are annotated in the Army Medical Protection System and recorded in the Service member's electronic medical record.
- Unit or MTF health care providers—
 - Train all unit medical personnel to recognize the signs of ECAST.
 - Train unit medical personnel and leaders on actions to take during an ECAST event.
 - Conduct an initial patient evaluation, provide treatment, and determine whether to call for urgent MEDEVAC, or immediately transport the individual to the nearest MTF.
- Discrete means of identifying SCT positive Service members are developed and added to the unit tactical SOP. Identification is like those used to identify Service members who have sustained cold and or heat related injuries.

SERVICE MEMBER RESPONSIBILITIES

- 18-17. Service members with a positive SCT screen are responsible for—
- Obtaining and wearing their red medical warning tags and unit SOP directed markings.
 - Seeking formal counseling includes educational materials and resources to aid in their ability to self-identify symptoms experienced during exercise and know when to seek immediate medical attention.
 - Gradually increasing exercise intensity when starting a training regimen or returning to training after an absence, such as an illness or leave.
 - Taking adequate time to recover between repetitive sprints or drills (for example, wind sprints or intervals).
- 18-18. Appropriate casualty response measures for suspected ECAST are to—
- Identify if the casualty has SCT (looking for a red identification tag or unit specific marking).
 - Check the casualty's pulse, breathing, and mental status.
 - Remove the casualty from the environment.
 - Cool the casualty (as indicated by weather conditions).
 - Notify nearest medical personnel to assess the casualty for a suspected ECAST event.
 - Call for urgent MEDEVAC or immediately transport the casualty to the nearest MTF.
 - Notify the receiving MTF that the Service member is suspected of experiencing an ECAST.

Note. Trained medical personnel should consider fluid and oxygen resuscitation.

18-19. Before returning to duty, Service members who have collapsed during exercise showed signs or symptoms of ECAST will (if not already completed) be tested for SCT. Tests for other diseases such as diabetes, electrolyte abnormalities, muscle damage, and heart and lung conditions may also be needed.

- 18-20. Service members will meet with a unit or MTF health care provider to discuss:
- The signs and symptoms of ECAST.
 - Family history.
 - Environmental factors.
 - Dietary supplements and medications that increase the risk of collapse.
 - The importance of good hydration.
 - The possibility of future ECAST episodes when exercising strenuously in high heat and humidity.

18-21. Service members must have no symptoms at rest and normal lab values before resuming exercise or training. The training must resume slowly and under a doctor's supervision.

FIRST RESPONDER, STRETCHER BEARER, MEDIC, CORPSMAN, AND LEADER TRAINING

18-22. The unit or MTF healthcare providers are tasked with training both first responders and medical personnel, such as combat medics or corpsmen, in the vital competencies of recognizing signs of ECAST. This training encompasses conducting rapid initial evaluations, providing immediate treatment, and making critical decisions regarding the need for urgent MEDEVAC or direct transport of the casualty to the nearest MTF.

18-23. Additionally, these healthcare providers also extend their training to unit leaders, equipping them with the necessary skills to identify the initial indicators of ECAST. This preparation is crucial for leaders to make informed decisions about whether to initiate an urgent MEDEVAC or to expedite the transportation of the affected individual to the closest medical facility, ensuring prompt and effective care.

Chapter 19

First Aid in a Chemical, Biological, Radiological, and Nuclear Environment

GENERAL INFORMATION FOR FIRST AID IN A CHEMICAL, BIOLOGICAL, RADIOLOGICAL, AND NUCLEAR ENVIRONMENT

19-1. Planning for large-scale combat operations must account for possible operations in a CBRN environment and allied response options. The threat of CBRN hazards can rapidly degrade unit effectiveness by forcing troops to wear protective clothing and by creating confusion and fear. Through training in CBRN protective and preventive procedures and first aid, units can maintain the desired level of combat effectiveness to meet mission requirements.

19-2. Because of the differences in presentation for CBRN casualties, specific triage and treatment recommendations along with evacuation priority considerations are addressed in the CPGs located on the deployed medicine website.

19-3. CBRN exposure is an enemy attack, and security will be the first priority to be coordinated. Triage for a CBRN event follows established triage principles, including doing the best for most people. Because of the complexity of combining CBRN and trauma casualties, it may be beneficial to assign experienced medical personnel to triage. Triage occurs for decontamination, for medical treatment, and for evacuation. Although assigning casualties to ambulatory vs. non-ambulatory decontamination lanes is a method of sorting, the optimal triage for decontamination determines who can tolerate delays in decontamination.

19-4. Decontamination triage may be based on extent of exposure and not on initial symptoms, as symptoms may be delayed. Those with greater exposure may benefit from rapid decontamination to increase their survivability and thus reduce the future burden on medical resources. Once decontaminated, patients are triaged according to medical priority. Decontamination is a medical intervention (it minimizes or eliminates the conversion of an external dose to an internal dose). Any casualty within an area suspected to contain a liquid chemical agent is automatically medically triaged as immediate until immediate decontamination has been accomplished. The casualty is then re-triaged. Except for blister agent casualties, those exposed only to vapor can tolerate delays in decontamination or may not even require thorough decontamination.

19-5. It is important to frequently reassess patients. Because many CBRN hazards have delayed symptoms, a casualty initially assessed to be minimal may become immediate as the hazard presents itself and process of decontamination can significantly change the casualty status. Additionally, actions during decontamination can significantly change the casualty status. It is critically important to monitor hypothermia, disruption of previously controlled bleeding, and other changes in casualty status that would change triage priority.

CLASSIFICATION OF CHEMICAL AND BIOLOGICAL AGENTS

19-6. Chemical agents are classified based on their properties, effects, and intended use. There are different classification systems depending on the context, such as their use in warfare, industry, or transportation. One widely used classification system for chemical agents is based on their potential harm to human health and the environment. Common chemical agent categories are:

- Adversary use: Persistent agents intended to cause casualties for more than 24 hours or restrict terrain. Nonpersistent agents that can lose their ability to cause casualties after 10-15 minutes.
- Hazard classification: Explosiveness (Class 1), flammability (Classes 3,4), toxicity (Class 6).
- Physiological action: Nerve, blood, blister, choking, and incapacitating.
- Physical state: solid, liquid, or vapor.

19-7. Different regulatory bodies and organizations may have their own specific classification systems and criteria for chemical agents, depending on the context and intended use. These classifications help in

identifying and managing potential hazards, ensuring appropriate handling, storage, transportation, and emergency response measures are in place.

19-8. Biological agents, also known as bioagents or biological hazards, are classified based on their potential to cause harm to living organisms, including humans. They can be categorized in various ways, considering factors such as their origin, mode of transmission, pathogenicity, and potential for use as weapons. Common classification systems for biological agents are:

- **Risk Group Classification:** Biological agents can be classified into different risk groups based on their level of pathogenicity and the risk they pose to human health. This classification system is commonly used in research laboratories and health care settings. The risk groups range from Risk Group 1 (low individual and community risk) to Risk Group 4 (high individual and community risk). The classification is determined by considering factors such as infectivity, severity of disease, available preventive measures, and efficacy of treatment.
- **Biosafety Level Classification:** This classification system is primarily used in laboratory settings to ensure appropriate containment measures and safety protocols are followed when handling biological agents. It categorizes agents into different biosafety levels based on their risk to laboratory personnel and the environment. The biosafety levels range from biosafety levels 1 (lowest level of risk) to biosafety levels 4 (highest level of risk). The classification is determined by factors such as pathogenicity, ease of transmission, and availability of effective preventive measures and treatments.
- **The U.S. Centers for Disease Control (also known as CDC) and Prevention Category A, B, and C Classification:** Developed by the Centers for Disease Control, is focused on potential bioterrorism agents. Biological agents are classified into three categories:
 - **Category A:** Agents that pose the highest risk to national security and public health. They have the potential for widespread dissemination, high mortality rates, and significant public panic. Examples include anthrax, smallpox, and Ebola virus.
 - **Category B:** Agents that are moderately easy to disseminate and cause moderate morbidity and low mortality rates. Examples include Q fever, Brucellosis, and ricin toxin.
 - **Category C:** Agents that have the potential for emerging threats and can be engineered for mass dissemination. Examples include Nipah virus, hantavirus, and emerging strains of influenza.

19-9. These classification systems help in identifying and managing biological hazards, implementing appropriate biosafety and biosecurity measures, developing effective response plans, and conducting research and surveillance activities. They are essential tools in ensuring the safe handling, containment, and prevention of the spread of biological agents. The classification of biological warfare (also known as BW) agents can vary slightly depending on the organization or country using the classification system. These classifications help guide efforts in preparedness, response planning, research, surveillance, and the development of countermeasures against potential biological warfare threats.

WARNING

Swallowing water or food contaminated with some chemical agents (i.e. nerve, blisters), and with some biological agents can be fatal.

CONDITIONS FOR MASKING WITHOUT ORDER OR ALARM

19-10. Once an attack with a chemical or biological agent is detected or suspected, you must STOP BREATHING and mask immediately. DO NOT WAIT to receive an order or alarm if you observe any of these indicators:

- Artillery missiles, rockets, unmanned ariel vehicles or other delivery systems that produce vapors, smoke, mists, and aerial sprays, near your position.
- Smoking or vaporizing from an unknown source is present or approaching.
- A suspicious odor, liquid, or solid is present.
- You are entering an area known or suspected of being contaminated.

- Casualties from an area where chemical or biological agents have reportedly been used enter your area.
- You have one or more of the following symptoms:
 - An unexplained runny nose.
 - A sudden unexplained headache.
 - A feeling of choking or tightness in the chest or throat.
 - Dimness of vision.
 - Irritation of the eyes.
 - Difficulty in or increased rate of breathing without obvious reasons.
 - Sudden feeling of depression.
 - Dread, anxiety, or restlessness.
 - Dizziness or light-headedness.
 - Slurred speech.
- Unexplained laughter or unusual behavior is noted in others.
- Numerous unexplained ill personnel.
- Service members collapsing without evident cause.
- Animals or birds exhibiting unusual behavior or suddenly dying.

19-11. For further information on individual protective measures, masking procedures, and other tasks related to surviving and operating in a CBRN environment, refer to ATP 3-11.32, ATP 4-02.7, ATP 4-02.84, and ATP 4-02.85.

FIRST AID IN A CHEMICAL ENVIRONMENT

19-12. Assessment and management of CBRN casualties is complex because it involves not only life-threatening symptoms (some of them have rapid onset) but also the threat of contamination of first responders. An added source of complexity is the combination of CBRN and penetrating trauma injuries. This complexity requires a stepwise approach that first rules in or rules out CBRN injuries so first responders can protect themselves before rendering aid. The next step integrates assessment and management of both CBRN and traumatic injuries, prioritizing the management of life-threats first. Chemical agents may be inhaled or ingested. Chemical agents are ingested when food or water contaminated by the agent is consumed or absorbed when the agent is exposed to the skin or eyes.

19-13. Service member's field protective mask and chemical protective clothing provide good protection from most chemical agents. The Skin Decontamination Kits are used to remove liquid chemical agents from exposed skin. The Individual Equipment Decontamination Kit is used to decontaminate personal equipment. Medications contained in the antidote treatment – nerve agent, auto-injector (ATNAA) kit help to counteract nerve agent poisoning. The effects of the antidote can be enhanced by taking pyridostigmine bromide tablets prior to being exposed to the nerve agent.

19-14. The field protective mask gives protection against biological and chemical warfare agents as well as radiological fallout. Service members should be able to do their field protective masks in nine seconds or less without the hood, and within 15 seconds with the hood.

19-15. The steps to donning a field protective mask in response to a chemical attack are to: stop breathing, put on (don) the mask, seal it properly, clear, and check it; resume breathing, give the alarm, and continue the mission. Keep your mask on until the "all clear" signal has been given.

Note. Keep your protective mask on until the area is no longer hazardous, and you are told to unmask.

19-16. If you begin to experience symptoms of nerve agent poisoning, immediately give yourself one ATNAA.

CAUTION

Do not inject a nerve agent antidote until you are sure you need it.

19-17. If your eyes and face become contaminated, you must immediately begin decontamination. You should try to get under cover before beginning immediate decontamination, but do not delay immediate decontamination to search for overhead cover. You need shelter to prevent further contamination while performing decontamination procedures on your face. If no overhead cover is available, put your poncho over your head before beginning the decontamination process. Then, you put on the remaining protective clothing. If vomiting occurs, the mask should be lifted momentarily and drained—with your eyes closed and while holding your breath—then replace, clear, and seal.

19-18. Decontaminate your skin, immediately, and clothing and equipment as soon as the mission permits.

19-19. You may come upon an unconscious chemical agent casualty who requires first aid. You should be able to successfully—

- Mask the unconscious casualty if they are unmasked.
- Inject them, if necessary, with all their auto injectors.
- Decontaminate their skin.
- Seek medical assistance.

CHEMICAL BIOLOGICAL RADIOLOGICAL NUCLEAR CASUALTY ASSESSMENT

19-20. Just as trauma casualties present differently (blunt, trauma, GSW, blasts, etc.), CBRN casualties also have varied presentations. CBRN casualties can be categorized by circumstances of exposure and presence or absence of trauma and CBRN effects. The acronym CRESS (Consciousness, Respirations, Eyes, Secretions, Skin) is employed to improve rapid identification of the type of chemical agent exposure.

- C - Consciousness (unconscious, seizures, depressed consciousness, agitation).
- R - Respirations (present or absent, labored, increased or decreased).
- E - Eyes (constricted, dilated, normal).
- S - Secretions (dry, normal, increased).
- S - Skin (diaphoretic, dry, hot, cyanosis).

Note. After using the CRESS algorithm to rule out or rule in a CBRN injury, and first responders doing PPE if present, the integrated assessment and management of TCCC and CBRN injuries can proceed. Additionally, NATO STANAG 2358 and *Joint Trauma System Clinical Practice Guide* contain additional details, figures, and tables.

19-21. The acronym (MARCHE)² which stands for (mask, antidotes, rapid spot decontamination, countermeasures, hypothermia prevention, extraction or evacuation) integrates the established TCCC MARCH algorithm with CBRN considerations. Combining these two approaches gives the acronym (MARCHE)² or “MARCHE-squared.” Like TCCC, (MARCHE)² can be broken into phases of care from highest threat (care under fire or active CBRN threat), to intermediate threat (tactical field care or warm zone casualty management), to lowest threat (PFC and or evacuation or care of decontaminated casualties in cold zone).

- M - Massive Hemorrhage.
- A - Airway.
- R - Respirations.
- C - Circulation.
- H - Hypothermia.
- M - Mask.
- A - Antidotes.
- R - Rapid spot decontamination.
- C - Countermeasures.

- H - Hypothermia Prevention.
- E - Extraction and Evacuation.

Note. For CASEVAC in a contaminated environment refer to ATP 3-11.33.

FIRST AID EQUIPMENT IN A CHEMICAL ENVIRONMENT

19-22. Service members may be issued the following materials to decontaminate and use as first aid for CBRN exposure. Service members must know how and when to use the items. Based on intelligence or known threats, each Service member may be issued the relevant medical CBRN defense materiel (table 19-1 on page 193).

Table 19-1. Initial Issue Medical Chemical, Biological, Radiological, and Nuclear Defense Materiel

<i>Nomenclature</i>	<i>Basis of Issue per Service member</i>
Antidote Treatment – Nerve Treatment Nerve Agent, Auto-Injector (ATNAA).	3 each.
Convulsant Antidote Nerve Agent (CANA) or Advanced Anticonvulsant System (AAS).	1 each.
Doxycycline 100mg tablets or Ciprofloxacin 500mg tablets.	15 days of supply or (bottle of 30 tablets).
Pyridostigmine Bromide Tablets 30mg. <i>Note:</i> Not issued unless expressly authorized by Office of The Surgeon General (OTSG).	42 tablets (on 2 blister packs).
Potassium Iodine tablets, (KI). <i>Note:</i> Not issued unless expressly authorized by Office of The Surgeon General (OTSG).	14 tablets on 1 strip.
Reactive Skin Decontamination Lotion (RSDL).	1 pouch of 3 packets.
Individual Soldier's Guide to Medical CBRN Defense Materiel.	1 each.
Legend: AAS advanced anticonvulsant system mg milligrams ATNAA antidote treatment – nerve treatment nerve agent, auto – injector OTSG office of the surgeon general CANA convulsant antidote nerve agent RSDL reactive skin decontamination lotion CBRN chemical, biological, radiological, and nuclear KI potassium iodine	

NERVE AGENTS

19-23. Nerve agents are among the deadliest chemical agents. They can be delivered by artillery shell, mortar shell, rocket, aircraft bomb, spray, or land mine. In general, nerve agents are colorless, odorless, and tasteless. There are multiple signs and symptoms of nerve agent poisoning, which can vary from mild to severe.

19-24. Nerve agents enter the body by inhalation, ingestion, and through the skin. Depending on the route of entry and the amount, nerve agents can produce injury or death within minutes. Nerve agents can achieve their effects with small amounts. Nerve Agent Pyridostigmine Pretreatment (NAPP) provides an effective pre-treatment for all Soldiers before exposure to nerve agents.

19-25. Service members will be issued the NAPP. NAPP tablet set (figure 19-2 on page 194) when the corps, division, or wing command determines there is a need. The instructions on the proper use of pyridostigmine will be provided at the time of issue. Additionally, instructions are on the tablet set. Although the pyridostigmine itself does not provide protection against nerve agents, it does enhance the efficiency of the medications contained in the ATNAA.

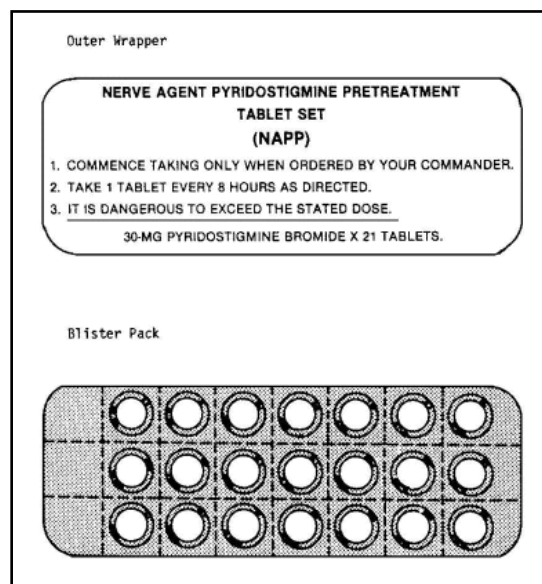


Figure 19-1. Nerve Agent Pyridostigmine Pretreatment Tablet Set

19-26. To be effective the NAPP Tablet Set must be taken within eight hours prior to exposure to nerve agents. Since the specific time of exposure is difficult to predict, units should assess the threat and direct Soldiers to initiate NAPP before they could be exposed. The NAPP consists of a blister pack containing 21 tablets to be taken over seven days. Each tablet contains 30mg pyridostigmine bromide. If a dose is missed, do not make it up. Do not take two tablets at one time because of a missed dose; merely start again with one tablet every eight hours. Taking more than one tablet at a time DOES NOT provide additional protection-in fact, it may be more hazardous if there is exposure to a nerve agent.

19-27. Service members are initially issued one NAPP when the chemical biological protective ensemble is expected to be donned. You are responsible for carrying out the NAPP and safeguarding it against loss. Service members will secure the blister pack in the sleeve or breast pocket of the chemical biological protective ensemble (or in another part of the ensemble, as directed by local SOP).

19-28. When the order to take pyridostigmine pretreatment is given, it should be taken as directed even if the protective mask is worn.

Note. Commanders must follow protocols for pyridostigmine bromide issuance and use.

WARNING

Pyridostigmine bromide is for pre-exposure prophylaxis only. Do not give to any patient who has been exposed to, presents, or has symptoms of organophosphate nerve agent poisoning.

19-29. Although no detrimental effects are expected at the recommended dosage, depending on the length of time and the amount of medication taken as well as individual physiological variations, some individuals may have contraindications for taking pyridostigmine bromide, while others may experience adverse reactions.

19-30. Signs and symptoms of overdose, adverse reactions, or side effects are:

- Abdominal cramps.
- Nausea and vomiting.
- Diarrhea.

- Blurred vision.
- Increased bronchial secretions.
- Cardiac arrhythmias and hypertension.
- Weaknesses, muscle cramps, and muscular twitching.
- Skin rash.

19-31. Since pyridostigmine bromide may increase bronchial secretions and aggravate bronchiolar constriction, caution should be used in its administration to personnel with bronchial asthma.

19-32. If any of the above signs or symptoms are present, medical personnel must be consulted as soon as possible.

Signs and Symptoms of Nerve Agent Poisoning

19-33. The signs and symptoms of nerve agent poisoning are grouped as MILD—those that you recognize and for which you can perform self-aid, and SEVERE—those that require buddy aid.

19-34. The signs and symptoms experienced after nerve agent exposure (table 19-2 on page 196) will determine which and how many auto injectors can be given to a Service member. Fourth generation agents and opioid exposures have very similar signs and symptoms and should be considered if the nerve agent treatment does not reverse or diminish signs and symptoms as expected after auto injector use.

Table 19-2. Nerve Agent Exposure, Signs and Symptoms, and Medical Countermeasures

Severity	Signs and Symptoms	Treatment
Mild	Unexplained runny nose. Unexplained sudden headache. Sudden drooling. Difficulty in seeing (dimness of vision and pinpoint pupils). Tightness in the chest or difficulty in breathing. Wheezing and coughing. Localized sweating and muscular twitching around the contaminated skin. Stomach cramps. Nausea with or without vomiting. Increased heart rate followed by a decreased heart rate.	Self-aid 1 - Antidote Treatment – Nerve Agent, and Auto Injector.
Severe	Confused behavior. Increased wheezing and increased dyspnea (difficulty in breathing). Temporarily stop breathing. Severely pinpoint pupils. Red eyes with tearing. Vomiting. Severe muscular twitching and general weakness. Involuntary urination and defecation. Convulsions. Unconsciousness. Respiratory failure. Increased or decreased heart rate.	Buddy aid 3 - Antidote Treatment – Nerve Agent, and Auto Injector. 1 - Convulsant Antidote – Nerve Agent. Note. Casualties with severe symptoms may be able to treat themselves and must receive prompt buddy aid, combat lifesaver aid, and prompt follow-on medical treatment if they are to survive.

First Aid for Nerve Agent Poisoning

19-35. Nerve agents are absorbed rapidly, and the effects are felt immediately upon entry into the body. You will be issued three ATNAAs and one convulsant antidote nerve agent (CANA) or advanced anticonvulsant system (AAS). Each ATNAA consists of a multi-chambered auto injector with the atropine and pralidoxime chloride in separate chambers. You will use the ATNAA in self-aid and buddy aid as issued.

19-36. If nerve agents are used, mission permitting, watch for persons needing nerve agent antidotes and immediately follow procedures. When you have the signs and symptoms of nerve agent poisoning, you should immediately put on the protective mask and then inject yourself with one set of the ATNAA. First aid for nerve agent poisoning consists of administering the ATNAA, CANA, or AAS.

Convulsant Antidote for Nerve Agent and Advanced Anticonvulsant System

19-37. Two anticonvulsant autoinjectors are available to control actively seizing patients. CANA (figure 19-2 on page 197) is the drug diazepam. AAS (figure 19-3 on page 197) is the drug midazolam. CANA and AAS are not intended for self-administration.

19-38. Diazepam or midazolam should be administered with the three ATNAA kits when the casualty's condition warrants the use of three ATNAA kits at the same time. Use either the CANA or the AAS to control an actively seizing patient. Do not administer both for the same series of three ATNAA kits.

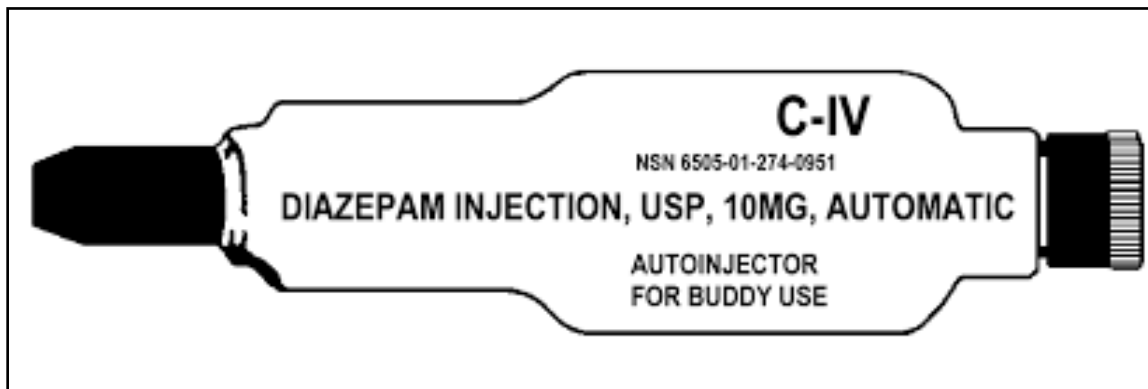


Figure 19-2. Convulsant Antidote Nerve Agent



Figure 19-3. Advanced Anticonvulsant System

Antidote Treatment – Nerve Agent, Auto-Injector

19-39. The ATNAA (figure 19-4) is a multi-chambered device with the atropine and pralidoxime chloride in separate chambers. Both antidotes will be administered through a single needle. The ATNAA provides atropine injections and pralidoxime chloride injections in separate chambers as sterile, pyrogen-free solutions for intramuscular injection.

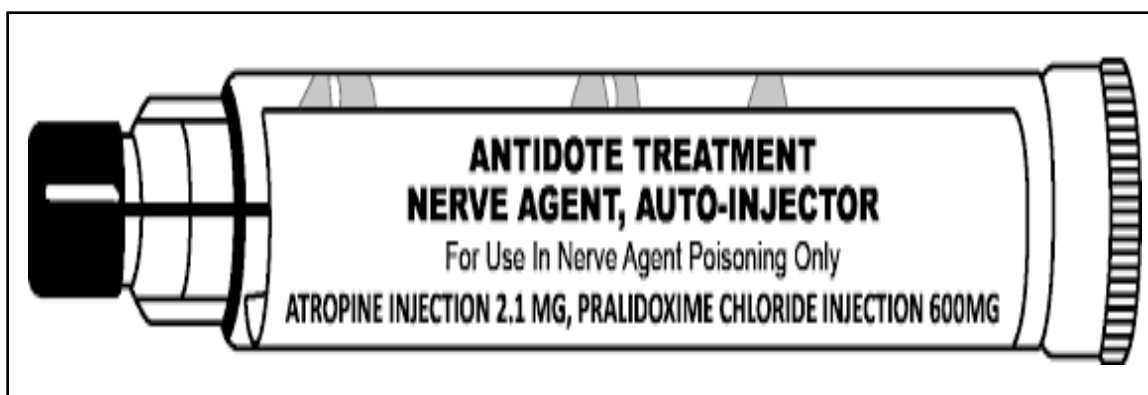


Figure 19-4. Antidote Treatment – Nerve Agent, Auto-Injector

19-40. The ATNAA is a specially designed unit for automatic self-aid or buddy aid administration by military personnel. When activated, the ATNAA sequentially administers atropine and pralidoxime chloride through a single needle. The recommended procedure is to inject the contents of the auto-injector into the muscles of an outer thigh or into the buttocks. When activated, each ATNAA dispenses:

- 2.1mg atropine in 0.7 milliliter of a sterile, pyrogen-free solution containing 12.47mg glycerin and not more than 2.8mg of phenol, citrate buffer, and water for injection. The potential hydrogen (pH) range is 4.0 – 5.0.

- 600mg of pralidoxime chloride in 2 milliliters of a sterile, pyrogen-free solution containing 40mg benzyl alcohol, 22.5mg glycine and water for injection. The pH is adjusted with hydrochloric acid. The pH range is 2.0 – 3.0.

19-41. The ATNAA should be self-aid or buddy aid administered by military personnel after donning a protective mask and hood but only if some or all the following mild symptoms of nerve agent exposure are present.

Reactive Skin Decontamination Lotion Kit

19-42. The RSDL kit (figure 19-5) is intended to remove or neutralize chemical warfare agents from the skin. The RSDL contains Dekon 139 and a small amount of 2,3-butanedione monoxime. These compounds are dissolved in a solvent composed of polyethylene glycol monomethyl ether and water. This solvent system is particularly important as it promotes the decontamination reaction by actively desorbing, retaining, and sequestering the chemical agent, while the active ingredient (Dekon 139) chemically reacts with, and rapidly neutralizes the vesicant chemical or the organophosphorus nerve agent. This reaction starts immediately, and neutralization is usually complete within two minutes.



Figure 19-5. Reactive Skin Decontamination Lotion Kit

19-43. The RSDL kit is the preferred method of decontamination. If a RSDL kit is not available soap and large amounts of water can be used for decontamination. Although soap and a large amount of water may be effective in physically removing chemical warfare agents, they do not neutralize chemical warfare agents. Soap and water can increase dermal absorption of mustard agents and can result in contaminated runoff that can affect others.

WARNING

For external use only. May be slightly irritating to the eyes. Keep decontaminating powder out of the eyes. Use water to wash the toxic agent out of the eyes.

Injection Site

19-44. The injection site for administering the antidotes is normally in the outer thigh muscle. The thigh injection site is the area about a hand's width below the hip joint (figure 19-6 step A on page 199). It is important that the injection be given into a large muscle area. If the individual is thinly built, then the injections should be administered into the upper outer quarter (quadrant) of the buttock (figure 19-6 step B on page 199). Injecting in the buttocks of a thinly built individual avoids injury to the thighbone.

WARNING

There is a nerve that crosses the buttocks; hitting this nerve can cause paralysis.

Service members that are thinly built, should inject themselves in the upper outer quadrant of the buttock to avoid this nerve.

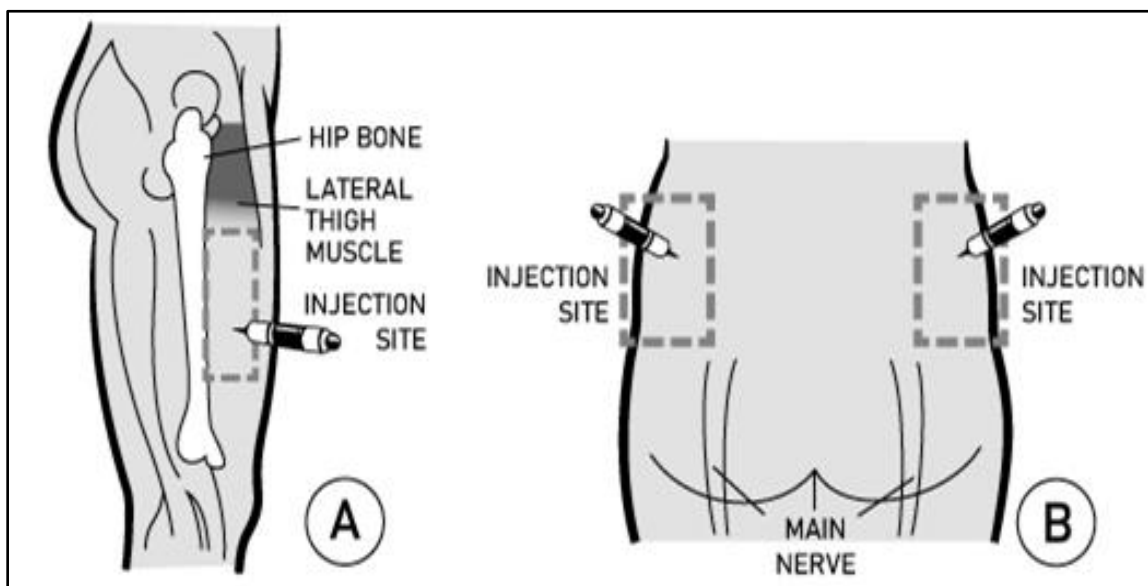


Figure 19-6. Injection site locations

19-45. If experiencing any or all of the MILD nerve agent symptoms, IMMEDIATELY self-administer one ATNAA kit and put on your protective mask. Follow the procedure given in table 19-3 on page 200. The ATNAA is carried in your protective mask carrier, pocket of the Mission-Oriented Protective Posture over garment, or other location as specified in your unit tactical SOP. In cold weather, the ATNAA should be stored in an inside pocket of the Service member's clothing to protect the antidote from freezing. A frozen ATNAA cannot be immediately used to provide the antidote, when needed; however, the ATNAA can still be used after complete thawing.

CAUTION

DO NOT cover or hold the needle end with your hand, thumb, or fingers—you might accidentally inject yourself. An accidental injection into the hand WILL NOT deliver an effective dose of the antidote, especially if the needle goes through the hand.

19-46. DO NOT self-administer another set of injections. If the Service member can walk without assistance, know who they are, and where they are, they may not need a second set of injections. Administering not needed additional injections may create a nerve-agent-antidote overdose, which could cause incapacitation or the inability to perform the mission or defend themselves.

Self-Administer Antidote Treatment – Nerve Agent, Auto Injector

19-47. If you experience any or all of the nerve agent MILD symptoms, you must IMMEDIATELY self-administer one ATNAA following the procedures given in table 19-3 on page 200.

Table 19-3. Self-Aid for Nerve Agent Poisoning

Antidote Treatment – Nerve Agent Autoinjector			
Step 1. Obtain one ATNAA.			
Step 2. Check injection site.			
Step 3. Hold ATNAA with dominant hand.			
Step 4. Grasp safety cap with non-dominant hand and remove from injector.			
Step 5. Clear hard objects from injection site.			
Step 6. Inject ATNAA at injection site applying even pressure to the injector. Hold in place for 10 seconds.			
Step 7. Bend needle of used injector by pressing on a hard surface to form a hook.			
Step 8. Attach used injector to blouse pocket flap of BDO and or JSLIST.			
Step 9. Massage injection site, mission permitting.			
Legend:			
ATNAA	antidote treatment – nerve treatment nerve agent, auto-injector	JSLIST	joint service lightweight integrated suit technology
BDO	battle dress overgarment		

Note. If you continue to have symptoms of nerve agent poisoning, seek someone else to check your symptoms and administer your remaining sets of injections, if required.

Buddy Evaluation and Buddy Aid

19-48. Service members may seek assistance after self-aid (self-administering) one ATNAA or may become incapacitated after self-aid. A teammate must evaluate the individual to determine if additional antidotes are required to counter the effects of the nerve agent. In addition, Service members may experience SEVERE symptoms of nerve agent poisoning; they will not be able to treat themselves. In either case, other Service members must perform buddy aid as quickly as possible. Before initiating buddy aid, determine if one set of MARK I auto injectors has already been used so that no more than three sets of the antidote are administered. Buddy aid also includes administering the CANA or AAS with the third ATNAA to prevent convulsions. Follow the procedures indicated in table 19-4 on page 201.

WARNING

Squat, DO NOT kneel, when masking the casualty or administering the nerve agent antidote to the casualty. Kneeling may force the chemical agent into or through your protective clothing.

CAUTION

DO NOT use your own ATNAA, CANA, or AAS on a casualty. If you use your own, you may not have any antidote if needed for self-aid.

WARNING

DO NOT inject into areas close to the hip, knee, or thighbone.

Table 19-4. Buddy Aid and Combat Lifesaver Aid for Nerve Agent Casualty

ATNAA	CANA or AAS
Step 1. Mask the casualty and position them on their side (swimmer's position).	Step 1. Obtain buddy's CANA or AAS.
Step 2. Position yourself near the casualty's thigh.	Step 2. Check injection site.
Step 3. Obtain buddy's three or remaining ATNAAs.	Step 3. Hold CANA or AAS in a closed fist with dominant hand.
Step 4. Check injection site.	Step 4. Grasp safety cap with non-dominant hand and remove from injector.
Step 5. Hold ATNAA in a closed fist with dominant hand.	Step 5. Clear hard objects from injection site.
Step 6. Grasp safety cap with non-dominant hand and remove from injector.	Step 6. Inject CANA or AAS at injection site by applying even pressure to the injector, not a jabbing motion. Hold in place for 10 seconds.
Step 7. Clear hard objects from injection site.	Step 7. Bend needle of injector by pressing on a hard surface to form a hook.
Step 8. Inject ATNAA at injection site by applying even pressure to the injector, not a jabbing motion. Hold in place for 10 seconds.	Step 8. Attach used injector to blouse pocket flap of BDO or JSLIST.
Step 9. Bend needle of injector by pressing on a hard surface to form a hook.	Step 9. Massage injection site, mission permitting.
Step 10. Attach all used injectors to blouse pocket flap of BDO or JSLIST.	
Step 11. Massage injection site, mission permitting.	
Legend: AAS advanced anticonvulsant system CANA convulsant antidote nerve agent ATNAA antidote treatment – nerve treatment nerve agent, auto injector JSLIST joint service lightweight integrated suit technology BDO battle dress overgarment	

19-49. Service members may seek assistance after self-aid (self-administering) one ATNAA or may become incapacitated after self-aid. A Service member must evaluate the individual to determine if additional antidotes are required to counter the effects of the nerve agent. In addition, Service members may experience SEVERE symptoms of nerve agent poisoning and will not be able to treat themselves. In either case, other Service members must perform buddy aid as quickly as possible. Before initiating buddy aid, determine if one ATNAA has already been used so that no more than three ATNAA are administered. Buddy aid also includes administering the CANA or AAS with the third ATNAA to prevent convulsions. Follow the procedures indicated in table 19-4.

WARNING

Squat, DO NOT kneel, when masking the casualty or administering the nerve agent antidotes to the casualty. Kneeling may force any chemical agent on your over garment into or through your protective clothing.

Note. If the casualty is thinly built, inject the antidote into the buttocks (figure 19-7 on page 202). Only inject the antidote into the upper outer portion of the casualty's buttocks. This avoids hitting the nerve that crosses the buttocks. Hitting this nerve can cause paralysis.

WARNING**DO NOT inject into areas close to the hip, knee, or thighbone.****Figure 19-7. Teammate injecting casualty's buttocks****Combat Lifesaver Assistance for Nerve Agent Poisoning**

19-50. The CLS must check to verify if the individual has received three sets of ATNAAs. If not, the CLS performs first aid as described for buddy aid above. If the individual has received the three sets of, ATNAA, the CLS may administer additional injections at approximately 15-minute intervals until atropinization is achieved (that is a heart rate above 90 beats per minute, reduced bronchial secretions, and reduced salivations). Administer additional atropines at intervals of 30 minutes to four hours to maintain atropinization or until the casualty is placed under the care of medical personnel. Check the heart rate by lifting the casualty's mask hood and feeling for a pulse at the carotid artery. Request medical assistance as soon as the tactical situation permits.

19-51. The CLS should administer additional CANA or AAS to casualties suffering from convulsions. Administer a second, and if needed, a third CANA or AAS at 5-to-10-minute intervals for a maximum of three injections (30mg diazepam or 30mg midazolam). Follow the steps and procedures described in paragraph 19-49 on page 201 for administering the CANA or AAS. DO NOT give more than two additional injections for a total of three (one self-aid plus two by the CLS).

BLISTER AGENTS

19-52. Blister agents (vesicants) include mustard, nitrogen mustards, lewisite, and other arsenicals, mixtures of mustards and arsenicals, and phosgene oxime. Blister agents may act on the eyes, mucous membranes, lungs, and skin. They burn and blister the skin or any other body parts they contact. Even relatively low doses may cause serious injury. Blister agents damage the respiratory tract (nose, sinuses, and windpipe) when inhaled and cause vomiting and diarrhea when absorbed. Lewisite and phosgene oxime cause immediate pain on contact. However, mustard agents are deceptive as there is little or no pain at the time of exposure. In some cases, signs of injury may not appear for several hours after exposure.

Protective Measures for Blister Agent Poisoning

19-53. Your protective mask with hood and protective over garment provides protection against blister agents. If it is known or suspected that blister agents are being used, STOP BREATHING, put on your mask and your protective over garment.

CAUTION

Large drops of liquid vesicants on the protective over garment ensemble may penetrate it if allowed to stand for an extended period. Remove large drops as soon as possible.

Signs and Symptoms of Blister Agent Poisoning

19-54. Immediate and intense pain upon contact with L, LH (lewisite and mustard) mixture, and phosgene oxime. No initial pain upon contact with mustard.

19-55. Inflammation and blisters (burns) resulting in tissue destruction. The severity of a chemical burn is directly related to the concentration of the agent and the duration of contact with the skin. The longer the agent is in contact with the tissue, the more serious the injury will be.

19-56. Vomiting and diarrhea. Exposure to high concentrations of vesicants may cause vomiting or diarrhea.

19-57. The blister agent vapors absorbed during ordinary field exposure will probably not cause enough internal body (systemic) damage to result in death. However, death may occur from prolonged exposure to high concentrations of vapor or from extensive liquid contamination over wide areas of the skin, particularly when decontamination is neglected or delayed.

First Aid Measures for Blister Agent Poisoning

19-58. Use your RSDL to decontaminate your skin and use water to flush contaminated eyes. Decontamination of vesicants must be done immediately (within one minute is best).

19-59. If blisters form, cover them loosely with a field dressing and secure the dressing.

CAUTION

Blisters are burns. DO NOT attempt to decontaminate the skin where blisters have formed, as the agent has already been absorbed.

19-60. If you receive blisters over a wide area of the body, you are considered seriously burned. Seek medical assistance immediately.

19-61. If vomiting occurs, the mask should be lifted momentarily and drained. Make sure, while doing this, the eyes are closed, and the breath is held. As soon as it is complete, replace, clear, and seal the mask.

19-62. Remember, if vomiting or diarrhea occurs after having been exposed to blister agents, seek medical assistance immediately.

CHOKING AGENTS (LUNG-DAMAGING AGENTS)

19-63. Chemical agents that attack lung tissue, primarily causing fluid buildup (pulmonary edema), are classified as choking agents (lung-damaging agents). This group includes phosgene, diphosgene (known as DP), chlorine (known as Cl), and chloropicrin (known as PS). Of these four agents, phosgene is the most dangerous and is more likely to be employed by the enemy in future conflict. Your protective mask gives adequate protection against choking agents.

Signs and Symptoms for Choking Agents

19-64. During and immediately after exposure to choking agents (depending on agent concentration and length of exposure), you may experience some or all the following signs and symptoms:

- Tears (lacrimation).
- Coughing.
- Choking.
- Tightness of chest.
- Nausea and vomiting.
- Headaches.

Self-Aid for Choking Agents

19-65. The protective mask should be put on immediately when any of the conditions described above exist. Another indication of a phosgene attack is an odor like newly mown hay; however, DO NOT rely upon odor as indication of a chemical attack.

19-66. If some phosgene is inhaled, normal combat duties should be continued unless there is difficulty in breathing, nausea, vomiting, or more than the usual shortness of breath during exertion. If any of the above symptoms occur and the mission permits, remain at quiet rest until MEDEVAC is accomplished.

Death from Choking Agents

19-67. With ordinary field exposure to choking agents, death will probably not occur. However, prolonged exposure to high concentrations of the vapor and neglect, or delay in masking can be fatal.

Cyanogen (Blood) Agents

19-68. Cyanogen agents interfere with proper oxygen utilization in the body. Hydrogen cyanide (AC) and cyanogen chloride are the primary agents in this group.

Protective Measures for Cyanogen Agents

19-69. The protective mask with a fresh filter gives adequate protection against field concentrations of cyanogen agent vapor. Protective over garments, as well as the mask, are needed when exposed to liquid AC hydrogen cyanide.

Signs and Symptoms for Cyanogen Agents

19-70. During and immediately after exposure to cyanogen agents (depending on agent concentration and length of exposure), you may experience some or all the following signs and symptoms:

- Tearing (lacrimation).
- Eye, nose, and throat irritation.
- Sudden stimulation of breathing (unable to hold breath).
- Nausea.
- Coughing.
- Tightness of chest.
- Headache.
- Light-headedness (dizziness).
- Unconsciousness.

First Aid for Cyanogen Agents

19-71. Hydrogen cyanide. During any chemical attack, if you get a sudden stimulation to breathe or detect an odor like bitter almonds, PUT ON YOUR MASK IMMEDIATELY. Speed is essential since this agent acts so rapidly that within a few seconds its effects will make it impossible for Service members to put on their mask by themselves. Stop breathing until the mask is on. This may be very difficult since the agent strongly stimulates respiration.

19-72. Cyanogen chloride. Put your mask on immediately if you experience any irritation of the eyes, nose, or throat. The nearest Service member should mask Service members who are unable to mask themselves.

CAUTION

Before attempting to help anyone else, make sure you appropriately mask yourself first.

Medical Assistance for Cyanogen Agents

19-73. If exposure to blood agents is suspected, immediate medical assistance is needed. Medical assistance for cyanogen agents is crucial because cyanogen is a highly toxic compound that poses significant health risks to humans. Cyanogen is a colorless gas that can be lethal when inhaled or when it encounters the skin or eyes.

INCAPACITATING AGENTS

19-74. An incapacitating agent is a chemical agent that produces temporary, disabling conditions, which persist for hours to days after exposure. Unlike riot control agents, which usually are momentary or fleeting in action, incapacitating agents have a persistent effect. It is likely that smoke-producing munitions or aerosols will disseminate such agents, thus making breathing their means of entry into the body. The protective mask is, therefore, essential.

19-75. There are no specific first aid measures to relieve the symptoms of incapacitating agents. Supportive first aid and physical restraint may be indicated. If the casualty is stuporous or comatose, be sure that respiration is unobstructed; then turn them on their side in case vomiting should occur. Complete cleansing of the skin with soap and water, this should be done as soon as possible; or RSDL can be used if washing is impossible. Remove weapons and other potentially harmful items from Service members who are suspected of having these symptoms. Harmful items include cigarettes, matches, medications, and small items that might be swallowed accidentally. Delirious (confused) persons have been known to attempt to eat items bearing only a superficial resemblance to food.

19-76. Incapacitating agents (anticholinergic drugs BZ type) may produce alarming dryness and coating of the lips and tongue; however, there is usually no danger of immediate dehydration. Fluids should be given sparingly, if at all, because of the danger of vomiting and because of the likelihood of temporary urinary retention due to paralysis of bladder muscles.

19-77. Immediate and vigorous cooling of the casualties is indicated if the body temperature is elevated and mucous membranes are dry, (as for heatstroke). The casualty can be cooled by spraying them with cool water and fanning; applying alcohol-soaked cloths and air circulation; and providing maximum exposure to air in a shaded area, along with maximum air circulation. Such cases are usually a result of anticholinergic poisoning.

CAUTION

DO NOT use ice for cooling the skin.

19-78. Rapid evacuation should be accomplished since medical treatment with the appropriate medication may be lifesaving. Reassurance and a firm, but friendly, attitude by individuals providing first aid will be beneficial if the casualty appears to comprehend what is being said. Conversation is a waste of time if the Service member is incoherent or cannot understand what is being said. In such cases, the less said, the better it is. These casualties will benefit more from prompt and vigorous restraint and evacuation to an MTF.

INCENDIARIES

19-79. Incendiaries can be grouped as white phosphorus (WP), thickened gasoline, metal, and oil and metal. You must learn to protect yourself against these incendiaries.

White Phosphorus

19-80. WP is primarily used as a smoke-producing agent but also serves an incendiary role for igniting field expedients and combustible materials. The burns caused by WP are typically multiple, deep, and vary in size. Once WP particles contact the skin or clothing, they will continue to burn until deprived of air, and they tend to adhere to any surface they contact, requiring physical removal.

19-81. In the event that WP particles ignite clothing, it's crucial to remove the contaminated clothing quickly to prevent the WP from burning through to the skin. If WP particles directly strike the skin, the immediate response should be to smother the flames using water, a wet cloth, or mud. It is important to note that since WP is soluble in oil, one should not use grease or oily substances to extinguish the flames as these can exacerbate the burning.

19-82. To prevent further combustion, keep the WP particles covered with wet material to exclude air. Removal of WP particles from the skin should be done cautiously using a wet cloth to brush them away and a knife, bayonet, stick, or other available tools to pick them out.

19-83. Once immediate first aid measures are taken, it is vital to seek medical assistance as soon as the mission allows us to address any burns or injuries caused by WP exposure properly.

Thickened Fuel Mixtures (Napalm)

19-84. Thickened fuel mixtures (napalm) tend to cling to clothing and body surfaces, thereby producing prolonged exposure and severe burns. The first aid for these burns is the same as for other heat burns. The heat and irritating gases given off by these combustible mixtures may cause lung damage, which must be treated by medical personnel.

Metal Incendiaries

19-85. Metal incendiaries pose special problems. Thermite particles on the skin should be immediately cooled with water and then removed. The first aid for these burns is the same as for other heat burns. Particles of magnesium on the skin burn quickly and deeply. Like other metal incendiaries, they must be removed. Ordinarily, medical personnel should do the complete removal of these particles as soon as possible. Immediate medical treatment is required.

Oil and Metal Incendiaries

19-86. Oil and metal incendiaries, upon contact with skin and clothing, produce effects similar to those caused by white phosphorus. These substances can cause severe burns that require immediate and proper first aid attention to prevent further damage and promote healing. For detailed guidance on administering first aid for burns resulting from exposure to incendiaries like oil and metal, see Chapter 12. This chapter provides comprehensive instructions on the steps to take in treating burns, including how to manage the injury before professional medical help is available.

FIRST AID IN A BIOLOGICAL ENVIRONMENT

19-87. Biological hazards can provide important advantages to adversaries who use them because of factors such as—

- Clandestine employment.
- Delayed onset of symptoms.
- Detection, identification, and verification difficulties.
- Agent persistence.
- Communicability.

19-88. Biological attacks can result in combat ineffectiveness by introducing disease-causing organisms such as bacteria or viruses into the force through aerosol release; food and water contamination; vectors such as ticks, fleas, mosquitoes, or body fluids from infected animals or humans. The occurrence of the signs and symptoms from a biological hazard may appear in a few minutes to several hours or days to weeks depending on the biological hazard, the Service member's susceptibility, and the amount of biological hazard that is inhaled, ingested, or deposited on the skin. Many biological warfare agents present the same signs and symptoms as the flu or common cold such as fever, cough, runny nose, and tiredness.

19-89. First aid or medical treatment is initiated after a disease is identified, depending on the seriousness of the disease. First aid measures are concerned with observable symptoms of the disease such as diarrhea or vomiting. For certain biological warfare hazards, Service members may be directed to take doxycycline or ciprofloxacin from the initial issue medical CBRN defense materiel.

TOXINS

19-90. Toxins have reportedly been used in past conflicts. Witnesses and survivors have described the employment of the T-2 mycotoxin as toxic rain (or yellow rain) because it was reported to have been released from aircraft as a yellow powder or liquid that covered ground, structures, vegetation, and people.

Signs and Symptoms of Toxin Poisoning

19-91. The occurrence of the symptoms from toxins may appear in a period of a few minutes to several hours depending on the toxin, the Service member's susceptibility, and the amount of toxin that is inhaled, ingested, or deposited on the skin. Symptoms from toxins usually involve the central nervous system but are often preceded by less prominent symptoms such as nausea, vomiting, diarrhea, cramps, stomach irritation, and burning sensation. Typical neurological symptoms often develop rapidly in severe cases; for example, visual disturbances, inability to swallow, speech difficulty, lack of muscle coordination, and sensory abnormalities (numbness of the mouth, throat, or extremities). Yellow rain (mycotoxins) also may have hemorrhagic symptoms, which could include any or all of the following:

- Dizziness.
- Severe itching or tingling of the skin.
- Formation of multiple, small, or hard blisters.
- Coughing up blood.
- Shock (which could result in death).

Self-Aid of Toxin Poisoning

- 19-92. Upon recognition of an attack employing toxins, you must immediately take the following actions:
- Stop breathing, put on your protective mask with hood, and then resume breathing. Next, put on your protective ensemble.
 - Should itching of the face become severe or unbearable, quickly—
 - Loosen the cap in your canteen or valve on your water source.
 - Take and hold a deep breath and lift your mask.
 - While holding your breath, close your eyes and flush your face with generous amounts of water.

CAUTION

DO NOT rub or scratch your eyes. Try not to let the water run onto your clothing or protective over garment.

- Put your protective mask back on, seat it properly, clear it, and check it for a seal; then resume breathing.
- Decontaminate your skin by bathing with soap and water as soon as the mission permits.
- Change clothing and decontaminate your protective mask using soap and water. Replace the filters if directed.

19-93. If vomiting occurs, the mask should be lifted momentarily and drained. While doing this, keep your eyes closed, and hold your breath. The mask is then replaced, cleared, and sealed.

Medical Assistance for Toxin Poisoning

19-94. If you think you might have been exposed to harmful substances, it's important to get medical help right away. Seeing a doctor quickly can ensure you get the care you need.

FIRST AID IN A RADIOLOGICAL AND NUCLEAR ENVIRONMENT

19-95. Three types of injuries may result from nuclear detonation. These are thermal, blast, and radiation injuries. Many times, the casualty will have a combination of these types of injuries. First aid for thermal and blast injuries is provided based on observable injuries, such as burns, hemorrhage, or fractures.

19-96. The signs and symptoms of radiation illness in the initial phase include the rapid onset of nausea, vomiting, and malaise (tiredness) with first aid measures focused on those observable symptoms and decontamination. If radioactive iodine is released, Service members may be directed to take potassium iodide from the initial issue medical CBRN defense materiel. Potassium iodide is only effective if taken within four hours before or after exposure to radiation from a nuclear or radiological incident. For additional information on first aid in a radiological and nuclear environment see paragraph 1-85 on page 16 or refer to DODM 3150.8, JP 3-11, or ATP 4-02.83.

Chapter 20

Combat and Operational Stress Control

OVERVIEW OF COMBAT AND OPERATIONAL STRESS CONTROL

20-1. Psychological first aid is just as crucial as physical first aid and is a familiar concept to many. Think back to when a simple comforting gesture from a parent soothed you as a child, or how supportive conversation with a friend helped you navigate through disappointment or grief. These actions are natural responses to emotional distress, similar to how we instinctively want to help someone who is physically hurt.

20-2. Just as we use bandages to cover wounds, we use supportive words and actions to help heal emotional wounds. Psychological first aid involves assisting those who are emotionally distressed, whether due to physical injuries, illness, or overwhelming stress. Signs of emotional distress, like overexcitement, severe fear, excessive worry, deep depression, or anger, can sometimes be less visible than physical injuries but are equally important to address.

20-3. These emotional reactions are indicators that stress is impacting someone's ability to cope effectively. When these symptoms become more pronounced, the need for psychological support becomes more urgent. Knowing how to offer this support is essential, as it can significantly aid in someone's recovery process, just as much as treating a physical ailment.

PSYCHOLOGICAL FIRST AID

20-4. You must know how to give psychological first aid to be able to help yourself, your team members, and your unit to keep performing the mission. Psychological first aid measures are simple and easy to understand. Your decision of what to do depends upon your ability to observe the Service member and understand their needs. Making the best use of resources requires ingenuity on your part. A stress reaction resulting in poor judgment can cause injury or even death to yourself or others on the battlefield. It can be even more dangerous if other personnel are affected by the judgment of an emotionally upset Service member. If it is detected early enough, the affected Service member stands a good chance of remaining in their unit as an effective member. If it is not detected early and if the Service member becomes more emotionally upset, they may become a threat to themselves and to others.

SITUATIONS REQUIRING PSYCHOLOGICAL FIRST AID

20-5. Psychological first aid (buddy aid) is most needed at the first sign that a Service member cannot perform the mission because of emotional distress. Stress is inevitable in combat, in hostage and terrorist situations, and in civilian disasters such as floods, hurricanes, or industrial accidents. Most emotional reactions to such situations are temporary, and the Service member can still carry on with encouragement. Painful or disruptive symptoms may last for minutes, hours, or days. However, if the stress symptoms are seriously disabling, they may be psychologically contagious and endanger not only the emotionally upset Service member but also the entire unit.

20-6. Sometimes people continue to function well during a disastrous event, but suffer from emotional scars, which impair their job performance or quality of life later. Painful memories and dreams may recur for months and years and still be considered a normal reaction. However, an individual needs treatment if the memories are so painful that the person must avoid all situations which arouse them, becomes socially withdrawn, show symptoms of anxiety, depression, or substance abuse. Experience with police, firefighters, emergency medical technicians, and others who deal with disasters have proved that the routine application of psychological first aid to all the participants, including those who have functioned well, greatly reduces the likelihood of future serious post-traumatic stress disorder(s).

INTERRELATIONSHIP OF PSYCHOLOGICAL AND PHYSICAL FIRST AID

20-7. Psychological first aid should go together with physical first aid. The discovery of a physical injury or cause for an inability to function does not rule out the possibility of a psychological injury (or vice versa). The person suffering from pain, shock, fear of serious injury, or fear of death does not respond well to joking, indifference, or fearful-tearful attention. Fear and anxiety may take as high a toll of on the Service member's strength as does the loss of blood.

GOALS OF PSYCHOLOGICAL FIRST AID

20-8. Psychological First Aid is aimed at supporting Service members in managing immediate stress reactions effectively, ensuring both the well-being of the individual and the operational readiness of the unit. The primary goals of Psychological First Aid are multifaceted, focusing on immediate mental health support, safety, and a swift return to duty.

20-9. The first and foremost goal of Psychological First Aid is to provide empathetic support to Service members experiencing stress reactions. By offering a supportive presence, responders can help individuals navigate their emotional responses effectively. This involves active listening, acknowledging their feelings without judgment, and providing reassurance. This supportive approach is crucial in helping Service members feel understood and less isolated in their experiences.

20-10. Another critical aim of Psychological First Aid is to prevent and control behaviors that could be harmful to the individual or others. This includes identifying signs of distress that might lead to harmful actions and implementing strategies to mitigate these risks. The ability to intervene early and effectively can prevent the escalation of stress reactions into more serious mental health issues, thereby maintaining the safety and cohesion of the unit.

20-11. Finally, Psychological First Aid seeks to return the Service member to duty as efficiently as possible after addressing their immediate psychological needs. This goal underscores the importance of resilience and rapid recovery within military operations, ensuring that personnel are mentally fit to resume their roles. Facilitating a quick return to duty not only helps maintain unit effectiveness but also promotes a sense of normalcy and purpose for the individual, which can be therapeutic in overcoming stress reactions.

RESPECT FOR OTHERS' FEELINGS

20-12. Accept the Service member you are trying to help without censorship or ridicule. Respect their right to their own feelings. Even though your feelings, beliefs, and behavior are different, do not blame or make light of them for the way they feel or act. Your purpose is to help them in this tough situation, not to be their critic. A person does not want to be upset and worried. When they seek help, they need and expect consideration of their fears, not abrupt dismissal, or ridicule.

20-13. Realize that people are the products of a wide variety of factors. All people do not react the same way to the same situations. Everyone has complex needs and motivations, both conscious and unconscious, that are uniquely their own. Often the one thing that finally causes the person to become overloaded by a stressful situation is not the stressor itself, but some other problem.

EMOTIONAL AND PHYSICAL DISABILITY

20-14. Accept emotional disability as being just as real as physical disability. If a Service member's ankle is seriously sprained in a fall, no one expects them to run right away. A Service member's emotions may be temporarily strained by the overwhelming stress of battle or another traumatic incident. Do not demand that they pull themselves together immediately and carry on without a break. Some individuals can pull themselves together immediately, but others cannot. The Service member whose emotional stability has been disrupted has a disability just as real as the Service member who has sprained their ankle. There is an unfortunate tendency in many people to regard as real only what they can see, such as a wound or bleeding. Some people tend to assume that damage involving a person's mind and emotions is just imagined, they are not sick or injured, and that they could overcome their trouble by using their will power.

20-15. The phrases "it is all in your head", "snap out of it", and "get control of yourself" are often used by people who believe they are being helpful. These terms are expressions of hostility because they show lack

of understanding. They only emphasize weakness and inadequacy. The above stated phrases are of no use in psychological first aid.

20-16. Every physically injured person has some emotional reaction to the fact that they are injured. It is normal for an injured person to feel upset. The more severe the injury, the more insecure and fearful they become, especially if the injury is to a body part that is highly valued. For example, an injury to the eyes or the genitals, even though relatively minor, is likely to be extremely upsetting. An injury to some other part of the body may be especially disturbing to an individual for their own reason. For example, an injury to the hand may be a terrifying blow to a surgeon, or an injury to the eye of a pilot.

20-17. An injured Service member can sometimes feel less secure, more anxious, and more afraid not only because of what has happened to them but also what they imagine may happen because of their injury. This fear and insecurity may cause them to be irritable, uncooperative, or unreasonable. As you help the Service member, always keep in mind that such behavior has little or nothing to do with you personally. They need your patience, reassurance, encouragement, and support.

COMBAT AND OPERATIONAL STRESS REACTION

20-18. Stress reaction is a temporary emotional disorder or inability to function, experienced by a previously normal Service member as a reaction to the overwhelming or cumulative stress of combat. Stress reaction gets better with reassurance, rest, physical replenishment, and activities that restore confidence. All Service members are likely to feel stressed under conditions of intense or prolonged stress. They may even become stress reaction casualties and are unable to perform their mission for hours or days. Other combat and operational stress reactions may result in negative behavior, but are not termed stress reaction, as they need treatment that is more intensive. These negative combat and operational stress reactions may result in misconduct and stress behaviors such as drug and alcohol abuse, criminal acts, looting, desertion, and self-inflicted wounds. These harmful combat and operational stress reactions can often be prevented by good psychological first aid. Service members who commit misconduct stress behaviors may require disciplinary action rather than medical treatment.

REACTIONS TO STRESS

20-19. Most Service members react to stressful incidents after the situation has passed. Service members feel some fear. This fear may be greater than they have experienced at any other time, or they may be more aware of their fear. In such a situation, they should not be surprised if they feel shaky or become sweaty, nauseated, or confused. These reactions are normal and are not a cause for concern. However, some reactions, either short- or long-term, will cause problems if left unchecked.

Emotional Reactions

20-20. The most obvious combat and operational stress reaction is inefficient performance. This can be demonstrated by—

- Slow thinking (or reaction time).
- Difficulty recognizing priorities and seeing what needs to be done.
- Difficulty getting started.
- Indecisiveness and having trouble focusing.
- Tendency to do familiar tasks and be preoccupied with familiar details. (This can reach the point in which the person is very passive, sitting or wandering, and not knowing what to do).

20-21. A less common reaction may be uncontrolled emotional outbursts; this can be demonstrated by crying, screaming, or laughing. Some Service members will react in the opposite way. They can show signs of being very withdrawn, silent, and try to isolate themselves from everyone. These Service members should be encouraged to remain with their assigned unit. Uncontrolled reactions may appear by themselves or in any combination. The person may be crying uncontrollably one minute and then laughing the next. In this state, the person is restless and cannot keep still. They may run about, apparently without purpose. Inside, they feel a great rage or fear and their physical acts may show this. In their anger, they may indiscriminately strike out at others.

Loss of Adaptability

20-22. In a desperate attempt to get away from the danger, which has overwhelmed them, a Service member may panic and become confused. Their mental ability may be so impaired they cannot think clearly or even follow simple commands. Their judgment may be faulty, and they may not be aware of their actions, such as standing up in their fighting position during an attack.

20-23. In other cases, overwhelming stress may produce symptoms that are often associated with head injuries. For example, the Service member may appear dazed or be found wandering around aimlessly. They may appear confused and disoriented and may seem to have a complete or partial loss of memory. In such cases, especially when no eyewitnesses can provide evidence that the Service member has NOT suffered a head injury, it is necessary for them to be rapidly evacuated. DO NOT allow the Service members to expose themselves to further personal danger until the cause of the problem has been determined.

Sleep Disturbance and Repetition of Dreams

20-24. A person who has been overwhelmed by stress, often has difficulty sleeping. The Service member may experience nightmares related to the stressors. Remember that nightmares, in themselves, are not considered abnormal when they occur soon after a period of intensive stress. As time passes, the nightmares usually become less frequent and less intense. In extreme cases, a Service member, even when awake, may think repeatedly of the incident, feel as though it is happening again, and act out parts of their stress repeatedly.

20-25. For some individuals, this repetitious re-experiencing of the stressful event may be necessary for eventual recovery; therefore, it should not be discouraged or viewed as abnormal. For the person re-experiencing the event, such reaction may be disruptive. The Service member needs to be encouraged to ventilate about the incident. Ventilation is a technique where the Service member is given the opportunity to talk extensively and often repetitiously about the experience.

Severe Stress or Stress Reaction

20-26. You do not need specialized training to recognize severe stress or stress reaction that will cause problems for the Service member, the unit, or the mission. Reactions that are less severe, however, are more difficult to detect. To determine whether a person needs help, you must observe them to see whether they are doing something meaningful, performing their duties, taking care of themselves, behaving in an unusual fashion, or acting out of character.

APPLICATION OF PSYCHOLOGICAL FIRST AID

20-27. The emotionally disturbed Service member has built a barrier against fear. They do this for their own protection, although they are probably not aware that they are doing it. If they find that they do not have to be afraid and that there are normal, understandable things about them, they will feel safer in dropping this barrier. Persistent efforts to make them realize that you want to understand them will be reassuring, especially if you remain calm. Nothing can cause an emotionally disturbed person to become even more fearful than feeling that others are afraid of them. Try to remain calm. Familiar things, such as a cup of coffee, the use of their name, attention to a minor wound, being given a simple job to do, or the sight of familiar people and activities, will add to their ability to overcome their fear. They may not respond well if you get excited, angry, or abrupt.

Ventilation

20-28. After the Service member becomes calmer, they are likely to have dreams about the stressful event. They also may think about it when they are awake or even repeat their personal reaction to the event. One benefit of this natural pattern is that it helps them master the stress by going over it just as one master the initial fear of parachuting from an aircraft by doing it repeatedly. Eventually, it is difficult to remember how frightening the event was initially. In giving first aid to the emotionally disturbed Service members, you should let them follow this natural pattern. Encourage them to talk. Be a good listener. Let them tell you, in their own words, what happened. If home front problems or worries have contributed to stress, it will help them to talk about these problems. Your patient listening will prove to the Service member that you are

interested in them, and by describing their personal problem, they can work at mastering their fear. If they become overwhelmed, suggest a cup of coffee or a break. Whatever you do, assure them that you will listen again as soon as they are ready. Try to help put the Service member's perception of what happened back into realistic perspective; but DO NOT argue about it.

Activity

20-29. A person who is emotionally disturbed as the result of a combat action is a casualty of anxiety and fear. The person may feel as though they are disabled because they have become temporarily overwhelmed by their anxiety. A good way to control fear is through activity. For example, almost all Service members experience a considerable sense of anxiety and fear while they are poised, awaiting the opening of a big offensive; but this is normally relieved, and they feel better once they begin to move into action. They take pride in effective performance and pleasure in knowing that they are a good Service member, perhaps being completely unaware that overcoming their initial fear was their first major accomplishment.

20-30. Useful activity can be very beneficial to Service members who are emotionally disturbed and not physically incapacitated. After you help a Service member get over their initial fear, help them to regain some self-confidence. Help them realize their job continues by finding them something useful to do. Encourage them to be active. Get them to help load trucks, clean up debris, or dig fighting positions. If possible, get them back to their usual duty. Seek out their strong points and help them apply these attributes. Avoid having them just sit around. You may have to provide directions by telling them what to do and where to do it. The instructions should be clear, simple, and should be repeated. A person who has panicked is likely to argue. Respect their feelings, but point out more immediate, obtainable, and demanding needs. Channel their excessive energy and, above all, DO NOT argue. If you cannot get them to become interested in doing profitable work, it may be necessary to enlist aid in controlling their over-activity before it results in more panic. Prevent the spread of such infectious feelings by restraining and segregating if necessary.

20-31. Involvement in an activity helps the Service member in three ways; they—

- Place their thoughts on something more productive.
- Have an outlet for their excessive tensions.
- Prove to them that they can do something useful.

Rest

20-32. There are times, particularly in combat, when physical exhaustion is a principal cause for emotional reactions. A unit sleep plan should be established and implemented. When possible, Service members should be given a safe and relatively comfortable area in which to sleep. Examples would be an area away from heavy traffic, noise, congestion, a place that is clean, dry, and protected from environmental conditions. The more uninterrupted sleep a Service member gets the better they will be able to function in the tactical environment.

Hygiene

20-33. Field hygiene is an important ingredient in a Service member's morale. A Service member who is lacking personal hygiene will not function as well as others who have had the opportunity to bathe and put on clean, dry clothing. During combat, unit leaders should stress the importance of personal hygiene. Good personal hygiene not only improves morale, but it is also a preventive measure against disease and disease and nonbattle injury.

Group Activity

20-34. You have probably already noticed that a person who works, faces danger, and handles serious problems better if they are a member of a closely-knit group. Each Service member in the team supports the other team members. Esprit de corps is built because the Service members have the same interests, goals, and mission. As a result, they are more productive; and are less worried because everyone is involved. This is one of the most effective tools you have in your psychological first aid bag. Getting the Service member back into the team or squad activities will re-establish their sense of belonging and security. This will go far toward making them a useful member of the unit.

REACTIONS AND LIMITATIONS

20-35. Up to this point, the discussion has been primarily about the feelings of the emotionally distressed Service member. What about your feelings towards them? Whatever the situation, you will have emotional reactions (conscious or unconscious) toward this Service member. Your reactions can either help or hinder your ability to help them. When you are tired or worried, you may very easily become impatient with them if they are unusually slow or exaggerated. You may even feel resentful toward them. Physical wounds can be seen and easily accepted. Emotional reactions are more difficult to accept as injuries. On the other hand, you tend to be overly sympathetic. Excessive sympathy for an incapacitated person can be as harmful as negative feelings in your relationship with them. They need strong help, but not your sorrow. To overwhelm them with pity may make them feel even more inadequate. You must expect your buddy to recover, to be able to return to duty, and to become a useful Service member again. This expectation should be displayed in your behavior and attitude as well as in what you say. If they can see your calmness, confidence, and competence, they will be reassured and will feel a sense of greater security. For more information on stress reactions see tables 20-1, table 20-2 on page 215 and table 20-3 on page 216.

Table 20-1. Mild Stress Reaction

<i>PHYSICAL SIGNS*</i>	<i>EMOTIONAL SIGNS*</i>
1. Trembling, tearful. 2. Jumpiness, nervousness. 3. Cold sweat, dry mouth. 4. Pounding heart, dizziness. 5. Insomnia, nightmares. 6. Nausea, vomiting, and diarrhea. 7. Fatigue. 8. Thousand-yard stare. 9. Difficulty thinking, speaking, and communicating.	1. Anxiety, indecisiveness. 2. Irritable, complaining. 3. Forgetful, unable to concentrate. 4. Easily startled by noise, movement. 5. Grief, tearful. 6. Anger, beginning to lose confidence in self and unit.
<i>SELF- AND BUDDY AID</i>	
1. Continue mission performance and focus on the immediate mission. 2. Expect Service members to perform the assigned duties. 3. Always remain calm; be directive, and in control. 4. Let Service member know their reaction is normal, and that there is nothing seriously wrong with them. 5. Keep the Service member informed of the situation, objectives, expectations, and support. Control rumors. 6. Build Service members' confidence, talk about succeeding. 7. Keep Service member productive (when not resting) through recreational activities, equipment, maintenance, and resupply. 8. Ensure Service members maintain good personal hygiene. 9. Ensure Service member eats, drinks, and sleeps as soon as possible. 10. Let the Service member talk about their feelings. Do not "put down" their feelings of grief or worry. Give practical advice and put emotions into perspective.	
* Most or all of these signs are present in mild stress reaction. They can be present in any normal Service member in combat, yet they can still do their job.	

Table 20-2. Serious Stress Reaction

<i>PHYSICAL SIGNS *</i>	<i>EMOTIONAL SIGNS *</i>
1. Constantly moves around. 2. Flinching or ducking at sudden sounds. 3. Shaking, trembling (whole body or arms). 4. Cannot use parts of the body, no physical reason (hands, arms, or legs). 5. Cannot see, hear, or feel (partial or complete loss of senses). 6. Physical exhaustion, crying. 7. Freezing under fire, or total immobility. 8. Vacant stares, staggers, and sways when they stand. 9. Panic running under fire.	1. Rapid and/or inappropriate talking. 2. Argumentative, reckless movements, or actions. 3. Inattentive to personal hygiene. 4. Indifferent to danger. 5. Memory loss. 6. Severe stuttering, mumbling, or cannot speak at all. 7. Insomnia, nightmares. 8. Seeing or hearing things that do not exist. 9. Rapid emotional shifts. 10. Social withdrawal. 11. Apathetic. 12. Hysterical outbursts. 13. Frantic or strange behavior.
<i>TREATMENT PROCEDURES **</i>	
1. If a Service member's behavior endangers the mission, self, or others, do whatever is necessary to control them. 2. If the Service member is upset, calmly talk them into cooperating. 3. If concerned about the Service member's reliability: • Unload their weapon. • Take weapons if seriously concerned. • Physically restrain them only when necessary for safety or transportation. 4. Reassure everyone that the signs are probably just stress reaction and will quickly improve. 5. If stress reaction signs continue: • Get the Service member to a safer place. • Do not leave the Service member alone, keep someone they know with them. • Notify the senior noncommissioned officer or officer. • Have the Service member examined by medical personnel. 6. Give the Service member easy tasks to do when they are not sleeping, eating, or resting. 7. Assure the Service member they will return to full duty in 24 hours; and return them to normal duty as soon as they are ready.	
* These signs are present in addition to the signs of mild stress reaction.	
** Do these procedures in addition to the self- and buddy aid care.	

Table 20-3. Preventive Measures to Combat and Operational Stress Reaction

<i>Preventive Measures</i>
1. Welcome new Service members into your team and get to know them quickly. If you are new, be active in making friends.
2. Be physically fit (strength, endurance, and agility).
3. Know and practice lifesaving self- and buddy aid.
4. Practice rapid relaxation techniques.
5. Help each other out when things are tough at home or in the unit.
6. Keep informed; ask your leader questions, ignore rumors.
7. Work together to give everyone food, water, shelter, hygiene, and sanitation.
8. Sleep when the mission and safety permit; let everyone get time to sleep. Follow these sleep rules:
• Sleep only in safe places and by standard operating procedure.
• If possible, sleep six to nine hours per day.
• Try to get at least four hours of sleep per day.
• Get good sleep before going on sustained operations.
• Take a nap when you can but allow time to wake up fully.
• Catch up on sleep after going without.

SUPPORTING AT-RISK SERVICE MEMBERS

20-36. Taking care of fellow Service members is a fundamental duty within the DOD, emphasizing the critical role of teamwork and the inherent value of each individual. Essential to this dynamic is the implementation of the battle-buddy concept, particularly when addressing COSR. Vigilance, appropriate intervention, and thorough follow-through are key elements in fostering a supportive environment.

20-37. The optimal initial response to Service member stress involves the caring and observant presence of battle buddies and leaders. The DOD supports this approach through a robust framework that extends from individual Service members to the highest levels of leadership, facilitating COSC. It is imperative that Service members recognize signs of distress among their peers, seek help, and provide consistent support.

20-38. In the context of handling COSRs, the "ACE" acronym—standing for Acknowledge, Care, and Escort—serves as a concise and memorable strategy. This method is crucial for both Service members and leaders in managing stress reactions effectively:

- **Acknowledge:** The first critical step is recognizing potential stress reactions in colleagues. This requires alertness to behavioral and emotional shifts that may suggest a Service member is struggling. Trusting one's instincts and taking any signs of distress seriously are essential in this initial phase.
- **Care:** After recognizing a colleague's distress, providing immediate support is vital. This support can include offering reassurance, listening with empathy, and creating a safe, non-judgmental environment for the individual to share their experiences. It's important to affirm that their reactions are normal responses to the stressful situations they have encountered.
- **Escort:** The final step involves guiding the affected Service member to suitable professional resources. This could mean directing them to a mental health expert, a chaplain, or another unit leader who can offer further support and advice. Ensuring the individual receives appropriate help is crucial and helps to mitigate any stigma associated with seeking mental health support.

20-39. The Acknowledge, Care, and Escort framework underscores the importance of timely intervention and the need to connect distressed Service members with suitable resources. It offers a practical guide for those who may lack extensive mental health training but are committed to assisting their colleagues.

20-40. The Acknowledge, Care, and Escort approach is a preliminary measure and does not substitute for professional mental health evaluation and treatment. Involving mental health professionals is crucial for a comprehensive assessment and care of Service members experiencing COSRs, ensuring they receive the full spectrum of necessary support.

Appendix A

First Aid Case and Kits, Authorized Medical Allowance List

JOINT FIRST AID KIT

A-1. This Service JFAK contains core components recommended by the CoTCCC and agreed upon by all Services. Table A-1 below provides a current list of equipment in the JFAK. These items are suitable for requisition as a JFAK for any Service or Army IFAK. Each Service maintains its individual JFAK using a national stock number that includes a bag which meets Service specific requirements and may contain additional items based on Service needs. The JFAK meets the USAF requirement and includes:

Table A-1. Joint First Aid Kit contents

<i>National Stock Number</i>	<i>Item</i>	<i>Number Per Kit</i>
6510-01-624-0840	Vented Chest Seal	1
6510-01-503-2117	Compressed Gauze	2
6510-01-598-8418	Bandage, Gauze	2
6515-01-632-0059	Gloves	2
6515-01-529-1187	Nasopharyngeal Airway/Lubricating Jelly Combo*	1
6515-01-630-5551	Eye Shield	2
5110-01-578-4889	Safety Cutting Hook	1
6515-01-521-7976	Tourniquet	2
Part number 35124	Mini marker	1
6510-01-549-0927	Combat Medic Tape	1
6510-01-562-3325	Hemostatic Dressing	2
6510-01-587-6579	Cravat, Triangular Dressing (Sterile)	1
	DD Form 1380	1
8465-01-633-7981	JFAK Bag	1
8465-01-633-9717	Tourniquet Pouch	2
Legend: DD department of defense JFAK joint first aid kit pr* pair Not provided in the Marines JFAK		

A-2. The Defense Medical Materiel Program Office is coordinating with the Joint Services on the transition of the IFAK to the JFAK. The U.S. Navy is currently the only Service utilizing needle chest decompression from the JFAK.

FIRST AID CASE AND KITS, AUTHORIZED MEDICAL ALLOWANCE LIST FOR THE UNITED STATES ARMY

A-3. The JFAK, described in paragraph A-1, does not meet all the U.S. Army requirements. The U.S. Army has identified Services specific first aid cases and kits consisting of the Service specific IFAK and the CLS Bag.

ARMY INDIVIDUAL FIRST AID KIT

A-4. The IFAK Generation I is issued to every Soldier at a Central Issue Facility for administering immediate self-aid and buddy aid. Every Soldier must be trained in the use of all contents of the IFAK. The IFAK does not contain a needle or Angio catheter for NDC. See figure A-1 on page 218 and table A-2 on page 218.



Figure A-1. Individual First Aid Kit, Generation I

Table A-2. Individual First Aid Kit Generation I Contents

<i>National Stock Number</i>	<i>Nomenclature</i>	<i>Number Per Kit</i>
8465-01-531-3647	100 Round Squad Automatic Weapon (SAW) Utility Pouch, MOLLE II	1
6515-01-521-7976	Combat Application Tourniquet	1
6510-01-460-0849	Bandage Kit, Elastic	1
6510-01-503-2117	Bandage, Gauze, 4 1/2IN, 100	1
6510-00-926-8883	Surgical Adhesive Tape, 2IN, 6	1
6515-01-180-0467	Nasopharyngeal Airway	1
6515-01-519-9161	Patient Exam Gloves, 100	4
6545-01-586-7691	Contents Kit, IFAK Resupply	1
6545-01-531-3147	Insert (Folding Panels with Cord)	1
6510-01-562-3325	Dressing, Combat Gauze	1
Legend: IFAK individual first aid kit MOLLE modular lightweight load-carrying equipment		

A-5. This IFAK includes two Combat Application TQs, which are stored in pouches that attach to the kit bag. The Generation II IFAK also contains a vented chest seal and eye shield. The Generation II IFAK is like the JFAK. Designers added DD Form 1380, June 2014, a permanent marker, and strap cutter. The kit weighs one pound and contains expendable medical items. See figure A-2 on page 219 and table A-3 on page 219.



Figure A-2. Individual First Aid Kit, Generation II

Table A-3. Individual First Aid Kit Generation II Contents

<i>National Stock Number</i>	<i>Nomenclature</i>	<i>Number Per Kit</i>
6545-01-584-1582	U.S. Army IFAK	1
6515-01-521-7976	Combat Application Tourniquet	2
6510-01-492-2275	Bandage Kit, Elastic	1
6510-01-503-2117	Bandage, Gauze, 4 ½IN, 100s	1
6510-00-926-8883	Surgical Adhesive Tape, 2IN, 6s	1
6515-01-180-0467	Nasopharyngeal Airway	1
6515-01-519-9161	Patient Exam Gloves, 100s	4
6510-01-562-3325	Dressing, Combat Gauze	1
4240-01-570-0319	Strap Cutter, Rescue	1
6510-01-549-0939	Bolin Chest Seal	1
6515-01-449-1016	Eye Shield, Fox	1
7520-00-312-6124	Marker, Tube Type	1
Legend:		
IFAK	individual first aid kit	s singles
IN	inches	U.S. United States

A-6. Due to constant changes, see the United States Army Medical Materiel Agency website for more information and updates on the IFAK, IFAK II, and their contents.

ARMY COMBAT LIFESAVER BAG

A-7. The U.S. Army CLS carries a small aid bag, also classified as a medical equipment set (MES). It contains supplies for controlling bleeding, relieving tension pneumothorax, and performing other procedures. The CLS bag authorization is contained in CTA 8-100, with unit basis of issue plan of one per: squad, crew, team as determined by the unit commander. For a current listing of the supplies found in the MES refer to the Army Medical Logistics Command website on the Medical Materiel Information Portal tab, which is managed by United States Army Medical Materiel Agency. The prescribed packing list can be located under unit assemblage code (UAC) 245# nomenclature of MES CLS. The # in the UAC code is a place holder as the last character is changed as the assemblage list is updated. At the time of publication, the current version of the MES CLS is UAC 245C. Table A-4 on page 220 is the listing of supplies for the MES CLS with UAC

245C. The most recent and active assemblage list should be used when inventorying and packing the MES CLS.

Table A-4. Combat lifesaver supplies for the United States Army

<i>Nomenclature</i>	<i>Unit of Issue</i>	<i>Quantity</i>
Adhesive Tape Surgical	PG	1
Airway, Nasopharyngeal	PG	2
Atropine Injection	EA	5
Bag, Combat Casualty Care	EA	1
Bandage, Elastic	PG	2
Bandage, Gauze	RO	2
Bandage, Emergency Trauma	EA	2
Bandage, Hemostatic	RO	2
Bandage, Triangular Cravat	EA	3
Bandage Kit	KT	1
Blanket, Heating	PG	1
Blanket, Survival	EA	1
Diazepam Injection	EA	5
Dressing, Chest Seal	PG	1
Dressing, Chest Seal	PG	1
Dressing, First Aid	EA	3
DD Form 1380	EA	5
Gloves, Pat Exam 25pr	PG	4
Gloves, Patient Exam	PG	4
Leash Shears, Trauma	EA	2
Light, Tactical Exam	EA	1
Marker, Laboratory	PG	1
Needle Decompression	EA	2
Pad, Isopropyl Alcohol	PG	1
Scissors, Bandage	EA	1
Shield Eye, Surgical	PG	4
Splint, Universal	EA	1
Tourniquet, Nonpneumatic	EA	2
Legend: DD department of defense pr pair EA each RO roll KT kit PG package		

FIRST AID CASE AND KITS, AUTHORIZED MEDICAL ALLOWANCE LIST FOR FIRST AID CASE AND KITS, AUTHORIZED MEDICAL ALLOWANCE LIST FOR UNITED STATES MARINE CORPS

A-8. The JFAK, described in paragraph A-1, does not meet all the USMC requirements. The USMC has identified specific first aid cases and kits consisting of the Service specific IFAK and the Service specific CLS bag. The JFAK, described in paragraph A-1, does not meet all the USMC requirements. The USMC has identified specific first aid cases and kits consisting of the service specific IFAK and the service specific CLS bag

UNITED STATES MARINE CORPS INDIVIDUAL FIRST AID KIT

A-9. The USMC IFAK provides individual Marines and Soldiers with the ability to perform self-aid for life-threatening arterial bleeding and common non-life-threatening injuries. It also provides the capability to treat non-potable water for personal consumption.

A-10. The IFAK is made up of a trauma kit for airway and bleeding injuries and an accessory kit containing supplies for non-life-threatening injuries and potable water disinfection.

A-11. Component inventories are provided with each IFAK. Control, reliability, readiness, issue, and recovery of the IFAK for the Marine expeditionary forces are the responsibility of the supporting individual issue facility. Control, reliability, readiness, issues, and recovery of the IFAK for units not supported by individual issue facility are the responsibility of the using unit.

A-12. United States Marine Corps Individual First Aid Kit. The USMC IFAK provides individual Marines and Sailors the ability to perform self-aid for life-threatening arterial bleeding and common non-life-threatening injuries. It also provides the capability to treat non-potable water for personal consumption.

A-13. The IFAK is made up of a trauma kit for airway and bleeding injuries and an accessory kit containing supplies for non-life-threatening injuries and potable water disinfection.

A-14. Component inventories are provided with each IFAK. Control, reliability, readiness, issue, and recovery of the IFAK for the Marine expeditionary forces are the responsibility of the supporting individual issue facility. Control, reliability, readiness, issues, and recovery of the IFAK for units not supported by individual issue facility are the responsibility of the using unit.

UNITED STATES MARINE CORPS COMBAT LIFESAVER BAG

A-15. The U.S. Marine Corps CLS carries a small aid bag. It contains supplies for controlling bleeding, relieving tension pneumothorax, and performing other procedures. Table A-5 is the listing of supplies for the MES CLS with UAC 245C. The most recent and active assemblage list should be used when inventorying and packing the MES CLS.

A-16. Table A-5. Combat lifesaver supplies for the USMC. The U.S. Marine Corps CLS carries a small aid bag. It contains supplies for controlling bleeding, relieving tension pneumothorax, and performing other procedures. Table A-5 on page 222 is the listing of supplies for the MES CLS with UAC 245C. The most recent and active list should be used when inventorying and packing the MES CLS.

Table A-5. Combat lifesaver supplies for the United States Marine Corps

<i>Nomenclature</i>	<i>Unit of Issue</i>	<i>Quantity</i>	
Alcohol Pad 200s	EA	8	
Bandage, Gauze	RO	8	
Bear Claw Glove Kit Large 25pr	PR	3	
Benchmade Strap Cutter	EA	1	
Blast Dressing	EA	1	
Burn Dressing Shaped into a 45x45x63 Triangular Bandage Sterilized	EA	3	
Tourniquet	EA	3	
Combat Eye Shield	EA	2	
Combat Gauze XL Hemostatic 4x4	EA	3	
Combat Life Saver Bag	EA	1	
Combat Medical Tape 50	EA	1	
Department of Defense Form 1380	EA	5	
Decompression Needle 14GA, Decompression Needle 14GA	EA	4	
Dressing Burn Burntec Hydrogel 4IN x 16IN	EA	2	
Dressing Kit, Chest Wound	EA	2	
H and H 30FR Nasopharyngeal Airway, Plastic, 30FR, Beveled Tip, 155 MM Long, Single-Use	EA	2	
Compression Emergency Trauma Dressing 8x10 Flat	EA	3	
SAM Splint	EA	1	
Surgi lube Packets 5 Gram 144s	EA	2	
Trauma Shears	EA	1	
Combat Lifesaver Trauma Bag-Coyote	EA	1	
Legend:			
EA	each	pr	pair
FR	French	S	singles
GA	gauge	SAM	Structural Aluminum Malleable
IN	inch	XL	extra large
MM	millimeter		

Appendix B

Rescue Equipment

WARRIOR AID AND LITTER KIT

B-1. The warrior aid and litter kit (WALK) (figure B-1) allows first aid treatment and immediate evacuation of one patient in a CASEVAC vehicle and treatment of multiple trauma patients at the POI. The total weight of the WALK can weigh approximately 29 pounds, 15 ounces. The WALK offers an option for a vehicle life-saving kit.



Figure B-1. Warrior aid and litter kit

B-2. The system holds a quad-fold litter (see figure B-2 on page 225) and a pocket system containing medical supplies. The WALK contains a hypothermia prevention kit to help stabilize patients during transport. The WALK also includes the following:

- Five pairs of nitrile trauma gloves.
- Two NPAs, size 28 French with lubricant.
- One casualty equipment bag.
- Two CoTCCC approved occlusive chest seals.
- Two combat application TQs.
- Six trauma dressings, six inches.
- Four rolls of rolled gauze, 4.5 inches by 4.1 yards.

- One abdominal emergency trauma dressing.
- Two moldable fracture splints.
- One pair of trauma shears, 7.25 inches.
- One roll surgical tape, two inches.
- One roll multipurpose tape.
- Six polycarbonate eye shields.
- Two TCCC cards.
- One aviation panel (Recognition, Orange).
- Four tie-down straps (universal litter).



Figure B-2. Quad-fold litter

B-3. The WALK fits into any standard or nonstandard CASEVAC asset and is useful as part of the prepositioning of medical supplies or cache on a forward operating base, CCP, or at a patrol base.

POLYMER FLEXIBLE LITTER SYSTEM

B-4. The CoTCCC approved polymer flexible litter system (figure B-3 on page 226) uses a sled design to allow the evacuator to continue returning fire as the casualty is evacuated. It is used for confined space, high angle, or technical rescue, and it provides casualty protection while allowing the aid and litter teams to have their hands free for weapons and security.

B-5. It is equipped for horizontal hoisting by helicopter or vertical hoisting in caves or industrial confined spaces. When the patient is packaged, the litter becomes rigid. It is rolled for storage in a backpack, which is included with the system. The litter system weighs approximately 17 pounds.



Figure B-3. Casualty evacuation using the polymer flexible litter system

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Glossary

SECTION I – ACRONYMS AND ABBREVIATIONS

AAS	advanced anticonvulsant system
ABCANZ	American, British, Canadian, Australian, and New Zealand
AFMAN	air force manual
AFTTP	air force tactics, techniques, and procedures
AR	army regulation
ATNAA	antidote treatment – nerve treatment nerve agent, auto-injector
ATP	army techniques publication
AVPU	alert, verbal, pain, and unresponsive
BAS	battalion aid station
BVM	bag valve mask
CANA	convulsant antidote nerve agent
CASEVAC	casualty evacuation
CBRN	chemical, biological, radiological, and nuclear
CCP	casualty collection point
CLS	combat lifesaver
CoTCCC	committee on tactical combat casualty care
CPG	clinical practice guideline
CPR	cardiopulmonary resuscitation
CUF	care under fire
CWMP	combat wound medication pack
DA	department of the army
DD	department of defense (form)
DOD	Department of Defense
DODI	Department of Defense Instruction
DODM	Department of Defense Manual
ECAST	exercise collapse associated with sickle cell trait
EMT	emergency medical treatment
FM	field manual
G6PD	glucose-6 phosphate dehydrogenase
HACE	high-altitude cerebral edema
HAPE	high-altitude pulmonary edema
HM	Hospital Corpsman
HSS	health service support
IAW	in accordance with
IFAK	individual first aid kit

JFAK	joint first aid kit
JP	joint publication
JTS	joint trauma system
KIA	killed in action
LZ	landing zone
MARCH	massive hemorrhaging, airway, respiration, circulation, and hypothermia
MCRP	marine corps reference publication
MCTP	marine corps tactical publication
MEDEVAC	medical evacuation
MES	medical equipment set
MG	milligrams
MIST	mechanism of injury, injury type, signs and symptoms, and treatments
MMPM	multimodal patient movement
MTF	medical treatment facility
NAPP	nerve agent pyridostigmine pretreatment
NATO	north atlantic treaty organization
NDC	needle decompression of the chest
NPA	nasopharyngeal airway
NTRP	navy tactical reference publication
PAWS	pain, antibiotics, wounds, splinting
PDD	pressure delivery device
PET	patient evacuation team
POI	point of injury
RSDL	reactive skin decontamination lotion
SCD	sickle cell disease
SCT	sickle cell trait
SOP	standard operating procedure
STANAG	standardization agreement
STP	soldier training publication
TB MED	technical bulletin medical
TC	training circular
TCCC	tactical combat casualty care
TFC	tactical field care
TQ	tourniquet
U.S.	United States
UAC	unit assemblage code
USA	United States Army
USAF	United States Air Force
USCG	United States Coast Guard
USMC	United States Marine Corps
USN	United States Navy

UV	ultraviolet
WALK	warrior aid and litter kit
WP	white phosphorus

SECTION II – TERMS

battalion aid station

The forward-most medically staffed treatment location organic to the maneuver battalion. (ATP 4-02.4).

***care under fire**

The first phase of care in tactical combat casualty care where very limited care is provided for the casualty while the enemy threat is still active. Also known as CUF or care under threat.

casualty

Any person who is lost to the organization by having been declared dead, duty status – whereabouts unknown, excused absence-whereabouts unknown, missing, ill, or injured. (JP 4-02).

casualty collection point

A location that may or may not be staffed, where casualties are assembled for evacuation to a medical treatment facility. (ATP 4-02.2).

casualty evacuation

(Joint) The unregulated movement of casualties aboard any vehicle. Also called CASEVAC. (JP 4-02).

(Army) The movement of casualties aboard nonmedical vehicles or aircraft without en route medical care. (ATP 4-02.13).

***casualty response**

The immediate and organized care, evacuation and reporting of those injured or killed in action.

***combat lifesaver**

(Joint) Nonmedical Department of Defense person who has received additional trauma training and equipment, providing enhanced medical treatment beyond self-aid/buddy aid. (JP 4-02).

(Army) A Soldier trained and equipped to provide enhanced first aid as a secondary mission, known as CLS.

***first aid**

The immediate lifesaving measures performed on casualties or military working animals.

first responder

Is anyone who provides initial and immediate treatment to self or others. (JP 4-02).

medical evacuation

The timely and effective movement of the wounded, injured, or ill to and between medical treatment facilities on dedicated and properly marked medical platforms with en route care provided by medical personnel. Also called MEDEVAC. (ATP 4-02.2).

medical treatment facility

(Joint) A facility established for the purpose of furnishing medical and/or dental care to eligible individuals. Also called MTF. (JP 4-02).

(Army) Any facility established for the purpose of providing medical treatment. (FM 4-02).

patient

(Joint) A sick, injured, wounded, or other person receiving medical care, dental care, or other medical treatment. (JP 4-02).

(Army) A patient is a wounded, ill and injured Soldier who receives medical care or treatment from medically trained personnel. (ATP 4-02.10).

tactical combat casualty care

The deliberate integration of tactics and medicine to treat casualties during combat. Also called TCCC. (JP 4-02).

***tactical evacuation care**

The third phase of care in tactical combat casualty care. Care provided during transportation from the battelefield to a medical treatment facility.

***tactical field care**

The second phase of care in tactical combat casualty care. Care provided once the threat has been neutralized, the scene is safe or the casualty has moved or been moved out of the immediate threat situation. Also known as TFC.

triage

Triage is the system of sorting and prioritizing casualties based on the tactical situation, mission, and available resources (FM 4-02).

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ATP 4-02.11

23 March 2026

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